Service

Workshop Manual

Crafter 2017 ➤

TGE 2017 ➤

Heating, air conditioner

Edition 10.2018

List of Workshop Manual Repair Groups

Repair Group

Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.

Contents

00 – Technical data

1 Safety information

(VRL012134; Edition 10.2018)

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1.1 Safety precautions when working on air conditioning systems

Risk of suffocation and poisoning from refrigerant

Coughing and nausea leading to suffocation and poisoning from refrigerant vapours possible.

– Never inhale refrigerant vapours.

– Only work on the refrigerant circuit and refrigerant tanks in well ventilated areas.

– Never work in or near cellars or other low lying areas.

– Switch on the extraction system.

Risk of freezing injury from refrigerant

When working on the air conditioning system, there is a risk of highly pressurised refrigerant escaping from the system. There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.

– Wear protective goggles.

– Extract refrigerant and open the refrigerant circuit immediately afterwards.

– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

Risk of damage to refrigerant lines

There is a risk of damage to the refrigerant lines due to rupture of the inner foil.

– Never bend refrigerant lines to a radius less than 100 mm.

1.2 Safety measures when working on vehicles with a start/stop system

Risk of injury due to unexpected motor start

If the vehicle's start/stop system is activated, the engine can start unexpectedly. A message in the dash panel insert indicates whether the start/stop system is activated.

– Deactivate start/stop system by switching off the ignition.

1.3 Safety precautions when using testers and measuring instruments during a road test

Risk of injury caused by unsecured testing and measuring instruments

When the front passenger airbag is triggered in an accident, insufficiently secured testing and measuring instruments become dangerous projectiles.

– Secure testing and measuring instruments on the rear seat.

or

– Have a second person operate the test and measuring equipment on the rear seat.

1.4 Safety precautions when working on the cooling system

Danger of scalding by hot coolant

On a warm engine, the cooling system is under high pressure. Danger of scalding by steam and hot coolant.

– Wear protective gloves.

– Wear protective goggles.

– Reduce excess pressure by covering cap of coolant expansion tank with cloths and opening it carefully.

1.5 Safety precautions when handling refrigerants

Risk of suffocation and poisoning from refrigerant

Coughing and nausea leading to suffocation and poisoning from refrigerant vapours possible.

– Never inhale refrigerant vapours.

– Only work on the refrigerant circuit and refrigerant tanks in well ventilated areas.

– Never work in or near cellars or other low lying areas.

– Switch on the extraction system.

Risk of freezing injury from refrigerant

When working on the air conditioning system, there is a risk of highly pressurised refrigerant escaping from the system. There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.

– Wear protective goggles.

– Extract refrigerant and open the refrigerant circuit immediately afterwards.

– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

1.6 Safety precautions when working on vehicles with auxiliary/supplementary heater

Danger of fire and explosion from auxiliary/supplementary heater

In fire and explosion-endangered areas, a spark or the high temperature of the auxiliary/supplementary heater can cause a fire or an explosion. Risk of burns.

– Switch the auxiliary/supplementary heater off in fire and explosion-endangered areas.

Danger of poisoning from exhaust fumes

auxiliary/supplementary heaters produce poisonous exhaust fumes. Risk of poisoning and injuries to respiratory system.

– In closed areas, only switch on the auxiliary/supplementary heated when it is exclusively connected to an exhaust extractor system.

– In closed areas without an exhaust extractor system, switch off the auxiliary/supplementary heater.

Risk of damage when starting the engine

If components of the fuel system or coolant circuit of the auxiliary/supplementary heater are removed or opened, there is a possibility of causing damage to the auxiliary/supplementary heater.

– Never attempt to start the engine when components are removed or opened.

Malfunction caused by air in the fuel supply system

Upon repair work on the fuel tank or the fuel delivery unit, air will be drawn in by the metering pump and conveyed to the auxiliary/

supplementary heater. Air in the fuel supply system may cause malfunction of the auxiliary/supplementary heater.

– Fill fuel take-off pipe with fuel.

Risk of accidents and risk of injury due to activated timer of auxiliary/supplementary heater

If the vehicle’s timer for the auxiliary/supplementary heater is active, the heater may switch on unexpectedly. This poses a risk of poisoning from exhaust gases, a risk of burns caused by hot auxiliary/supplementary heater components as well as a risk of fire and explosions caused by high temperatures.

– Deactivate timer for auxiliary/supplementary heater.

1.7 Safety precautions when working on SCR system

Risk of injury from reducing agent

The reducing agent may cause eye and skin irritations as well as injuries to the respiratory system and poisoning.

– Wear protective goggles.

– Wear protective gloves.

– Wear protective work clothing.

– Make sure that a sufficient amount of fresh air is supplied. In closed areas, switch on the exhaust gas extractor system.

Risk of damage caused by reducing agent

If reducing agent gets onto trim panels and body components, the reducing agent may solidify after a while and thus damage the surface.

– Make sure that no reducing agent gets onto the trim panels or body components.

– If any reducing agent does get onto surfaces, remove it immediately using water and a cloth.

Risk of injury from reducing agent

The reducing agent may cause eye and skin irritations as well as injuries to the respiratory system and poisoning.

– Wear protective goggles.

– Wear protective gloves.

– Wear protective work clothing.

– Make sure that a sufficient amount of fresh air is supplied. In closed areas, switch on the exhaust gas extractor system.

Risk of damage caused by reducing agent

If reducing agent gets onto trim panels and body components, the reducing agent may solidify after a while and thus damage the surface.

– Make sure that no reducing agent gets onto the trim panels or body components.

– If any reducing agent does get onto surfaces, remove it immediately using water and a cloth.

When removing and installing components of the SCR system, note the following:

♦ The reducing agent tank must be empty when working on the SCR system. Information on when the reducing agent tank has to be emptied can be obtained from the respective description of work. Drain reducing agent tank ⇒ Rep. gr. 26 ; SCR system (selective catalytic reduction); Draining reducing agent tank .

Automatic return of reducing agent

• After the ignition has been switched off, the reducing agent is returned from the metering line leading to the reducing agent injector - N474- back to the reducing agent tank.

• Before any work is done in this area, it is necessary to wait until all the reducing agent has been returned to the tank; this can take up to 10 minutes after the ignition is switched off.

• Similarly, it is not permissible to disconnect the battery until all the reducing agent has been returned to the tank, i.e. 10 minutes after the ignition is switched off ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and connecting battery .

2 General information

⇒ “2.1 Type plates”, page 6

⇒ “2.2 Notes concerning odours in air conditioned vehicles”, page 7

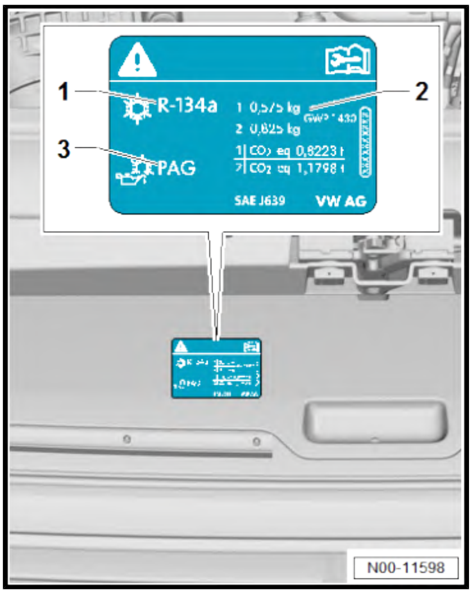
2.1 Type plates

⇒ “2.1.1 Type plate, refrigerant R134a”, page 6

⇒ “2.1.2 Type plate, refrigerant R134a, rest of world”, page 6

⇒ “2.1.3 Type plate, refrigerant R1234yf”, page 7

2.1.1 Type plate, refrigerant R134a

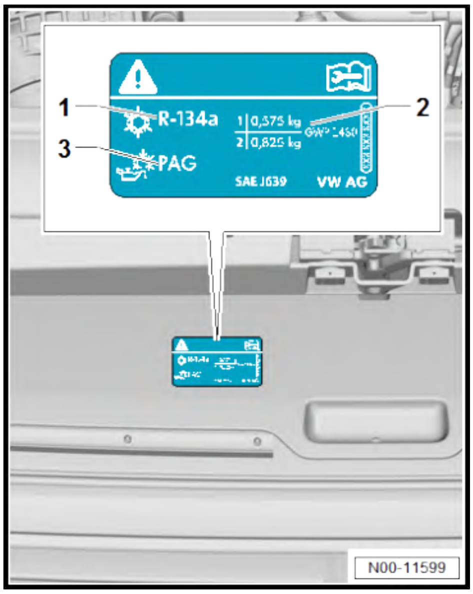


1 - Name of refrigerant

2 - Capacity, refrigerant ⇒ page 13

3 - Designation of refrigerant oil ⇒ page 13

2.1.2 Type plate, refrigerant R134a, rest of world



1 - Name of refrigerant

2 - Capacity, refrigerant ⇒ page 13

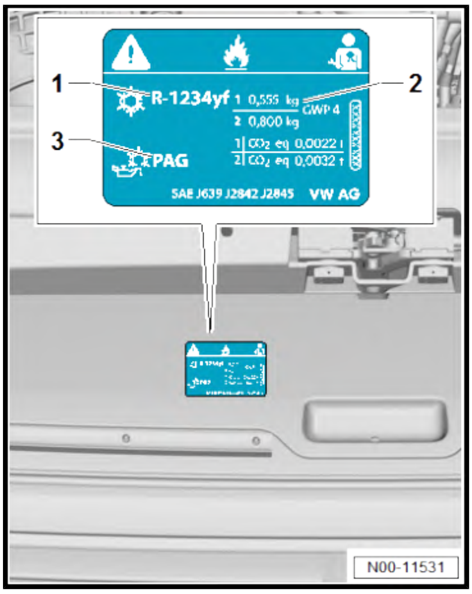
3 - Designation of refrigerant oil ⇒ page 13

2.1.3 Type plate, refrigerant R1234yf

1 - Name of refrigerant

2 - Capacity, refrigerant ⇒ page 13

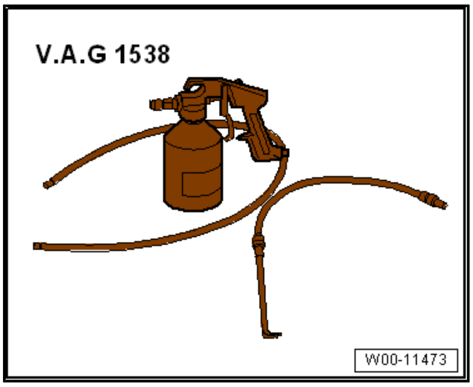
3 - Designation of refrigerant oil ⇒ page 14



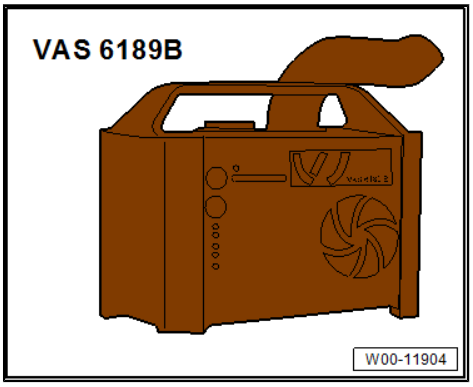
2.2 Notes concerning odours in air conditioned vehicles

Special tools and workshop equipment required

- ♦ Suction feed spray gun - V.A.G 1538- together with probe for evaporator cleaning, straight - V.A.G 1538/5- or probe for evaporator cleaning, curve - V.A.G 1538/6-



- ♦ Ultrasonic air conditioner cleaning unit - VAS 6189B-



Should the ventilation emit unpleasant odours, the evaporator should be cleaned.

For cleaning the evaporator, ultrasonic A/C cleaner - VAS 6189B- or suction feed spray gun - V.A.G 1538- together with probe for evaporator cleaning, straight - V.A.G 1538/5- or probe for evaporator cleaning, curved - V.A.G 1538/6- have been tested and approved.

Instructions on cleaning the evaporator are supplied with the equipment.

3 Repair instructions

3.1 Rules for cleanliness

Even small amounts of contamination/soiling can lead to defects. Therefore, observe the following rules for cleanliness when working on the air conditioning system:

- Seal open pipes and connections immediately with clean plugs, for example from the engine bung set - VAS 6122- .
- Place removed parts on a clean surface and cover them over. Use only lint-free cloths.
- If repair work cannot be performed immediately, carefully cover or seal components.
- Install only clean parts; do not remove new parts from packaging until immediately before installing. Do not use parts that have been stored outside their packaging (e.g. in tool boxes).
- If system is open, do not work with compressed air.
- Protect disconnected electrical connections from dirt and water and only reconnect in dry condition.

3.2 General information

- The engine control unit has a self-diagnosis capability. Before carrying out repairs and for fault finding, first read event memory.
- For trouble-free operation of electrical components, a voltage of at least 11.5 volts is necessary.
- Do not use sealants containing silicone. Particles of silicone drawn into the engine will not be burnt in the engine and damage the Lambda probe.
- Vehicles are fitted with a crash fuel shut-off circuit. It reduces the danger of a fire in a crash as the fuel pump is switched off via the fuel pump relay.
- The system also improves the starting characteristics of the engine. When the driver door is opened, the fuel pump is activated for 2 seconds to develop pressure in the fuel system.

3.3 General repair instructions

Refer to the following sections for relevant procedures:

- Repairing wiring harnesses and connectors ⇒ Electrical system, General information; Rep. gr. 97 ; Wiring harness and connector repair .

- Releasing and dismantling connector housings ⇒ Electrical system, General information; Rep. gr. 97 ; Wiring harness and connector repair; Releasing and dismantling contact housings .
- Cleaning contact surfaces ⇒ Electrical system, General information; Rep. gr. 97 ; Repair case - VAS 6410- .
- Vehicle diagnostic, testing and information systems ⇒ Electrical system, General information; Rep. gr. 97 ; Vehicle diagnostic, testing and information systems .

3.4 Contact corrosion

Contact corrosion can occur if unsuitable connecting elements (screws, bolts, nuts, washers, etc.) are used.

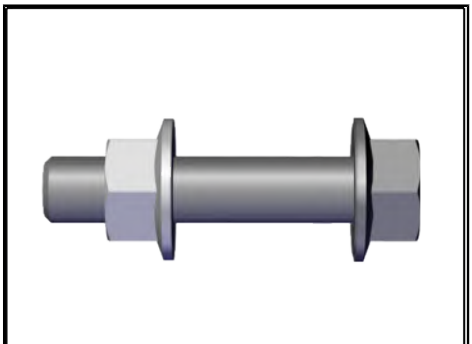
For this reason, only connecting elements with a special surface coating have been fitted.

In addition, rubber, plastic and adhesives are made of non-conductive materials.

If there is any doubt about the suitability of parts, a general rule is to use new parts ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

3.5 Nuts and bolts

- Specified torques given are for unoiled nuts, bolts and screws.
- Always renew self-locking nuts and bolts.
- Always renew nuts and bolts with turning further angle.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3.6 Routing and attachment of lines

Risk of damage to lines

Lines may become damaged by moving or hot components.

– Route lines in their original positions.

– Make sure there is sufficient clearance between the lines and moving as well as hot components.

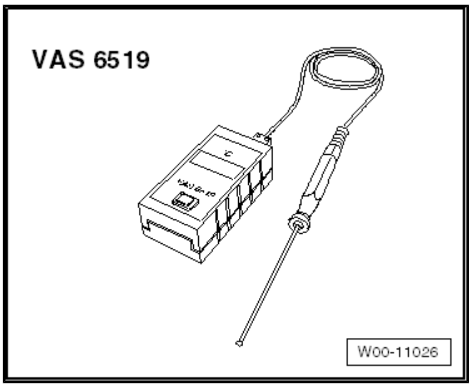
3.7 Fitting radiator and condensers

Even if installed correctly, the radiator, the condenser and the charge air cooler may have small dents in their fins. This does not mean that these components have been damaged. It is not permissible to renew radiators, charge air coolers or condensers only because of such minor dents.

3.8 Checking cooling output

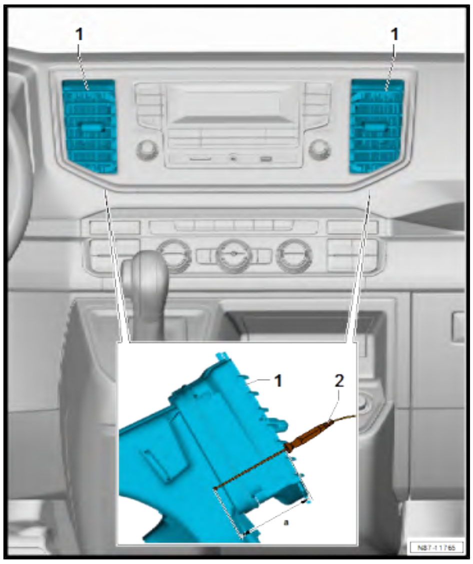
Special tools and workshop equipment required

- Temperature gauge - VAS 6519-



Test procedure

- Close bonnet.
- Use temperature gauge - VAS 6519- to measure the ambient temperature.
- Close all doors and windows.
- Open all dash panel vents.
- Set air conditioning system temperature to coldest setting.
- Set air distribution to centre position.
- Hold temperature probe -2- of temperature gauge - VAS 6519- approx. 20 mm in area shown -a- of vents -1-.
- Switch on air conditioning system using A/C button.
- Start engine.
- Raise engine speed to 2000 rpm and maintain.



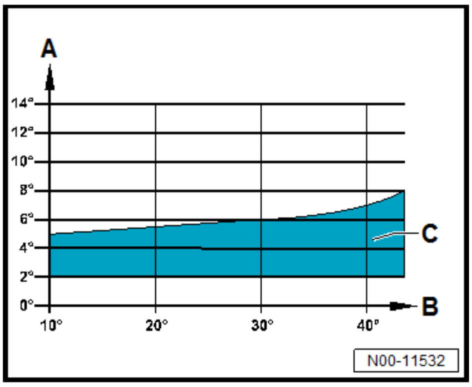
Evaluation for vehicles with one evaporator

After 5 minutes, the temperature of the air blown out of the dash panel vents on the driver side must lie within permitted tolerances depending on the ambient temperature (see diagram).

A - Temperature of air blown out of dash panel vents on driver side

B - Ambient temperature

C - Permitted tolerance range

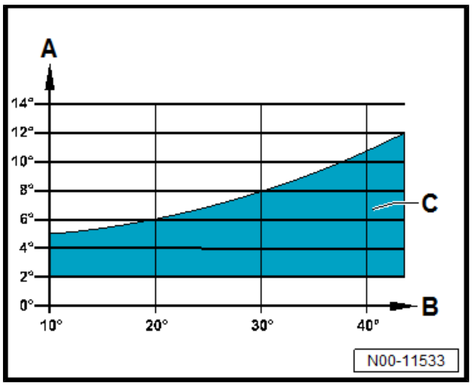


Evaluation for vehicles with two evaporators

A - Temperature of air blown out of vents

B - Ambient temperature

C - Permitted tolerance range



3.9 Working on refrigerant circuit

Risk of freezing injury from refrigerant

When working on the air conditioning system, there is a risk of highly pressurised refrigerant escaping from the system. There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.

– Wear protective goggles.

– Extract refrigerant and open the refrigerant circuit immediately afterwards.

– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

Note

- *The refrigerant circuit must be purged with refrigerant R134a / R1234yf under the following conditions:*
- *If dirt or other contamination is in the circuit.*
- *The search for faults in the electrical system, the vacuum system and the air ducts did not turn up any fault.(Moisture is in the refrigerant circuit and builds up pressure.)*
- *If the refrigerant circuit has been left open for longer than normally required for repairs (e.g. following an accident).*
- *When pressure and temperature measurements in the refrigerant circuit indicate that there is moisture in the refrigerant circuit.*
- *There is doubt about the amount of refrigerant oil in the refrigerant circuit.*
- *The air conditioner compressor has to be exchanged because of internal damage (e.g. noisy or lack of power).*

For details regarding the procedure of purging with refrigerant R134a / R1234yf, refer to the following:

Vehicles with R134a refrigerant

- ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Cleaning refrigerant circuit of contaminants, commercial vehicles

Vehicles with R1234yf refrigerant

- ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Cleaning the refrigerant circuit

For notes on testers and tools for repair work on climate-controlled vehicles, refer to the following:

Vehicles with R134a refrigerant

- ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Testers and tools

Vehicles with R1234yf refrigerant

- ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 00 ; Testers and tools

For information on opening the refrigerant circuit, refer to the following:

Vehicles with R134a refrigerant

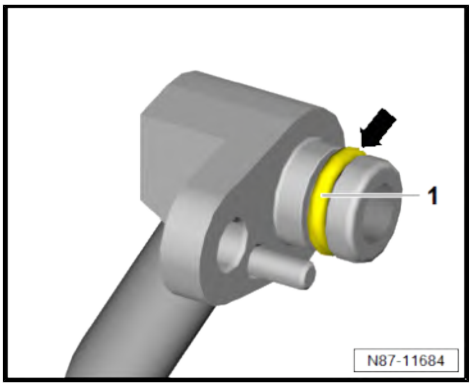
In some cases, it is no longer necessary on air conditioning systems with R134a refrigerant to renew the desiccant bag each time the refrigerant circuit is opened ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Renewing components .

Vehicles with R1234yf refrigerant

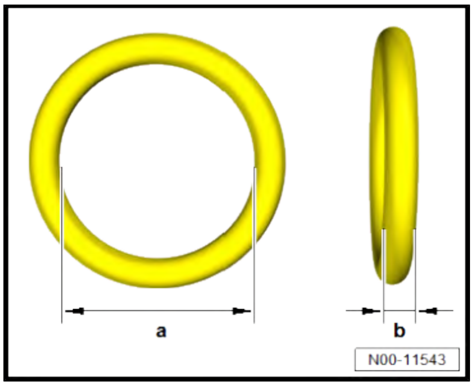
In some cases, it is no longer necessary on air conditioning systems with refrigerant R1234yf to renew the desiccant bag each time the refrigerant circuit is opened ⇒ Air conditioning system with refrigerant R1234yf; Rep. gr. 87 ; Refrigerant circuit; Renewing components .

3.10 Refrigerant circuit seals

- Renew seals after removal.
- Moisten seals with refrigerant oil before installing.
- Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.
- Make sure connections of refrigerant lines in question are clean.
- Install only seals resistant to refrigerant R134a/R1234yf and respective refrigerant oil ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



The dimensions -a- and -b- depend on the fitting location of the seal ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



4 Technical data

4.1 Refrigerant capacity

4.1.1 Capacities for refrigerant R134a

Total capacity
0.575 kg ± 0.015 kg or 575 g ± 15 g
Vehicles with second evaporator 0.825 kg ± 0.015 kg or 825 g ± 15 g

4.1.2 Capacities for refrigerant R1234yf

Total capacity
0.555 kg ± 0.015 kg or 555 g ± 15 g
Vehicles with second evaporator 0.800 kg ± 0.015 kg or 800 g ± 15 g

4.2 Refrigerant oil capacities

4.2.1 Refrigerant oil capacities, refrigerant R134a

Note

- *Refrigerant oils from containers which have been open for a longer period of time are unusable.*
- *As refrigeration oil is extremely hygroscopic, open containers must be immediately re-sealed after use to prevent the ingress of moisture.*

Air conditioner compressor, Delphi

Part number for air conditioner compressor from Delphi	Required refrigerant oil	Total amount of oil in refrigerant circuit *)
5Q0 820 803 J	-G 052 154 A2- or -G 052 300 A2-	110 ml ± 10 ml

1) This volume of refrigerant oil is contained in a replacement air conditioner compressor and corresponds to the total capacity.

Air conditioner compressor, Denso

Part number for Denso air conditioner compressor	Required refrigerant oil	Total amount of oil in refrigerant circuit *)
5Q0 820 803 F	-G 052 300 A2-	110 ml ± 10 ml

2) This volume of refrigerant oil is contained in a replacement air conditioner compressor and corresponds to the total capacity.

Air conditioner compressor, Mahle

Part number for air conditioner compressor from Mahle	Required refrigerant oil	Total amount of oil in refrigerant circuit *)
7C0 820 803	-G 052 154 A2- or -G 052 300 A2-	110 ml

3) This volume of refrigerant oil is contained in a replacement air conditioner compressor and corresponds to the total capacity.

Sanden air conditioner compressors

Part numbers for Sanden air conditioner compressors	Required refrigerant oil	Total amount of oil in refrigerant circuit *)
5Q0 820 803 H	-G 052 154 A2-	75 ml ± 10 ml
5Q0 820 803 L	-G 052 154 A2-	75 ml ± 10 ml
5Q0 820 803 M	-G 052 154 A2-	75 ml ± 10 ml
7E0 820 803 M	-G 052 154 A2-	60 ml **)
7E0 820 803 S	-G 052 154 A2-	60 ml **)
7E0 820 803 T	-G 052 154 A2-	60 ml ± 10 ml **)

4) This volume of refrigerant oil is contained in a replacement air conditioner compressor and corresponds to the total capacity.

5) **) On vehicles with a second evaporator there are additional refrigerant oil volumes in the refrigerant circuit.

Part numbers for Sanden air conditioner compressors	Additional refrigerant oil volume in refrigerant circuit
7E0 820 803 M	80 ml ± 10 ml
7E0 820 803 S	80 ml ± 10 ml
7E0 820 803 T	80 ml ± 10 ml

4.2.2 Refrigerant oil capacities, refrigerant R1234yf

Note

- *Refrigerant oils from containers which have been open for a longer period of time are unusable.*
- *As refrigeration oil is extremely hygroscopic, open containers must be immediately re-sealed after use to prevent the ingress of moisture.*

Air conditioner compressor, Denso

Part number for Denso air conditioner compressor	Required refrigerant oil	Total amount of oil in refrigerant circuit *)
7LA 816 803	-G 055 535 M2-	110 ml ± 10 ml **)

6) This volume of refrigerant oil is contained in a replacement air conditioner compressor and corresponds to the total capacity.

7) **) On vehicles with a second evaporator there are additional refrigerant oil volumes in the refrigerant circuit.

Part number for Denso air conditioner compressor	Additional refrigerant oil volume in refrigerant circuit
7LA 816 803	60 ml ± 10 ml

Sanden air conditioner compressor

Part numbers of Sanden air conditioner compressor	Required refrigerant oil	Total amount of oil in refrigerant circuit *)
7E0 816 803 F	-G 055 535 M2-	60 ml ± 10 ml **)

8) This volume of refrigerant oil is contained in a replacement air conditioner compressor and corresponds to the total capacity.

9) **) On vehicles with a second evaporator there are additional refrigerant oil volumes in the refrigerant circuit.

Part numbers of Sanden air conditioner compressor	Additional refrigerant oil volume in refrigerant circuit
7E0 816 803 F	80 ml ± 10 ml

4.3 Oil distribution

The refrigerant oil located in the sump of the air conditioner compressor before the air conditioner system is switched on for the first time distributes itself through the refrigerant circuit as follows:

- ♦ Air conditioner compressor approx. 50 %
- ♦ Condenser approx. 20 %
- ♦ Suction hose approx. 10 %
- ♦ Evaporator approx. 20 %

80 – Heating

1 Heater

Described in this chapter are only the air conditioning system scopes that deviate from the norm.

Information regarding the identical repair descriptions is here ⇒ “5 Front heater and air conditioning unit”, page 131 .

⇒ “1.1 Assembly overview – heater unit”, page 16

⇒ “1.2 Assembly overview - second heater unit”, page 18

⇒ “1.3 Removing and installing rear warm air blower V47 ”, page 19

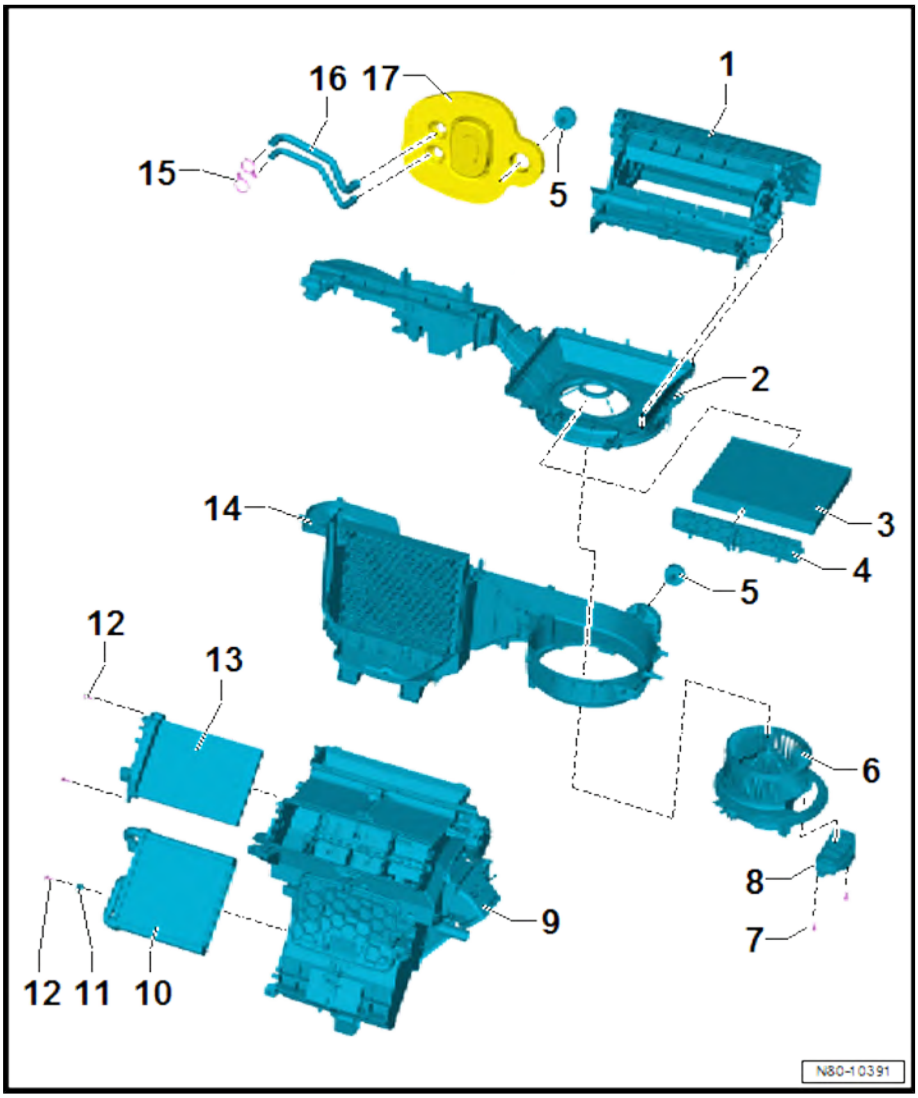
⇒ “1.4 Removing and installing warm air blower control unit J350 ”, page 22

⇒ “1.5 Removing and installing heat exchanger for second heater unit”, page 23

⇒ “1.6 Removing and installing second heater unit”, page 24

⇒ “1.7 Dismantling and assembling second heater unit”, page 26

1.1 Assembly overview – heater unit



1 - Air intake duct

❑ Removing and installing ⇒ page 189

2 - Heater housing

❑ Upper part

❑ Dismantling and assembling ⇒ page 142

3 - Dust and pollen filter

❑ Different versions ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

❑ Replacement interval ⇒ Maintenance tables or ⇒ maintenance test lists for MAN TGE

❑ Removing and installing ⇒ page 144

4 - Dust and pollen filter cover

5 - Bracket

❑ Qty. 2

6 - Fresh air blower - V2-

❑ Removing and installing ⇒ page 146

7 - Bolts

❑ Qty. 2

❑ Specified torque ⇒ Item 9 (page 132)

8 - Fresh air blower control unit - J126-

❑ Removing and installing ⇒ page 146

9 - Air distribution housing

❑ Removing and installing ⇒ page 143

10 - Heat exchanger

❑ Removing and installing ⇒ page 147

11 - Heat exchanger bracket

12 - Bolts

❑ Qty. 3

❑ Specified torque ⇒ Item 9 (page 132)

13 - Auxiliary air heater element - Z35-

❑ Checking ⇒ page 134

❑ Removing and installing ⇒ page 135

14 - Heater housing

❑ Lower part

❑ Dismantling and assembling ⇒ page 142

15 - Clamps

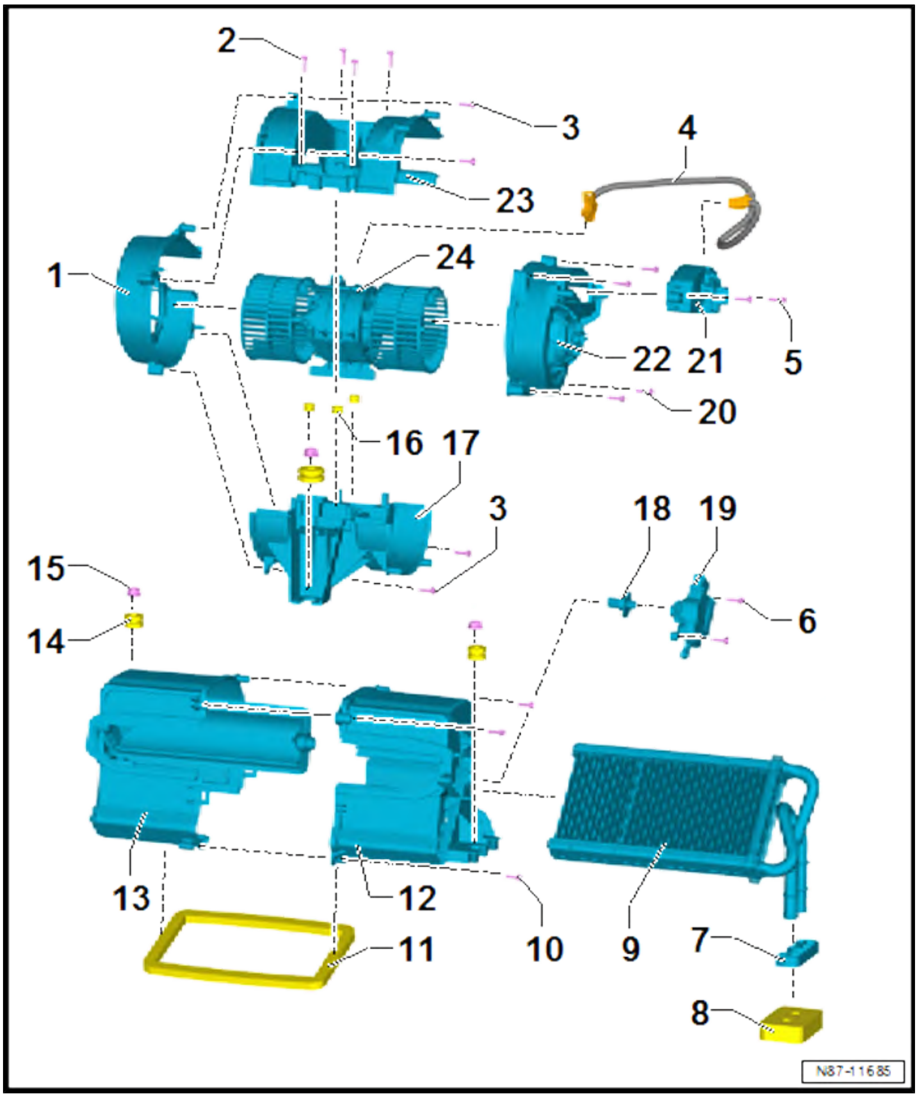
❑ Qty. 2

❑ Specified torque ⇒ Item 16 (page 133)

16 - Coolant pipes on heat exchanger
- ❑ Removing and installing ⇒ page 148

17 - Seal
- ❑ For sealing and insulation

1.2 Assembly overview - second heater unit



1 - Blower housing
- ❑ Rear
- ❑ Dismantling and assembling ⇒ page 26

2 - Bolts
- ❑ Qty. 4
- ❑ 1.5 Nm

3 - Bolts
- ❑ Qty. 4
- ❑ 1.5 Nm

4 - Adapter line

5 - Bolts
- ❑ Qty. 2
- ❑ 1.5 Nm

6 - Bolts
- ❑ Qty. 2
- ❑ 1.5 Nm

7 - Bracket
- ❑ For heat exchanger coolant pipes
- ❑ Check for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

8 - Seal
- ❑ Renew after removal
- ❑ For heat exchanger coolant pipes

9 - Heat exchanger
- ❑ Removing and installing ⇒ page 23

10 - Bolts
- ❑ Qty. 4
- ❑ 1.5 Nm

11 - Seal
- ❑ Renew after removal
- ❑ For air outlet

12 - Heat exchanger housing
- ❑ Left
- ❑ Dismantling and assembling ⇒ page 26

13 - Heat exchanger housing
- ❑ Right
- ❑ With temperature flap
- ❑ Dismantling and assembling ⇒ page 26

14 - Flexible joints
- ❑ Renew after removal
- ❑ Qty. 3
- ❑ Together with sleeves

15 - Nuts
- ❑ Qty. 3
- ❑ 12 Nm

16 - Flexible joints
- ❑ Renew after removal
- ❑ Qty. 3

17 - Blower housing
- ❑ Bottom
- ❑ Dismantling and assembling ⇒ page 26

18 - Adapter
- ❑ For rear temperature flap control motor - V137-
- ❑ Remains on rear temperature flap control motor - V137- during removal

19 - Rear temperature flap motor - V137-
- ❑ Removing and installing ⇒ page 31

20 - Bolts
- ❑ Qty. 4
- ❑ 1.5 Nm

21 - Warm air blower control unit - J350-
- ❑ Removing and installing ⇒ page 22

22 - Blower housing
- ❑ Front
- ❑ Dismantling and assembling ⇒ page 26

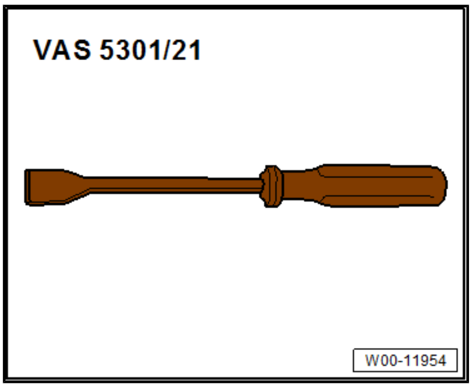
23 - Blower housing
- ❑ Top
- ❑ Dismantling and assembling ⇒ page 26

24 - Rear warm air blower - V47-
- ❑ Removing and installing ⇒ page 19

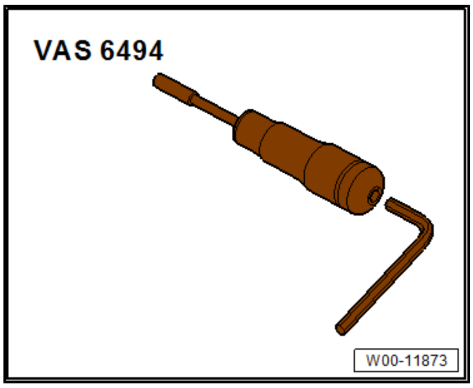
1.3 Removing and installing rear warm air blower - V47-

Special tools and workshop equipment required

- Scraper - VAS 5301/21-



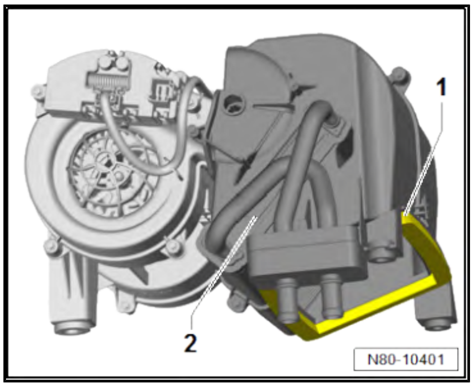
- Torque screwdriver - VAS 6494-



Removing

– Remove heater unit ⇒ page 24 .

– Remove rear temperature flap control motor - V137- ⇒ page 31 .



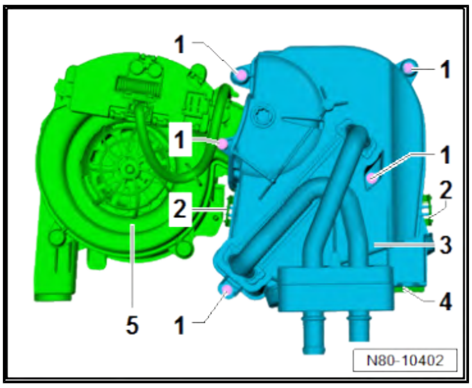
Note

The air outlet seal -1- cannot be removed intact.

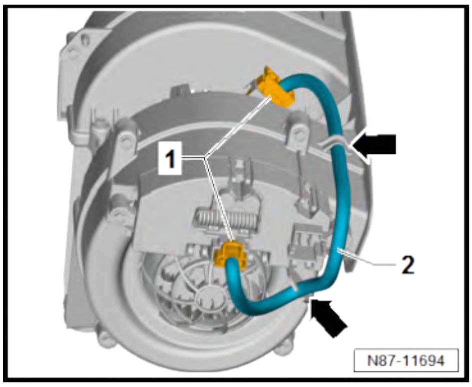
– Remove air outlet seal -1- from heat exchanger housing -2- using scraper - VAS 5301/21- .

– Remove bolts -1-.

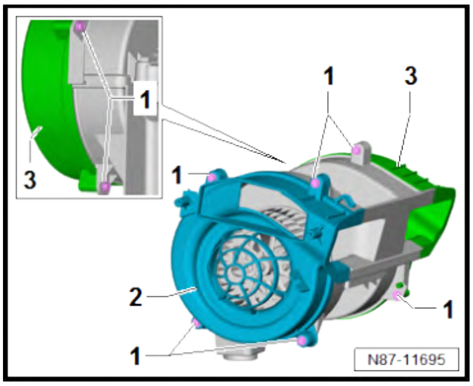
– Release clips -2-. While doing so, detach left heat exchanger housing -3- from right heat exchanger housing -4- sufficiently to allow hot air blower housing -5- to be removed.



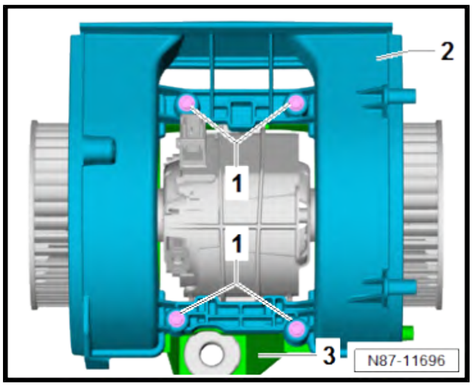
– Separate electrical connectors -1-.
– Unclip adapter cable -2- in area marked by -arrows- and remove.



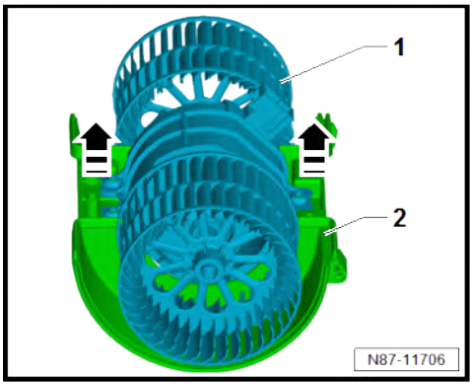
– Unscrew bolts -1-.
– Remove left blower housing -2- and right blower housing -3- from middle section.



– Unscrew bolts -1-.
– Separate top blower housing -2- and bottom blower housing -3-.

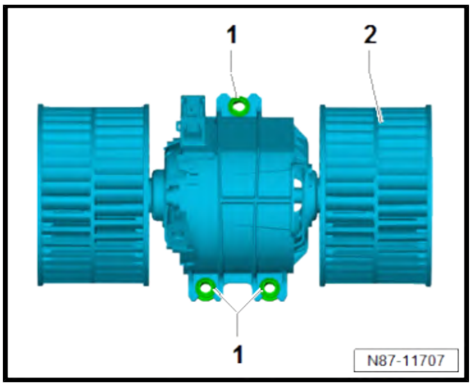


– Remove rear warm air blower - V47- -1- in direction of -arrow- from bottom blower housing -2-.

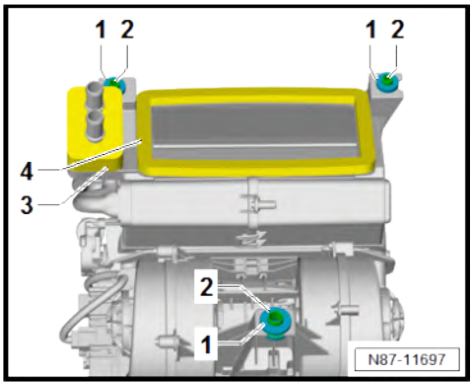


Installing

Install in reverse order of removal, observing the following:



– Renew flexible joints -1- on rear warm air blower - V47- -2- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



– Renew flexible joints -1- together with sleeves -2- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Renew seal -3- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Clean bonding surface of air outlet seal -4- on heat exchanger housing.

– Renew air outlet seal -4- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

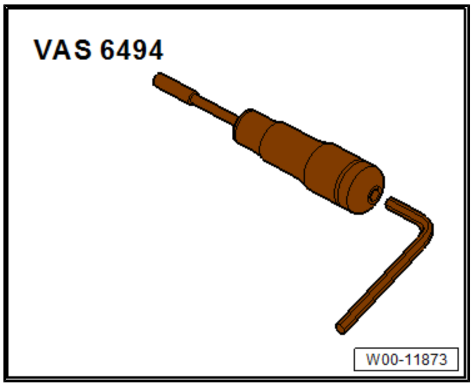
Specified torques

♦ ⇒ “1.2 Assembly overview - second heater unit”, page 18

1.4 Removing and installing warm air blower control unit - J350-

Special tools and workshop equipment required

♦ Torque screwdriver - VAS 6494-



Removing

Vehicles with single seat on front passenger side

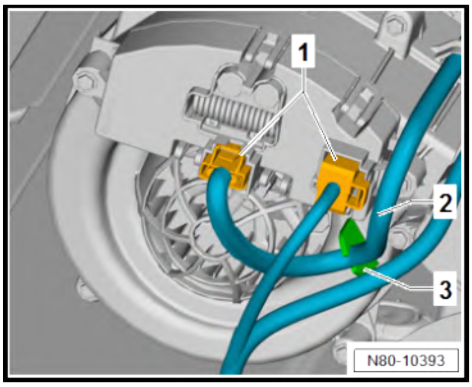
– Remove front passenger seat ⇒ General body repairs, interior; Rep. gr. 72 ; Front seats; Removing and installing front seat .

Vehicles with bench seat on front passenger side

– Fold up seat padding.

– Remove storage compartment ⇒ General body repairs, interior; Rep. gr. 72 ; Front seats; Removing and installing storage compartment .

Continued for all vehicles



⚠ CAUTION

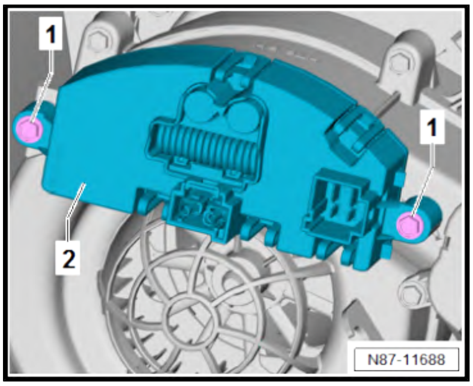
Risk of burns when touching hot cooling surface of control unit.

Risk of burns to the hands.

– Wear protective gloves.

– Separate electrical connectors -1-.

– Unclip wiring harness -2- from retainer -3-.



– Unscrew bolts -1-. While doing so, detach warm air blower control unit - J350- -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

♦ ⇒ “1.2 Assembly overview - second heater unit”, page 18

1.5 Removing and installing heat exchanger for second heater unit

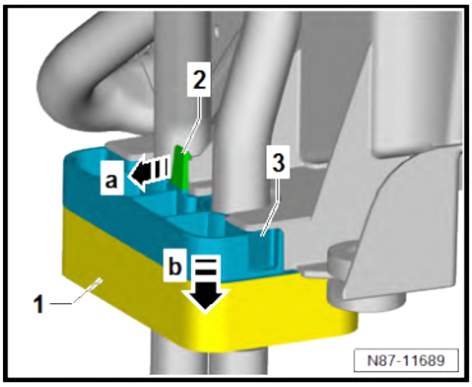
Removing

– Remove heater unit ⇒ page 24 .

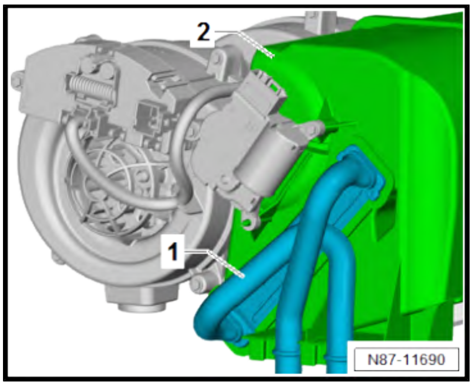
– Push seal -1- off coolant pipes.

– Release retaining tab -2- in direction of -arrow a-.

– Remove bracket -3- in direction of -arrow b-.

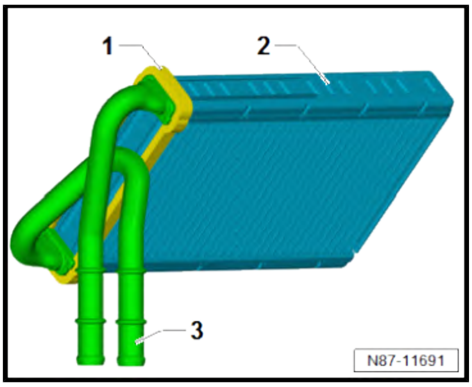


– Pull heat exchanger -1- out of heater unit -2-.



Installing

Install in reverse order of removal, observing the following:



– Check foam seal -1- on heat exchanger -2- for damage. Renew heat exchanger -2- if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

Note

An improperly bonded or damaged seal -1- could roll up when the heat exchanger -2- is pushed into the heater unit.

– Check coolant pipes -3- for damage and contamination. Renew heat exchanger -2- if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check heater duct for contamination and clean if necessary.

– Push heat exchanger into heater unit as far as stop.

– Check coolant pipe retainer for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

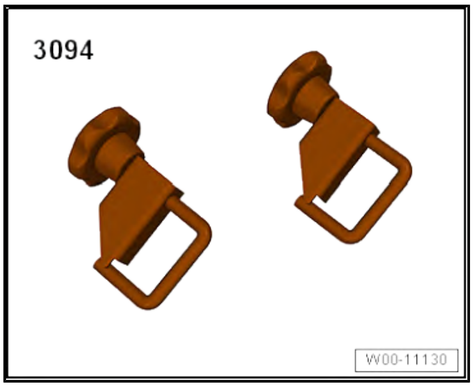
Specified torques

♦ ⇒ “1.2 Assembly overview - second heater unit”, page 18

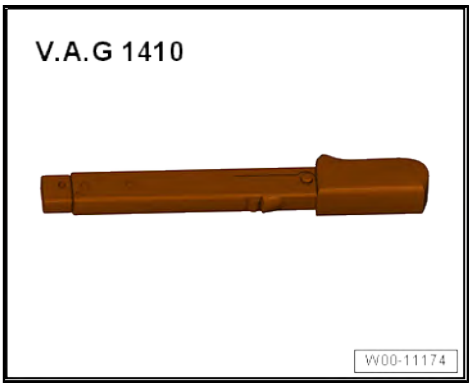
1.6 Removing and installing second heater unit

Special tools and workshop equipment required

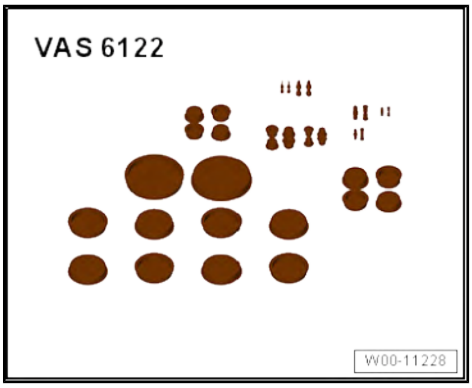
♦ Hose clamps to 25 mm - 3094-



♦ Torque wrench - V.A.G 1410-



♦ Set of plugs for engine - VAS 6122-



♦ Drip tray for workshop hoist - VAS 6208-

Removing

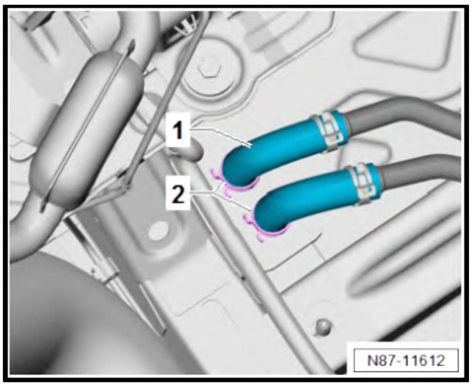
CAUTION

On a warm engine, the cooling system is under high pressure. Danger of scalding by steam and hot coolant.

Skin and other parts of the body may be scalded.

– Wear protective gloves.
– Wear protective goggles.
– Reduce excess pressure by covering cap of coolant expansion tank with cloths and opening it carefully.

– Remove front right underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .
– Place drip tray for workshop hoist - VAS 6208- underneath.



- Mark connections of coolant hoses -1- on heat exchanger.
- Clamp off coolant hoses -1- with hose clips, up to 25 mm - 3094- .
- Loosen hose clip -2-.
- Pull off coolant hoses -1- from heat exchanger.
- Seal off open lines and connections with clean plugs from engine bung set - VAS 6122- .

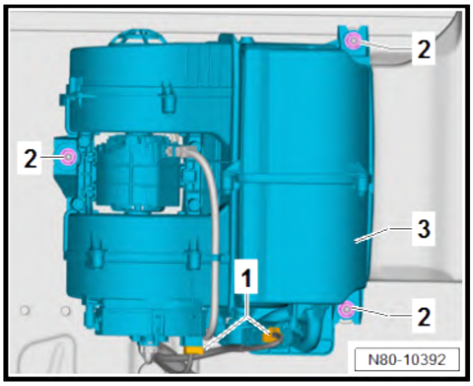
Vehicles with single seat on front passenger side

- Remove seat box of front seat ⇒ General body repairs, interior; Rep. gr. 72 ; Front seats; Removing and installing front seat .

Vehicles with bench seat on front passenger side

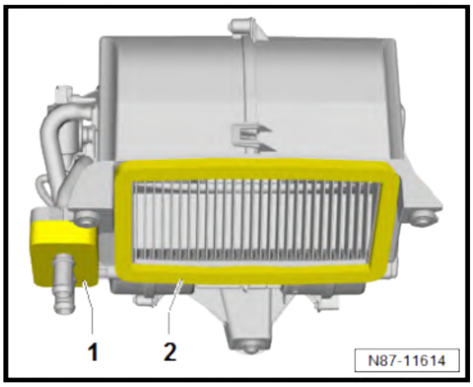
- Fold up seat padding.
- Remove second heater unit cover ⇒ General body repairs, interior; Rep. gr. 72 ; Front seats; Removing and installing front seat; Removing and installing cover of second heater unit .

Continued for all vehicles



- Place waterproof foil and absorbent paper over floor covering in area of heat exchanger.
- Separate electrical connectors -1-.
- Unscrew nuts -2-.
- Remove heater unit -3- upwards.

Installing



Install in reverse order of removal, observing the following:

- Ensure seals -1- and -2- are seated correctly and check for damage; renew if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
- Fill up with coolant ⇒ Rep. gr. 19 ; Cooling system/coolant; Draining and filling coolant .
- Check connection between coolant hoses and heat exchanger for leaks.
- Renew flexible joints together with sleeves ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
- Check function of second heater unit.

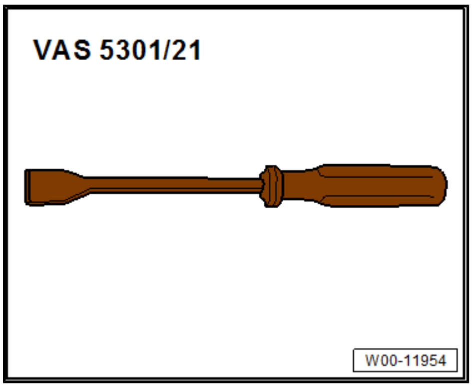
Specified torques

- ⇒ “1.2 Assembly overview - second heater unit”, page 18

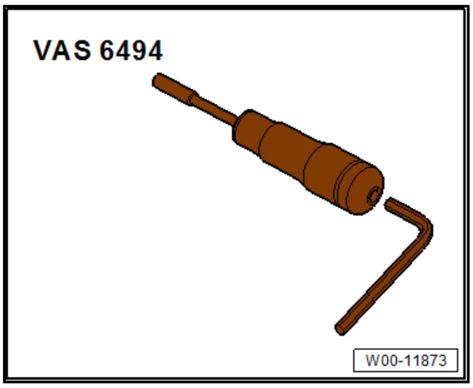
1.7 Dismantling and assembling second heater unit

Special tools and workshop equipment required

- Scraper - VAS 5301/21-



- Torque screwdriver - VAS 6494-



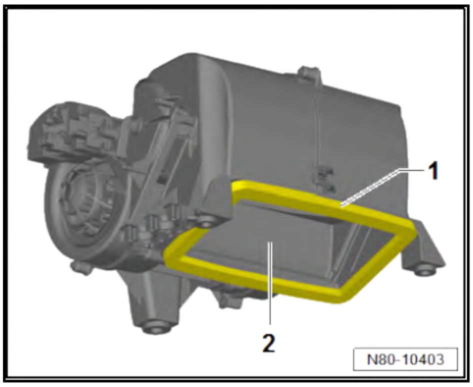
Dismantling

– Remove heat exchanger ⇒ page 23 .

– Remove rear temperature flap control motor - V137- ⇒ page 31 .

Note

The air outlet seal -1- cannot be removed intact.

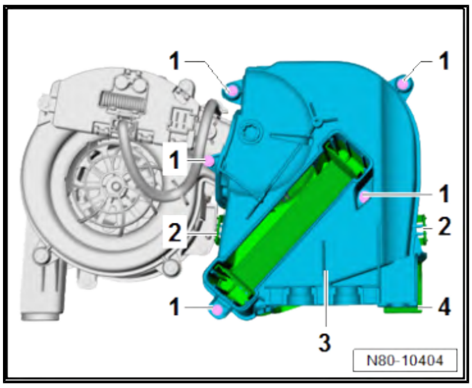


– Remove air outlet seal -1- from heat exchanger housing -2- using scraper - VAS 5301/21- .

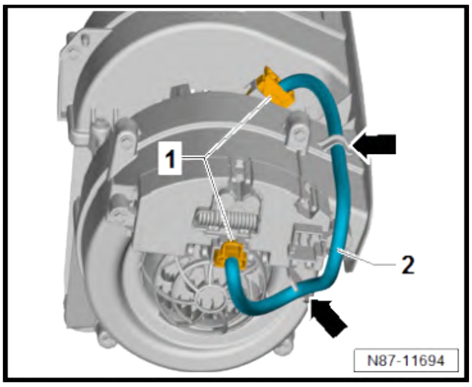
– Remove bolts -1-.

– Release clips -2-. When doing this, separate left heat exchanger housing -3- and right heat exchanger housing -4-.

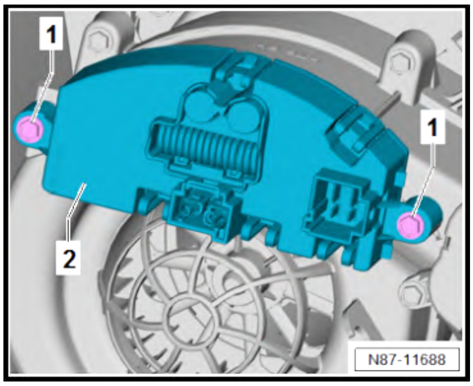
– Release clips -2-. When doing this, separate left heat exchanger housing -3- and right heat exchanger housing -4-. When doing this, remove hot air blower housing.



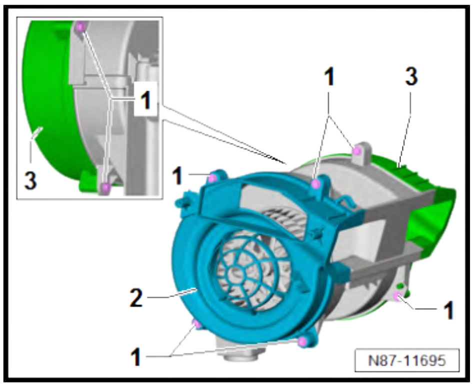
– Separate electrical connectors -1-.
– Unclip adapter cable -2- in area marked by -arrows- and remove.



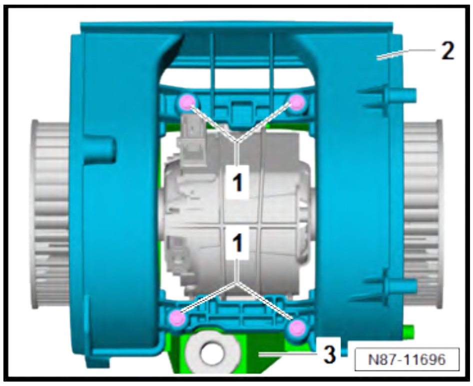
– Unscrew bolts -1-. While doing so, detach warm air blower control unit - J350- -2-.

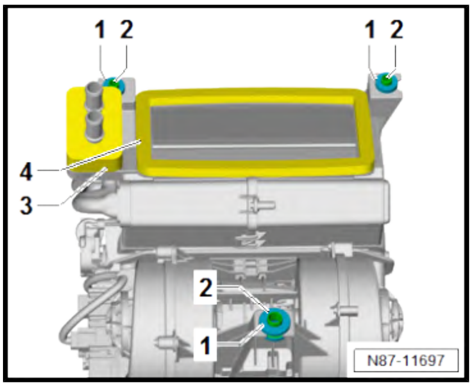


– Unscrew bolts -1-.
– Remove left blower housing -2- and right blower housing -3- from middle section.



– Unscrew bolts -1-.
– Separate top blower housing -2- and bottom blower housing -3-.





Installing

Assembly is carried out in reverse sequence; note the following:

– Renew flexible joints -1- together with sleeves -2- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Renew seal -3- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Clean bonding surface of air outlet seal -4- on heat exchanger housing.

– Renew air outlet seal -4- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

Specified torques

- ⇒ “1.2 Assembly overview - second heater unit”, page 18

2 Control motors

Described in this chapter are only the air conditioning system scopes that deviate from the norm.

Information regarding the identical repair descriptions is here ⇒ “4 Control motors”, page 113 .

⇒ “2.1 Overview of fitting locations - front control motors”, page 30

⇒ “2.2 Removing and installing rear temperature flap control motor V137 ”, page 31

2.1 Overview of fitting locations - front control motors

⇒ “2.1.1 Overview of fitting locations - front control motors, LHD vehicles”, page 30

⇒ “2.1.2 Overview of fitting locations - front control motors, RHD vehicles”, page 31

2.1.1 Overview of fitting locations - front control motors, LHD vehicles

1 - Air distribution flap control motor - V428-

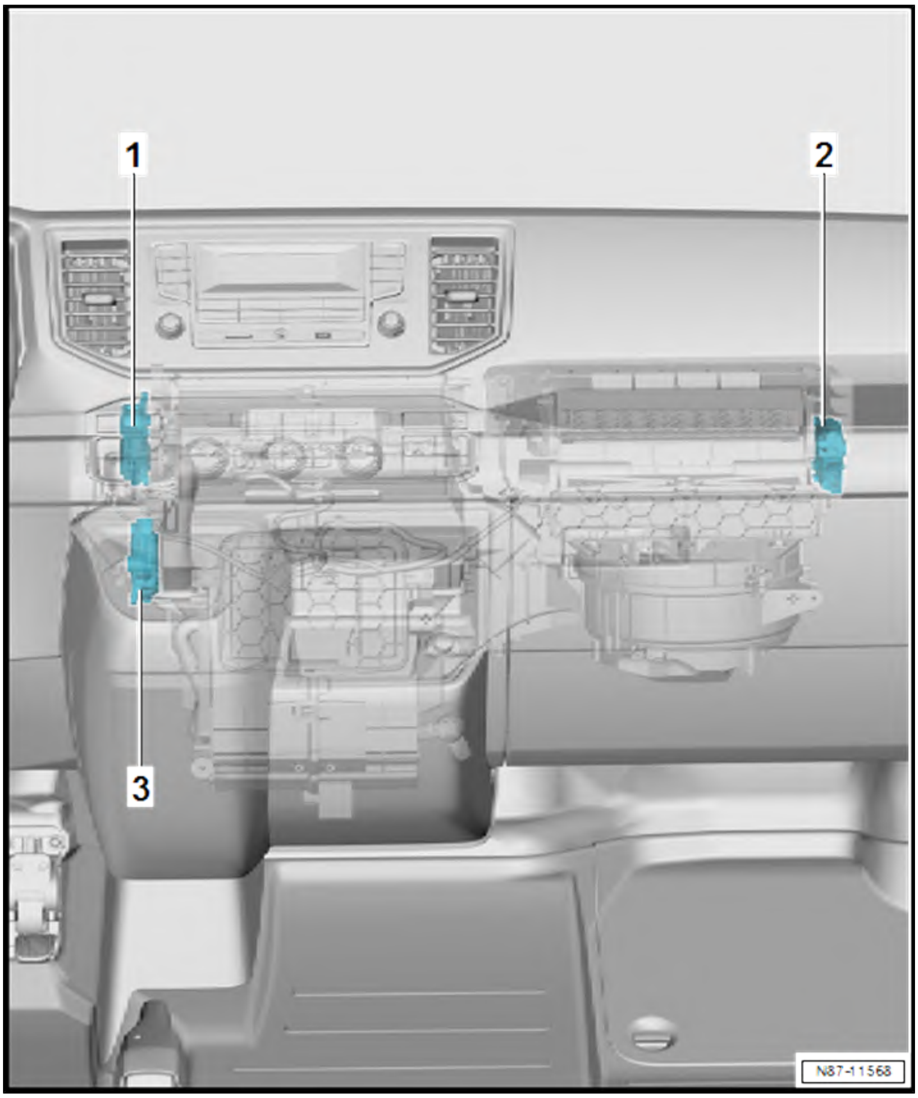
- ❑ With potentiometer for air distribution flap control motor - G645-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 129

2 - Air recirculation flap control motor - V113-

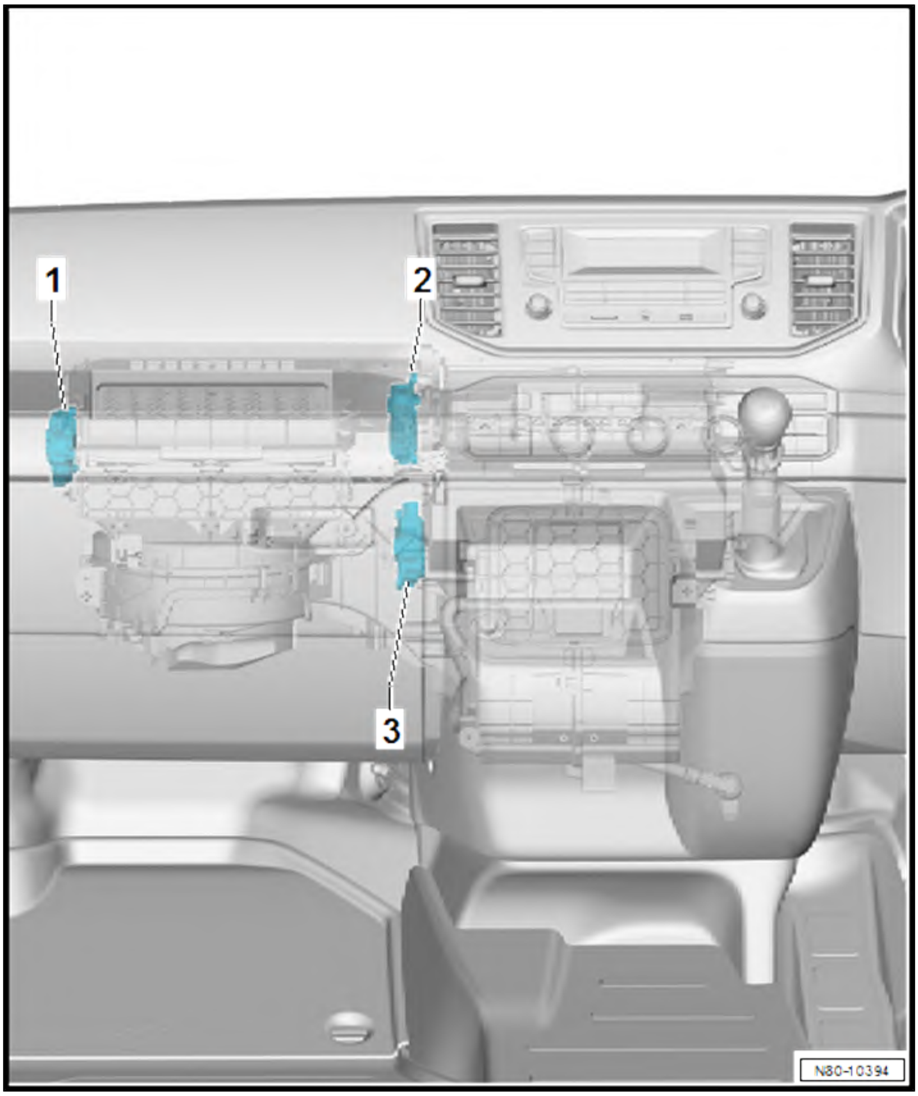
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 122

3 - Temperature flap control motor - V68-

- ❑ With potentiometer for temperature flap control motor - G92-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 119



2.1.2 Overview of fitting locations - front control motors, RHD vehicles



1 - Air recirculation flap control motor - V113-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 122

2 - Air distribution flap control motor - V428-
- ❑ With potentiometer for air distribution flap control motor - G645-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 129

3 - Temperature flap control motor - V68-
- ❑ With potentiometer for temperature flap control motor - G92-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 119

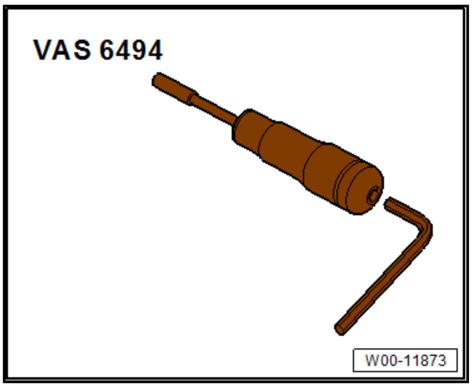
2.2 Removing and installing rear temperature flap control motor - V137-

Note

Only on vehicles with second heater unit.

Special tools and workshop equipment required

- Torque screwdriver - VAS 6494-



- Vehicle diagnostic tester

Removing

Vehicles with single seat on front passenger side

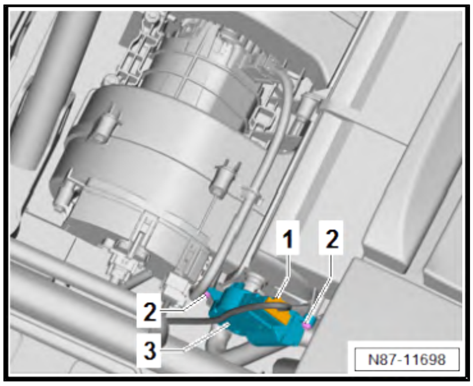
– Remove front passenger seat ⇒ General body repairs, interior; Rep. gr. 72 ; Front seats; Removing and installing front seat .

Vehicles with bench seat on front passenger side

– Fold up seat padding.

– Remove storage compartment ⇒ General body repairs, interior; Rep. gr. 72 ; Front seats; Removing and installing storage compartment .

Continued for all vehicles



– Disconnect connector -1-.

– Unscrew bolts -2-.

– Detach rear temperature flap control motor - V137- -3- from heater unit.

Installing

Install in reverse order of removal, observing the following:

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

Specified torques

- ⇒ “1.2 Assembly overview - second heater unit”, page 18

3 Air duct

Described in this chapter are only the air conditioning system scopes that deviate from the norm.

Information regarding the identical repair descriptions is here ⇒ “7 Air duct”, page 173 .

⇒ “3.1 Removing and installing air ducts”, page 33

3.1 Removing and installing air ducts

⇒ “3.1.1 Removing and installing air duct on second heater unit”, page 33

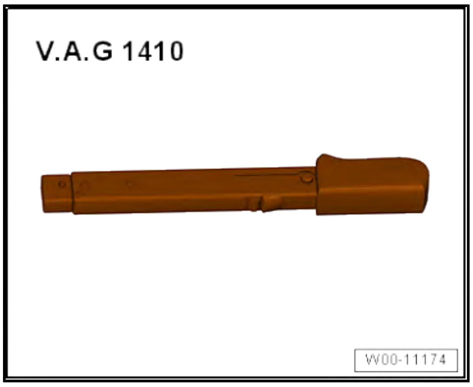
⇒ “3.1.2 Removing and installing floor vents in passenger compartment”, page 33

⇒ “3.1.3 Cleaning floor vents in passenger compartment”, page 34

3.1.1 Removing and installing air duct on second heater unit

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1410-



Removing

- – Remove second heater unit ⇒ page 24 .
- – Detach floor covering on left in passenger compartment.
- – Detach floor covering on right in passenger compartment.

Vehicles with Climatronic

- – Separate electrical connector on rear vent temperature sender - G174- .

Continued for all vehicles

- – Remove air duct together with rear vent temperature sender - G174- through prepared opening in floor covering on right in passenger compartment.

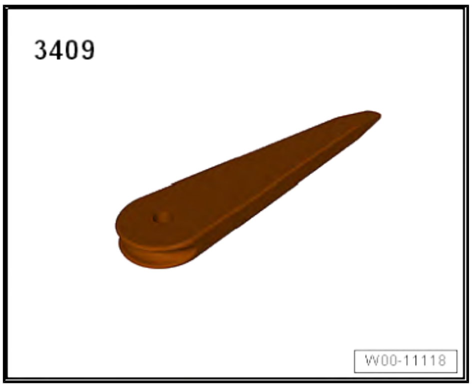
Installing

Install in reverse order of removal.

3.1.2 Removing and installing floor vents in passenger compartment

Special tools and workshop equipment required

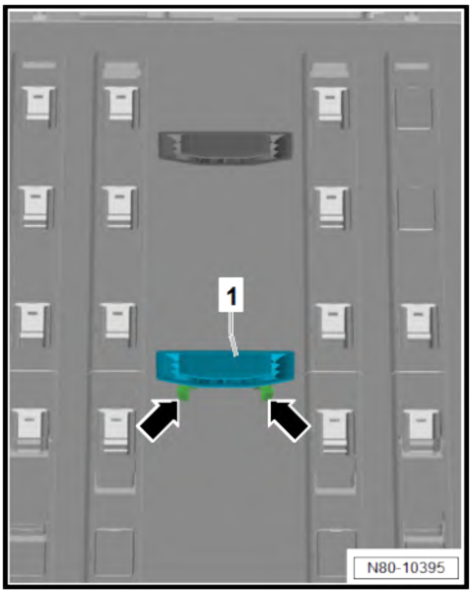
- Removal wedge - 3409-



Removing

– Release vent -1- in area marked with -arrows- using removal wedge - 3409- .

– Swivel out vent -1- from floor covering and remove.



Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs on vent -1- for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

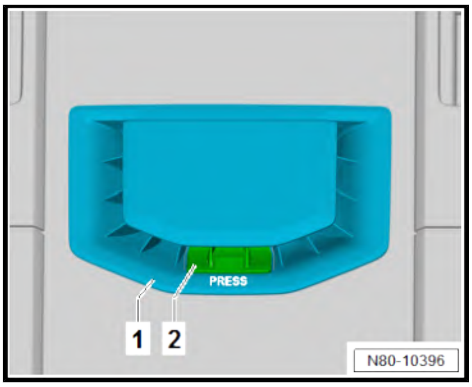
– Check vent -1- and air duct for dirt, clean if necessary ⇒ page 34 .

3.1.3 Cleaning floor vents in passenger compartment

Sequence of operations

– Open flap on vent -1- by pressing button -2-.

– Check vent -1- and air duct for dirt, clean if necessary.



4 Operating and display unit

Described in this chapter are only the air conditioning system scopes that deviate from the norm.

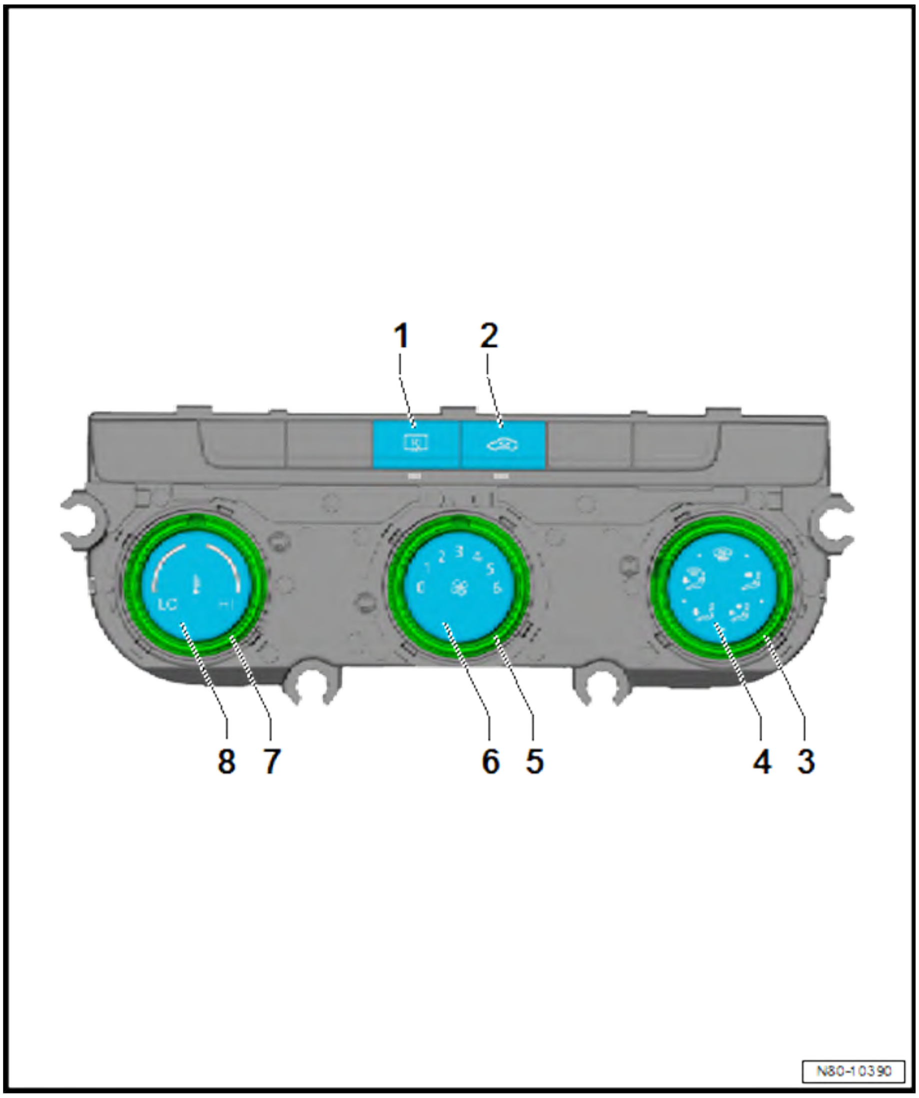
Information regarding the identical repair descriptions is here
⇒ “8 Operating and display unit”, page 193 .

⇒ “4.1 Overview of operating and display unit”, page 35

4.1 Overview of operating and display unit

Note

- *Additional display and operating unit 1 - E857- with heater control unit - J65-*
- *A warning lamp in the instrument panel controls will indicate that the selected function is active.*



1 - Heated rear window button
- ❑ Works only while engine is running and switches off automatically after 10 minutes at the latest

2 - Air recirculation button
- ❑ Pressing the air recirculation button will prevent polluted air from entering the interior.

3 - Rotary knob for setting air distribution

4 - Display for air distribution

5 - Rotary knob for setting blower speed
- ❑ Anticlockwise: reduces blower speed
- ❑ Clockwise: increases speed

6 - Display for blower speed

7 - Rotary knob for setting temperature
- ❑ Anticlockwise: lowers temperature
- ❑ Clockwise: increases temperature

8 - Temperature setting display

5 Other controlling and regulating components

Described in this chapter are only the air conditioning system scopes that deviate from the norm.

Information regarding the identical repair descriptions is here ⇒ “9 Other controlling and regulating components”, page 199 .

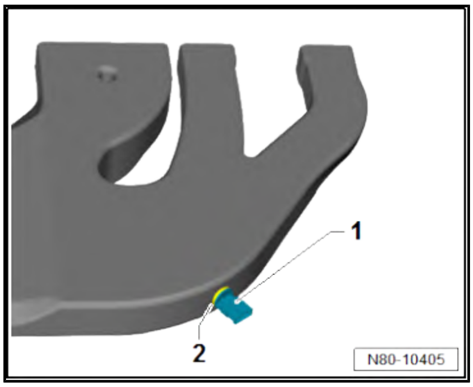
⇒ “5.1 Removing and installing rear vent temperature sender G174 ”, page 36

5.1 Removing and installing rear vent temperature sender - G174-

Removing

– Remove air duct on second heater ⇒ page 33 .

– Turn rear vent temperature sender - G174- -1- by 90°. Remove when doing this.



Installing

Install in reverse order of removal, observing the following:

– Check seal -2- for damage; renew seal if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

87 – Air conditioning system

1 Overview of fitting locations - air conditioning system

⇒ “1.1 Overview of fitting locations - components not located in passenger compartment”, page 37

⇒ “1.2 Overview of fitting locations - components located in front section of passenger compartment”, page 42

⇒ “1.3 Overview of fitting locations - components located in rear section of passenger compartment”, page 44

1.1 Overview of fitting locations - components not located in passenger compartment

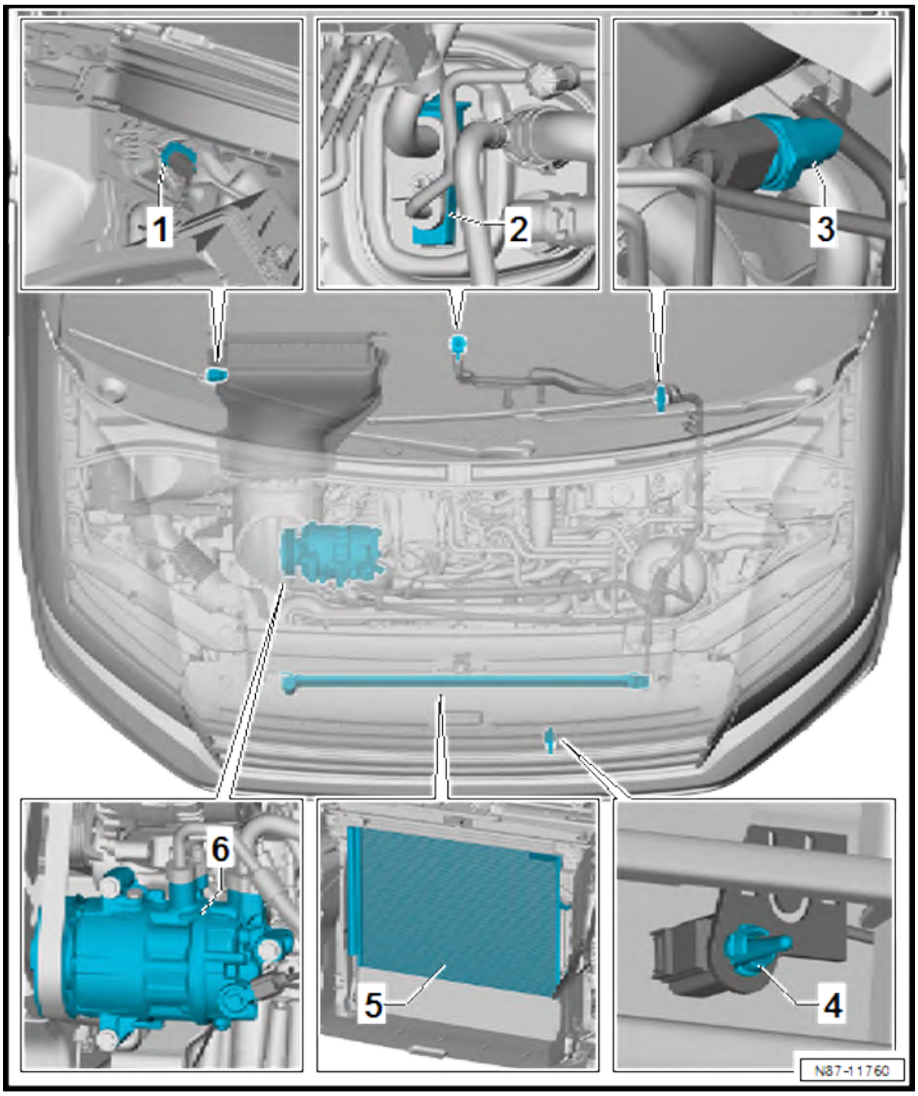
⇒ “1.1.1 Overview of fitting locations - components not located in passenger compartment, transverse engine, left-hand drive vehicles”, page 37

⇒ “1.1.2 Overview of fitting locations - components not located in passenger compartment, transverse engine, right-hand drive vehicles”, page 39

⇒ “1.1.3 Overview of fitting locations - components not located in passenger compartment, straight engine, left-hand drive vehicles”, page 40

⇒ “1.1.4 Overview of fitting locations - components not located in passenger compartment, straight engine, right-hand drive vehicles”, page 41

1.1.1 Overview of fitting locations - components not located in passenger compartment, transverse engine, left-hand drive vehicles



1 - Air quality sensor - G238-
- ❑ Only on vehicles with Climatronic, to calendar week 45/2017
- ❑ Principle of operation ⇒ page 200
- ❑ Removing and installing ⇒ page 201

2 - Expansion valve
- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

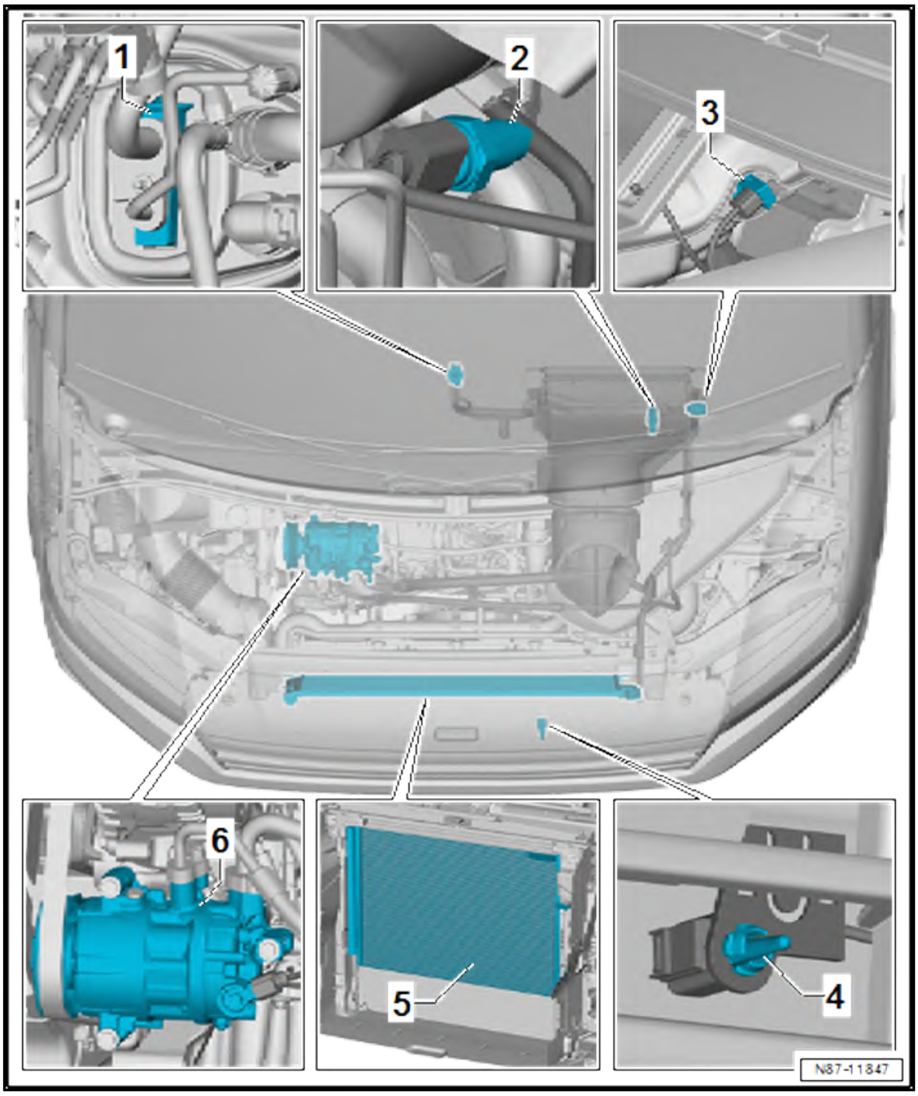
3 - Pressure sender for refrigerant circuit - G805-
- ❑ Removing and installing ⇒ page 54

4 - Ambient temperature sensor - G17-
- ❑ Removing and installing ⇒ page 202

5 - Condenser
- ❑ With desiccant bag/desiccant cartridge
- ❑ Assembly overview ⇒ page 54
- ❑ Removing and installing ⇒ page 62

6 - Air conditioner compressor
- ❑ Assembly overview ⇒ page 100
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104

1.1.2 Overview of fitting locations - components not located in passenger compartment, transverse engine, right-hand drive vehicles



1 - Expansion valve

- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

2 - Pressure sender for refrigerant circuit - G805-

- ❑ Removing and installing ⇒ page 54

3 - Air quality sensor - G238-

- ❑ Only on vehicles with Climatronic, to calendar week 45/2017
- ❑ Principle of operation ⇒ page 200
- ❑ Removing and installing ⇒ page 201

4 - Ambient temperature sensor - G17-

- ❑ Removing and installing ⇒ page 202

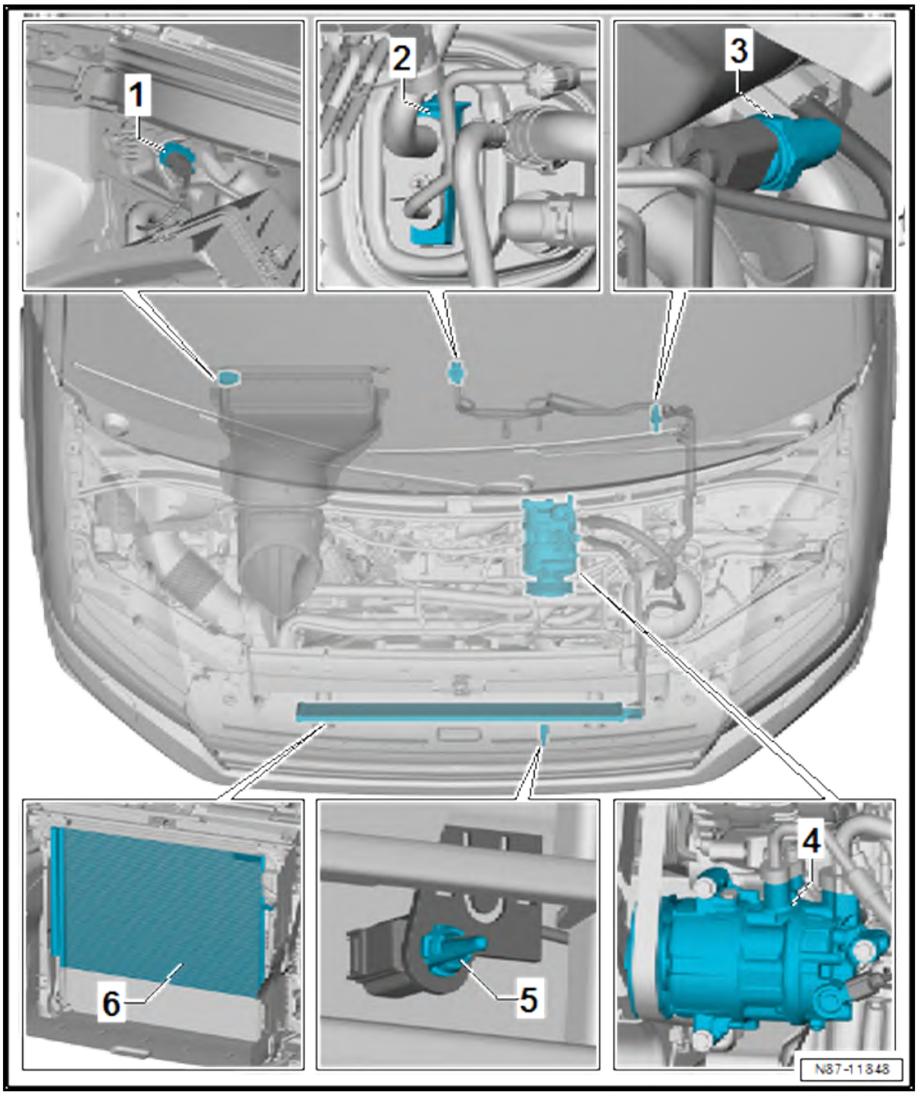
5 - Condenser

- ❑ With desiccant bag/desiccant cartridge
- ❑ Assembly overview ⇒ page 54
- ❑ Removing and installing ⇒ page 62

6 - Air conditioner compressor

- ❑ Assembly overview ⇒ page 100
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104

1.1.3 Overview of fitting locations - components not located in passenger compartment, straight engine, left-hand drive vehicles



1 - Air quality sensor - G238-

- ❑ Only on vehicles with Climatronic, to calendar week 45/2017
- ❑ Principle of operation ⇒ page 200
- ❑ Removing and installing ⇒ page 201

2 - Expansion valve

- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

3 - Pressure sender for refrigerant circuit - G805-

- ❑ Removing and installing ⇒ page 54

4 - Air conditioner compressor

- ❑ Assembly overview ⇒ page 100
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104

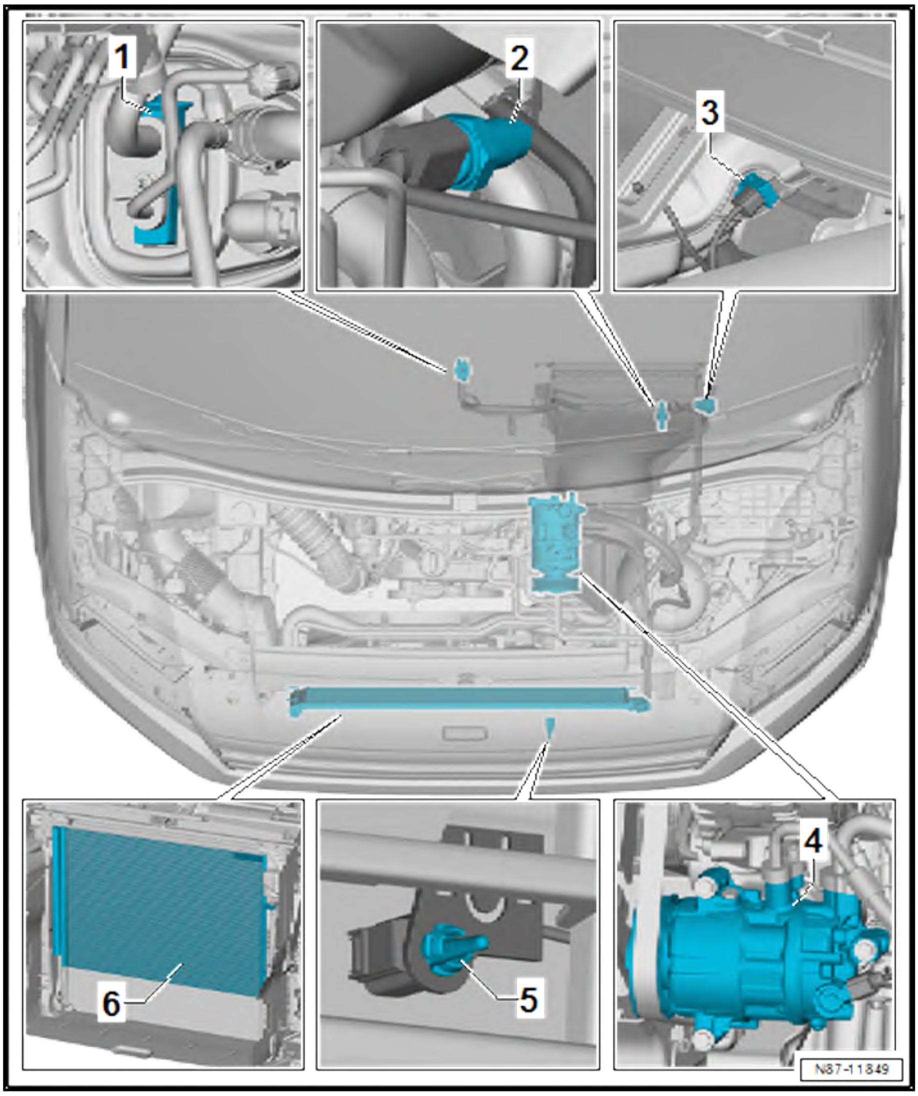
5 - Ambient temperature sensor - G17-

- ❑ Removing and installing ⇒ page 202

6 - Condenser

- ❑ With desiccant bag/desiccant cartridge
- ❑ Assembly overview ⇒ page 54
- ❑ Removing and installing ⇒ page 62

1.1.4 Overview of fitting locations - components not located in passenger compartment, straight engine, right-hand drive vehicles



1 - Expansion valve

- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

2 - Pressure sender for refrigerant circuit - G805-

- ❑ Removing and installing ⇒ page 54

3 - Air quality sensor - G238-

- ❑ Only on vehicles with Climatronic, to calendar week 45/2017
- ❑ Principle of operation ⇒ page 200
- ❑ Removing and installing ⇒ page 201

4 - Air conditioner compressor

- ❑ Assembly overview ⇒ page 100
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104

5 - Ambient temperature sensor - G17-

- ❑ Removing and installing ⇒ page 202

6 - Condenser

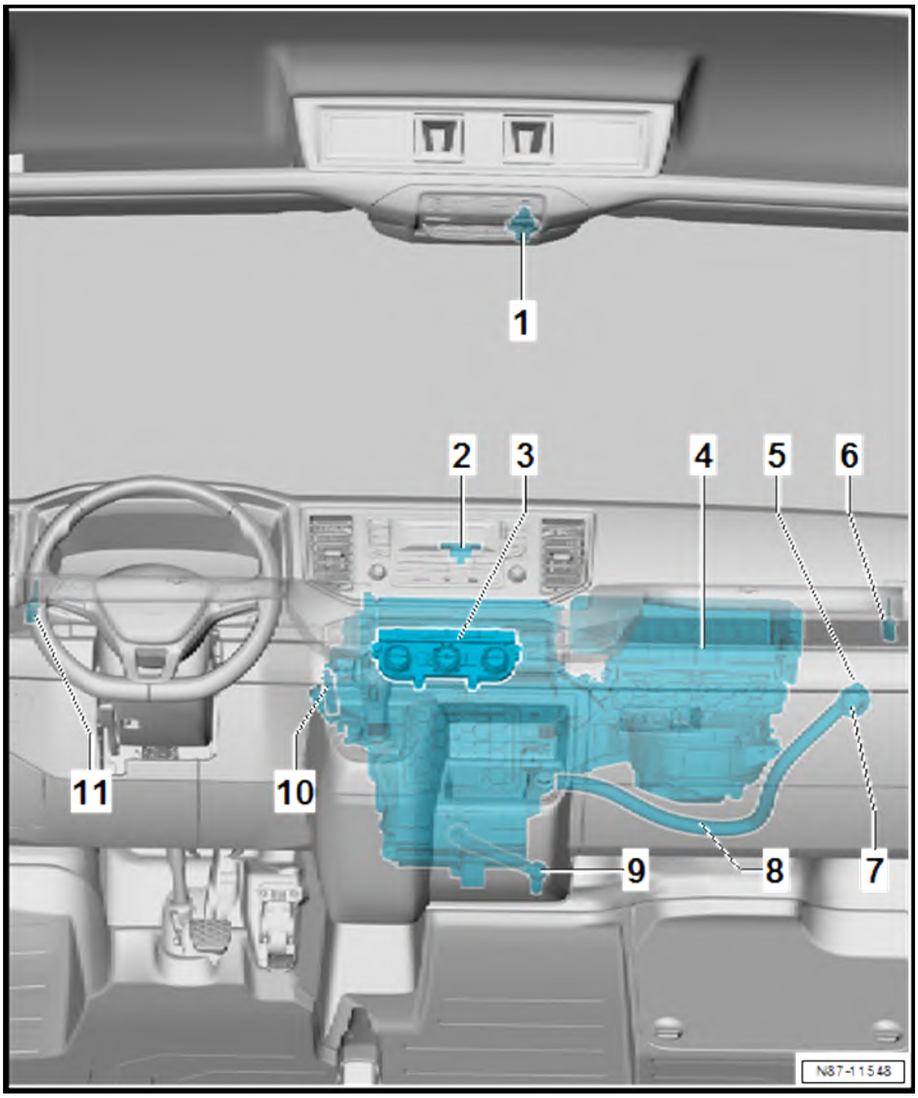
- ❑ With desiccant bag/desiccant cartridge
- ❑ Assembly overview ⇒ page 54
- ❑ Removing and installing ⇒ page 62

1.2 Overview of fitting locations - components located in front section of passenger compartment

⇒ "1.2.1 Overview of fitting locations - components inside of front passenger compartment, left-hand drive vehicles", page 42

⇒ "1.2.2 Overview of fitting locations - components inside of front passenger compartment, right-hand drive vehicles", page 43

1.2.1 Overview of fitting locations - components inside of front passenger compartment, left-hand drive vehicles



1 - Humidity sender for air conditioning system - G260-

- ❑ Removing and installing ⇒ page 202

2 - Sunlight penetration photosensor - G107-

- ❑ Removing and installing ⇒ page 199

3 - Operating and display unit

- ❑ Overview of operating and display units ⇒ page 193
- ❑ Removing and installing ⇒ page 196

4 - Heater and air conditioning unit

- ❑ Assembly overview ⇒ page 131
- ❑ Removing and installing ⇒ page 136
- ❑ Dismantling and assembling ⇒ page 142

5 - Connection

- ❑ For cold-air hose, glove compartment
- ❑ Removing and installing ⇒ page 190

6 - Front right chest vent temperature sensor - G386-

- ❑ Only vehicles with Climatronic.
- ❑ On vehicles with Climatic, apertures in air duct system are sealed with plugs
- ❑ Removing and installing ⇒ page 204

7 - Valve

- ❑ For cold-air hose, glove compartment
- ❑ Removing and installing ⇒ page 191

8 - Cold-air hose, glove compartment

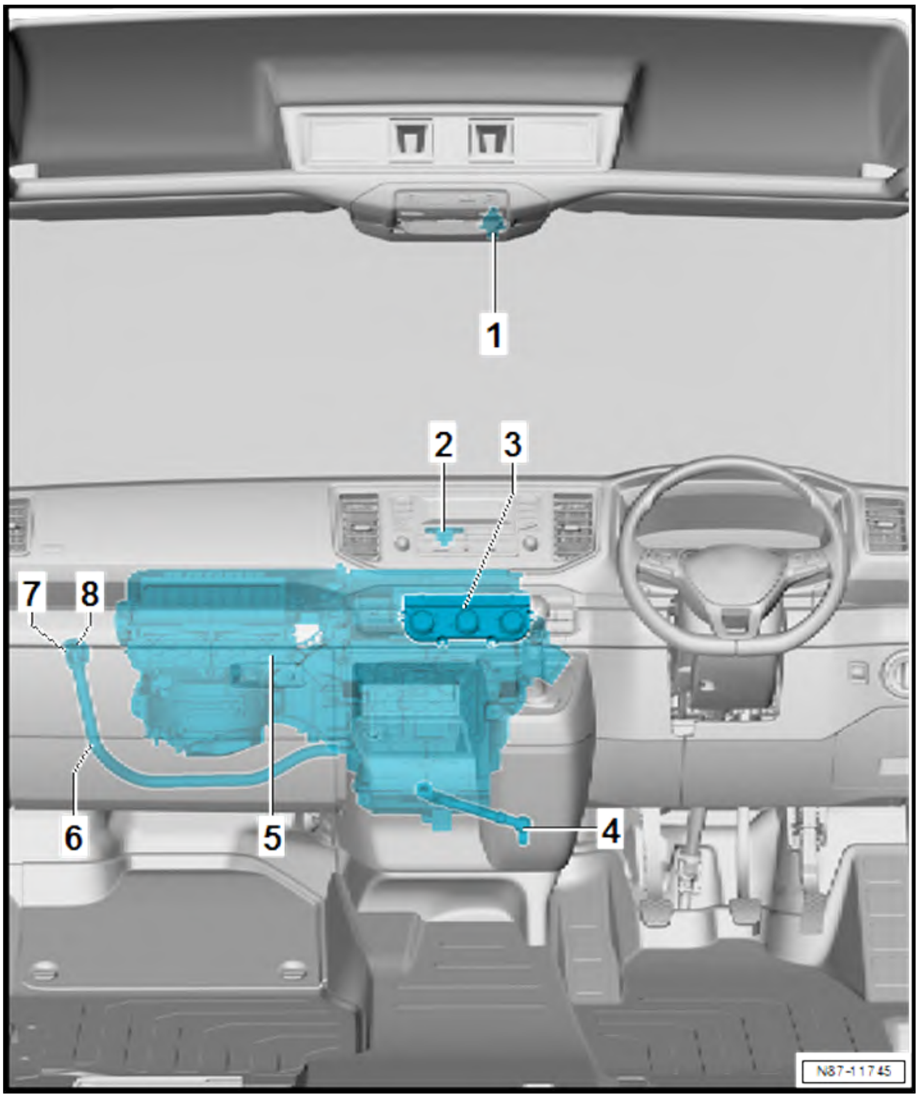
- ❑ Removing and installing ⇒ page 190

9 - Condensation drainage
- ❑ Removing and installing ⇒ page 153
- ❑ Checking ⇒ page 154

10 - Footwell vent temperature sender - G192-
- ❑ Removing and installing ⇒ page 202

11 - Front left chest vent temperature sensor - G385-
- ❑ Only vehicles with Climatronic.
- ❑ On vehicles with Climatic, apertures in air duct system are sealed with plugs
- ❑ Removing and installing ⇒ page 203

1.2.2 Overview of fitting locations - components inside of front passenger compartment, right-hand drive vehicles



1 - Humidity sender for air conditioning system - G260-
- ❑ Removing and installing ⇒ page 202

2 - Sunlight penetration photosensor - G107-
- ❑ Removing and installing ⇒ page 199

3 - Operating and display unit
- ❑ Overview of operating and display units ⇒ page 193
- ❑ Removing and installing ⇒ page 196

4 - Condensation drainage
- ❑ Removing and installing ⇒ page 153
- ❑ Checking ⇒ page 154

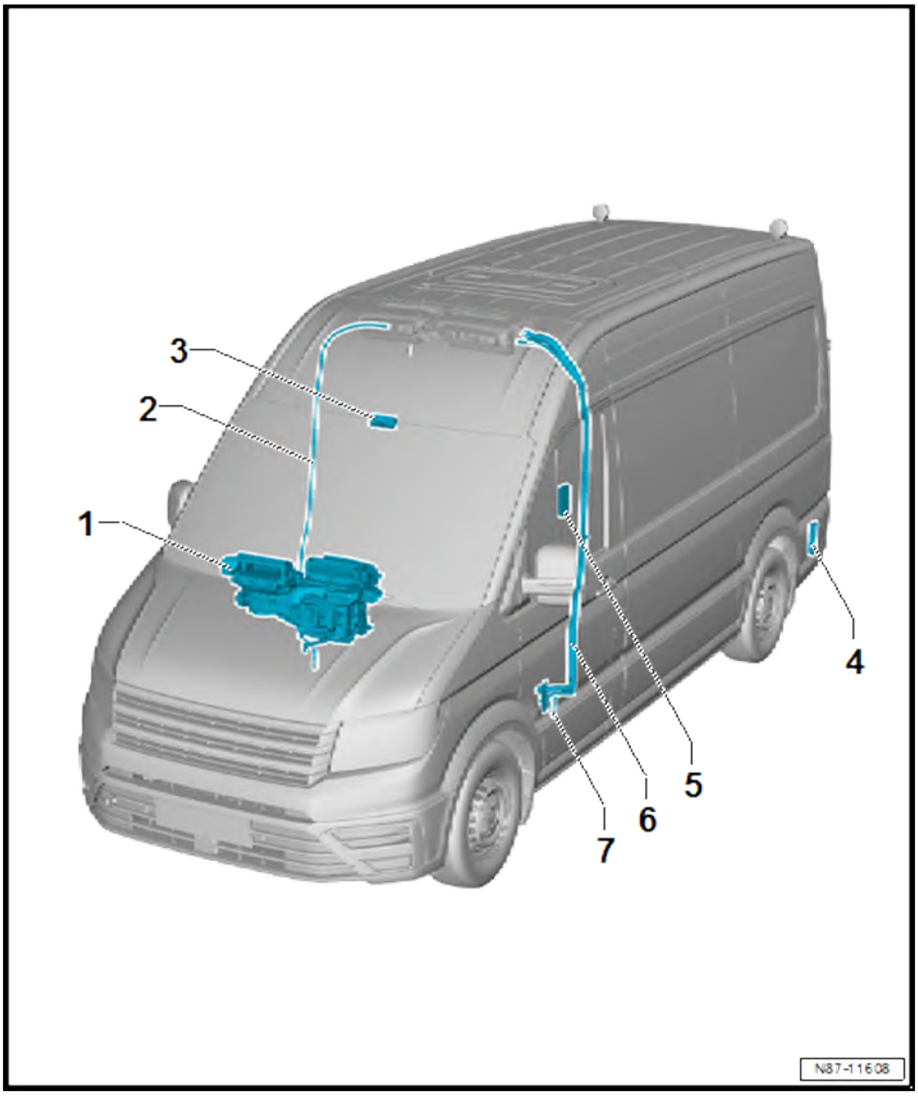
5 - Heater and air conditioning unit
- ❑ Assembly overview ⇒ page 131
- ❑ Removing and installing ⇒ page 136
- ❑ Dismantling and assembling ⇒ page 142

6 - Cold-air hose, glove compartment
- ❑ Removing and installing ⇒ page 190

7 - Valve
- ❑ For cold-air hose, glove compartment
- ❑ Removing and installing ⇒ page 191

8 - Connection
- ❑ For cold-air hose, glove compartment
- ❑ Removing and installing ⇒ page 190

1.3 Overview of fitting locations - components located in rear section of passenger compartment



1 - Heater and air conditioning unit

- ❑ Assembly overview ⇒ page 131
- ❑ Removing and installing ⇒ page 136
- ❑ Dismantling and assembling ⇒ page 142

2 - Condensation drain hose

- ❑ Right
- ❑ Removing and installing ⇒ page 170
- ❑ Checking ⇒ page 172

3 - Operating and display unit in roof console

- ❑ Overview ⇒ page 196 .
- ❑ Removing and installing ⇒ page 198

4 - Forced ventilation in passenger compartment

- ❑ Left
- ❑ Removing and installing ⇒ page 182
- ❑ Checking ⇒ page 184

5 - Forced ventilation in passenger compartment

- ❑ Right
- ❑ Removing and installing ⇒ page 182
- ❑ Checking ⇒ page 184

6 - Connection for refrigerant lines on underbody - expansion valve for air conditioner refrigerant - N697-

- ❑ Removing and installing ⇒ page 93

7 - Condensation drain hose

- ❑ Left
- ❑ Removing and installing ⇒ page 169
- ❑ Checking ⇒ page 172

2 Refrigerant circuit

2.1 System overview - refrigerant circuit

2.1.1 System overview - refrigerant circuit, vehicles with one evaporator, transverse engine

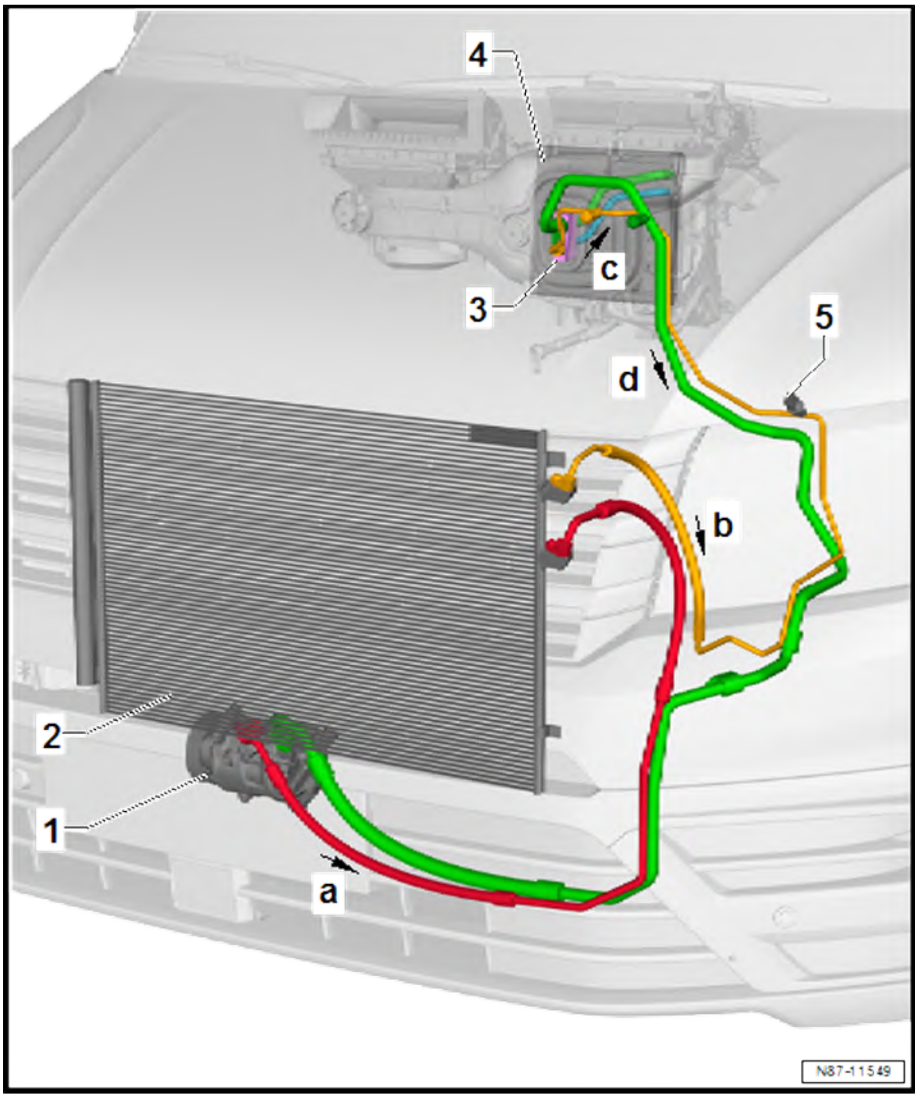
1 - Air conditioner compressor
- ❑ Assembly overview ⇒ page 100
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104

2 - Condenser
- ❑ With desiccant bag/desiccant cartridge
- ❑ Assembly overview ⇒ page 54
- ❑ Removing and installing ⇒ page 62

3 - Expansion valve
- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

4 - Evaporator
- ❑ Removing and installing ⇒ page 134

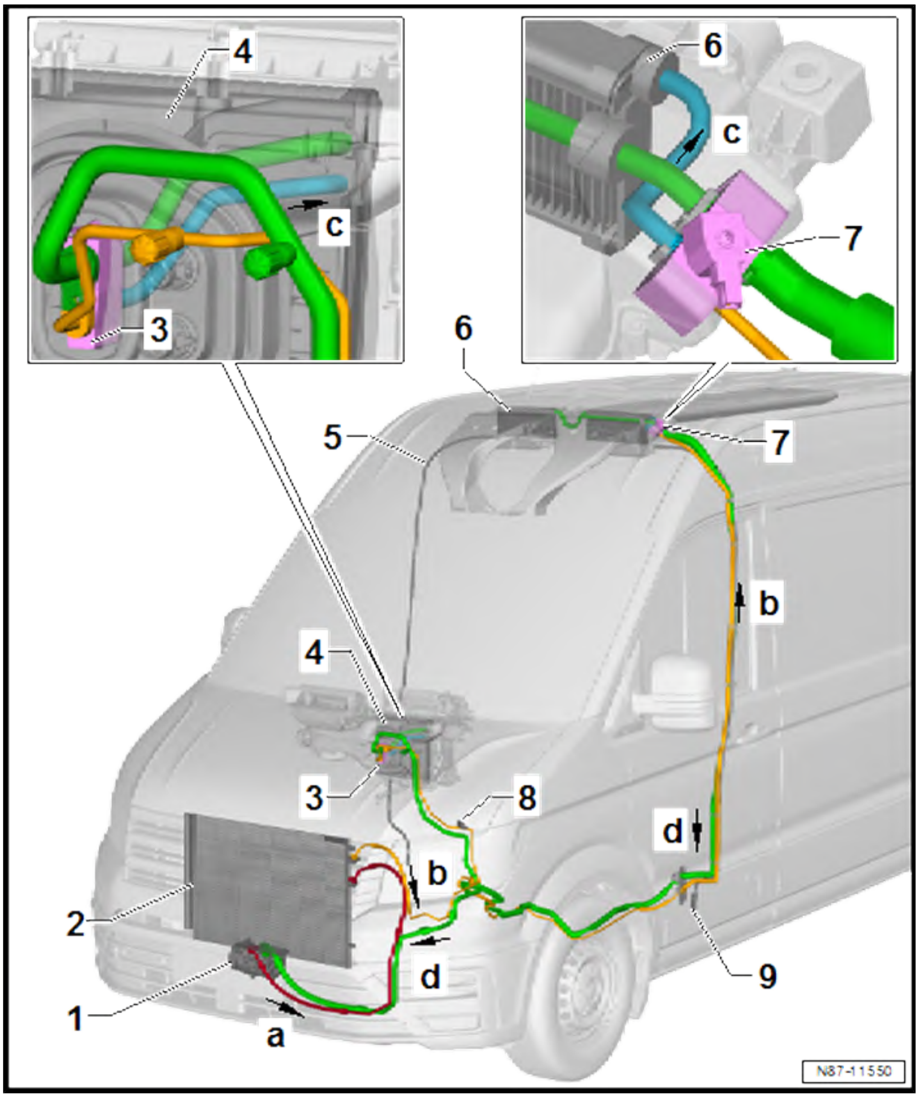
5 - Pressure sender for refrigerant circuit - G805-
- ❑ Removing and installing ⇒ page 54



Directions of flow and assembly statuses in high and low-pressure area

- ♦ -Arrow a- - high pressure, gaseous (red)
- ♦ -Arrow b- - high pressure, liquid (yellow)
- ♦ -Arrow c- - low pressure, gaseous (blue)
- ♦ -Arrow d- - low pressure, liquid (green)

2.1.2 System overview - refrigerant circuit, vehicles with two evaporators, transverse engine



1 - Air conditioner compressor
- ❑ Assembly overview ⇒ page 100
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104

2 - Condenser
- ❑ With desiccant bag/desiccant cartridge
- ❑ Assembly overview ⇒ page 54
- ❑ Removing and installing ⇒ page 62

3 - Expansion valve
- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

4 - Evaporator
- ❑ Removing and installing ⇒ page 134

5 - Condensation drain hose
- ❑ Right
- ❑ Removing and installing ⇒ page 170
- ❑ Checking ⇒ page 172

6 - Evaporator
- ❑ Removing and installing ⇒ page 157

7 - Expansion valve for air conditioner refrigerant - N697-
- ❑ Removing and installing ⇒ page 59

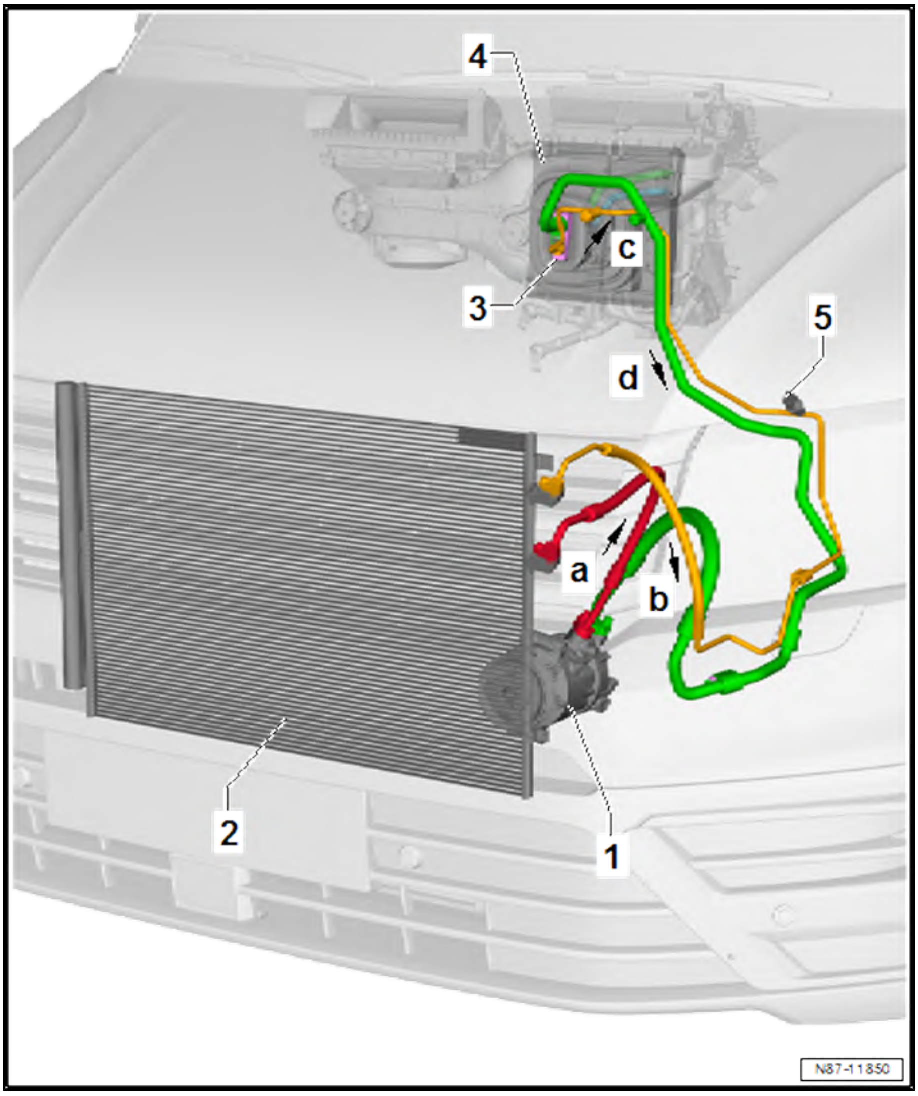
8 - Pressure sender for refrigerant circuit - G805-
- ❑ Removing and installing ⇒ page 54

9 - Condensation drain hose
- ❑ Left
- ❑ Removing and installing ⇒ page 169
- ❑ Checking ⇒ page 172

Directions of flow and assembly statuses in high and low-pressure area

- ♦ -Arrow a- - high pressure, gaseous (red)
- ♦ -Arrow b- - high pressure, liquid (yellow)
- ♦ -Arrow c- - low pressure, gaseous (blue)
- ♦ -Arrow d- - low pressure, liquid (green)

2.1.3 System overview - refrigerant circuit, vehicles with one evaporator, straight engine



1 - Air conditioner compressor
- ❑ Assembly overview ⇒ page 100
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104

2 - Condenser
- ❑ With desiccant bag/desiccant cartridge
- ❑ Assembly overview ⇒ page 54
- ❑ Removing and installing ⇒ page 62

3 - Expansion valve
- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

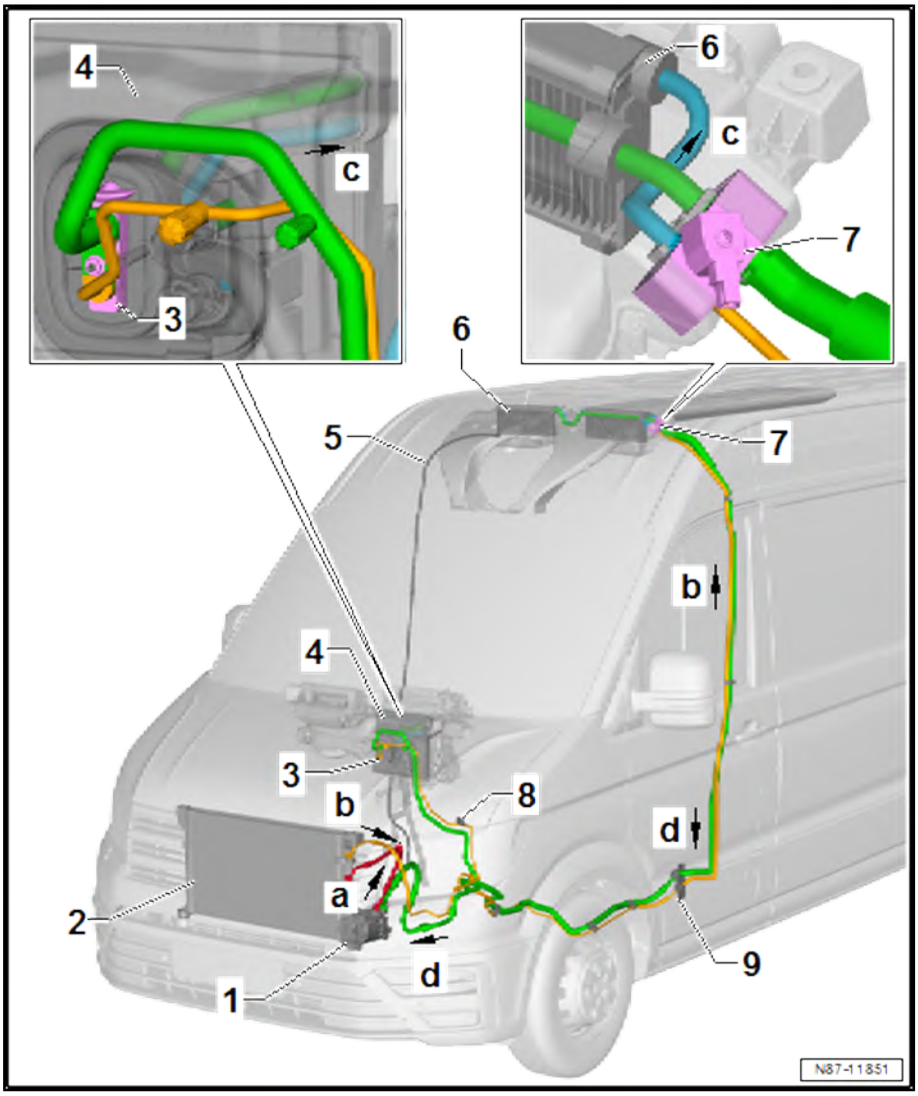
4 - Evaporator
- ❑ Removing and installing ⇒ page 134

5 - Pressure sender for refrigerant circuit - G805-
- ❑ Removing and installing ⇒ page 54

Directions of flow and assembly statuses in high and low-pressure area

- ♦ -Arrow a- - high pressure, gaseous (red)
- ♦ -Arrow b- - high pressure, liquid (yellow)
- ♦ -Arrow c- - low pressure, gaseous (blue)
- ♦ -Arrow d- - low pressure, liquid (green)

2.1.4 System overview - refrigerant circuit, vehicles with two evaporators, straight engine



1 - Air conditioner compressor

- ❑ Assembly overview ⇒ page 100
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104

2 - Condenser

- ❑ With desiccant bag/desiccant cartridge
- ❑ Assembly overview ⇒ page 54
- ❑ Removing and installing ⇒ page 62

3 - Expansion valve

- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

4 - Evaporator

- ❑ Removing and installing ⇒ page 134

5 - Condensation drain hose

- ❑ Right
- ❑ Removing and installing ⇒ page 170
- ❑ Checking ⇒ page 172

6 - Evaporator

- ❑ Removing and installing ⇒ page 157

7 - Expansion valve for air conditioner refrigerant - N697-

- ❑ Removing and installing ⇒ page 59

8 - Pressure sender for refrigerant circuit - G805-

- ❑ Removing and installing ⇒ page 54

9 - Condensation drain hose

- ❑ Left
- ❑ Removing and installing ⇒ page 169
- ❑ Checking ⇒ page 172

Directions of flow and assembly statuses in high and low-pressure area

- ♦ -Arrow a- - high pressure, gaseous (red)
- ♦ -Arrow b- - high pressure, liquid (yellow)
- ♦ -Arrow c- - low pressure, gaseous (blue)
- ♦ -Arrow d- - low pressure, liquid (green)

2.2 Assembly overview - refrigerant lines

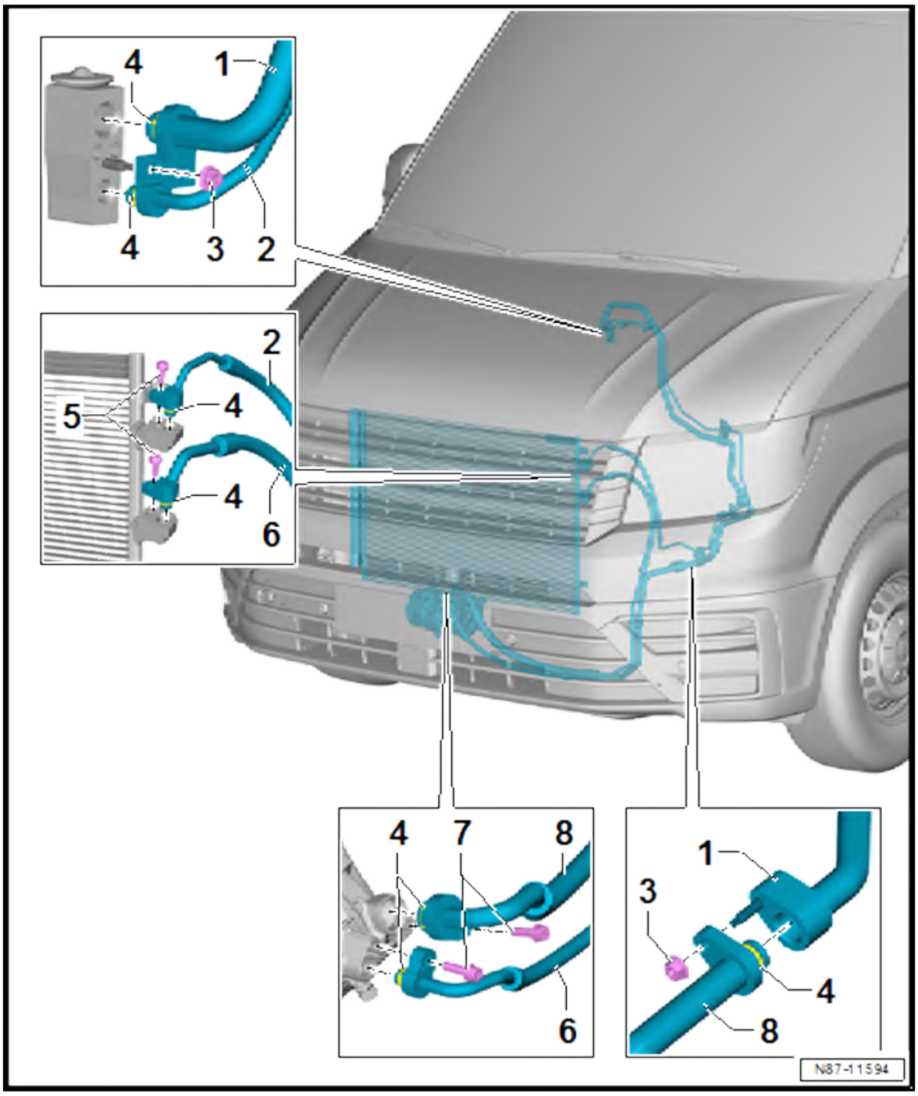
⇒ “2.2.1 Assembly overview - refrigerant lines, vehicles with one evaporator, transverse engine”, page 50

⇒ “2.2.2 Assembly overview - refrigerant lines, vehicles with two evaporators, transverse engine”, page 51

⇒ “2.2.3 Assembly overview - refrigerant lines, vehicles with one evaporator, straight engine”, page 52

⇒ “2.2.4 Assembly overview - refrigerant lines, vehicles with two evaporators, straight engine”, page 53

2.2.1 Assembly overview - refrigerant lines, vehicles with one evaporator, transverse engine



1 - Top refrigerant line from evaporator to air conditioner compressor

- ❑ Removing and installing ⇒ page 83

2 - Refrigerant line from condenser to evaporator

- ❑ Removing and installing ⇒ page 79

3 - Nuts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 3 (page 51)

4 - Oil seals

- ❑ Renew after removal
- ❑ Qty. 8
- ❑ Moisten with refrigerant oil before installation

5 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 10 (page 51)

6 - Refrigerant line from air conditioner compressor to condenser

- ❑ Removing and installing ⇒ page 75

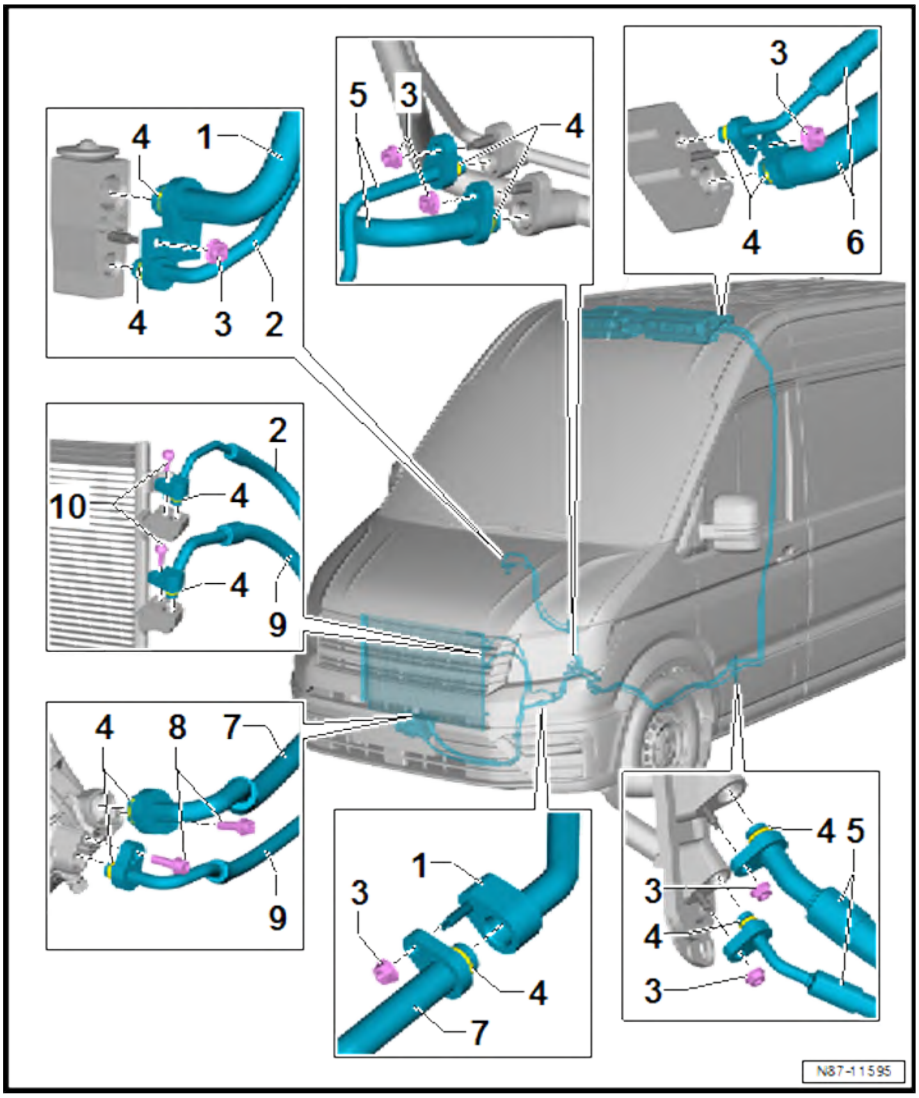
7 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 4 (page 100)

8 - Bottom refrigerant line from evaporator to air conditioner compressor

- ❑ Removing and installing ⇒ page 87

2.2.2 Assembly overview - refrigerant lines, vehicles with two evaporators, transverse engine



1 - Top refrigerant line from evaporator to air conditioner compressor

- ❑ Removing and installing ⇒ page 83

2 - Refrigerant line from condenser to evaporator

- ❑ Removing and installing ⇒ page 79

3 - Nuts

- ❑ Qty. 7
- ❑ 8 Nm

4 - Oil seals

- ❑ Renew after removal
- ❑ Qty. 13
- ❑ Moisten with refrigerant oil before installation

5 - Refrigerant lines on underbody

- ❑ Removing and installing ⇒ page 90

6 - Connection for refrigerant lines on underbody to expansion valve for air conditioner refrigerant - N697-

- ❑ Removing and installing ⇒ page 93

7 - Bottom refrigerant line from evaporator to air conditioner compressor

- ❑ Removing and installing ⇒ page 87

8 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 4 (page 100)

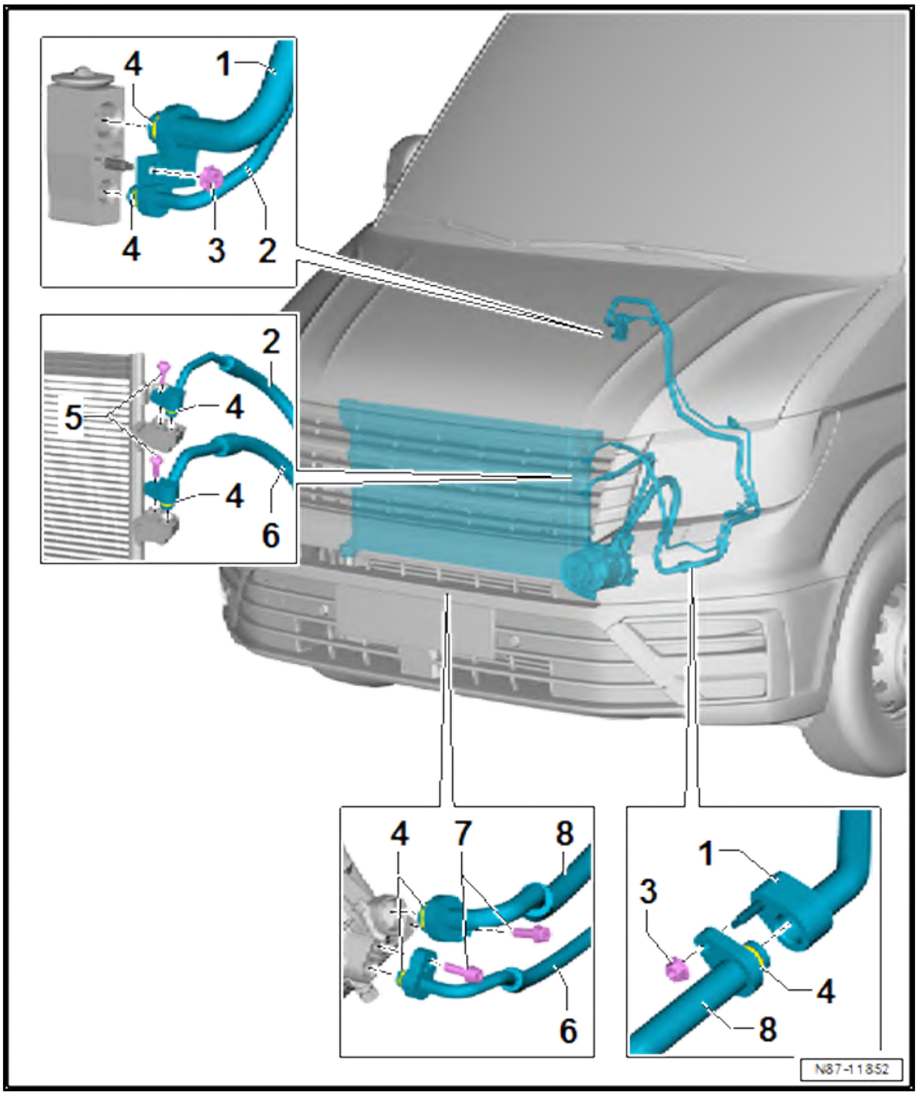
9 - Refrigerant line from air conditioner compressor to condenser

- ❑ Removing and installing ⇒ page 75

10 - Bolts

- ❑ Qty. 2
- ❑ 8 Nm

2.2.3 Assembly overview - refrigerant lines, vehicles with one evaporator, straight engine



1 - Top refrigerant line from evaporator to air conditioner compressor

- ❑ Removing and installing ⇒ page 83

2 - Refrigerant line from condenser to evaporator

- ❑ Removing and installing ⇒ page 79

3 - Nuts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 3 (page 51)

4 - Oil seals

- ❑ Renew after removal
- ❑ Qty. 8
- ❑ Moisten with refrigerant oil before installation

5 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 10 (page 51)

6 - Refrigerant line from air conditioner compressor to condenser

- ❑ Removing and installing ⇒ page 75

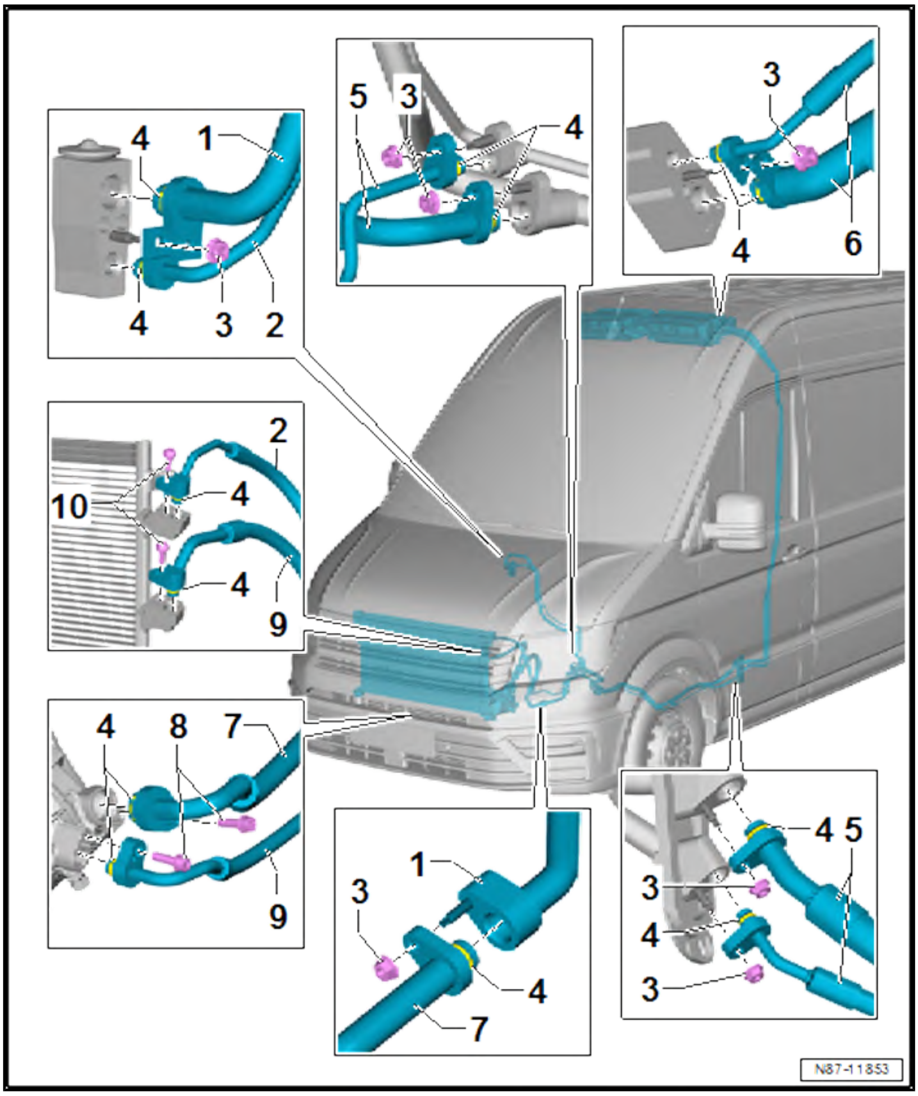
7 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 4 (page 100)

8 - Bottom refrigerant line from evaporator to air conditioner compressor

- ❑ Removing and installing ⇒ page 87

2.2.4 Assembly overview - refrigerant lines, vehicles with two evaporators, straight engine



1 - Top refrigerant line from evaporator to air conditioner compressor

- ❑ Removing and installing ⇒ page 83

2 - Refrigerant line from condenser to evaporator

- ❑ Removing and installing ⇒ page 79

3 - Nuts

- ❑ Qty. 7
- ❑ Specified torque ⇒ Item 3 (page 51)

4 - Oil seals

- ❑ Renew after removal
- ❑ Qty. 13
- ❑ Moisten with refrigerant oil before installation

5 - Refrigerant lines on underbody

- ❑ Removing and installing ⇒ page 90

6 - Connection for refrigerant lines on underbody to expansion valve for air conditioner refrigerant - N697-

- ❑ Removing and installing ⇒ page 93

7 - Bottom refrigerant line from evaporator to air conditioner compressor

- ❑ Removing and installing ⇒ page 87

8 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 4 (page 100)

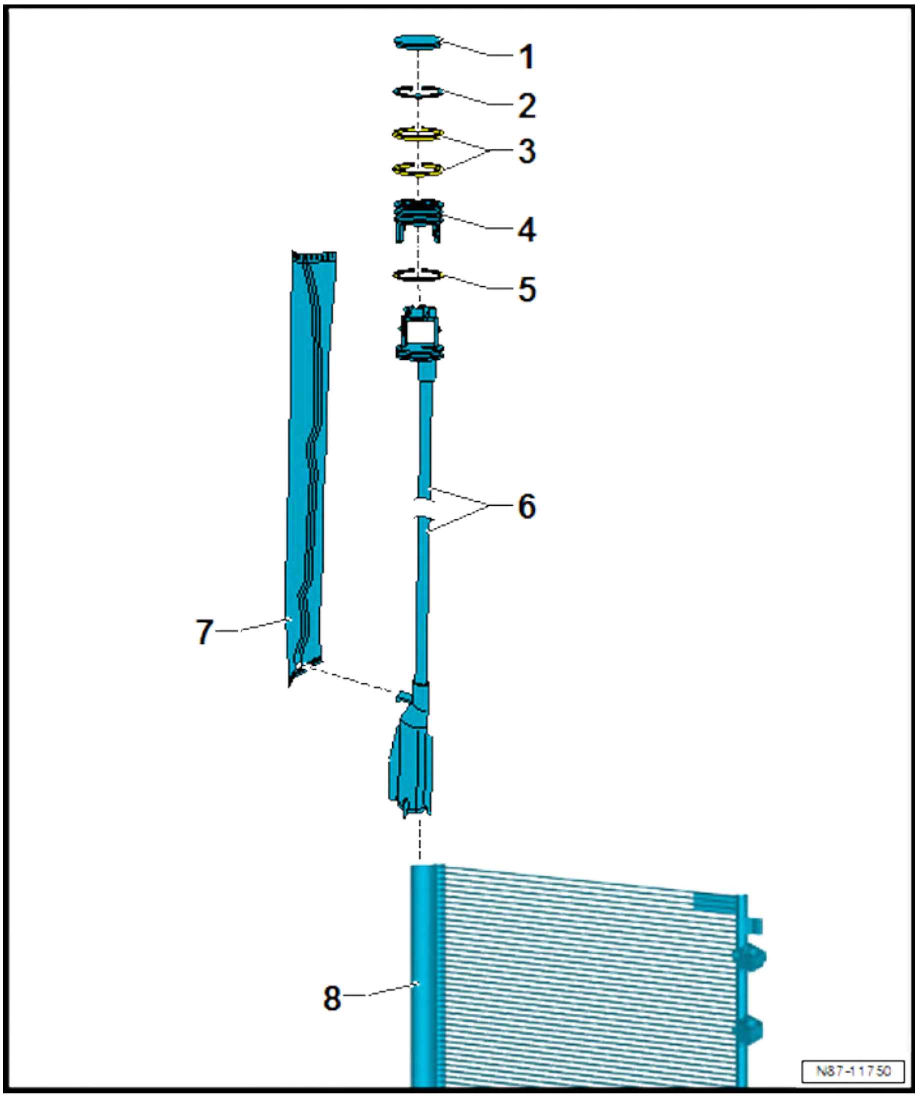
9 - Refrigerant line from air conditioner compressor to condenser

- ❑ Removing and installing ⇒ page 75

10 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Item 10 (page 51)

2.3 Assembly overview - condenser



1 - Cap
- ❑ Renew after removal
- ❑ Not available as a separate part
- ❑ 4.5 Nm

2 - Circlip
- ❑ Renew after removal
- ❑ Not available as a separate part

3 - Oil seals
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ Not available as a separate part

4 - Cap
- ❑ Renew after removal
- ❑ Not available as a separate part

5 - Seal
- ❑ Renew after removal
- ❑ Not available as a separate part

6 - Desiccant bag carrier
- ❑ With integrated strainer

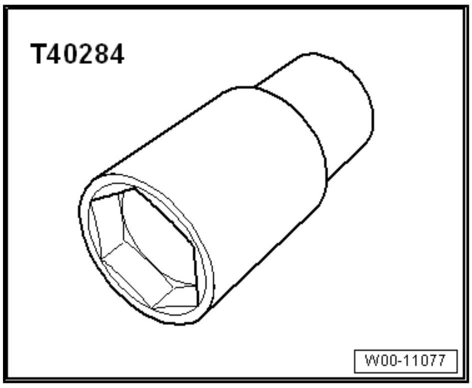
7 - Desiccant bag
- ❑ Renew after removal
- ❑ Not available as a separate part
- ❑ Removing and installing ⇒ page 68
- ❑ Attached to desiccant bag carrier

8 - Receiver
- ❑ Integrated in condenser

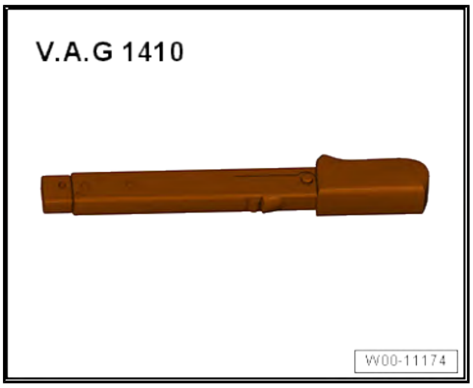
2.4 Removing and installing refrigerant circuit pressure sender - G805-

Special tools and workshop equipment required

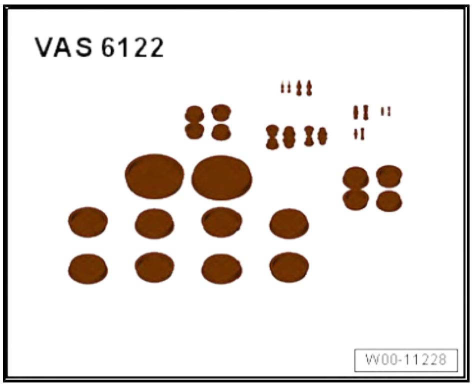
◆ Socket 24 mm - T40284-

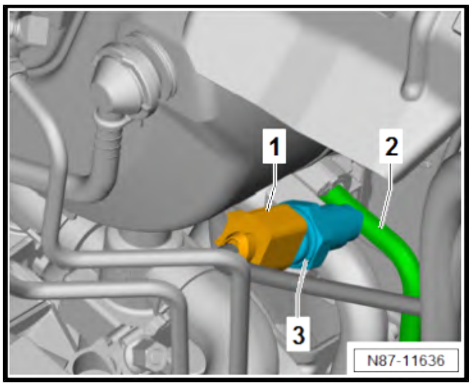


◆ Torque wrench - V.A.G 1410-



◆ Set of plugs for engine - VAS 6122-





Removing

– Separate connector -1-.

⚠ CAUTION
Risk of freezing injury caused by escaping pressurised refrigerant. If handled incorrectly, union could break off and refrigerant could escape. There is a risk of injury to the skin and parts of the body due to freezing. – Wear protective gloves. – Wear protective goggles. – Counterhold refrigerant lines using a suitable tool.

– Counterhold refrigerant line -2- using a suitable tool. While doing so, detach pressure sender for refrigerant circuit - G805- -3- using socket 24 mm - T40284- .

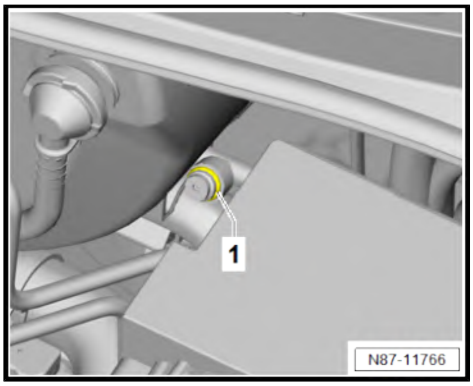
⚠ CAUTION
Risk of freezing injury caused by escaping pressurised refrigerant. There is a risk of injury to the skin and parts of the body due to freezing. – Wear protective gloves. – Wear protective goggles. – If, when detaching the pressure sender, refrigerant escapes from the refrigerant line for longer than 1 second, tighten the pressure sender and renew the non-return valve that is defective. – Extract refrigerant and open the refrigerant circuit immediately afterwards. – If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .

– While doing so, detach pressure sender for refrigerant circuit - G805- -3- using socket, 24 mm - T40284- .

– Check if it is necessary to purge the refrigerant circuit ⇒ page 98 .

Installing



Install in reverse order of removal, observing the following:

– Moisten new seal -1- with refrigerant oil before installing.

Specified torques

Component	Specified torque
Pressure sender for refrigerant circuit - G805- -3-.	8 Nm

2.5 Removing and installing expansion valve

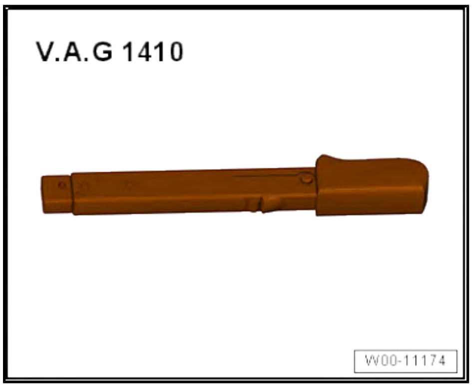
⇒ “2.5.1 Removing and installing expansion valve”, page 57

⇒ “2.5.2 Removing and installing expansion valve for air conditioner system N697 ”, page 59

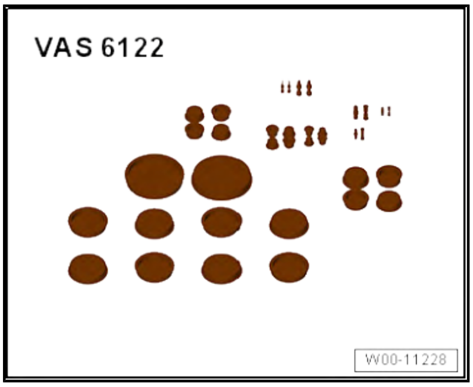
2.5.1 Removing and installing expansion valve

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1410-



- ♦ Set of plugs for engine - VAS 6122-



Removing

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

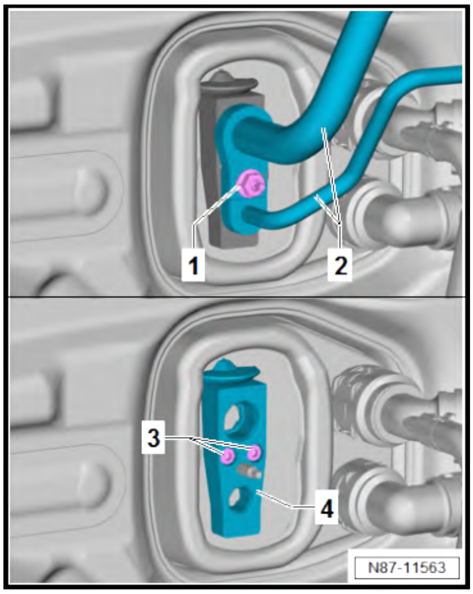
CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.
– Wear protective goggles.
– Extract refrigerant and open the refrigerant circuit immediately afterwards.
– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .
– Unscrew nut -1-.



NOTICE

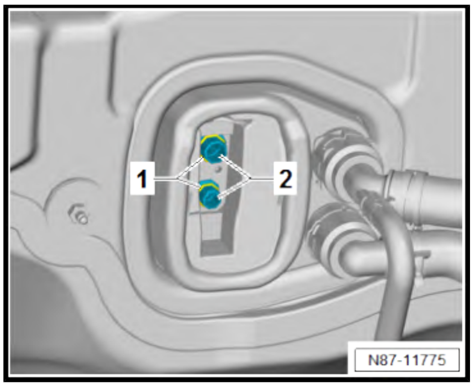
Risk of damage to refrigerant lines from rupture of inner foil.

– Never bend refrigerant lines to a radius less than 100 mm.

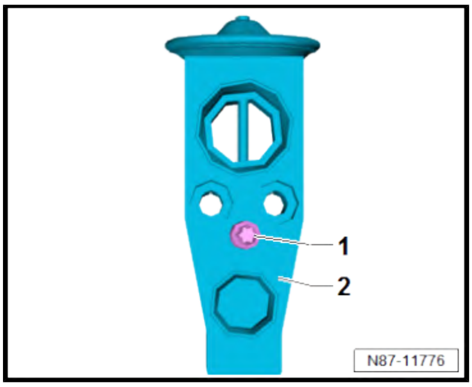
– Remove refrigerant lines -2- and lay them to one side.
– Unscrew bolts -3-.
– Remove expansion valve -4-.
– Check if it is necessary to purge the refrigerant circuit ⇒ page 98 .

Installing

Install in reverse order of removal, observing the following:



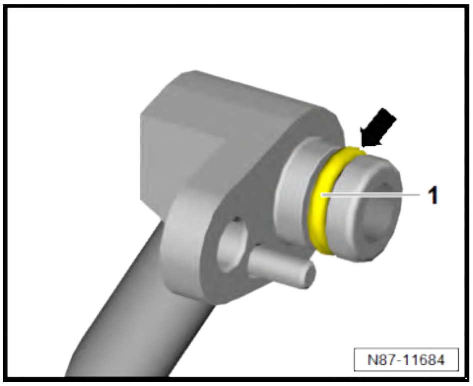
– Check connecting pipe -2- for soiling and damage, clean or renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Moisten new seals -1- with refrigerant oil before installing.
– Press new seals -1- on connecting pipes -2-.
– Check refrigerant lines for soiling and damage, clean or renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



Procedure for installing new expansion valve

– Check stud -1- for damage; renew stud -1- if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Screw stud -1- in new expansion valve -2- and tighten ⇒ page 59 .



Continued for all vehicles

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

Specified torques

♦ ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

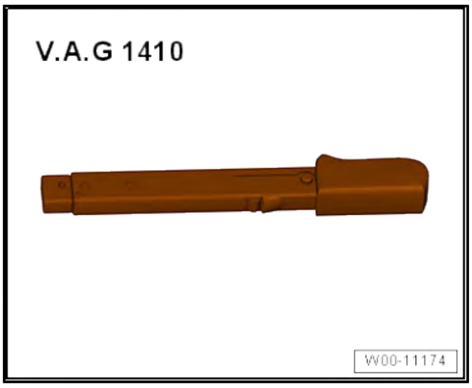
♦ ⇒ “5.2 Assembly overview - evaporator housing”, page 133

Component	Specified torque
Stud -1-	8 Nm

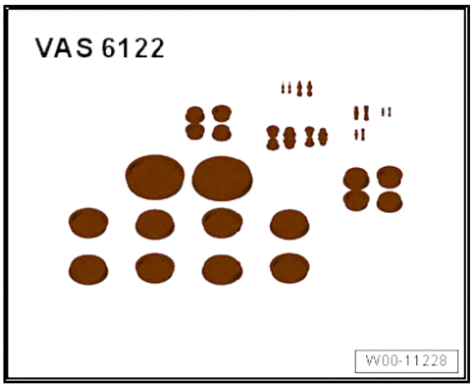
2.5.2 Removing and installing expansion valve for air conditioner system - N697-

Special tools and workshop equipment required

- Torque wrench - V.A.G 1410-



- Set of plugs for engine - VAS 6122-



Removing

– Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.

– Wear protective goggles.

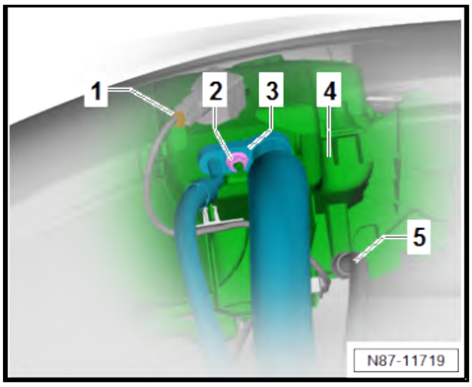
– Extract refrigerant and open the refrigerant circuit immediately afterwards.

– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .

– Disconnect connector -1-.

– Unscrew nut -2-.

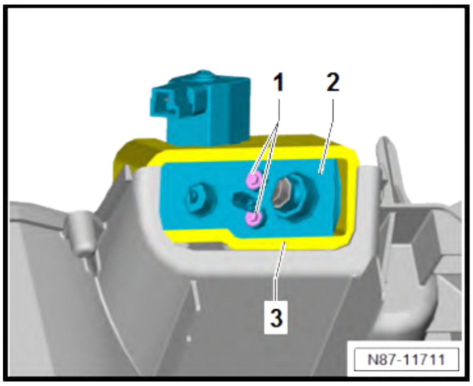


NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

– **Never bend refrigerant lines to a radius less than 100 mm.**

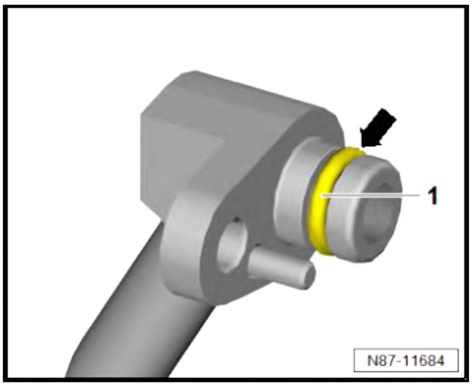
– Detach refrigerant lines -3- together with condensation drain hose -5- from evaporator housing -4-, and lay them aside.



– Unscrew bolts -1-.

– Remove expansion valve for air conditioner refrigerant - N697- -2- together with seal -3- from evaporator.

– Check if it is necessary to purge the refrigerant circuit ⇒ page 98 .



Installing

Install in reverse order of removal, observing the following:

– Check seal -3- of expansion valve for air conditioner refrigerant - N697- -2- for damage and renew if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Prior to installation of expansion valve for air conditioner refrigerant - N697- , moisten new oil seals with refrigerant oil.

– Moisten new seals with refrigerant oil before installing refrigerant line.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

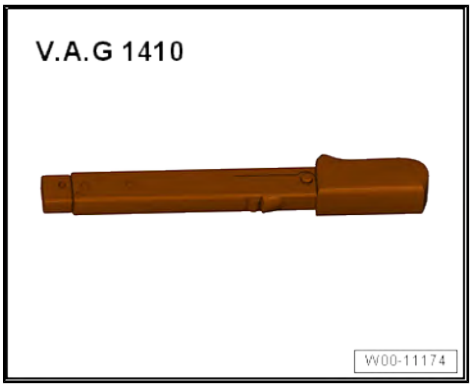
Specified torques

♦ ⇒ “6.1 Assembly overview - heater and air conditioning unit”, page 155

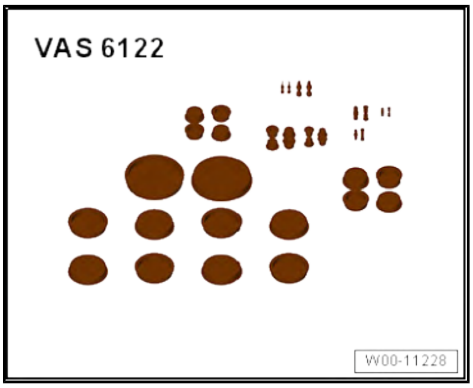
2.6 Removing and installing condenser

Special tools and workshop equipment required

- Torque wrench - V.A.G 1410-



- Set of plugs for engine - VAS 6122-



Removing

Vehicles without radiator blind

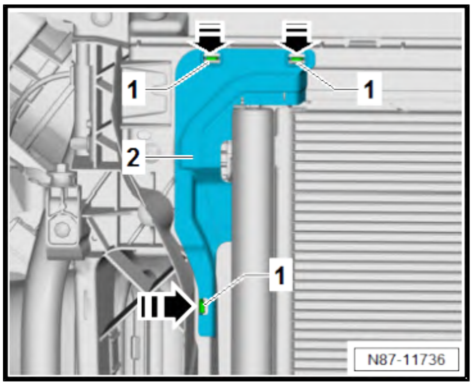
– Remove front bumper cover ⇒ General body repairs, exterior; Rep. gr. 63 ; Front bumper; Removing and installing bumper cover .

Vehicles with radiator blind

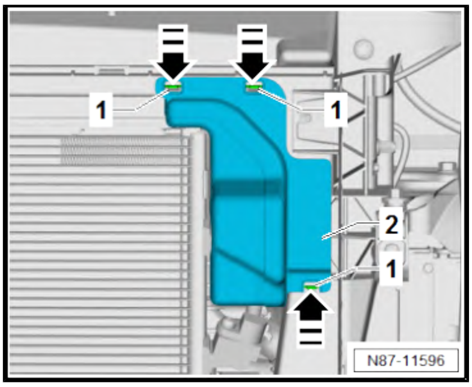
– Remove radiator blind ⇒ Rep. gr. 19 ; Radiator/radiator fan; Removing and installing radiator blind .

Continued for all vehicles

– Release locking lugs -1- in direction of -arrow-. While doing this, detach air deflector -2-.

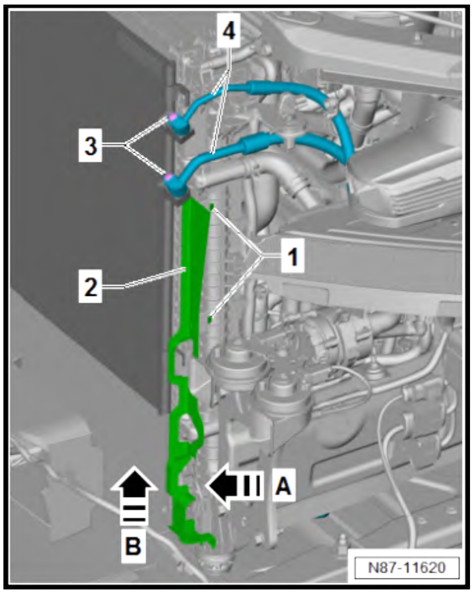


– Release locking lugs -1- in direction of -arrow-. While doing this, detach air deflector -2-.



– Release locking lugs -1-. While doing so, press air deflector -2- in direction of -arrow A-.

– Remove air deflector -2- in direction of -arrow B-.



Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

– **Wear protective gloves.**
– **Wear protective goggles.**
– **Extract refrigerant and open the refrigerant circuit immediately afterwards.**
– **If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.**

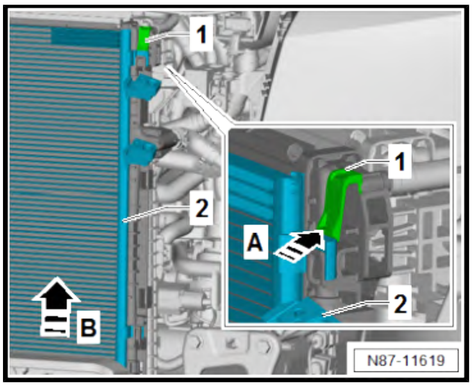
– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .

– Unscrew bolts -3-.

– Remove refrigerant lines -4-.

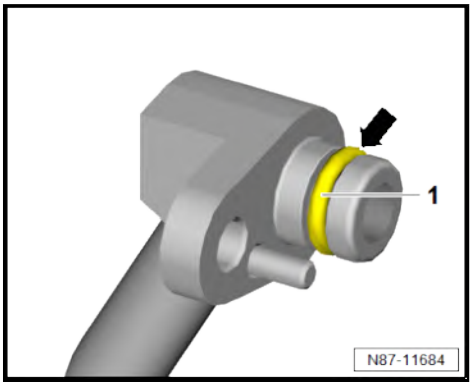
Note

Only the left side is described as follows. The description for the right side follows the same pattern.



– Press locking lug -1- in -direction of arrow A-. While doing this, remove condenser -2- from mountings in direction of -arrow B-.

– Check if it is necessary to purge the refrigerant circuit ⇒ page 98 .



Installing

Install in reverse order of removal, observing the following:

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– Never start the engine if the refrigerant circuit is empty.

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

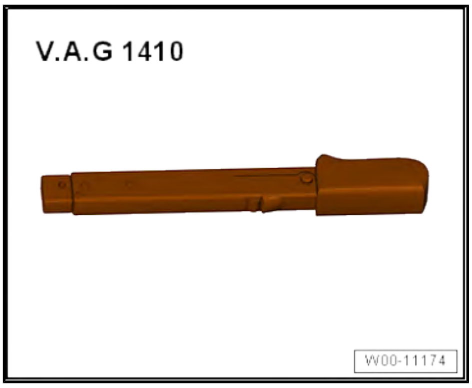
Specified torques

♦ ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

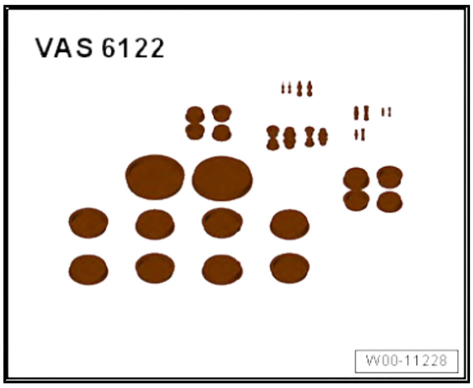
2.7 Separating and connecting refrigerant lines on condenser

Special tools and workshop equipment required

- Torque wrench - V.A.G 1410-



- Set of plugs for engine - VAS 6122-



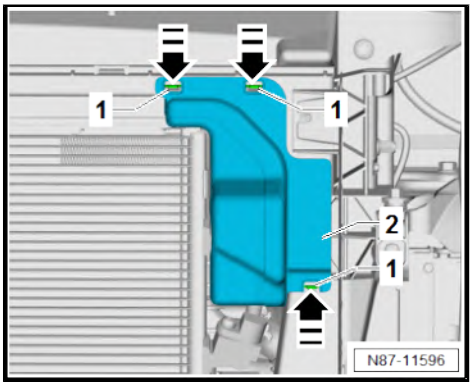
Disconnecting

Vehicles without radiator blind

– Remove front bumper cover ⇒ General body repairs, exterior; Rep. gr. 63 ; Front bumper; Removing and installing bumper cover .

Vehicles with radiator blind

– Remove radiator blind ⇒ Rep. gr. 19 ; Radiator/radiator fan; Removing and installing radiator blind .



Continued for all vehicles

– Release locking lugs -1- in direction of -arrow-. While doing this, detach air deflector -2-.

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

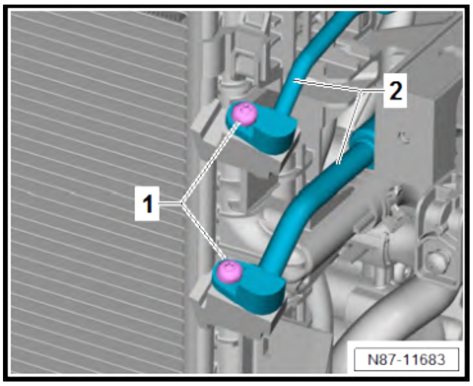
CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.
– Wear protective goggles.
– Extract refrigerant and open the refrigerant circuit immediately afterwards.
– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .



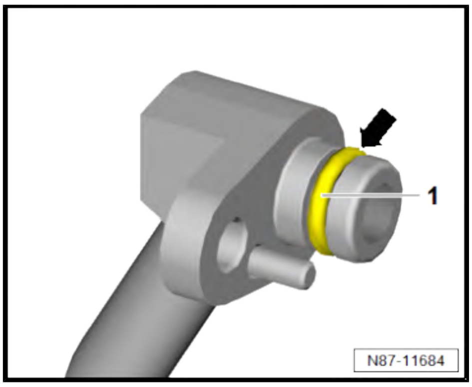
– Unscrew bolts -1-.

– Remove refrigerant lines -2-.

NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

– Never bend refrigerant lines to a radius less than 100 mm.



Connecting

Connect in reverse order of removal, noting the following:

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

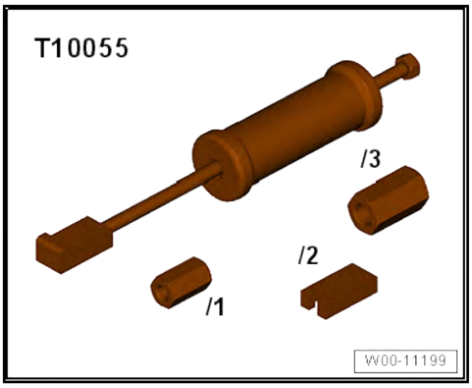
Specified torques

♦ ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

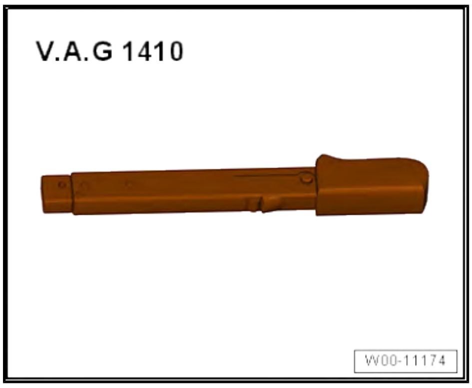
2.8 Removing and installing desiccant bag or cartridge

Special tools and workshop equipment required

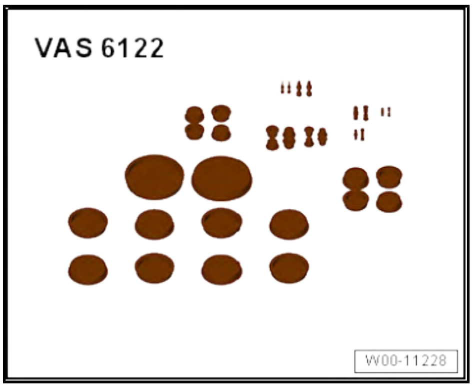
♦ Puller - T10055-



- Torque wrench - V.A.G 1410-



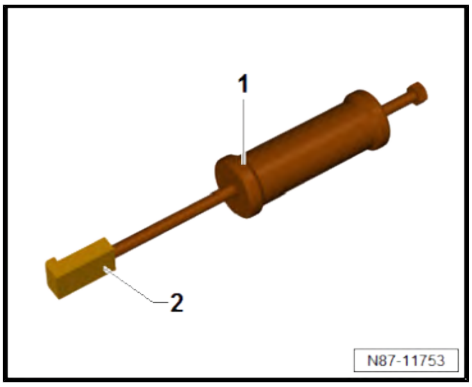
- Set of plugs for engine - VAS 6122-



- Long-nose pliers

Prepare special tool as follows

– If a socket -2- is fitted to puller - T10055- -1-, unscrew it.



Removing

– Remove front bumper cover ⇒ General body repairs, exterior; Rep. gr. 63 ; Front bumper; Removing and installing bumper cover .

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

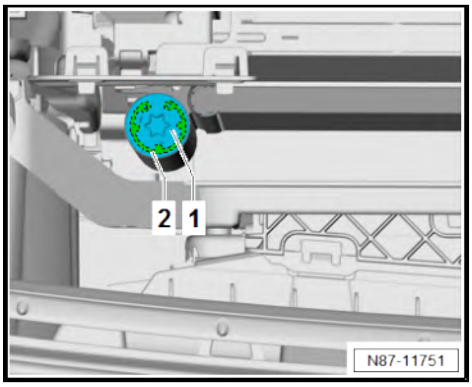
There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.

– Wear protective goggles.

– Extract refrigerant and open the refrigerant circuit immediately afterwards.

– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.



– Unscrew deflector cap -1-.

– Remove circlip -2-.

Vehicles with second evaporator

– Check display on air conditioner service station .

• Pressure above 0 bar: evacuate refrigerant circuit again ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station or ⇒ Air conditioning systems with R1234yf refrigerant - General information; Rep. gr. 87 ; Working with air conditioner service station; Evacuating refrigerant circuit .

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

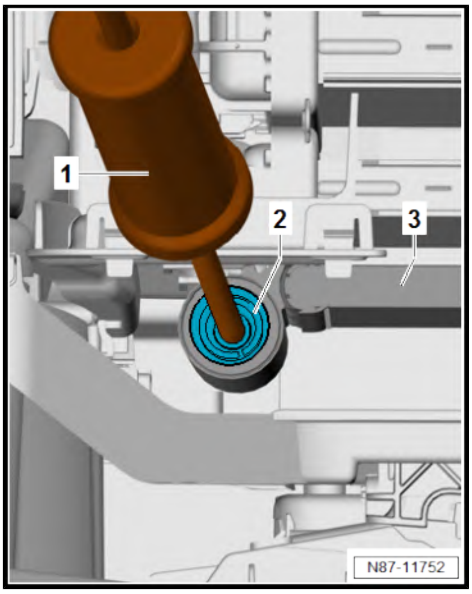
There is a risk of injury to the skin and parts of the body due to freezing.

– **Wear protective gloves.**

– **Wear protective goggles.**

– **Extract refrigerant and open the refrigerant circuit immediately afterwards.**

– **If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.**



Continued for all vehicles

– Screw prepared puller - T10055- -1- into cap -2-.

– Pull cap -2- together with desiccant bag/desiccant cartridge out of condenser -3- using puller - T10055- -1-.

– Should desiccant bag remain in condenser -3-, use long-nose pliers to pull it out of condenser -3-.

– Immediately seal open receiver with clean plugs from engine bung set - VAS 6122- .

– Check if it is necessary to purge the refrigerant circuit ⇒ page 98 .

Installing

Install in reverse order of removal, observing the following:

– Renew desiccant bag ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check cap -2- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

– Clean upper area of inner surface of receiver, then moisten area with refrigerant oil.

– Moisten new seals with refrigerant oil before installing.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– Never start the engine if the refrigerant circuit is empty.

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

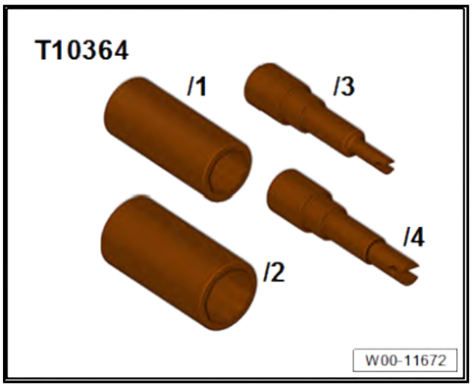
Specified torques

♦ ⇒ “2.3 Assembly overview - condenser”, page 54

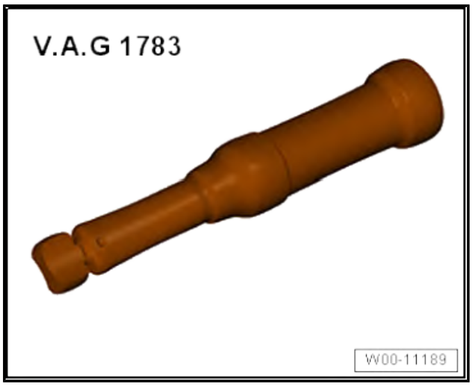
2.9 Removing and installing evacuating and charging valves on low and high-pressure side

Special tools and workshop equipment required

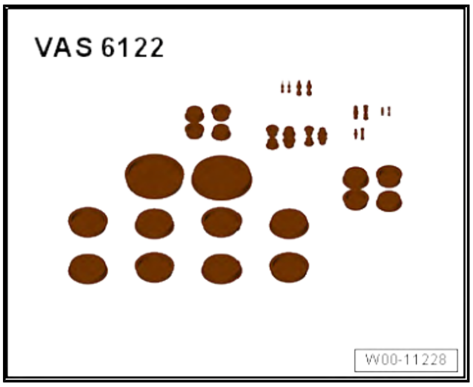
- ♦ Adapter set for service connections - T10364-



- ♦ Torque wrench - V.A.G 1783-



- ♦ Set of plugs for engine - VAS 6122-



Removing

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

CAUTION

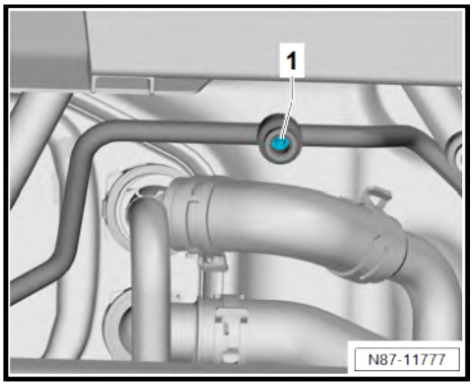
Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

- **Wear protective gloves.**
- **Wear protective goggles.**
- **Extract refrigerant and open the refrigerant circuit immediately afterwards.**
- **If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.**

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .

– Unscrew valve core -1- with adapter set for service connections - T10364- .



Installing

Install in reverse order of removal, observing the following:

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

- **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

Specified torques

Component	Specified torque
Valve insert -1-	2.5 Nm

2.10 Commissioning of air conditioning system after filling refrigerant circuit

NOTICE

Risk of damage to air conditioner compressor. Refrigerant oil may accumulate in the compression chamber of a removed air conditioner compressor.

- **After installing a new air conditioner compressor or adding new refrigerant oil, fully turn the air conditioner compressor 10 times by hand before the poly V-belt is fitted.**

- Switch on ignition.
- Set fresh air blower speed to at least level 3 on operating and display unit.
- Set temperature to max. cold on operating and display unit.
- Set air distribution to torso on operating and display unit.
- Start engine with air conditioner compressor switched off (version with magnetic clutch).
- Following idling speed stabilisation, switch on air conditioning system using A/C button on operating and display unit.
- Allow engine to run for at least 2 minutes at less than 1500 rpm with air conditioning system switched on.

NOTICE

Risk of damage to the air conditioner compressor or air conditioner service station.

Opening the valves with the air conditioning system switched on can cause a short circuit between the high pressure and low pressure side.

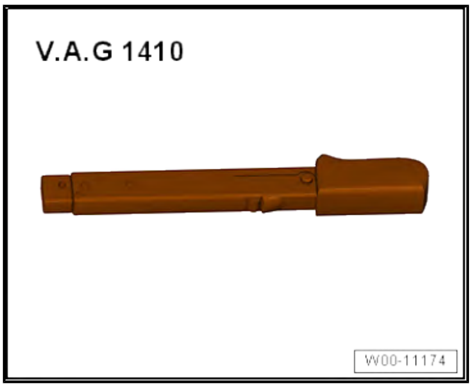
- **Never open valves on the high pressure or low pressure side with the air conditioning system switched on.**

- If necessary, check pressures in refrigerant circuit using air conditioner service station .
- Switch off engine.
- Turn out handwheel on quick-release coupling adapter.
- Detach the charging hose from the refrigerant circuit.
- Screw protective caps back on.

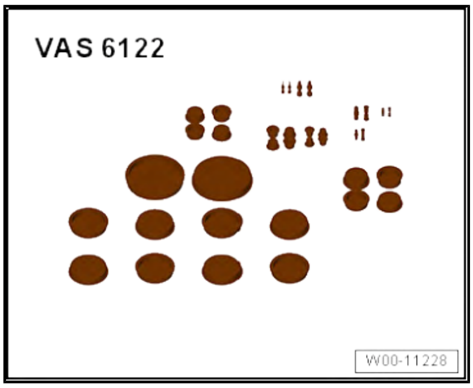
2.11 Removing and installing refrigerant line from air conditioner compressor to condenser

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1410-



♦ Set of plugs for engine - VAS 6122-



Removing

Vehicles without radiator blind

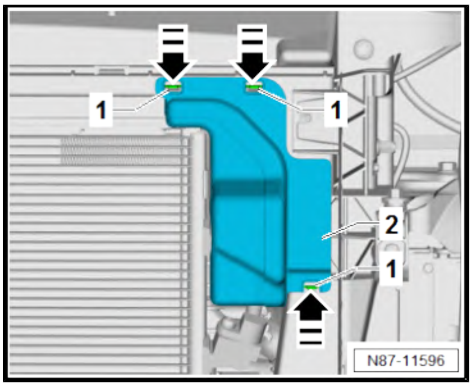
– Remove front bumper cover ⇒ General body repairs, exterior; Rep. gr. 63 ; Front bumper; Removing and installing bumper cover .

Vehicles with radiator blind

– Remove radiator blind ⇒ Rep. gr. 19 ; Radiator/radiator fan; Removing and installing radiator blind .

Continued for all vehicles

– Release locking lugs -1- in direction of -arrow-. While doing this, detach air deflector -2- in direction of travel.



Vehicles with second alternator

– Remove alternator for auxiliary drive ⇒ Rep. gr. 13 ; Auxiliary drive; Removing and installing 2nd alternator .

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

– **Wear protective gloves.**

– **Wear protective goggles.**

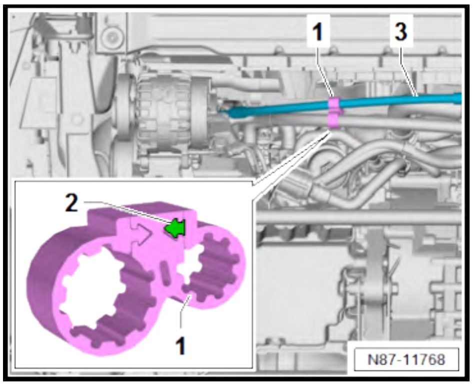
– **Extract refrigerant and open the refrigerant circuit immediately afterwards.**

– **If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.**

Vehicles with toothed clip ring

– Open toothed clip ring -1- by laterally pushing tab indicated by arrow -2- out of centre mounting.

– Remove refrigerant line -3- from locking ring -1-.

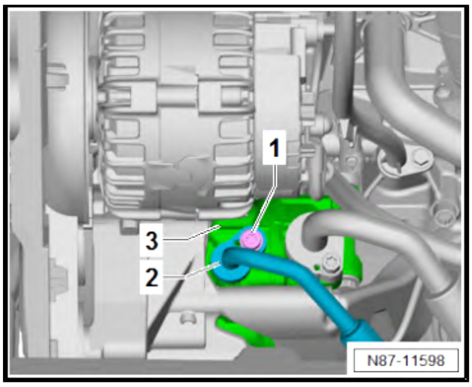


Continued for all vehicles

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .

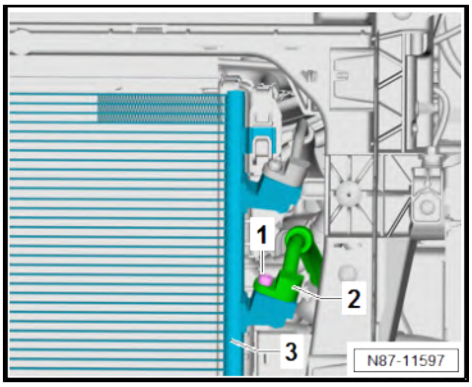
– Unscrew bolt -1-.

– Remove refrigerant line -2- from air conditioner compressor -3-.



– Unscrew bolt -1-.

– Remove refrigerant line -2- from condenser -3-.



NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

– **Never bend refrigerant lines to a radius less than 100 mm.**

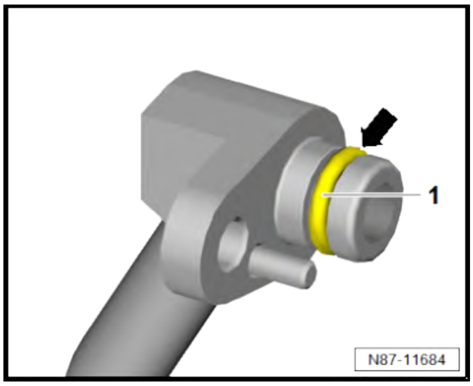
– Unclip refrigerant line -2- from retainer on longitudinal member and remove downwards.

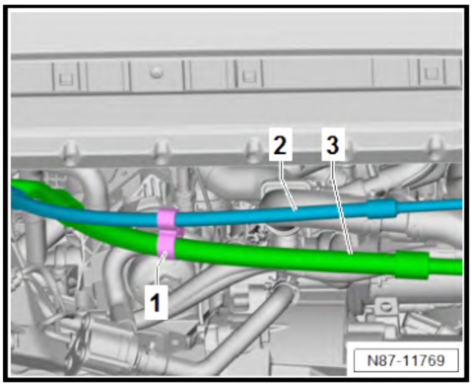
Installing

Install in reverse order of removal, observing the following:

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.





Note

If not yet present, fit serrated ring clip -1- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check serrated ring clip -1- for damage; renew if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Make sure that locking ring -1- is fitted at narrowest point between refrigerant lines -2- and -3-.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

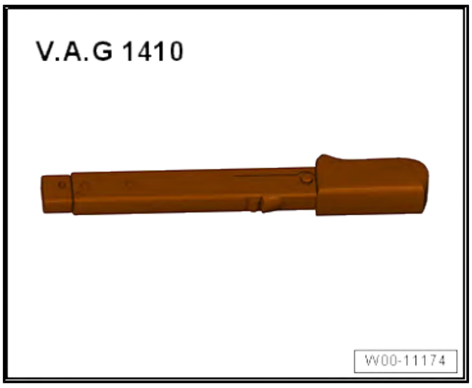
Specified torques

♦ ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

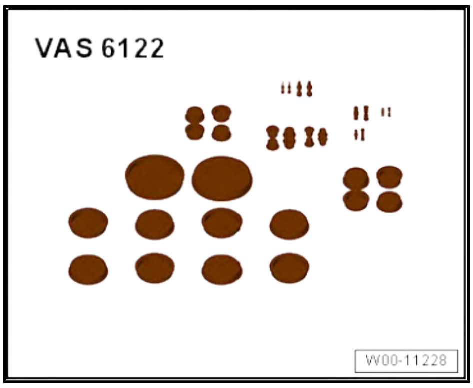
2.12 Removing and installing refrigerant line from air conditioner compressor to evaporator

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1410-



- ♦ Set of plugs for engine - VAS 6122-



Removing

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

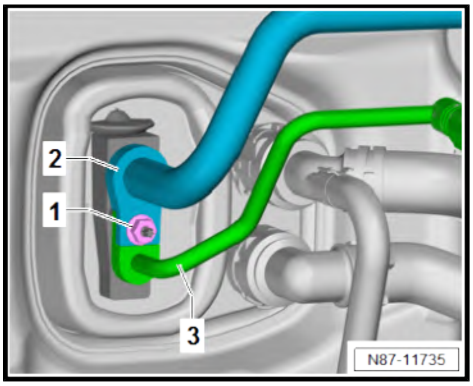
⚠ CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.
– Wear protective goggles.
– Extract refrigerant and open the refrigerant circuit immediately afterwards.
– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .



– Unscrew nut -1-.
– Remove refrigerant line -2-.
– Remove refrigerant line -3-.
– Place refrigerant lines -2- and -3- to one side.

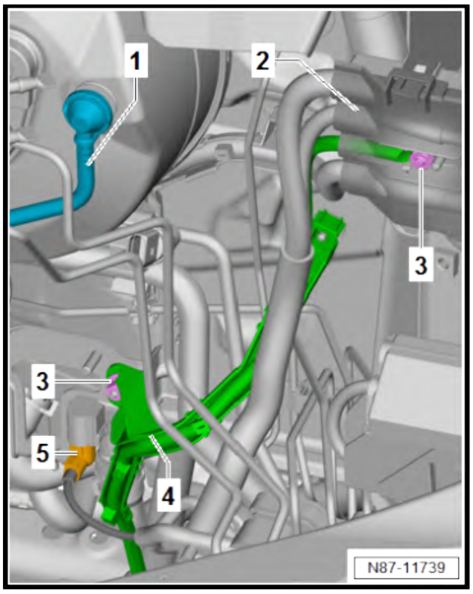
Vehicles without radiator blind

– Remove front bumper cover ⇒ General body repairs, exterior; Rep. gr. 63 ; Front bumper; Removing and installing bumper cover .

Vehicles with radiator blind

– Remove radiator blind ⇒ Rep. gr. 19 ; Radiator/radiator fan; Removing and installing radiator blind .

Continued for all vehicles



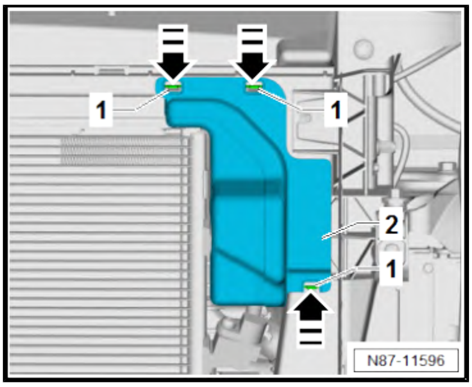
– Remove battery tray ⇒ Electrical system; Rep. gr. 27 ; Battery; Removing and installing battery tray .
– Pull vacuum line -1- off brake servo.
– Fold open cover for jump-start connection - TV32- -2-.
– Unscrew nuts -3-.
– Pull wire duct -4- off stud together with wiring harness.

Vehicles with auxiliary coolant heater and / or auxiliary heater

– Disconnect electrical connector -5-.

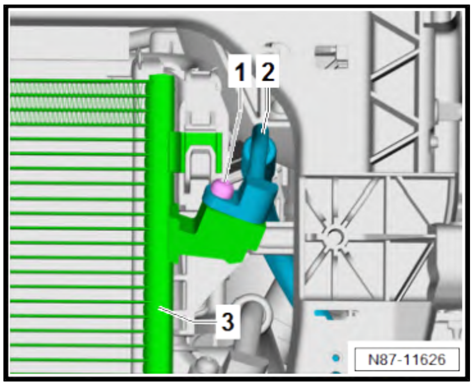
Continued for all vehicles

– Release locking lugs -1- in direction of -arrow-. While doing this, detach air deflector -2-.



– Unscrew bolt -1-.

– Remove refrigerant line -2- from condenser -3-.



Vehicles with second evaporator

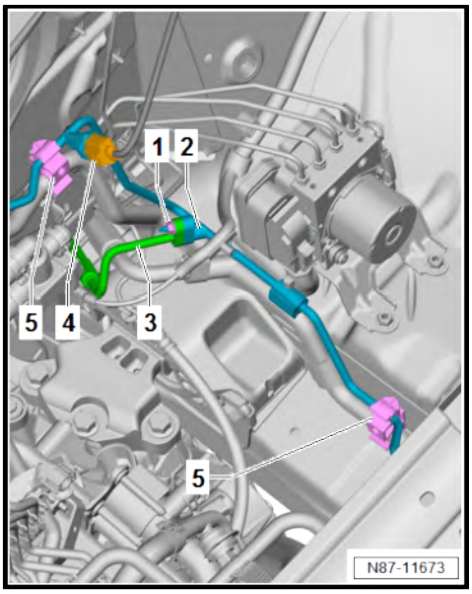
– Unscrew nut -1-.

– Disconnect refrigerant lines -2- and -3-.

Continued for all vehicles

– Separate electrical connector -4-.

– Unclip refrigerant line -2- from retainer -5-.

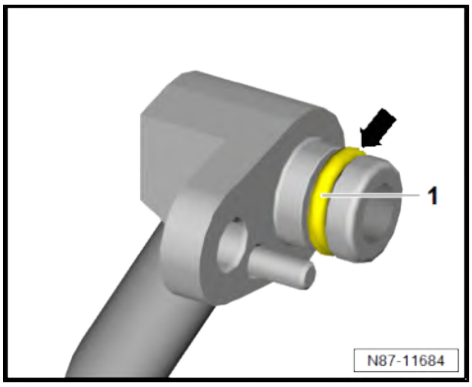


NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

– Never bend refrigerant lines to a radius less than 100 mm.

– Remove refrigerant line -2- upwards.



Installing

Install in reverse order of removal, observing the following:

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

Specified torques

♦ ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

2.13 Removing and installing refrigerant line from evaporator to air conditioner compressor

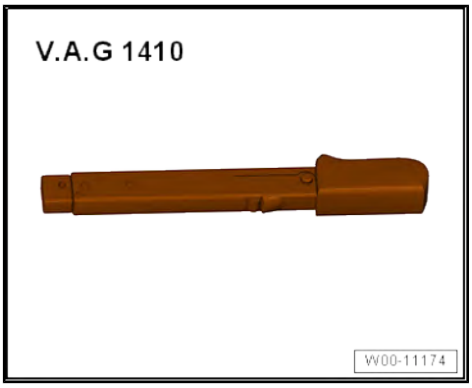
⇒ “2.13.1 Removing and installing top refrigerant line from evaporator to air conditioner compressor”, page 83

⇒ “2.13.2 Removing and installing bottom refrigerant line from evaporator to air conditioner compressor”, page 87

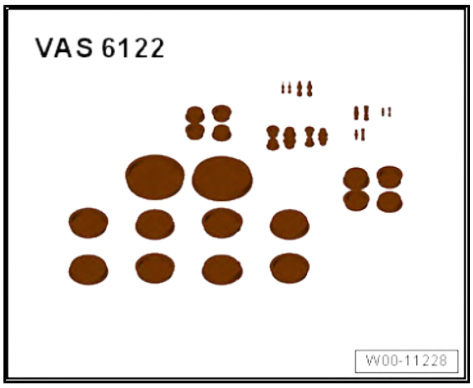
2.13.1 Removing and installing top refrigerant line from evaporator to air conditioner compressor

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1410-



♦ Set of plugs for engine - VAS 6122-



Removing

– Remove battery tray ⇒ Electrical system; Rep. gr. 27 ; Battery; Removing and installing battery tray .

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

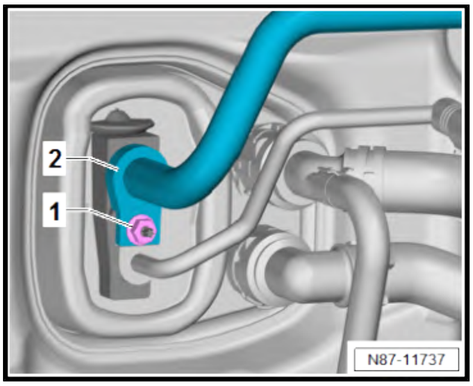
CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

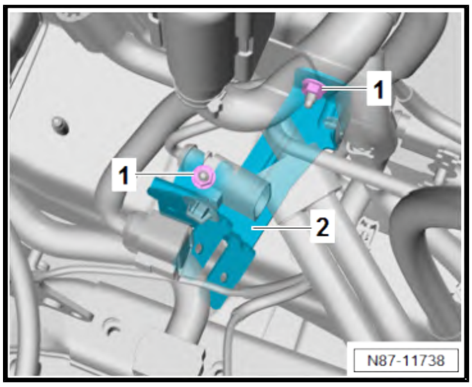
There is a risk of injury to the skin and parts of the body due to freezing.

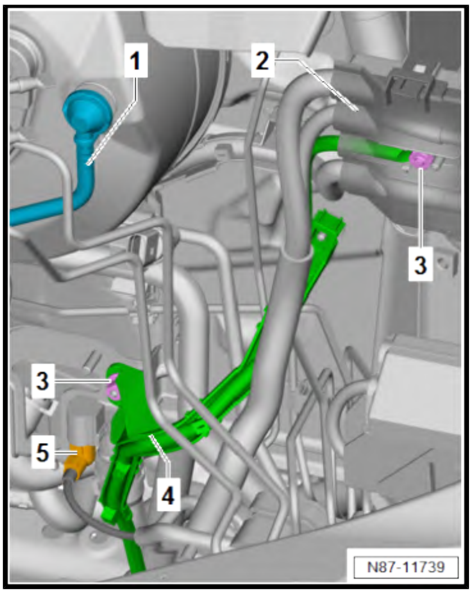
– Wear protective gloves.
– Wear protective goggles.
– Extract refrigerant and open the refrigerant circuit immediately afterwards.
– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .
– Unscrew nut -1-.
– Detach and lay aside refrigerant line -2-.



– Unscrew nuts -1-.
– Pull bracket -2- in direction of engine off stud.

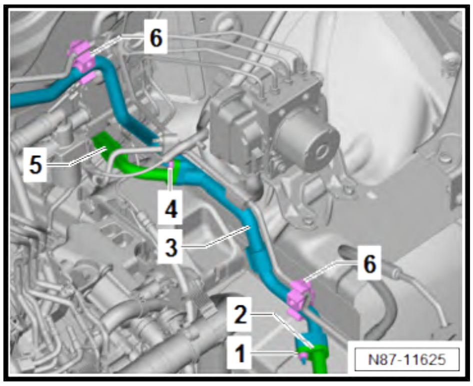




– Pull vacuum line -1- off brake servo.
– Fold open cover for jump-start connection - TV32- -2-.
– Unscrew nuts -3-.
– Pull wire duct -4- off stud together with wiring harness.

Vehicles with auxiliary coolant heater and / or auxiliary heater

– Disconnect electrical connector -5-.



Continued for all vehicles

– Unscrew nut -1-.
– Disconnect refrigerant lines -2- and -3-.

Vehicles with second evaporator

– Unscrew nut -4-.
– Disconnect refrigerant lines -3- and -5-.

Continued for all vehicles

– Unclip refrigerant line -3- from retainer -6-.

NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

– Never bend refrigerant lines to a radius less than 100 mm.

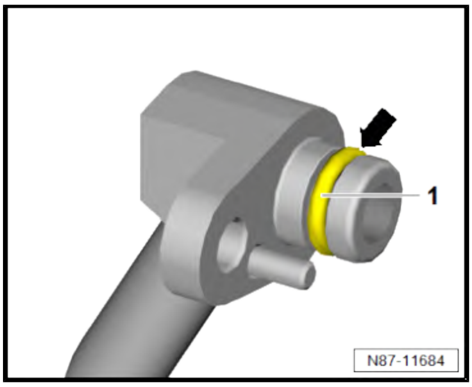
– Remove refrigerant line -3- upwards.

Installing

Install in reverse order of removal, observing the following:

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.



NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

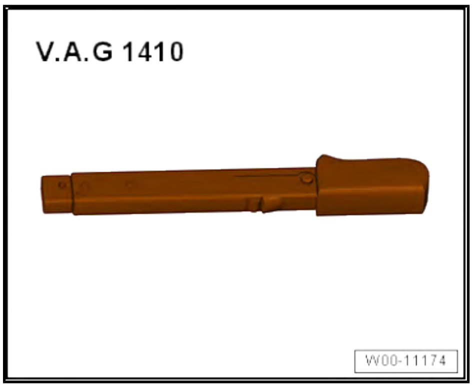
Specified torques

♦ ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

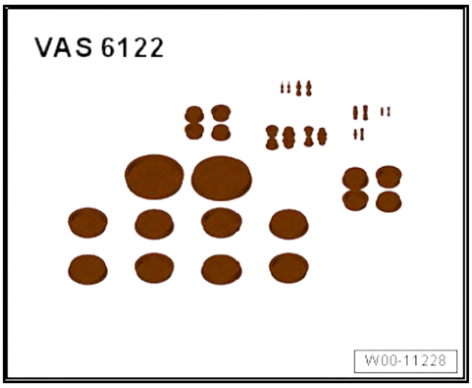
2.13.2 Removing and installing bottom refrigerant line from evaporator to air conditioner compressor

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1410-



♦ Set of plugs for engine - VAS 6122-



Removing

– If fitted, remove noise insulation ⇒ General body repairs, exterior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation .

Vehicles with second alternator

– Remove alternator for auxiliary drive ⇒ Rep. gr. 13 ; Auxiliary drive; Removing and installing 2nd alternator .

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

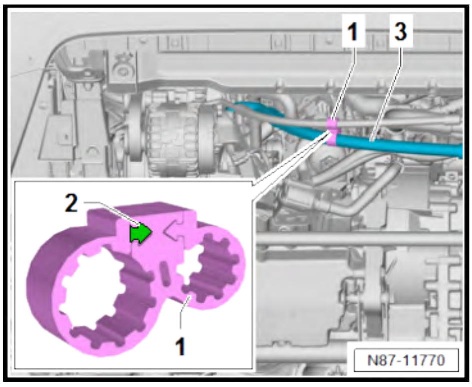
There is a risk of injury to the skin and parts of the body due to freezing.

– **Wear protective gloves.**
– **Wear protective goggles.**
– **Extract refrigerant and open the refrigerant circuit immediately afterwards.**
– **If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.**

Vehicles with toothed clip ring

– Open toothed clip ring -1- by laterally pushing tab indicated by arrow -2- out of centre mounting.

– Remove refrigerant line -3- from locking ring -1-.

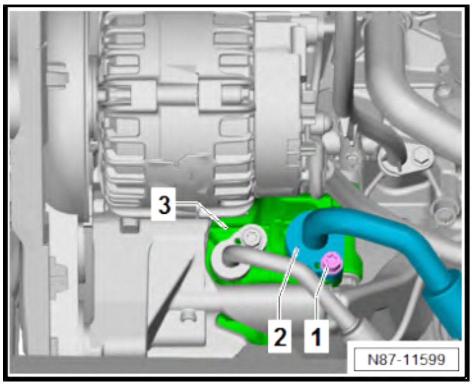


Continued for all vehicles

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .

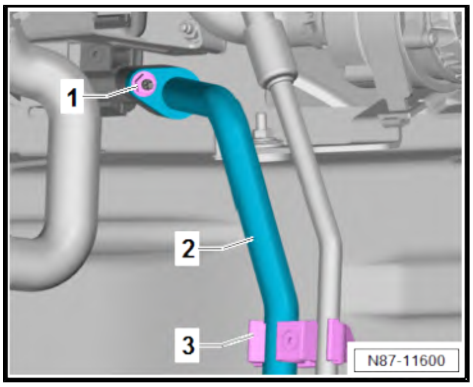
– Unscrew bolt -1-.

– Remove refrigerant line -2- from air conditioner compressor -3-.



– Unscrew nut -1-.

– Unclip refrigerant line -2- from retainer on longitudinal member -3-.



NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

– Never bend refrigerant lines to a radius less than 100 mm.

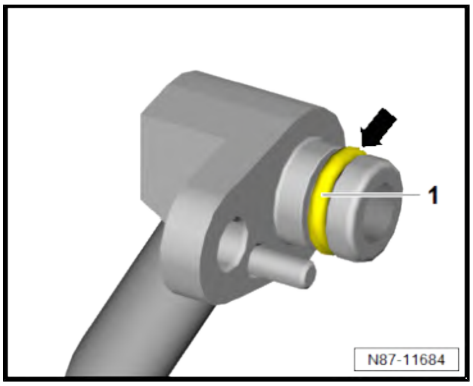
– Remove refrigerant line -2- from coupling point downwards.

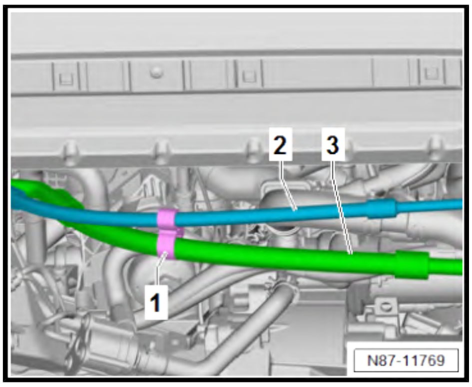
Installing

Install in reverse order of removal, observing the following:

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.





Note

If not yet present, fit serrated ring clip -1- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check serrated ring clip -1- for damage; renew if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Make sure that locking ring -1- is fitted at narrowest point between refrigerant lines -2- and -3-.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– Never start the engine if the refrigerant circuit is empty.

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

Specified torques

- ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

2.14 Removing and installing refrigerant lines to heater and air conditioning unit

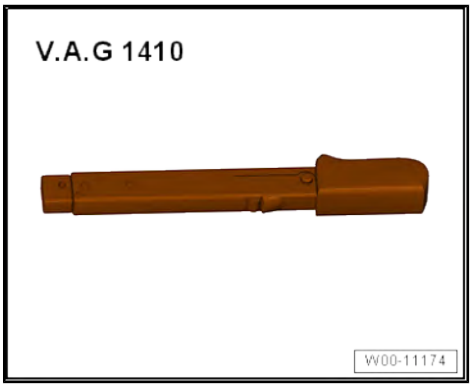
⇒ “2.14.1 Removing and installing refrigerant lines on underbody”, page 90

⇒ “2.14.2 Removing and installing refrigerant lines at connection on underbody to expansion valve for air conditioner refrigerant N697 ”, page 93

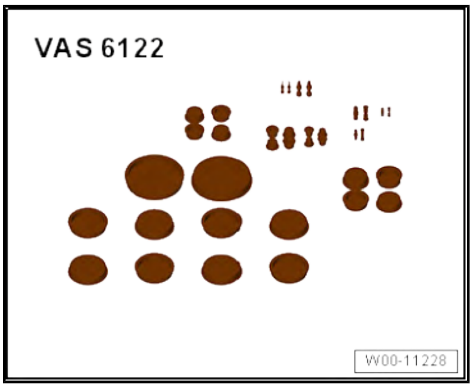
2.14.1 Removing and installing refrigerant lines on underbody

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1410-



♦ Set of plugs for engine - VAS 6122-



Removing

– Remove reducing agent tank ⇒ Rep. gr. 26 ; SCR system (selective catalytic reduction); Removing and installing reducing agent tank .

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

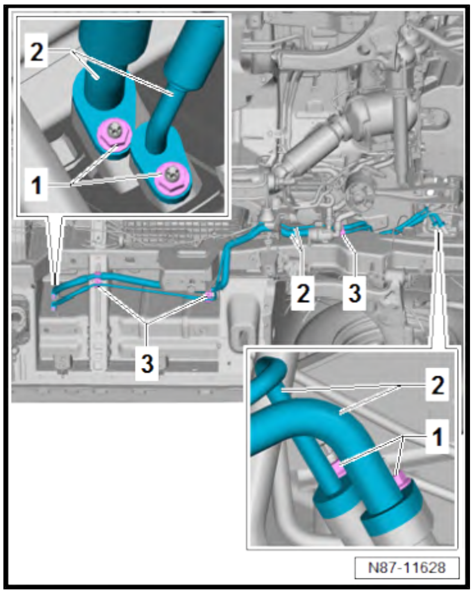
CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.
– Wear protective goggles.
– Extract refrigerant and open the refrigerant circuit immediately afterwards.
– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .
– Unscrew nuts -1-.

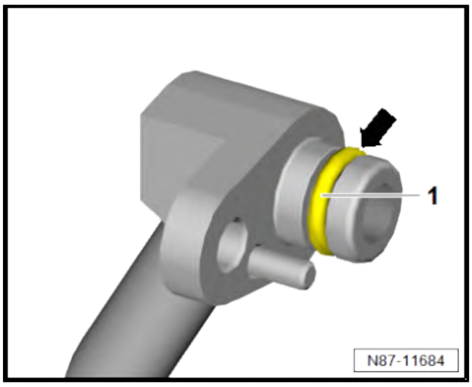


NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

– Never bend refrigerant lines to a radius less than 100 mm.

– Unclip refrigerant lines -2- from retainer -3- and remove downwards.



Installing

Install in reverse order of removal, observing the following:

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

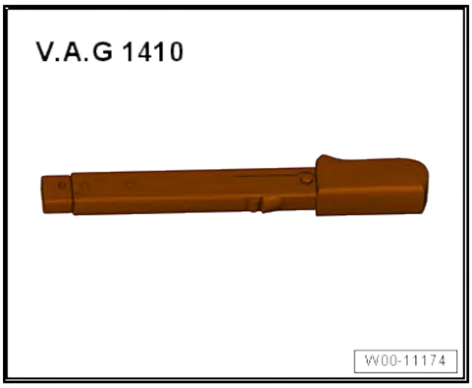
Specified torques

♦ ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

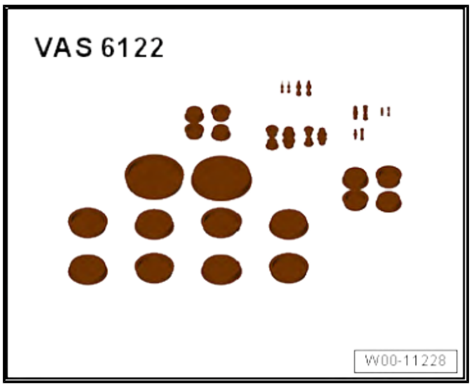
2.14.2 Removing and installing refrigerant lines at connection on underbody to expansion valve for air conditioner refrigerant - N697-

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1410-



♦ Set of plugs for engine - VAS 6122-

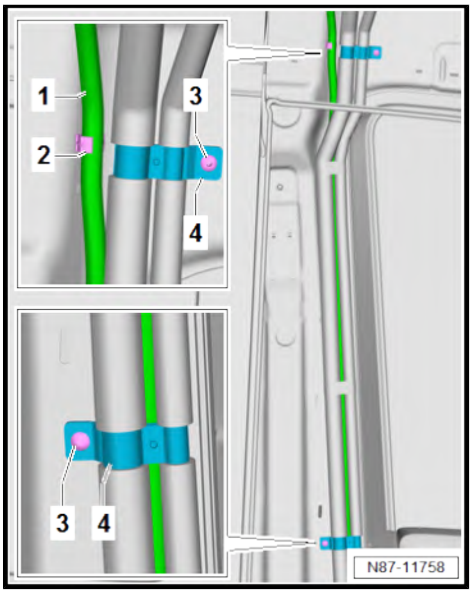


Removing

– Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .

Vehicles with curtain airbag

– Remove left-hand curtain airbag with igniter ⇒ General body repairs, interior; Rep. gr. 69 ; Curtain airbags; Removing and installing curtain airbag with igniter .



Continued for all vehicles

– Remove left step ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing step; Removing and installing front step .

– Unclip condensation drain hose -1- from hose retainer -2-.

– Remove spreader rivets -3- for refrigerant line retainer -4-.

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

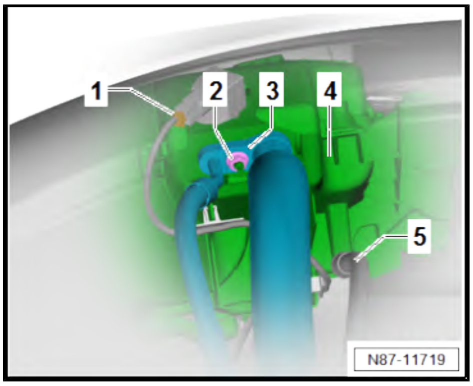
CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

- **Wear protective gloves.**
- **Wear protective goggles.**
- **Extract refrigerant and open the refrigerant circuit immediately afterwards.**
- **If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.**

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .



– Disconnect connector -1-.

– Unscrew nut -2-.

NOTICE

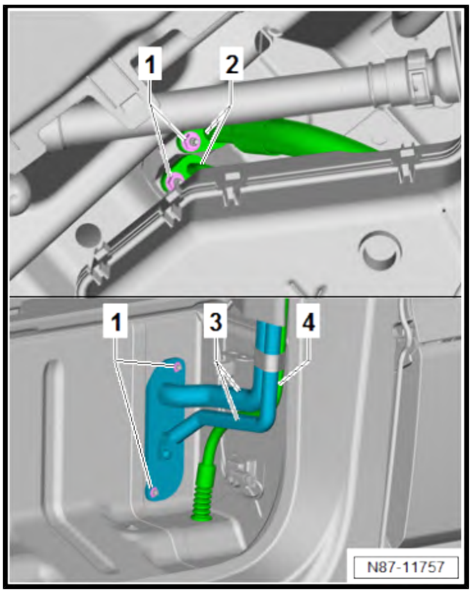
Risk of damage to refrigerant lines from rupture of inner foil.

- **Never bend refrigerant lines to a radius less than 100 mm.**

– Pull refrigerant lines -3- off evaporator housing -4- together with condensation drain hose -5-.

– Move clear hose connecting rear window washer system to refrigerant lines -3-.

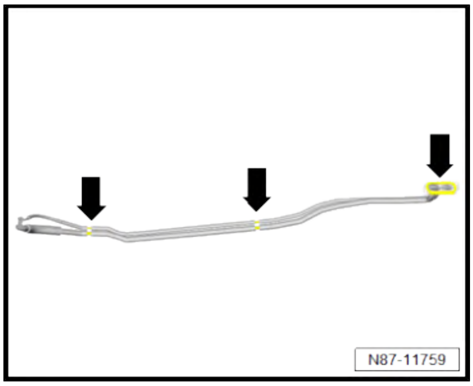
– Unscrew nuts -1-.
– Pull off refrigerant lines -2-.
– Remove refrigerant lines -3- from vehicle body together with condensation drain hose -4-.



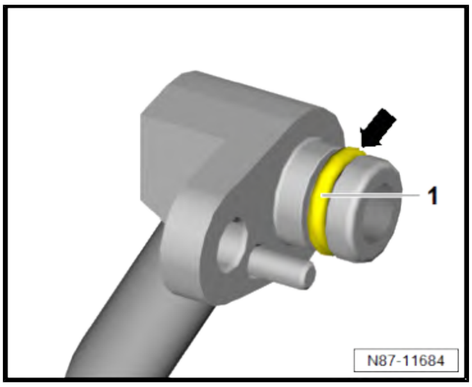
Installing

Install in reverse order of removal, observing the following:

– Check condensation drain hose ⇒ page 154 .
– Renew spreader rivets for retainer for refrigerant lines.
– Check seals and insulation in area marked by -arrows- for damage; renew if necessary ⇒ Electronic parts catalogue (ET-KA) or ⇒ webMANTIS for MAN TGE.

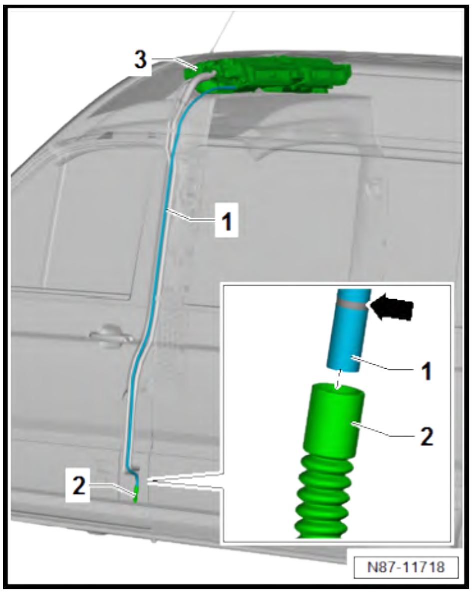


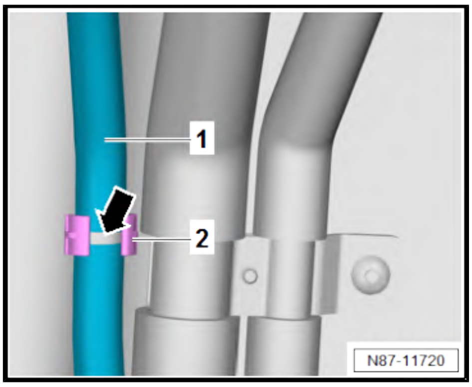
– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.
– Moisten new seals with refrigerant oil before installing refrigerant line.



– Insert condensation drain hose -1- up to marking -arrow- on water drain valve -2-.

– Push condensation drain hose -1- to stop on evaporator housing -3-.





Note

The condensation drain hose -1- must be secured with clips in area of narrow marking -arrow- on hose retainer -2-.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

Specified torques

♦ ⇒ “2.2 Assembly overview - refrigerant lines”, page 50

Component	Specified torque
Nuts -1- in area of step to outside of vehicle	8 Nm

2.15 When is it necessary to purge the refrigerant circuit with refrigerant

Note

♦ *In general, the refrigerant circuit must always be purged if a component of the refrigerant circuit had to be renewed due to an internal damage.*

♦ *If the necessary steps only comprise removal and installation, the refrigerant circuit needs to be sealed with plugs from engine bung set - VAS 6122- immediately, and the plugs should not be removed until immediately prior to installation.*

♦ *The following table gives several examples of possible scenarios.*

For detailed information, refer to the workshop manual for the corresponding refrigerant.

Vehicles with R134a refrigerant

⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Cleaning refrigerant circuit of contaminants, commercial vehicles

Vehicles with R1234yf refrigerant

- ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Cleaning the refrigerant circuit

Vehicles with R134a refrigerant

- ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Renewing components

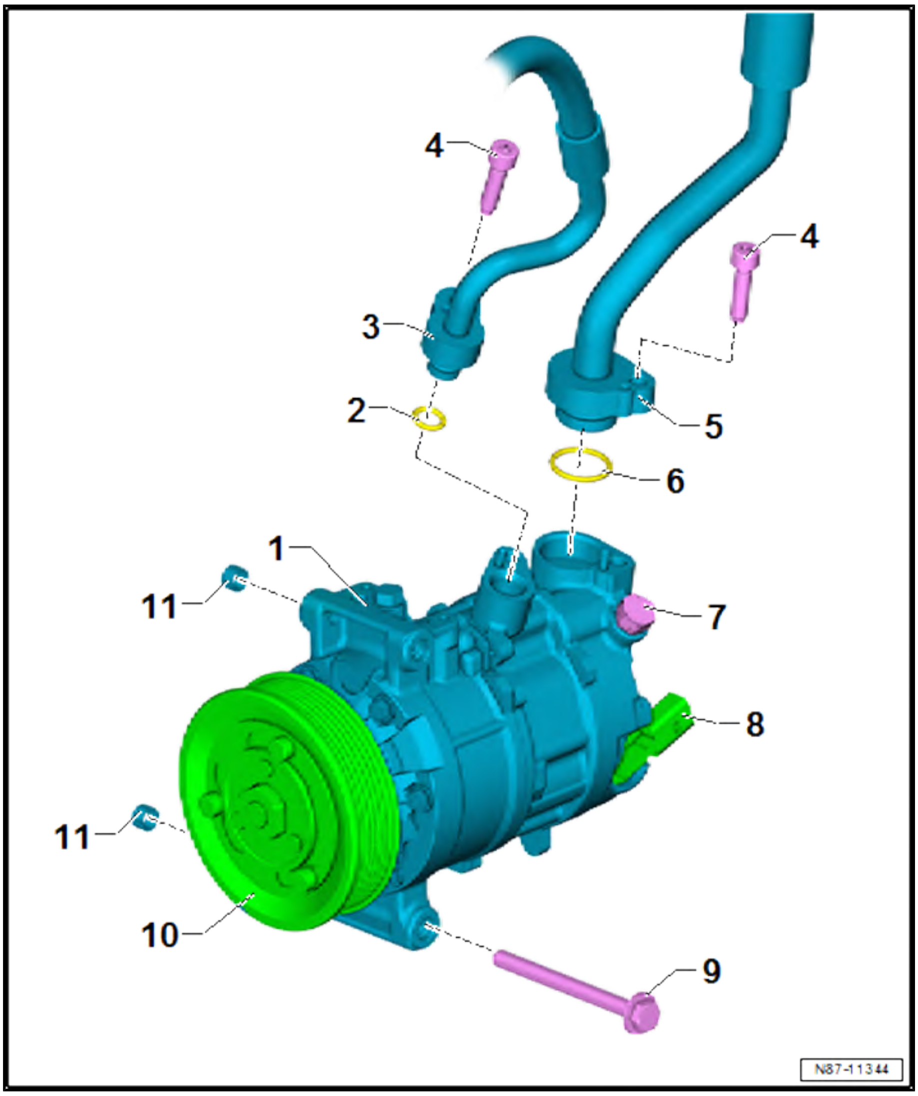
Vehicles with R1234yf refrigerant

- ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Renewing components

Necessary work steps and possible scenarios	Does the refrigerant circuit need to be purged?	
	yes	no
Removing and installing air conditioner compressor		X
Renewing air conditioner compressor due to internal damage	X	
Renewing air conditioner compressor due to external damage		X
Renewing air conditioner compressor due to external damage (new manufacturer)	X	
Removing and installing air conditioner compressor regulating valve - N280-		X
Removing and installing refrigerant circuit pressure sender - G805-		X
Removing and installing expansion valve		X
Removing and installing condenser		X
Removing and installing desiccant bag or cartridge		X
Renewing condenser due to external damage leading to loss of refrigerant	X	
Doubt regarding amount of refrigerant oil in refrigerant circuit or regarding the manufacturer	X	
Refrigerant circuit has been left open for longer than normally required for repairs (e.g. following an accident)	X	
During evacuation of a leak-tight refrigerant circuit, vacuum display is not constant on gauge (moisture in refrigerant circuit, which generates vapour pressure)	X	

3 Air conditioner compressor

3.1 Assembly overview - drive unit of air conditioner compressor



1 - Air conditioner compressor

- ❑ Different versions ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE
- ❑ Removing from and installing on bracket ⇒ page 101
- ❑ Removing and installing ⇒ page 104
- ❑ Removing and installing 2nd air conditioner compressor ⇒ Rep. gr. 13 ; Auxiliary drive; Removing and installing 2nd air conditioner compressor

2 - Seal

- ❑ Renew after removal
- ❑ Moisten with refrigerant oil before installation

3 - Refrigerant line

- ❑ High-pressure side
- ❑ Removing and installing ⇒ page 75

4 - Bolts

- ❑ Qty. 2
- ❑ 20 Nm

5 - Refrigerant line

- ❑ Low-pressure side
- ❑ Removing and installing ⇒ page 87

6 - Seal

- ❑ Renew after removal
- ❑ Moisten with refrigerant oil before installation

7 - High-pressure safety valve

- ❑ Protects refrigerant circuit from excessive pressure

- ❑ Checking ⇒ page 109

8 - Air conditioner compressor regulating valve - N280-

- ❑ Renew after removal
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 110
- ❑ Observe ⇒ Electronic parts catalogue (ETKA)

9 - Bolts

- ❑ Qty. 3
- ❑ 25 Nm

10 - Pulley

- ❑ With overload protection

11 - Dowel sleeves

- ❑ Qty. 2

3.2 Removing air conditioning compressor from and installing on bracket

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-

- ♦ Welding wire

Removing

Note

Instead of the poly V-belts being removed completely, they are just removed from the air conditioner compressor pulley.

– Relieve tension from poly V-belt ⇒ Rep. gr. 13 ; Cylinder block on pulley side; Removing and installing poly V-belt .

– Remove poly V-belt from air conditioner compressor pulley.

Vehicles with second alternator

– Remove alternator for auxiliary drive ⇒ Rep. gr. 13 ; Auxiliary drive; Removing and installing 2nd alternator .

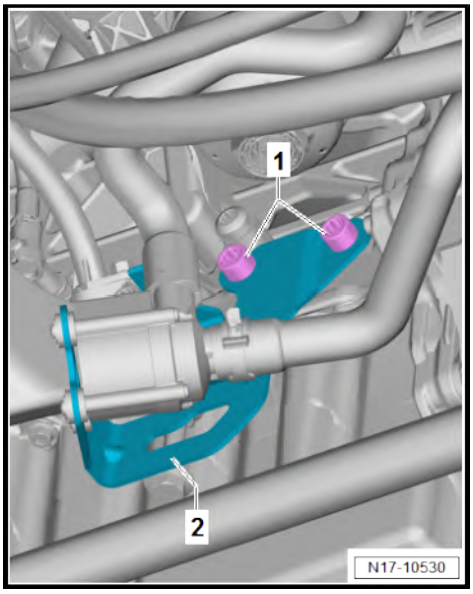
Vehicles with second air conditioner compressor

– Remove air conditioner compressor for auxiliary drive ⇒ Rep. gr. 13 ; Auxiliary drive; Removing and installing 2nd air conditioner compressor .

Continued for vehicles with second alternator or second air conditioner compressor

– Unscrew bolts -1-.

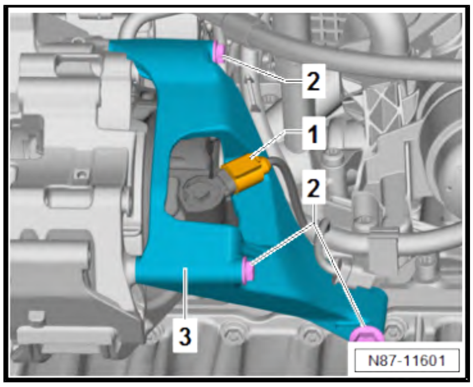
– Remove bracket -2- and lay aside.



– Disconnect connector -1-.

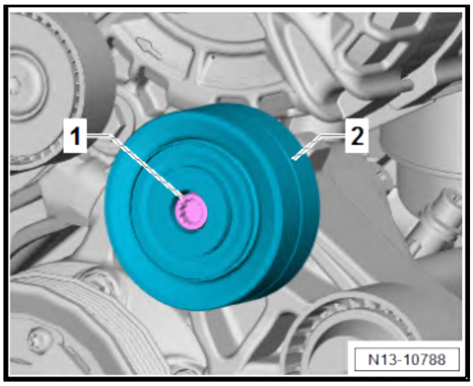
– Unscrew bolts -2-.

– Remove retainer -3-.

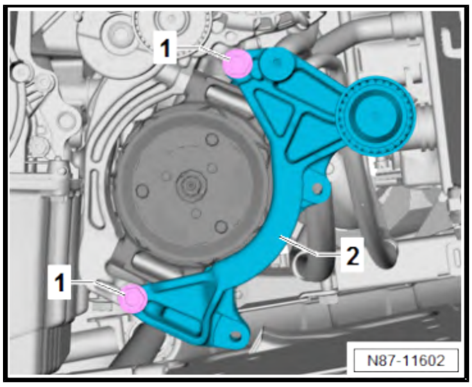


– Unscrew bolt -1-.

– Remove idler pulley -2-.



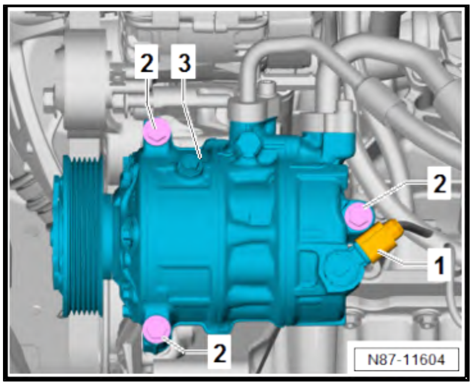
– Unscrew bolts -1-. While doing this, detach bracket -2-.



Continued for all vehicles

– Disconnect connector -1-.

– Unscrew bolts -2-. While doing so, detach air conditioner compressor -3- from bracket.

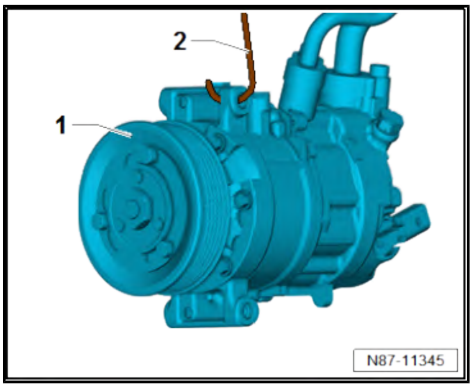


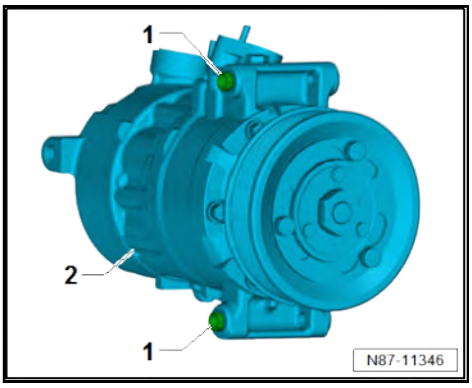
NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

– Never bend refrigerant lines to a radius less than 100 mm.

– Secure air conditioner compressor -1- to body with suitable material (e.g. welding wire -2-).





Installing

Install in reverse order of removal, observing the following:

– Thoroughly clean contact surfaces on air conditioner compressor -2- and ancillary bracket.

– Insert dowel sleeves -1- into air conditioner compressor -2-.

NOTICE

Risk of damage to air conditioner compressor. Refrigerant oil may accumulate in the compression chamber of a removed air conditioner compressor.

– **After installing a new air conditioner compressor or adding new refrigerant oil, fully turn the air conditioner compressor 10 times by hand before the poly V-belt is fitted.**

Specified torques

♦ ⇒ “3.1 Assembly overview - drive unit of air conditioner compressor”, page 100

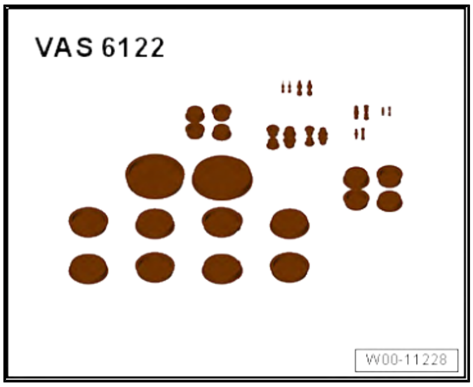
♦ Power take-off; Assembly overview - power take-off ⇒ Rep. gr. 13 ; Power take-off; Assembly overview - power take-off

3.3 Removing and installing air conditioner compressor

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1331-

♦ Set of plugs for engine - VAS 6122-



♦ Vehicle diagnostic tester

Removing

Note

- *If only removal of the ancillary bracket is planned, the refrigerant circuit remains unopened* *⇒ page 101* *.*
- *Instead of the poly V-belts being removed completely, they are just removed from the air conditioner compressor pulley.*

– Relieve tension from poly V-belt ⇒ Rep. gr. 13 ; Cylinder block on pulley side; Removing and installing poly V-belt .

– Remove poly V-belt from air conditioner compressor pulley.

Vehicles with second alternator

– Remove alternator for auxiliary drive ⇒ Rep. gr. 13 ; Auxiliary drive; Removing and installing 2nd alternator .

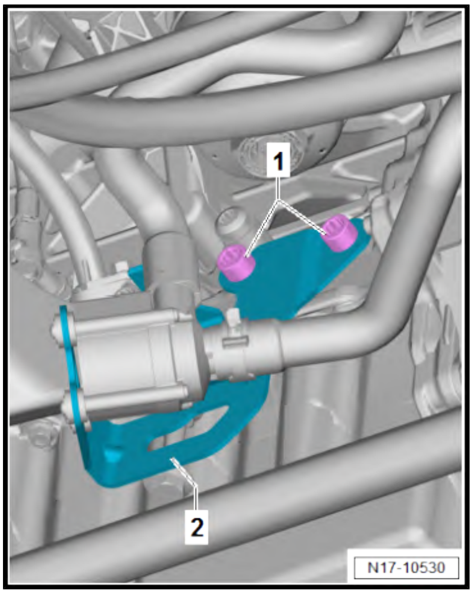
Vehicles with second air conditioner compressor

– Remove air conditioner compressor for auxiliary drive ⇒ Rep. gr. 13 ; Auxiliary drive; Removing and installing 2nd air conditioner compressor .

Continued for vehicles with second alternator or second air conditioner compressor

– Unscrew bolts -1-.

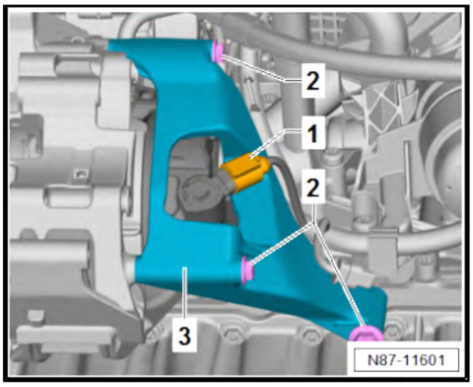
– Remove bracket -2- and lay aside.

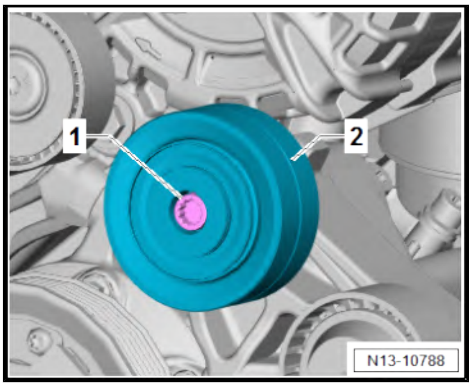


– Disconnect connector -1-.

– Unscrew bolts -2-.

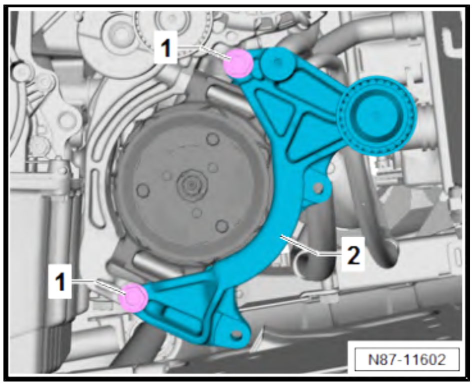
– Remove retainer -3-.





– Unscrew bolt -1-.

– Remove idler pulley -2-.



– Unscrew bolts -1-. While doing this, detach bracket -2-.

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

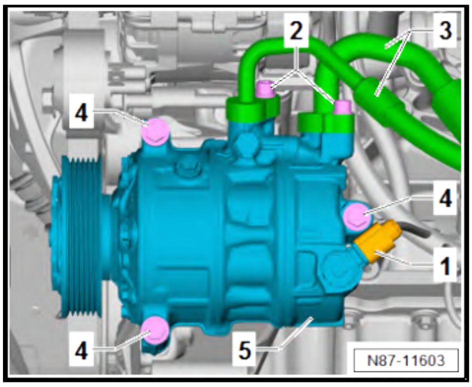
– Wear protective gloves.

– Wear protective goggles.

– Extract refrigerant and open the refrigerant circuit immediately afterwards.

– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .



- Disconnect connector -1-.
- Unscrew bolts -2-.

NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

- **Never bend refrigerant lines to a radius less than 100 mm.**

- Remove refrigerant lines -3-.
- Unscrew bolts -4-. While doing so, detach air conditioner compressor -5- from bracket.
- Check if it is necessary to purge the refrigerant circuit ⇒ page 98 .

Installing

Install in reverse order of removal, observing the following:

Procedure for installing new air conditioner compressor

- Determine refrigerant oil volume in removed air conditioner compressor by draining oil into measuring beaker.

Note

Collect refrigerant oil from the new air conditioner compressor in a suitable container for reuse.

- Drain refrigerant oil from new air conditioner compressor.
- Fill exact same amount of refrigerant oil into the new, emptied air conditioner compressor as that determined in the removed air conditioner compressor.

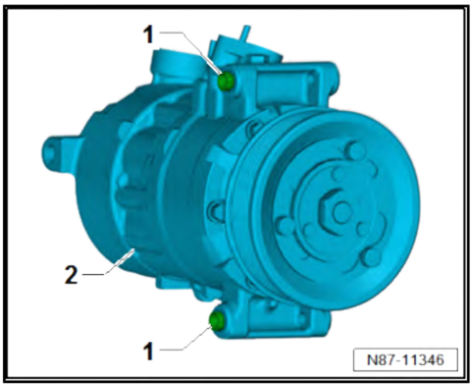
Vehicles with R134a refrigerant

♦ ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Renewing components

Vehicles with R1234yf refrigerant

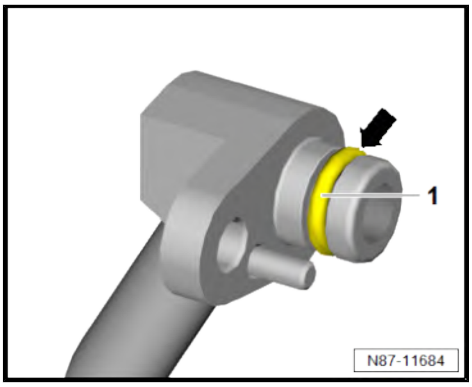
♦ ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Renewing components

Continued for all vehicles



- Thoroughly clean contact surfaces on air conditioner compressor -2- and ancillary bracket.
- Insert dowel sleeves -1- into air conditioner compressor -2-.

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.



– Moisten new seals with refrigerant oil before installing refrigerant line.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

Procedure for installing new air conditioner compressor

– Use ⇒ Vehicle diagnostic tester to run-in air conditioner compressor.

Note

♦ *The LED of the A/C button flashes while the air conditioner compressor runs-in.*

♦ *The end of air conditioner compressor run-in is indicated by the LED of the A/C button going out or staying lit.*

Continued for all vehicles

– Check operation of heater and air conditioning system.

Specified torques

♦ ⇒ “3.1 Assembly overview - drive unit of air conditioner compressor”, page 100

♦ Power take-off; Assembly overview - power take-off ⇒ Rep. gr. 13 ; Power take-off; Assembly overview - power take-off

3.4 Removing and installing 2nd air conditioner compressor

The procedure for removal and installation can be found under ⇒ Rep. gr. 13 ; Auxiliary drive; Removing and installing 2nd air conditioner compressor .

3.5 Checking high-pressure safety valve on air conditioner compressor

Test sequence

– If fitted, remove noise insulation ⇒ General body repairs, exterior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation .

CAUTION

Risk of freezing injury caused by refrigerant. The high-pressure safety valve releases refrigerant when the engine is running and the pressure in the refrigerant circuit is too high.

There is a risk of injury on hands and other parts of the body due to freezing.

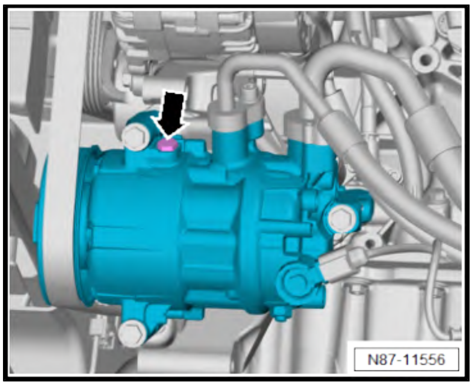
– Wear protective gloves.

– Wear protective goggles.

– Switch off engine.

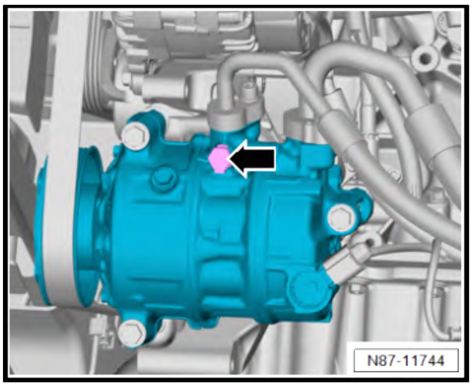
Checking high-pressure safety valve on air conditioner compressor, manufacturer: Delphi or Mahle

The high-pressure safety valve -arrow- indicates whether the valve has been opened. An attached adhesive plate is then pushed out.



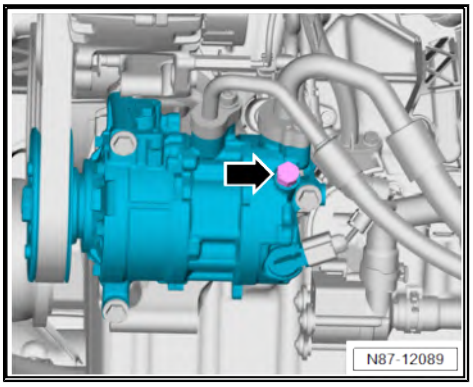
Checking high-pressure safety valve on air conditioner compressor, manufacturer: Sanden

The high-pressure safety valve -arrow- indicates whether the valve has been opened. An attached adhesive plate is then pushed out.



Checking high-pressure safety valve on air conditioner compressor, manufacturer: Denso

The high-pressure safety valve -arrow- indicates whether the valve has been opened. An attached adhesive plate is then pushed out.

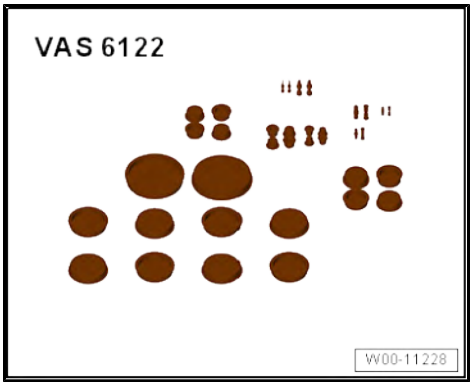


3.6 Removing and installing air conditioner compressor regulating valve - N280-

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1331-

♦ Set of plugs for engine - VAS 6122-



Note

The removal and installation procedure applies only for vehicles with an air conditioner compressor made by Sanden on which the air conditioner compressor regulating valve - N280- can be replaced ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

Removing

– If fitted, remove noise insulation ⇒ General body repairs, exterior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation .

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

– Wear protective gloves.
– Wear protective goggles.
– Extract refrigerant and open the refrigerant circuit immediately afterwards.
– If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .

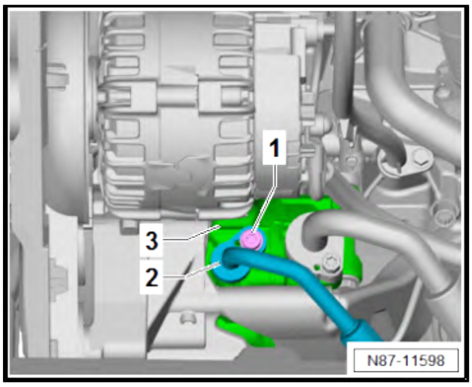
Note

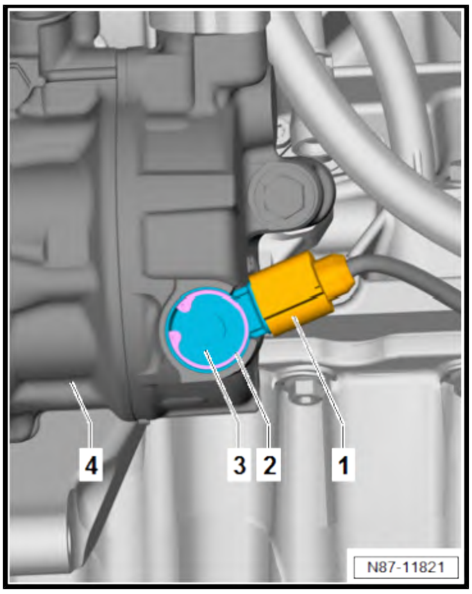
♦ *If there is excessive pressure in the air conditioner compressor, refrigerant and refrigerant oil could escape.*

♦ *If there is a vacuum in the air conditioner compressor, dirt could be drawn in.*

♦ *Before removing the air conditioner compressor regulating valve - N280- ensure pressure is equalised.*

– Unscrew bolt -1-.
– Remove refrigerant line -2- from air conditioner compressor -3-.





– Clean area around retaining ring -2- and air conditioner compressor regulating valve - N280- -3- of contaminants.
– Disconnect connector -1-.
– Remove retaining ring -2-.
– Pull air conditioner compressor regulating valve - N280- -3- out of air conditioner compressor -4-.
– Check if it is necessary to purge the refrigerant circuit ⇒ page 98 .

Installing

Install in reverse order of removal, observing the following:

– Visually check for swarf through open area of air conditioner compressor.

Note

Should swarf be discovered during the visual inspection, the necessary components must be renewed and the refrigerant circuit must be flushed with refrigerant.

– Clean mounting of air conditioner compressor regulating valve - N280- -3- on air conditioner compressor -4- with a lint-free cloth.
– Clean groove for retaining ring -2- on air conditioner compressor -4- with lint-free cloth.
– Renew flexible joints -2- on supplementary blower for air conditioning - N280- -3- ⇒ Electronic parts catalogue (ETKA) , or ⇒ webMANTIS for MAN TGE.
– Prior to installation of air conditioner compressor regulating valve - N280- -3-, moisten new oil seals with refrigerant oil.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– Never start the engine if the refrigerant circuit is empty.

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .
– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .
– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

Specified torques

♦ ⇒ “3.1 Assembly overview - drive unit of air conditioner compressor”, page 100

4 Control motors

4.1 Overview of fitting locations - front control motors

4.1.1 Overview of fitting locations - control motors at front, LHD vehicles with Climatic

1 - Air distribution flap control motor - V428-

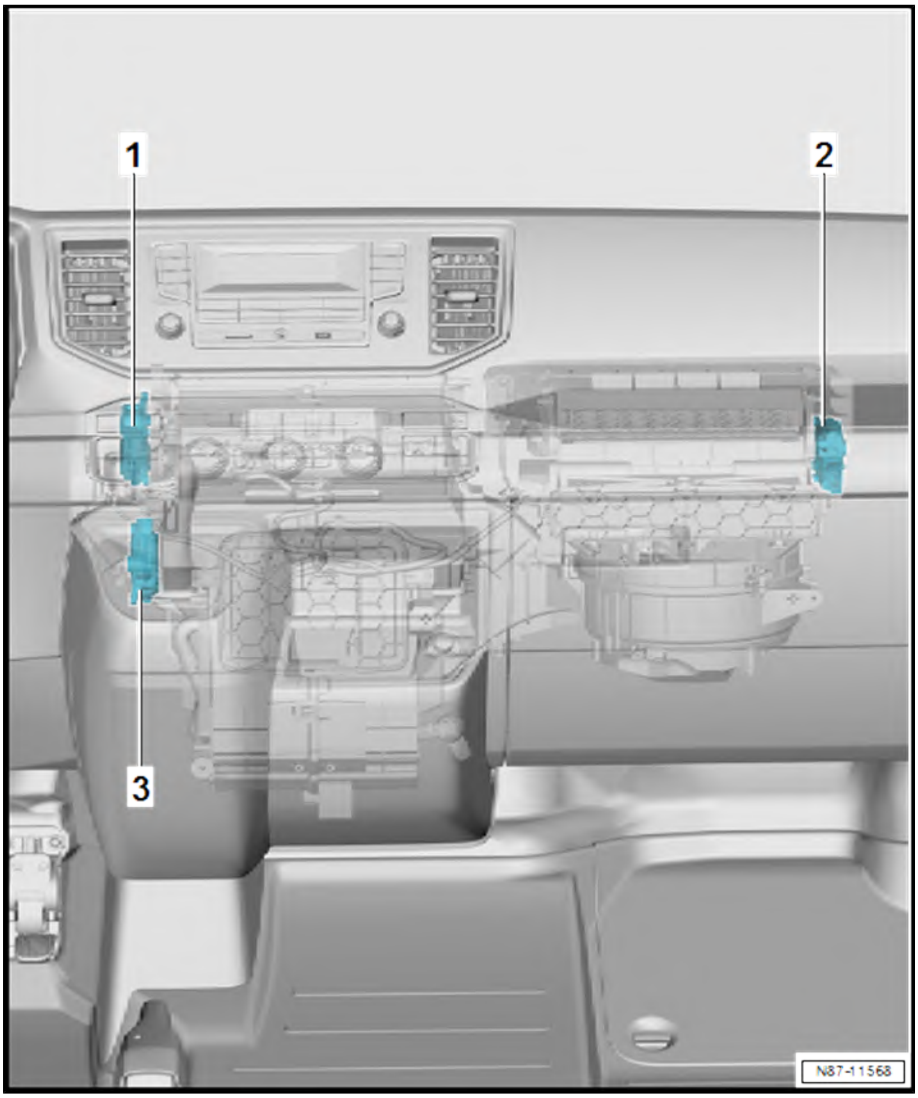
- ❑ With potentiometer for air distribution flap control motor - G645-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 129

2 - Temperature flap control motor - V68-

- ❑ With potentiometer for temperature flap control motor - G92-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 119

3 - Air recirculation flap control motor - V113-

- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 122



4.1.2 Overview of fitting locations - control motors at front, RHD vehicles with Climatic

1 - Temperature flap control motor - V68-

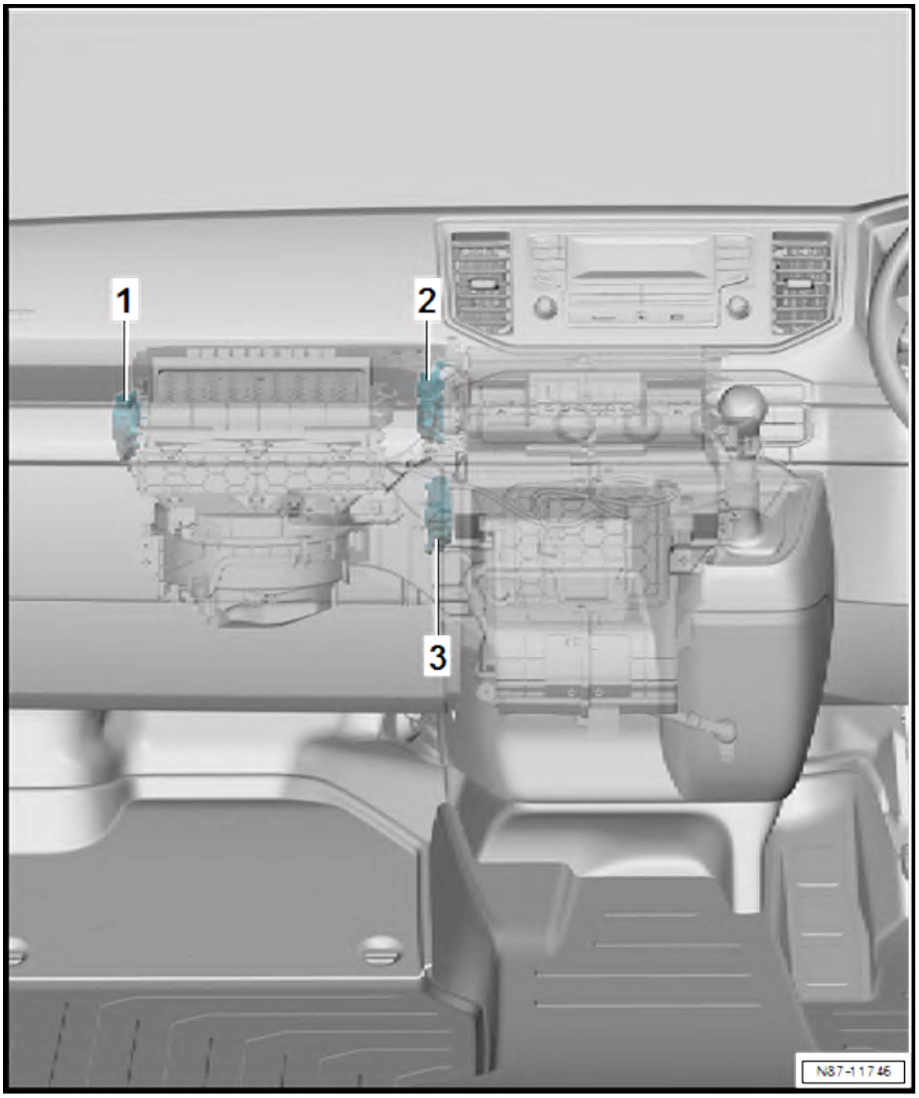
- ❑ With potentiometer for temperature flap control motor - G92-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 119

2 - Air distribution flap control motor - V428-

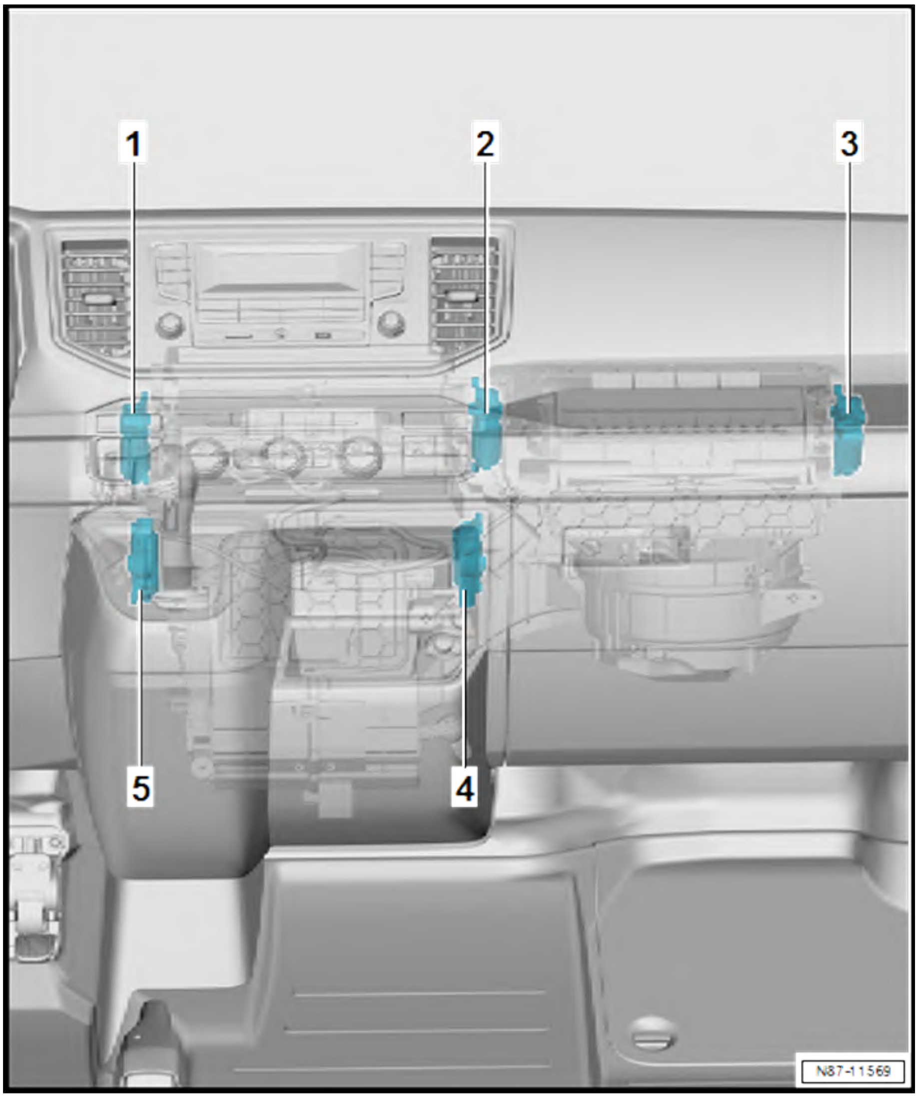
- ❑ With potentiometer for air distribution flap control motor - G645-
- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 129

3 - Air recirculation flap control motor - V113-

- ❑ Assembly overview ⇒ page 117
- ❑ Removing and installing ⇒ page 122



4.1.3 Overview of fitting locations - control motors at front, vehicles with Climatronic



1 - Front air distribution flap control motor - V426- with potentiometer for front air distribution flap control motor - G642-

- ❑ With potentiometer for front air distribution flap control motor - G642-
- ❑ Assembly overview ⇒ page 119
- ❑ Removing and installing ⇒ page 128

2 - Defroster flap control motor - V107-

- ❑ With potentiometer for defroster flap control motor - G135-
- ❑ Assembly overview ⇒ page 119
- ❑ Removing and installing ⇒ page 121

3 - Fresh air/air recirculation, air flow flap control motor - V425-

- ❑ With potentiometer for fresh air/recirculated air and air flow flap control motor - G644-
- ❑ Assembly overview ⇒ page 119
- ❑ Removing and installing ⇒ page 126

4 - Right temperature flap control motor - V159-

- ❑ With potentiometer for right temperature flap control motor - G221-
- ❑ Assembly overview ⇒ page 119
- ❑ Removing and installing ⇒ page 125

5 - Left temperature flap control motor - V158-

- ❑ With potentiometer for left temperature flap control motor - G220-
- ❑ Assembly overview ⇒ page 119
- ❑ Removing and installing ⇒ page 124

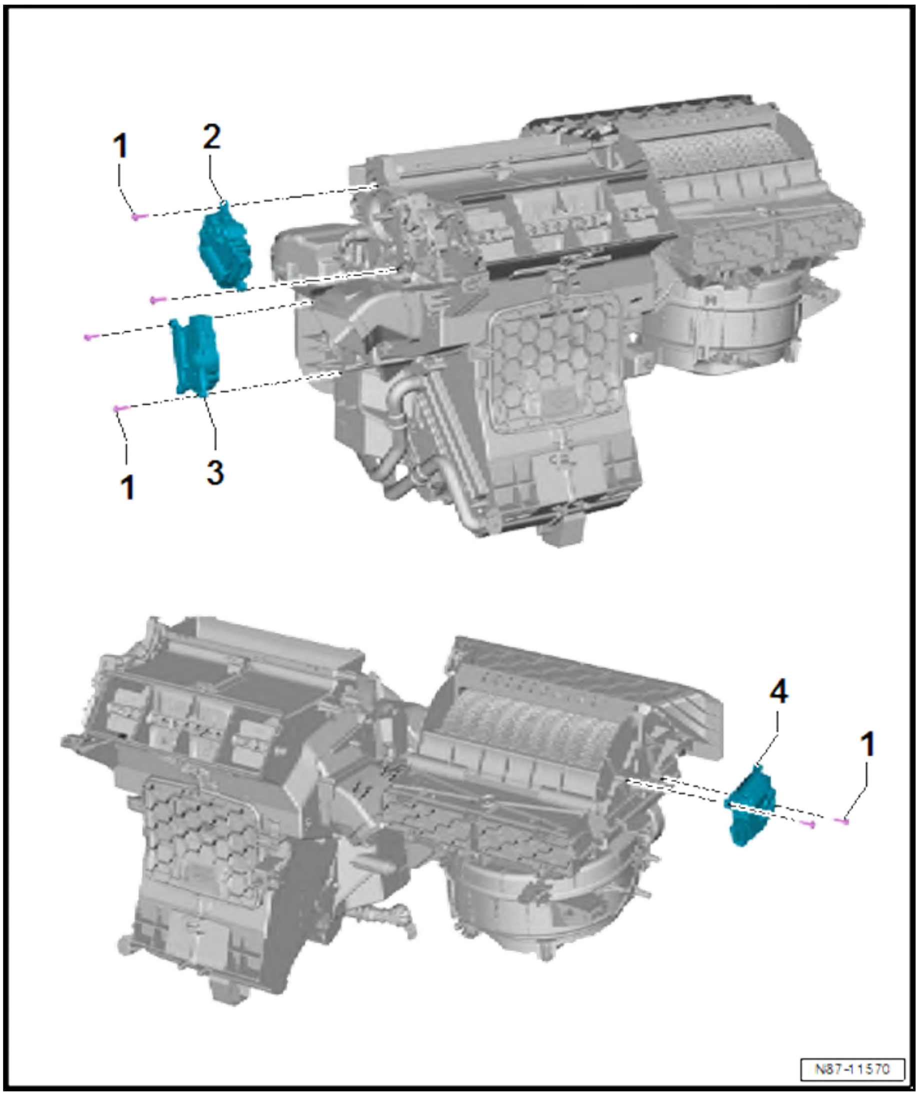
4.2 Assembly overview - front control motors

⇒ “4.2.1 Assembly overview - control motors at front, LHD vehicles with Climatic”, page 117

⇒ “4.2.2 Assembly overview - control motors at front, RHD vehicles with Climatic”, page 118

⇒ “4.2.3 Assembly overview - control motors at front, vehicles with Climatronic”, page 119

4.2.1 Assembly overview - control motors at front, LHD vehicles with Climatic



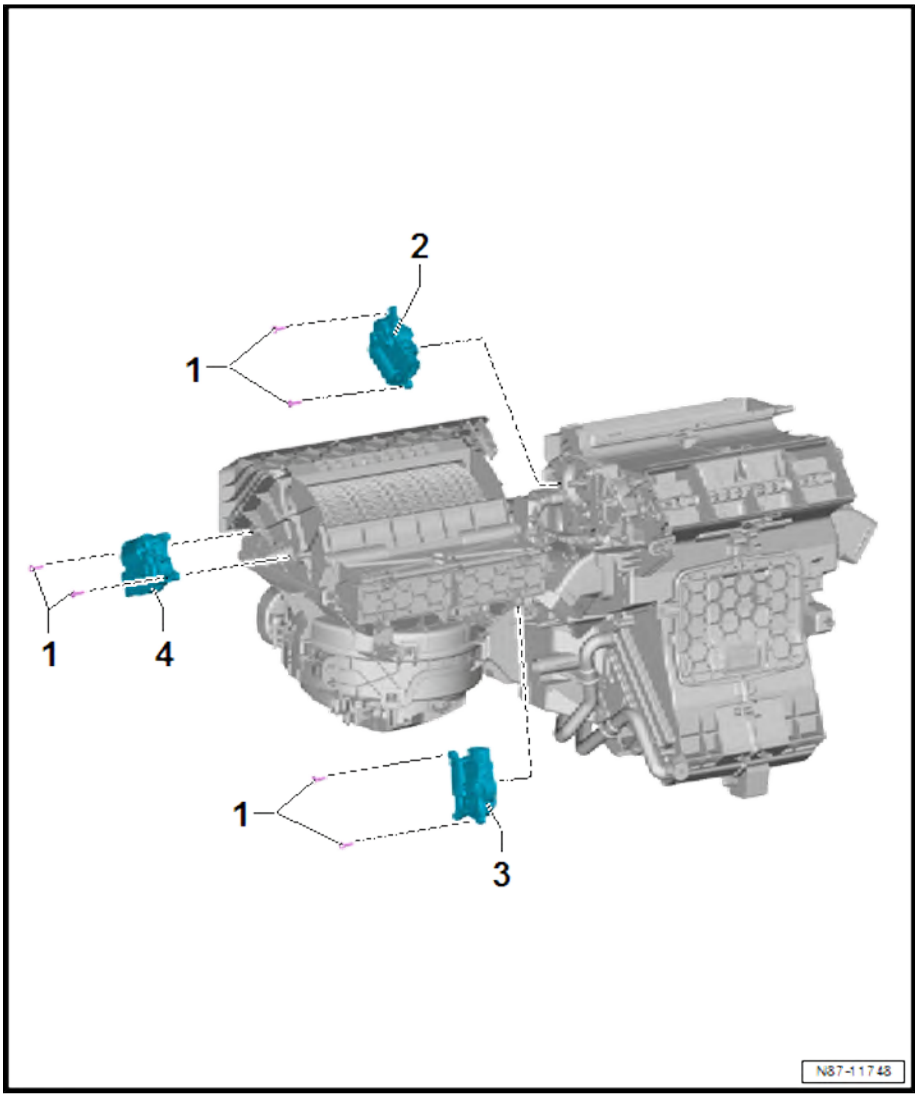
1 - Bolts
- ❑ Qty. 6
- ❑ 1.5 Nm

2 - Air distribution flap control motor - V428-
- ❑ With potentiometer for air distribution flap control motor - G645-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 129

3 - Temperature flap control motor - V68-
- ❑ With potentiometer for temperature flap control motor - G92-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 119

4 - Air recirculation flap control motor - V113-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 122

4.2.2 Assembly overview - control motors at front, RHD vehicles with Climatic



1 - Bolts

- ❑ Qty. 6
- ❑ 1.5 Nm

2 - Air distribution flap control motor - V428-

- ❑ With potentiometer for air distribution flap control motor - G645-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 129

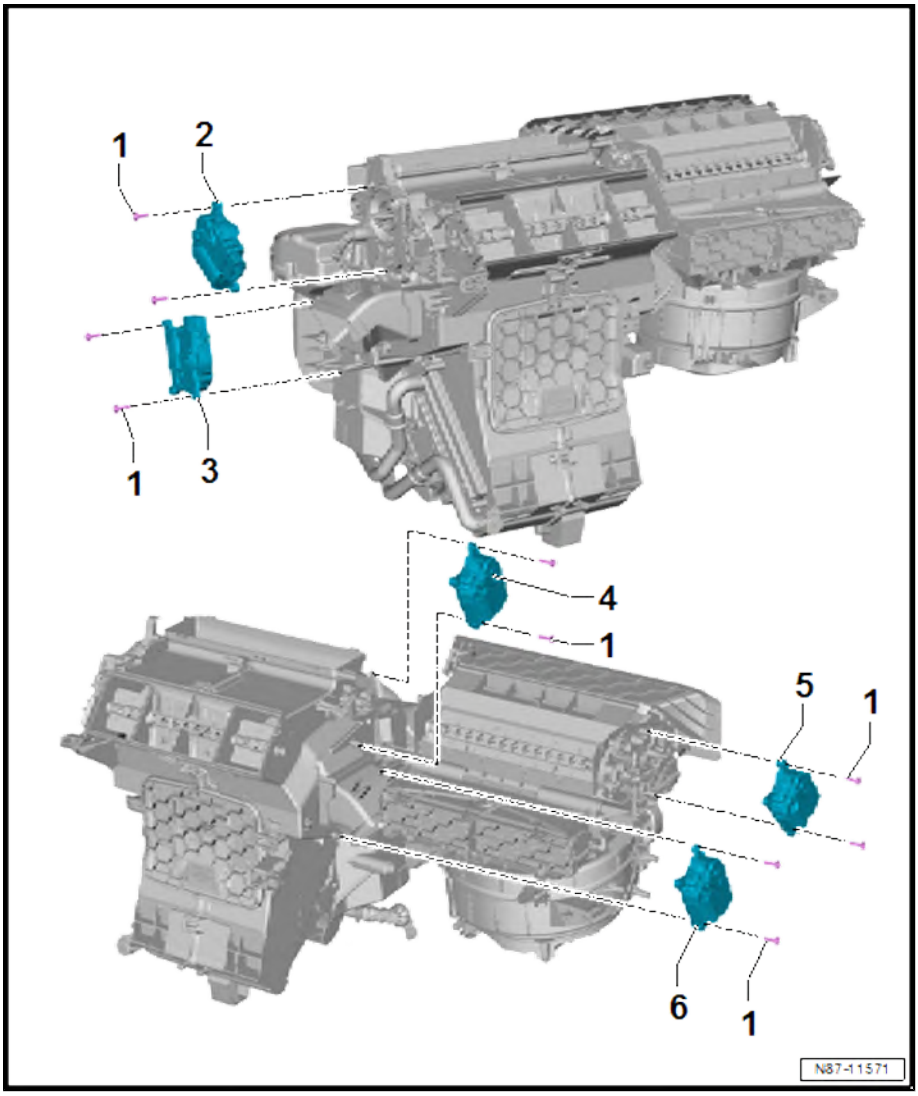
3 - Temperature flap control motor - V68-

- ❑ With potentiometer for temperature flap control motor - G92-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 119

4 - Air recirculation flap control motor - V113-

- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 122

4.2.3 Assembly overview - control motors at front, vehicles with Climatronic



1 - Bolts

- ❑ Qty. 10
- ❑ 1.5 Nm

2 - Front air distribution flap control motor - V426- with potentiometer for front air distribution flap control motor - G642-

- ❑ With potentiometer for front air distribution flap control motor - G642-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 128

3 - Left temperature flap control motor - V158-

- ❑ With potentiometer for left temperature flap control motor - G220-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 124

4 - Defroster flap control motor - V107-

- ❑ With potentiometer for defroster flap control motor - G135-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 121

5 - Fresh air/air recirculation, air flow flap control motor - V425-

- ❑ With potentiometer for fresh air/recirculated air and air flow flap control motor - G644-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 126

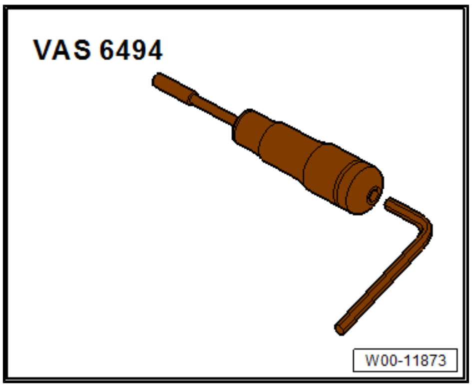
6 - Right temperature flap control motor - V159-

- ❑ With potentiometer for right temperature flap control motor - G221-
- ❑ Check with ⇒ Vehicle diagnostic tester
- ❑ Removing and installing ⇒ page 125

4.3 Removing and installing temperature flap control motor - V68-

Special tools and workshop equipment required

- Torque screwdriver - VAS 6494-



- Vehicle diagnostic tester

Removing

– Remove footwell vent on driver side ⇒ page 176 .

Right-hand drive vehicles

– Remove footwell vent on front passenger side ⇒ page 177 .

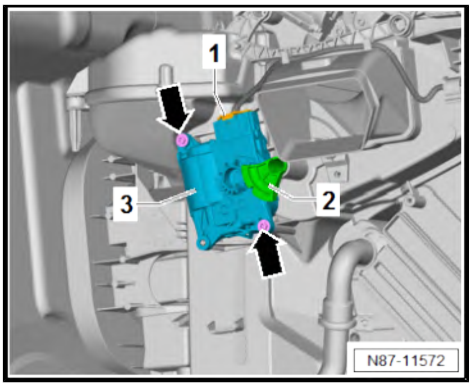
Continued for all vehicles

– Pull off actuating lever -2- from air distribution housing.

– Unscrew bolts -arrows-.

– Remove temperature flap control motor - V68- -3- from air distribution housing.

– Disconnect connector -1-.

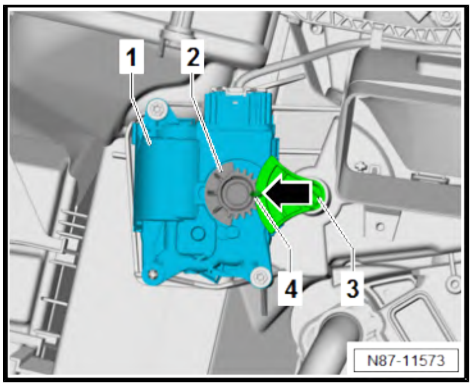


Installing

Install in reverse order of removal, observing the following:

– Check operation of flaps and hinge mechanism before fitting.

– Make sure levers and shafts are properly fitted in the mounts.



– Mount temperature flap control motor - V68- -1- on air distribution housing.

– Insert actuating lever -3- into air distribution housing. Gear wheel on actuating lever -3- must engage in gear wheel -2- on temperature flap control motor - V68- -1-.

• Long tooth -4- must engage in recess -arrow- on actuating lever -3-.

Note

If gear wheel -2- on temperature flap control motor - V68- -1- is not aligned with gear wheel on actuating lever -3- for temperature flap, turn mounting in temperature flap control motor - V68- .

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

– Check operation of heater and air conditioning system.

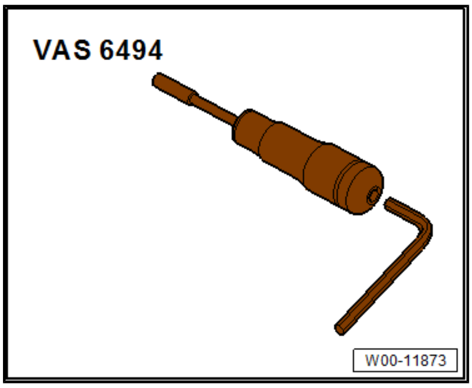
Specified torques

♦ ⇒ “4.2 Assembly overview - front control motors”, page 117

4.4 Removing and installing defroster flap control motor - V107-

Special tools and workshop equipment required

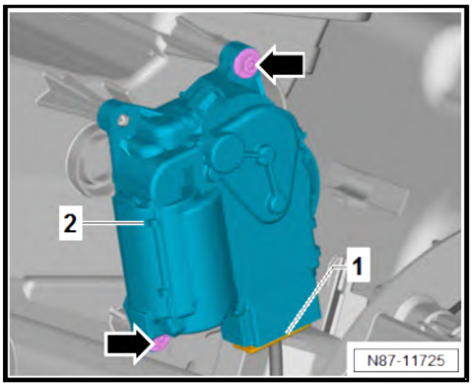
♦ Torque screwdriver - VAS 6494-



♦ Vehicle diagnostic tester

Removing

– Remove dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .



– Disconnect connector -1-.

– Unscrew bolts -arrows-.

– Remove defroster flap control motor - V107- -2- from heater and air conditioning unit.

Installing

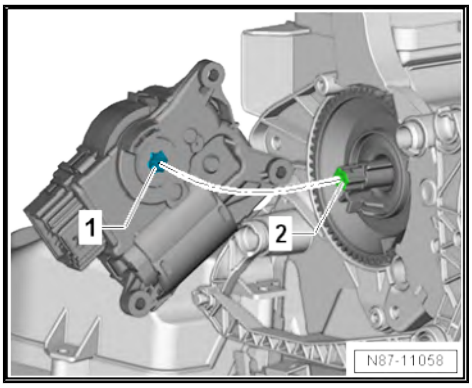
Install in reverse order of removal, observing the following:

– Check operation of flaps and hinge mechanism before fitting.

– Make sure levers and shafts are properly fitted in the mounts.

Note

- *Mounting -1- on defroster flap control motor - V107- can be fitted on relay lever -2- in only one position.*
- *If mounting -1- on defroster flap control motor - V107- and relay lever -2- are not aligned, turn mounting in defroster flap control motor - V107- .*



– Install defroster flap control motor - V107- on heater and air conditioning unit.

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

– Check operation of heater and air conditioning system.

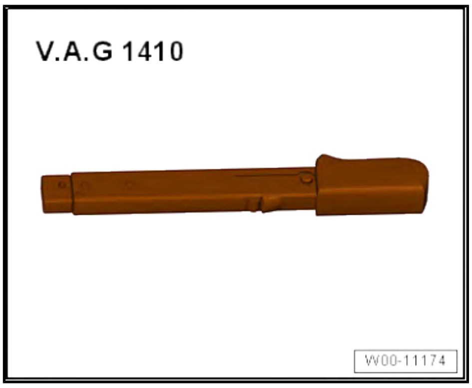
Specified torques

- ⇒ "4.2 Assembly overview - front control motors", page 117
- Dash panel; Assembly overview - dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Assembly overview - dash panel .

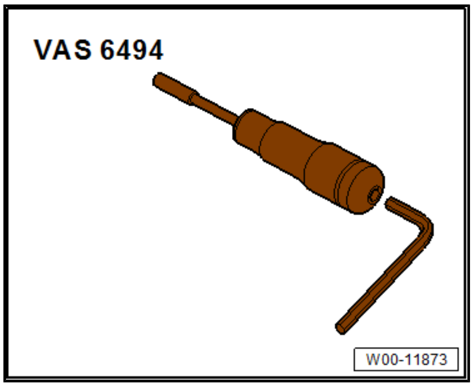
4.5 Removing and installing air recirculation flap control motor - V113-

Special tools and workshop equipment required

- Torque wrench - V.A.G 1410-



- Torque screwdriver - VAS 6494-

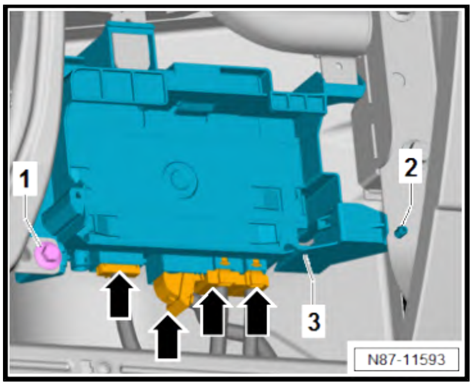


- Vehicle diagnostic tester

Removing

– Remove glove compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing glove compartment .

Vehicles with special vehicle control unit - J608-



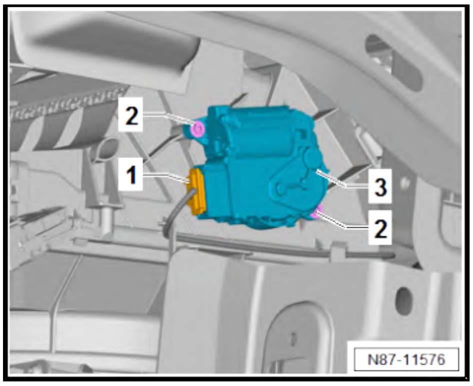
– Disconnect electrical connectors -arrows-.

– Unscrew bolt -1-.

– Press together spreader clip -2-. While doing so, push special vehicle control unit - J608- -3- in direction of engine.

– Remove bracket for special vehicle control unit - J608- -3- in direction of footwell.

Continued for all vehicles



– Disconnect connector -1-.

– Unscrew bolts -2-.

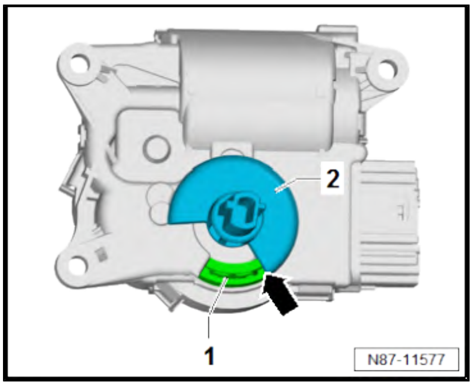
– Remove air recirculation flap control motor - V113- -3- from air intake duct.

Installing

Install in reverse order of removal, observing the following:

– Check operation of flaps and hinge mechanism before fitting.

– Make sure levers and shafts are properly fitted in the mounts.



Note

- *Fresh air flap must be set to closed position to allow for installation of air recirculation flap control motor - V113- .*
- *Actuating shaft -2- must be at stop -1- -arrow-.*
- *If actuating shaft -2- of air recirculation flap control motor - V113- is not against fresh air mode stop -1-, turn mounting in control motor.*

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

– Check operation of heater and air conditioning system.

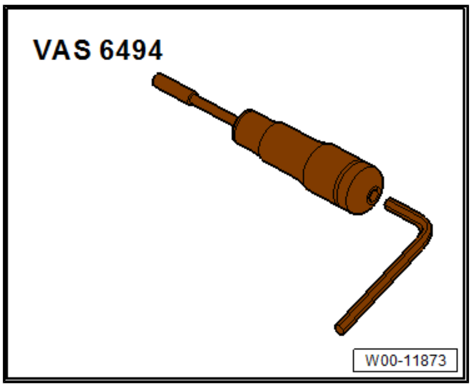
Specified torques

- ⇒ “4.2 Assembly overview - front control motors”, page 117
- Compartments/covers; Assembly overview - glove compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Assembly overview - glove compartment .
- Overview of fitting locations - control units ⇒ Electrical system; Rep. gr. 97 ; Control units; Overview of fitting locations - control units

4.6 Removing and installing left temperature flap control motor - V158-

Special tools and workshop equipment required

- ♦ Torque screwdriver - VAS 6494-



- ♦ Vehicle diagnostic tester

Removing

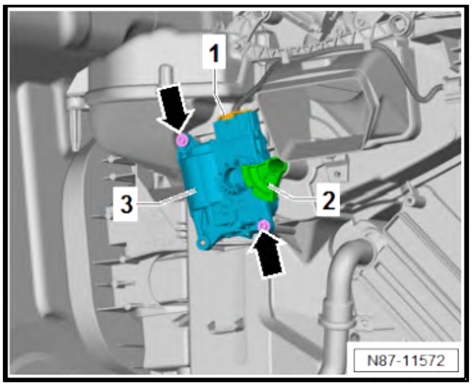
– Remove footwell vent on driver side ⇒ page 176 .

– Pull off actuating lever -2- from air distribution housing.

– Unscrew bolts -arrows-.

– Remove left temperature flap control motor - V158- -3- from air distribution housing.

– Disconnect connector -1-.



Installing

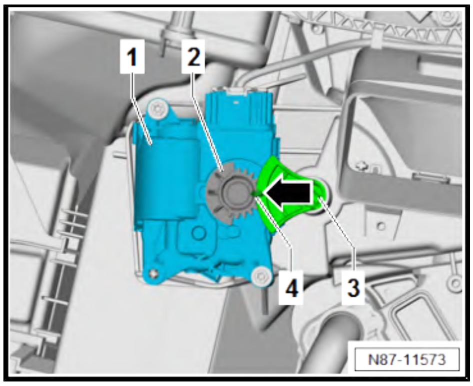
Install in reverse order of removal, observing the following:

– Check operation of flaps and hinge mechanism before fitting.

– Make sure levers and shafts are properly fitted in the mounts.

– Fit left temperature flap control motor - V158- -1- on air distribution housing. While doing this, make sure that gear wheel -2- of control motor engages in gear wheel on actuating lever -3-.

• Long tooth -4- must engage in recess -arrow- on actuating lever -3-.



Note

If gear wheel -2- on left temperature flap control motor - V158- -1- is not aligned with gear wheel on actuating lever -3- for temperature flap, turn mounting in left temperature flap control motor - V158- .

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

– Check operation of heater and air conditioning system.

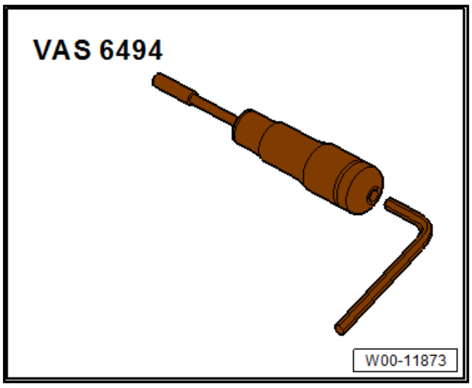
Specified torques

- ♦ ⇒ “4.2 Assembly overview - front control motors”, page 117

4.7 Removing and installing right temperature flap control motor - V159-

Special tools and workshop equipment required

- ♦ Torque screwdriver - VAS 6494-



- ♦ Vehicle diagnostic tester

Removing

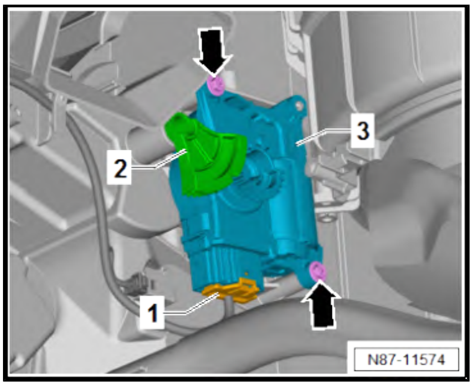
– Remove footwell vent on front passenger side ⇒ page 177 .

– Disconnect connector -1-.

– Pull off actuating lever -2- from air distribution housing.

– Unscrew bolts -arrows-.

– Remove right temperature flap control motor - V159- -3- from air distribution housing.



Installing

Install in reverse order of removal, observing the following:

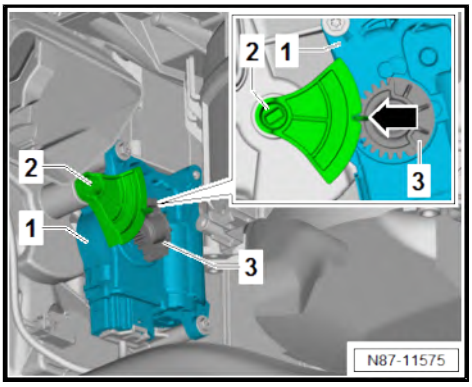
– Check operation of flaps and hinge mechanism before fitting.

– Make sure levers and shafts are properly fitted in the mounts.

– Mount right temperature flap control motor - V159- -1- on air distribution housing.

– Insert actuating lever -2- into air distribution housing. Make sure that gear wheel -3- on actuating lever -2- engages in gear wheel -3- on right temperature flap control motor - V159- -1-.

• Long tooth must engage in recess -arrow- on actuating lever -2-.



Note

If gear wheel -3- on right temperature flap control motor - V159- -1- is not aligned with gear wheel on actuating lever -2- for temperature flap, turn mounting in right temperature flap control motor - V159- .

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

– Check operation of heater and air conditioning system.

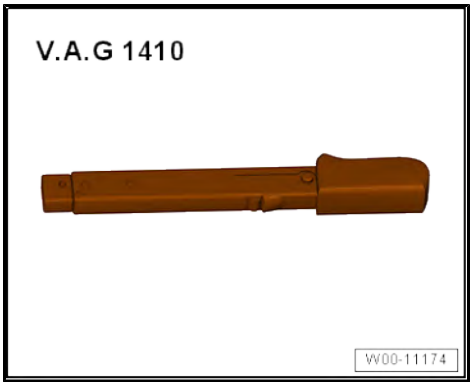
Specified torques

♦ ⇒ “4.2 Assembly overview - front control motors”, page 117

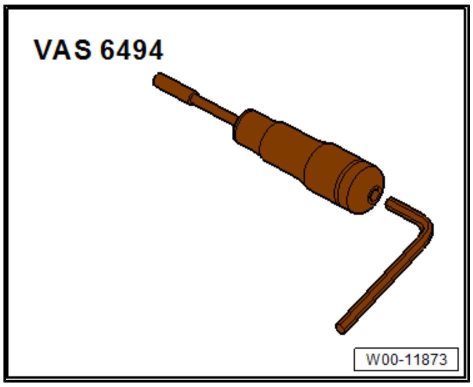
4.8 Removing and installing fresh air/air recirculation, air flow flap control motor - V425-

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1410-



♦ Torque screwdriver - VAS 6494-

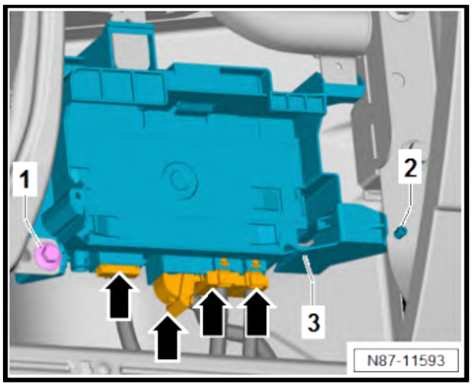


♦ Vehicle diagnostic tester

Removing

– Remove glove compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing glove compartment .

Vehicles with special vehicle control unit - J608-



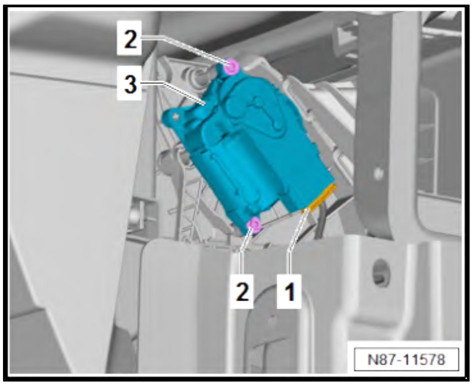
– Disconnect electrical connectors -arrows-.

– Unscrew bolt -1-.

– Press together spreader clip -2-. While doing so, push special vehicle control unit - J608- -3- in direction of engine.

– Remove bracket for special vehicle control unit - J608- -3- in direction of footwell.

Continued for all vehicles



– Disconnect connector -1-.

– Unscrew bolts -2-.

– Remove fresh air/air recirculation air flow flap control motor - V425- -3- from air intake duct.

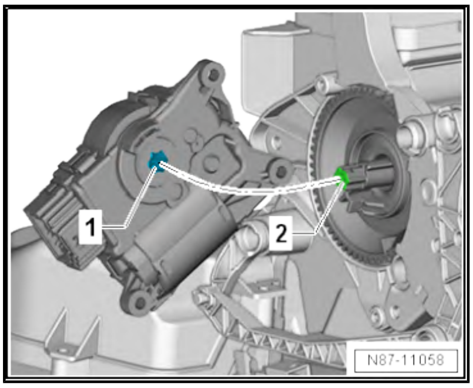
Installing

Install in reverse order of removal, observing the following:

– Check operation of flaps and hinge mechanism before fitting.

– Make sure levers and shafts are properly fitted in the mounts.

Note



♦ *Fresh air flap must be set to closed position to allow for installation of fresh air/air recirculation, air flow flap control motor - V425- .*

♦ *Mounting -1- on fresh air/air recirculation, air flow flap control motor - V425- can be fitted on relay lever -2- in only one position.*

♦ *If mounting -1- on fresh air/air recirculation, air flow flap control motor - V425- and relay lever -2- are not aligned, turn mounting in fresh air/air recirculation, air flow flap control motor - V425- .*

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

– Check operation of heater and air conditioning system.

Specified torques

♦ ⇒ “4.2 Assembly overview - front control motors”, page 117

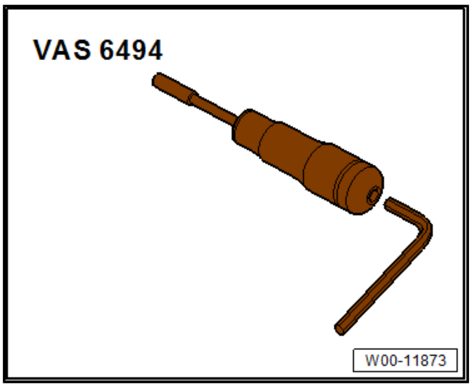
♦ Compartments/covers; Assembly overview - glove compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Assembly overview - glove compartment .

♦ Control units; Assembly overview - data bus diagnostic interface ⇒ Electrical system; Rep. gr. 97 ; Control units; Assembly overview - data bus diagnostic interface

4.9 Removing and installing front air distribution flap control motor - V426-

Special tools and workshop equipment required

- ♦ Torque screwdriver - VAS 6494-



- ♦ Vehicle diagnostic tester

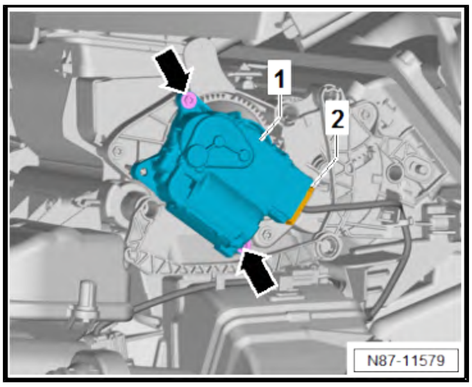
Removing

– Remove dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .

– Unscrew bolts -arrows-.

– Remove front air distribution flap control motor - V426- -1- from air distribution housing.

– Separate electrical connector -2-.

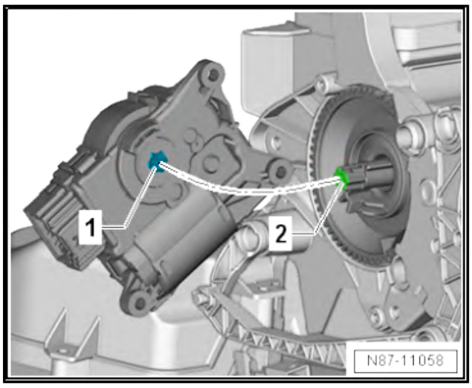


Installing

Install in reverse order of removal, observing the following:

– Check operation of flaps and hinge mechanism before fitting.

– Make sure levers and shafts are properly fitted in the mounts.



Note

- *Mounting -1- on front air distribution flap control motor - V426- can be fitted on relay lever -2- in only one position.*
- *If mounting -1- on front air distribution flap control motor - V426- and relay lever -2- are not aligned, turn mounting in front air distribution flap control motor - V426- .*

– Mount front air distribution flap control motor - V426- on air distribution housing.

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

– Check operation of heater and air conditioning system.

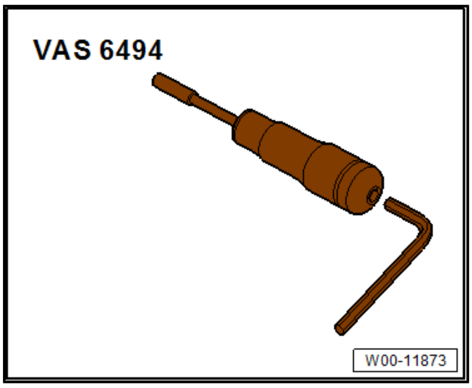
Specified torques

- ⇒ “4.2 Assembly overview - front control motors”, page 117
- Dash panel; Assembly overview - dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Assembly overview - dash panel .

4.10 Removing and installing air distribution flap control motor - V428-

Special tools and workshop equipment required

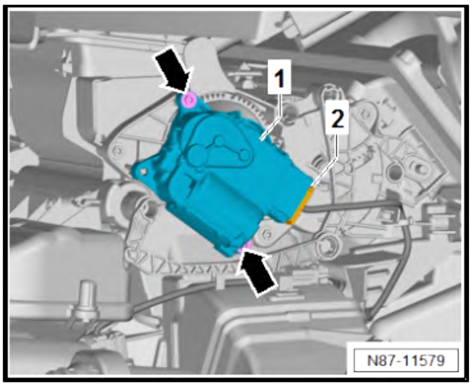
- Torque screwdriver - VAS 6494-



- Vehicle diagnostic tester

Removing

– Remove dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .

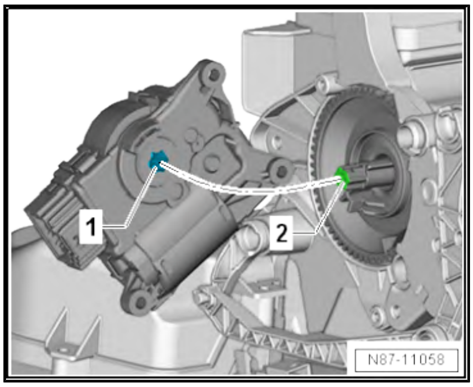


– Unscrew bolts -arrows-.
– Remove front air distribution flap control motor - V428- -1- from air distribution housing.
– Separate electrical connector -2-.

Installing

Install in reverse order of removal, observing the following:

– Check operation of flaps and hinge mechanism before fitting.
– Make sure levers and shafts are properly fitted in the mounts.



Note

- *Mounting -1- on front air distribution flap control motor - V428- can be fitted on relay lever -2- in only one position.*
- *If mounting -1- on front air distribution flap control motor - V428- and relay lever -2- are not aligned, turn mounting in front air distribution flap control motor - V428- .*

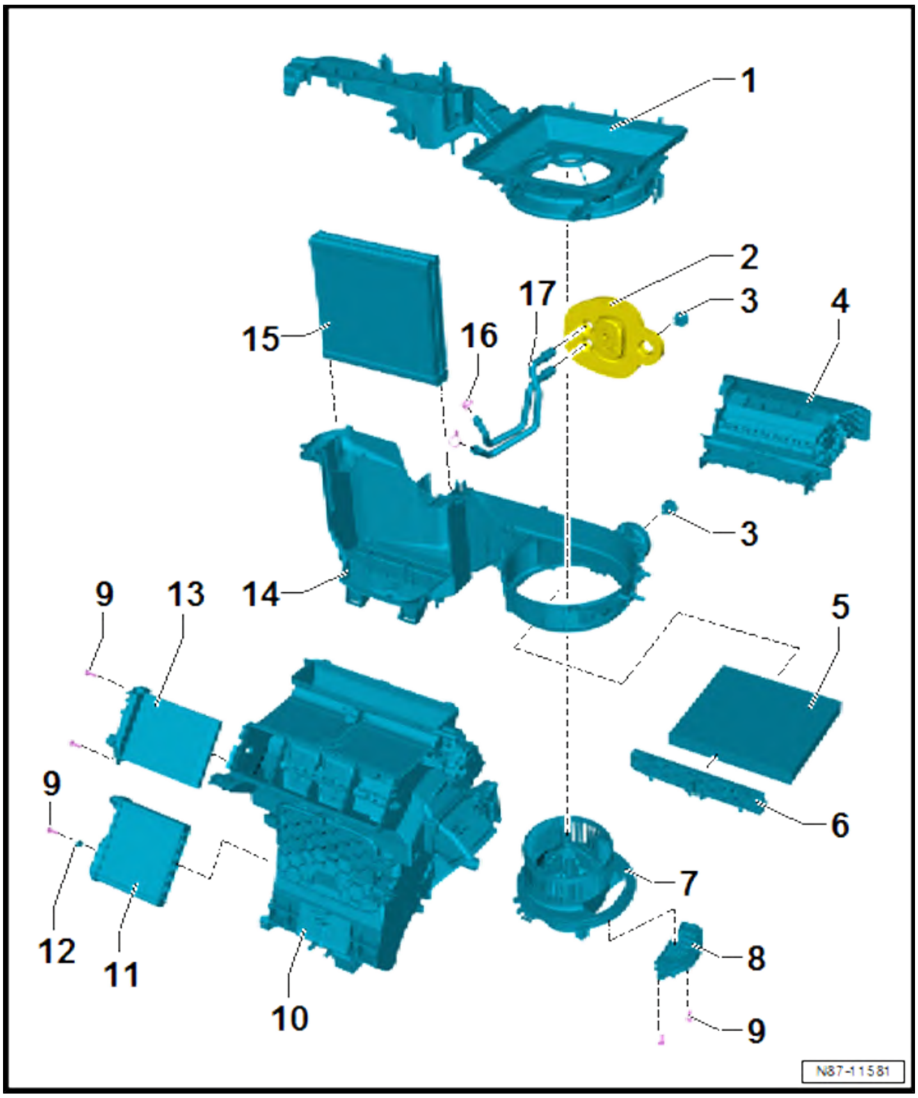
– Mount front air distribution flap control motor - V428- on air distribution housing.
– Initiate basic setting using ⇒ Vehicle diagnostic tester.
– Check operation of heater and air conditioning system.

Specified torques

- ⇒ “4.2 Assembly overview - front control motors”, page 117
- Dash panel; Assembly overview - dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Assembly overview - dash panel .

5 Front heater and air conditioning unit

5.1 Assembly overview - heater and air conditioning unit



1 - Evaporator housing
- ❑ Upper part
- ❑ Assembly overview ⇒ page 133
- ❑ Dismantling and assembling ⇒ page 142

2 - Seal
- ❑ For sealing and insulation

3 - Bracket
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 141

4 - Air intake duct
- ❑ Removing and installing ⇒ page 189

5 - Dust and pollen filter
- ❑ Different versions ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE
- ❑ Replacement interval ⇒ Maintenance tables or ⇒ maintenance test lists for MAN TGE
- ❑ Removing and installing ⇒ page 144

6 - Cover
- ❑ For dust and pollen filter

7 - Fresh air blower - V2-
- ❑ Removing and installing ⇒ page 146

8 - Fresh air blower control unit - J126-
- ❑ Removing and installing ⇒ page 146

9 - Bolts
- ❑ Qty. 5
- ❑ 1.5 Nm

10 - Air distribution housing
- ❑ Removing and installing ⇒ page 143

11 - Heat exchanger
- ❑ Removing and installing ⇒ page 147

12 - Heat exchanger bracket

13 - Auxiliary air heater element - Z35-
- ❑ Checking ⇒ page 134
- ❑ Removing and installing ⇒ page 135

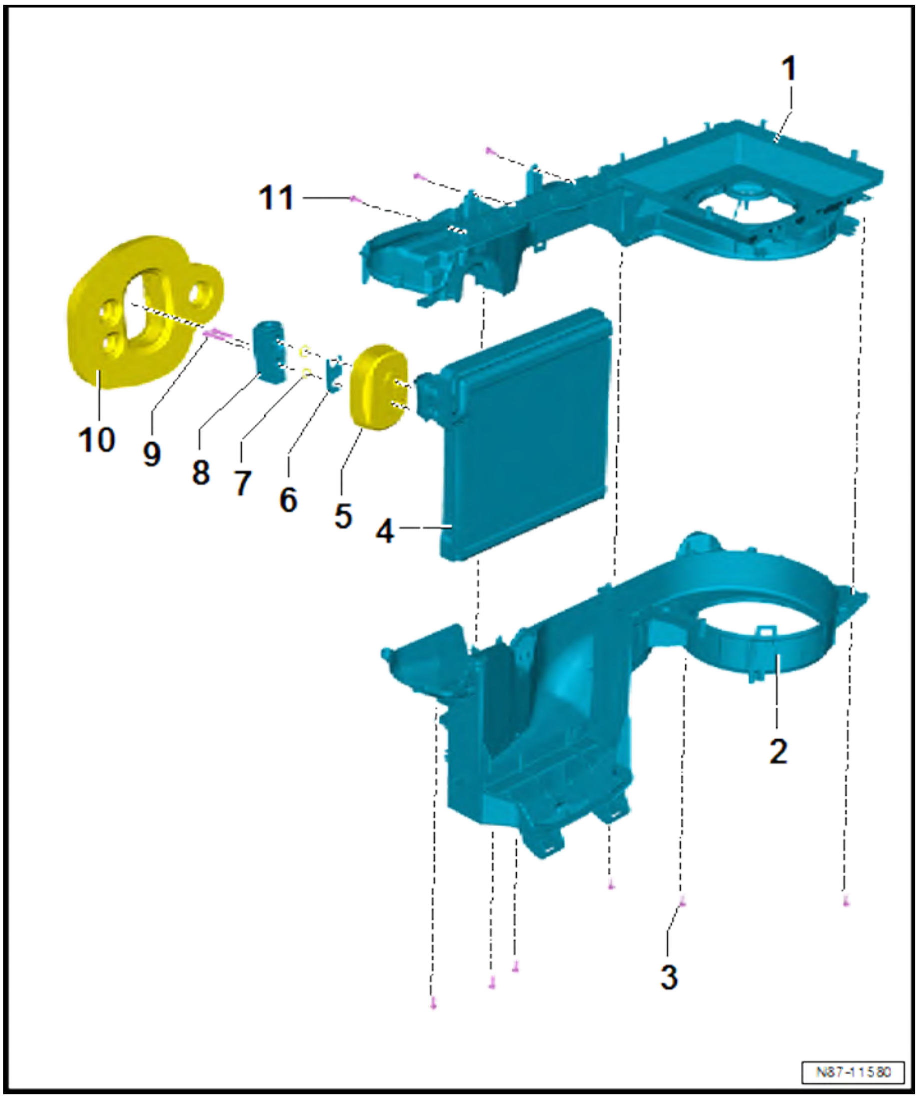
14 - Evaporator housing
- ❑ Lower part
- ❑ Assembly overview ⇒ page 133
- ❑ Dismantling and assembling ⇒ page 142

15 - Evaporator
- ❑ Check foam seals for damage and proper attachment
- ❑ Removing and installing ⇒ page 134

16 - Clamps
- ❑ Qty. 2
- ❑ 6 Nm

17 - Coolant pipes on heat exchanger
- ❑ Removing and installing ⇒ page 148

5.2 Assembly overview - evaporator housing



1 - Evaporator housing
- ❑ Upper part
- ❑ Dismantling and assembling ⇒ page 142

2 - Evaporator housing
- ❑ Lower part
- ❑ Dismantling and assembling ⇒ page 142

3 - Bolts
- ❑ Qty. 8
- ❑ 1.5 Nm

4 - Evaporator
- ❑ Check foam seals for damage and proper attachment
- ❑ Removing and installing ⇒ page 134

5 - Seal
- ❑ For sealing and insulation

6 - Retaining clip

7 - Oil seals
- ❑ Renew after removal
- ❑ Qty. 2

8 - Expansion valve
- ❑ Note stud when removing ⇒ page 59
- ❑ Removing and installing ⇒ page 57

9 - Bolts
- ❑ Qty. 2
- ❑ 10 Nm

10 - Seal
- ❑ For sealing and insulation

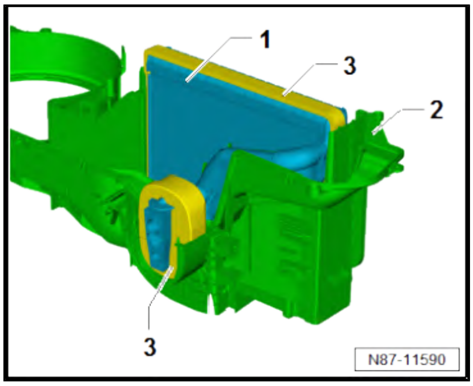
11 - Bolts
- ❑ Qty. 3
- ❑ 1 Nm

5.3 Removing and installing evaporator

Removing

– Dismantling heater and air conditioning unit ⇒ page 142 .

– Pull out evaporator -1- from bottom part of air conditioner and heater -2-.



Installing

Install in reverse order of removal, observing the following:

– Check seals -3- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Clean evaporator housing.

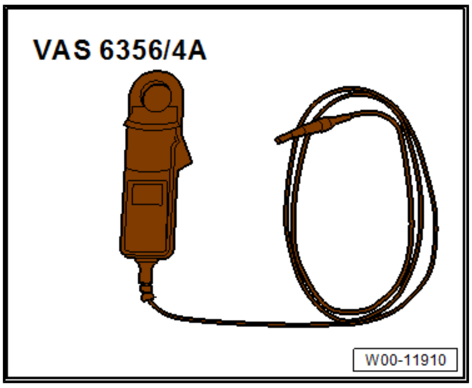
– Check the condensation drain ⇒ page 154 .

5.4 Checking auxiliary air heater element -Z35-

Special tools and workshop equipment required

♦ Vehicle diagnostic tester

♦ 100A pick-up clamp - VAS 6356/4A-



Test sequence

Test conditions:

- Ambient temperature lower than 5°C
- Coolant temperature less than 80°C
- Passenger compartment temperature approx. 20°C
- Battery voltage higher than 11 V
- Alternator load no greater than 50%
- Engine speed greater than 450 rpm
- Turn rotary knob for interior temperature to maximum heat output (on vehicles with Climatronic select highest level).

– Remove trim in centre footwell ⇒ General body repairs, interior; Rep. gr. 68 ; Centre console; Removing and installing centre console trim in front footwell .

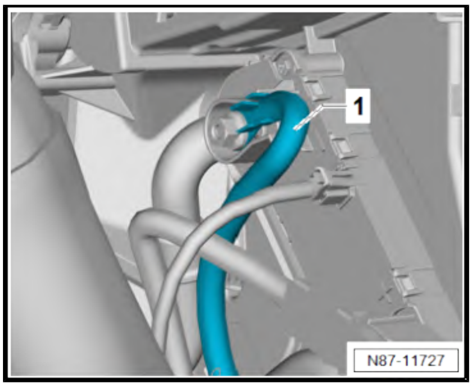
CAUTION

Risk of burns on metal surfaces of auxiliary air heater element. The heater elements may become very hot.

Risk of burns to the skin.

– Wear protective gloves.

– Never touch the metal surfaces of the heater element.



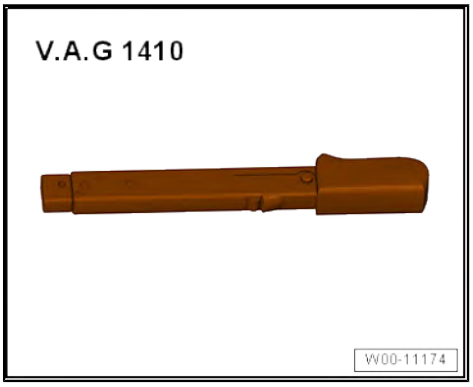
– Measure current draw with ⇒ Vehicle diagnostic tester and current clamp 100A - VAS 6356/4A- at earth wire -1-.

Heating output	Current draw
Low heating output	30 A
Medium heating output	60 A
High heating output	80 A

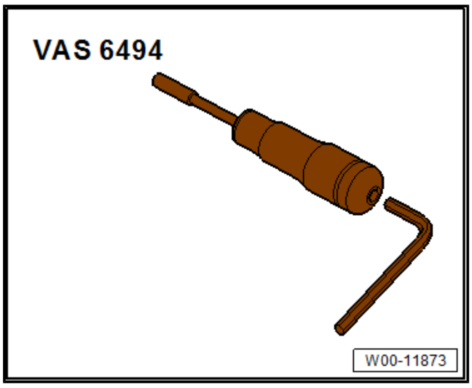
5.5 Removing and installing auxiliary air heater element - Z35-

Special tools and workshop equipment required

◆ Torque wrench - V.A.G 1410-



◆ Torque screwdriver - VAS 6494-



Removing

– Remove central tube for dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Central tube for dash panel; Removing and installing central tube for dash panel .

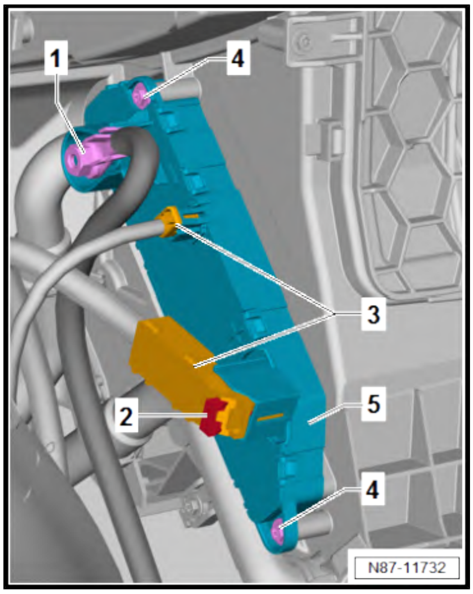
– Push dash panel wiring harness aside in direction of travel.

CAUTION

Risk of burns on metal surfaces of auxiliary air heater element. The heater elements may become very hot.

Risk of burns to the skin.

– Wear protective gloves.
– Never touch the metal surfaces of the heater element.



– Unscrew nut -1-.
– Release locking element -2- in direction of vehicle exterior.
– Separate electrical connectors -3-.
– Unscrew bolts -4-.
– Pull out auxiliary air heater element - Z35- -5- from heater and air conditioning unit.

Installing

Install in reverse order of removal, observing the following:

Specified torques

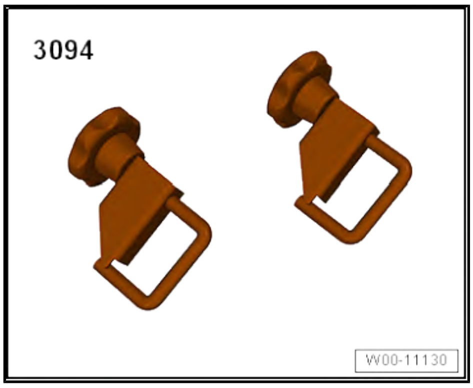
♦ ⇒ “5.1 Assembly overview - heater and air conditioning unit”, page 131

Component	Specified torque
Nut -1- for earth wire	9 Nm

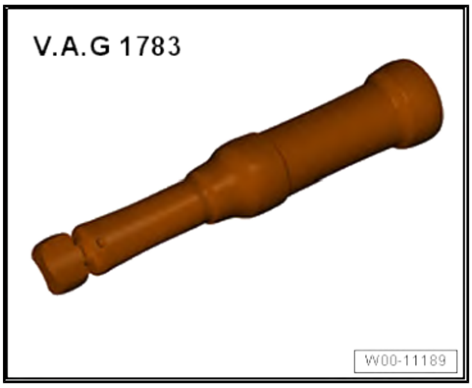
5.6 Removing and installing heater and air conditioning unit

Special tools and workshop equipment required

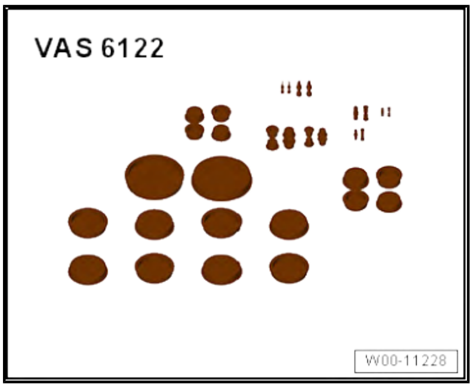
♦ Hose clamps to 25 mm - 3094-



- Torque wrench - V.A.G 1783-



- Set of plugs for engine - VAS 6122-



- Drip tray for workshop hoist - VAS 6208-

- Compressed air gun

Removing

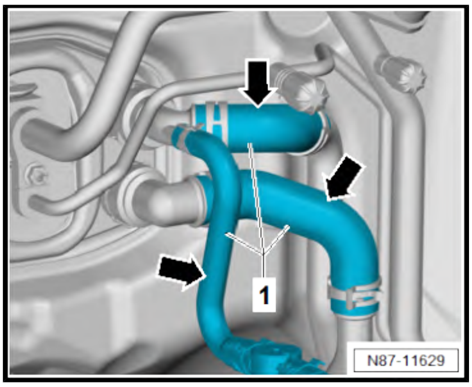
CAUTION

On a warm engine, the cooling system is under high pressure. Danger of scalding by steam and hot coolant.

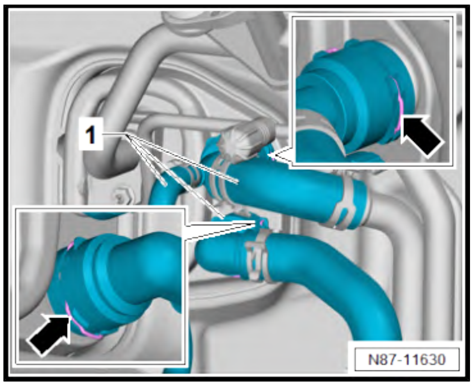
Skin and other parts of the body may be scalded.

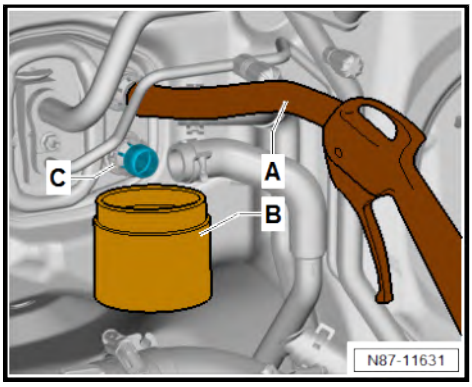
- Wear protective gloves.
- Wear protective goggles.
- Reduce excess pressure by covering cap of coolant expansion tank with cloths and opening it carefully.

- Place drip tray for workshop hoist - VAS 6208- underneath.
- Clamp off coolant hoses -1- with hose clamps, up to 25 mm - 3094- -arrows-.



- Lift retaining clips -arrows- to release plug-in connectors.
- Pull off coolant hoses -1- from heat exchanger.





– Push a suitable hose -A- onto upper connection of heat exchanger.

– Hold suitable container -B- under lower connection leading to heat exchanger -C-.

– Carefully blow out the remaining coolant into the container -B- through the hose from the heat exchanger using a pneumatic gun -A-.

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

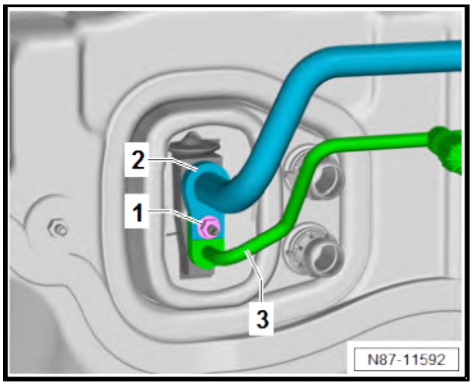
CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

There is a risk of injury to the skin and parts of the body due to freezing.

- **Wear protective gloves.**
- **Wear protective goggles.**
- **Extract refrigerant and open the refrigerant circuit immediately afterwards.**
- **If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.**

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .



– Unscrew nut -1-.

NOTICE

Risk of damage to refrigerant lines from rupture of inner foil.

- **Never bend refrigerant lines to a radius less than 100 mm.**

– Remove refrigerant line -2-.

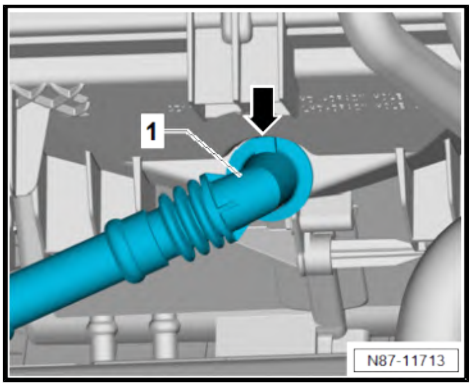
– Remove refrigerant line -3-.

– Place refrigerant lines -2- and -3- to one side.

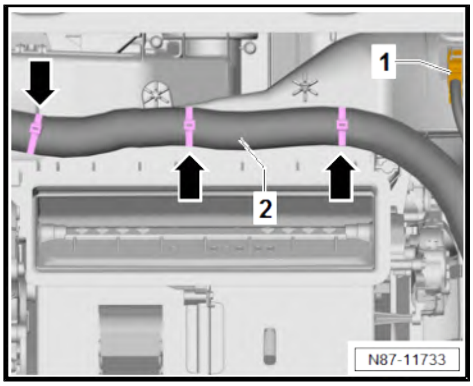
– Remove central tube for dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Central tube for dash panel; Removing and installing central tube for dash panel .

– Cover floor covering in front area of vehicle with waterproof foil and absorbent paper.

– If present, cut through cable ties on condensation drain hose -1-.
– Pull off condensation drain hose -1- from heater and air conditioning unit in area marked by -arrow-.



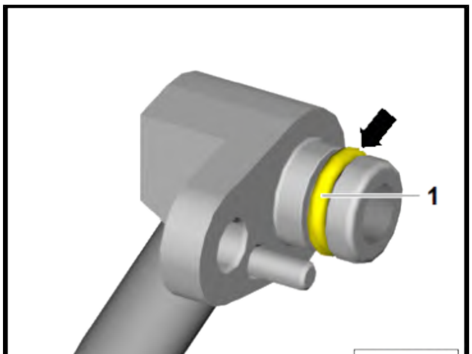
– Disconnect connector -1-.
– Unclip wiring harness -2- in area indicated by -arrows-.



Note

The following work steps must be performed by a second mechanic.

– Pull off heater and air conditioning unit from retainers on plenum chamber bulkhead and remove.



N87-11684

Installing

Install in reverse order of removal, observing the following:

– Ensure proper seating of seal -1- in groove -arrow- of respective refrigerant line.

– Moisten new seals with refrigerant oil before installing refrigerant line.

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Fill up with coolant ⇒ Rep. gr. 19 ; Cooling system/coolant; Draining and filling coolant .

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

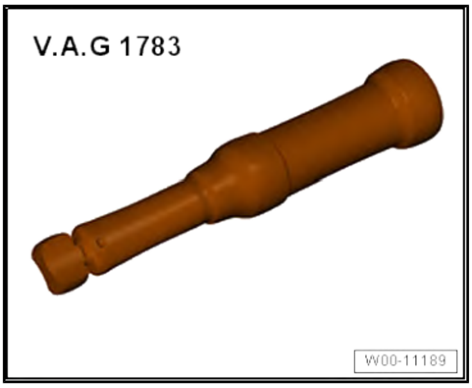
Specified torques

- ⇒ “2.2 Assembly overview - refrigerant lines”, page 50
- ⇒ “5.1 Assembly overview - heater and air conditioning unit”, page 131
- ⇒ “5.2 Assembly overview - evaporator housing”, page 133
- Central tube for dash panel; Assembly overview - central tube for dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Central tube for dash panel; Assembly overview - central tube for dash panel

5.7 Removing and installing holder for heater and air conditioning unit

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1783-



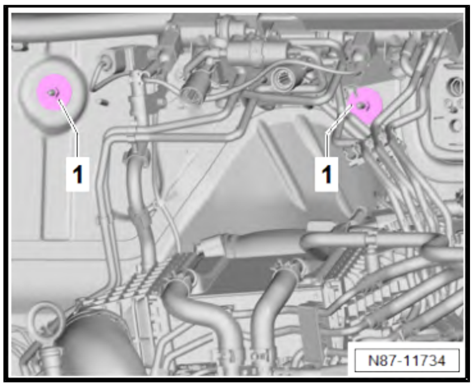
Removing

– Remove heater and air conditioning unit ⇒ page 136 .

– If present, fold perforation in bulkhead insulation to one side.

– Unscrew nuts -1-.

– Remove holder for heater and air conditioning unit in interior.



Installing

Install in reverse order of removal, observing the following:

– Check holder for heater and air conditioning unit for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Insert holder for heater and air conditioning unit in heater and air conditioning unit.

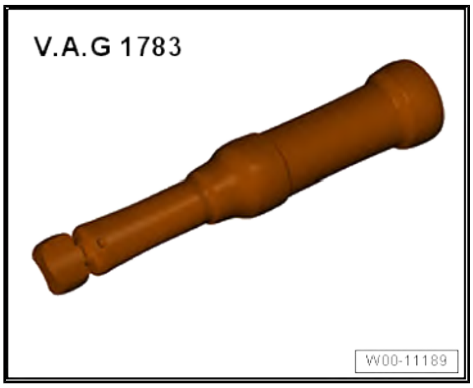
Specified torques

Component	Specified torque
Nuts -1-	4.5 Nm

5.8 Dismantling and assembling heater and air conditioning unit

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1783-

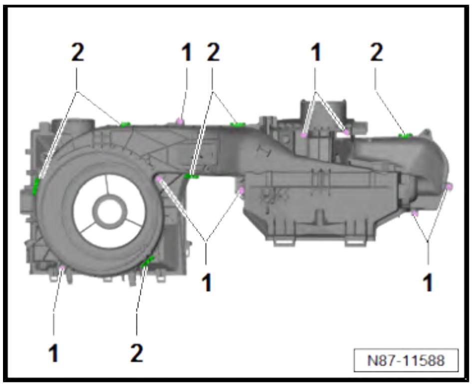


Dismantling

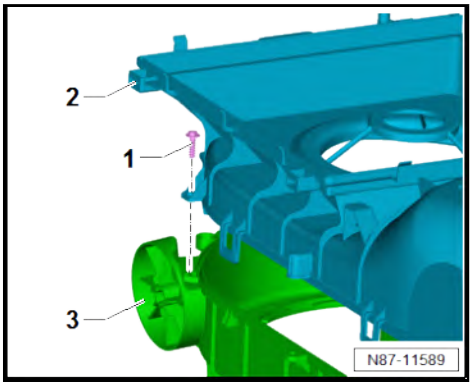
– Remove fresh air blower control unit - J126- ⇒ page 146 .

– Remove air distribution housing ⇒ page 143 .

– Remove air intake ⇒ page 189 .



– Unscrew bolts -1-.
– Release clips -2-.



– Unscrew bolt -1-.
– Remove upper part of heater and air conditioning unit -2- from lower part -3-.

Assembly

Assembly is carried out in reverse sequence; note the following:

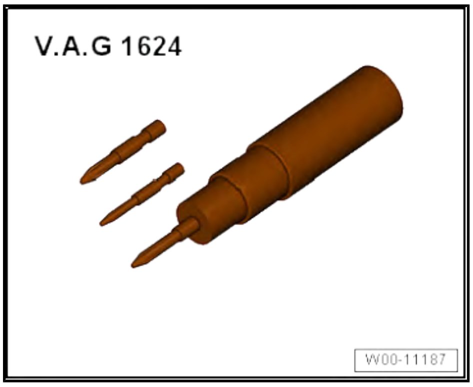
Specified torques

- ⇒ “5.2 Assembly overview - evaporator housing”, page 133

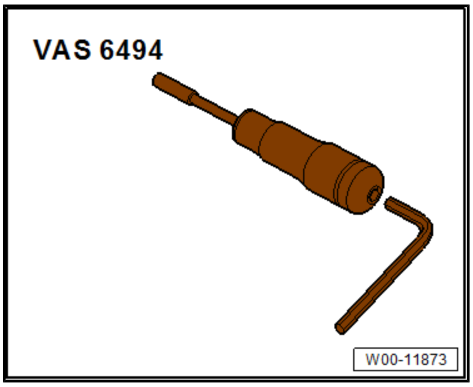
5.9 Removing and installing air distribution housing

Special tools and workshop equipment required

- Torque screwdriver - V.A.G 1624-



- Torque screwdriver - VAS 6494-



- Vehicle diagnostic tester

Removing

Vehicles with air conditioning system

– Remove temperature flap control motor - V68- ⇒ page 119 .

Vehicles with Climatronic

– Remove left temperature flap control motor - V158- ⇒ page 124 .

– Remove right temperature flap control motor - V159- ⇒ page 125 .

Continued for all vehicles

– Remove coolant pipes on heat exchanger ⇒ page 148 .

– On vehicles with air conditioning system, separate the following electrical connector:

♦ Air distribution flap control motor - V428-

– On vehicles with Climatronic, separate the following electrical connector:

♦ Front air distribution flap control motor - V426-

♦ Defroster flap actuator - V107-

♦ Evaporator temperature sensor - G308- ,

♦ Footwell vent temperature sender - G192-

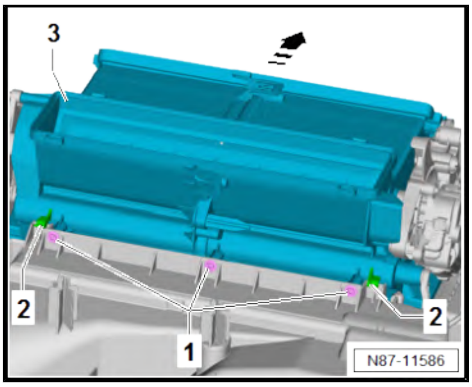
Continued for all vehicles

– Lay aside wiring harness on air distribution housing.

– Unscrew bolts -1-.

– Release locking lugs -2-.

– Remove air distribution housing -3- from evaporator housing in direction of -arrow-.



Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs -2- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Initiate basic setting using ⇒ Vehicle diagnostic tester.

– Check operation of heater and air conditioning system.

Specified torques

♦ ⇒ “5.2 Assembly overview - evaporator housing”, page 133

5.10 Removing and installing dust and pollen filter

Special tools and workshop equipment required

♦ Cover plate - T10532-

Removing

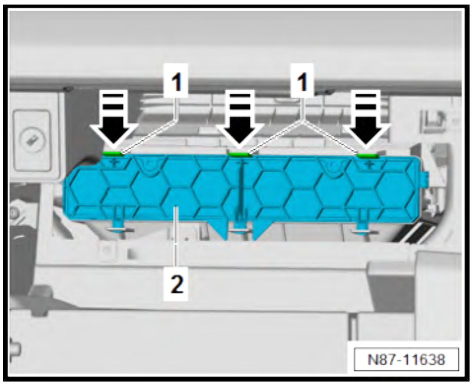
– Move glove compartment lid to service position ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments and covers; Glove compartment lid, service position .

Vehicles with control unit 1 for information electronics - J794-

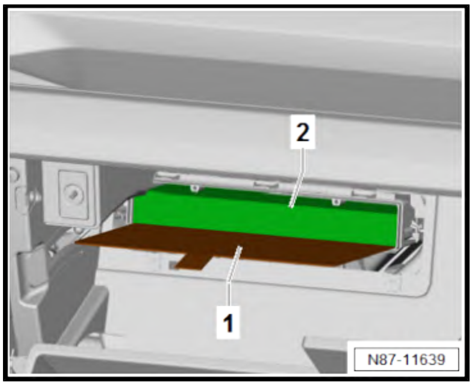
– Remove installation frame for control unit 1 for information electronics - J794- ⇒ General body repairs, interior; Rep. gr. 68 ; Storage compartments/covers; Removing and installing installation frame for information electronics .

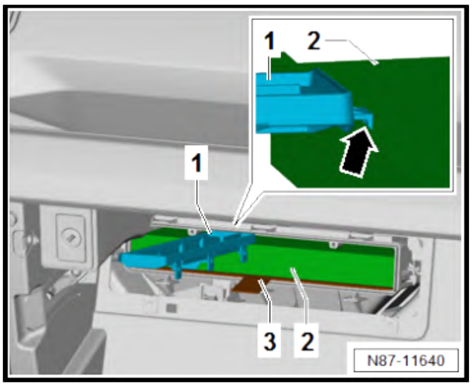
Continued for all vehicles

– Release locking lugs -1- in direction of -arrow-. While doing this, remove cover -2-.



– Push cover plate - T10532- -1- under dust and pollen filter -2-.





– Engage cover -1- with hook -arrow- in dust and pollen filter -2-.

NOTICE

Risk of damage to fresh air blower caused by dirt dropping out of the dust and pollen filter when the cover plate is not seated properly.

– **Carefully pull out dust and pollen filter.**

– **Keep dust and pollen filter covered by cover plate.**

– Pull out dust and pollen filter -2- with cover -1-. Counterhold cover plate - T10532- -3- when doing so.

Installing

Install in reverse order of removal, observing the following:

– Check intake duct for contamination and clean if necessary.

– Observe installation position of dust and pollen filter.

5.11 Removing and installing fresh air blower - V2-

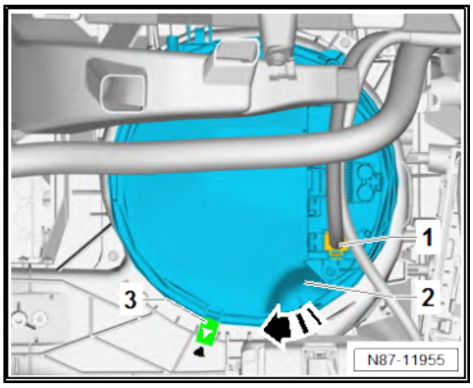
Removing

CAUTION

Risk of burns when touching hot cooling surface of control unit.

Risk of burns to the hands.

– **Wear protective gloves.**



– Separate electrical connector -1-.

– Remove fresh air blower - V2- -2- from evaporator housing lower part by lifting securing tab -3- and turning in direction of -arrow- at the same time.

Installing

Install in reverse order of removal, observing the following:

– Check evaporator housing lower part for contamination and clean if necessary.

NOTICE

Improper handling may damage the fresh air blower.

Imbalance leading to customer complaints may occur during operation.

– **Avoid applying excessive pressure to the fan wheel.**

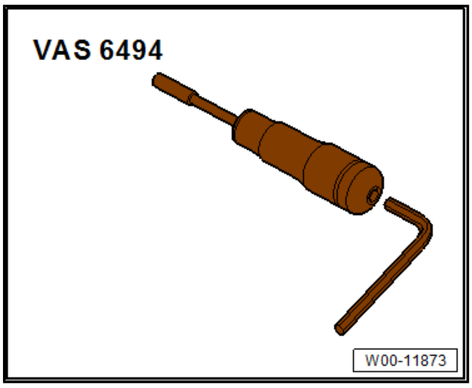
– **Never change position of the balancing weights on fan wheel.**

– Ensure that arrow on locking tab -3- and arrow on bottom part of evaporator housing align.

5.12 Removing and installing fresh air blower control unit - J126-

Special tools and workshop equipment required

♦ Torque screwdriver - VAS 6494-



Removing

– Remove fresh air blower - V2- ⇒ page 146 .

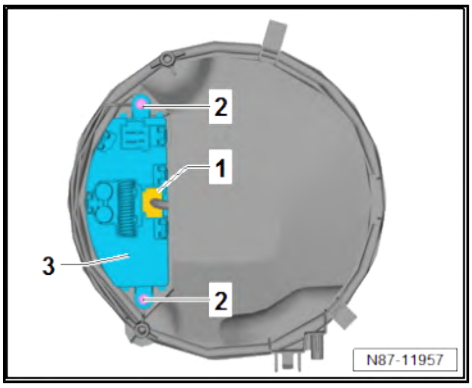
– Place fresh air blower - V2- on a clean foam underlay.

CAUTION

Risk of burns when touching hot cooling surface of control unit.

Risk of burns to the hands.

– Wear protective gloves.



– Separate electrical connector -1-.

– Unscrew bolt -2-.

– Remove fresh air blower control unit - J126- -3-.

Installing

Install in reverse order of removal, observing the following:

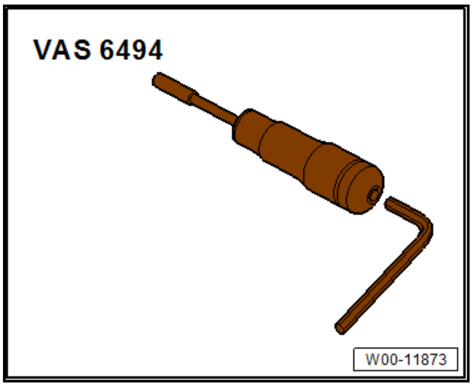
Specified torques

♦ ⇒ “5.1 Assembly overview - heater and air conditioning unit”, page 131

5.13 Removing and installing heat exchanger

Special tools and workshop equipment required

♦ Torque screwdriver - VAS 6494-

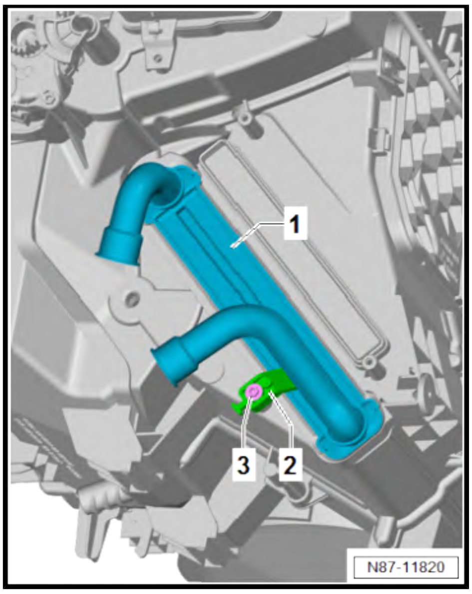


Removing

– Remove coolant pipes on heat exchanger ⇒ page 148 .

– Unscrew bolt -3-. While doing this, detach bracket -2-.

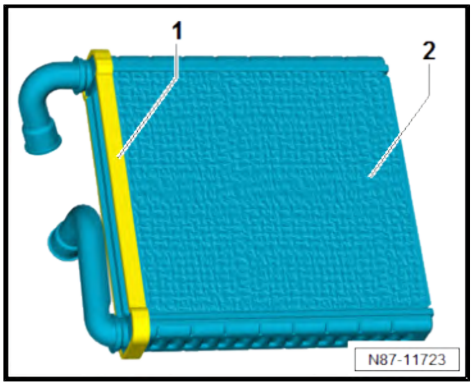
– Pull out heat exchanger -1- from heater and air conditioning unit.



Installing

Install in reverse order of removal, observing the following:

– Check foam seal -1- on heat exchanger -2- for damage. Renew heat exchanger -2- if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



Note

♦ *An improperly bonded or damaged seal -1- could roll up when the heat exchanger -2- is pushed into the heater unit.*

♦ *If the foam seal -1- is damaged or improperly fitted, cold air can flow past the heat exchanger -2-.*

– Check heater duct for contamination and clean if necessary.

– Push heat exchanger into heater and air conditioning unit as far as stop.

– Check connection between coolant pipes and heat exchanger for leaks.

Specified torques

♦ ⇒ “5.1 Assembly overview - heater and air conditioning unit”, page 131

5.14 Removing and installing coolant pipes on heat exchanger

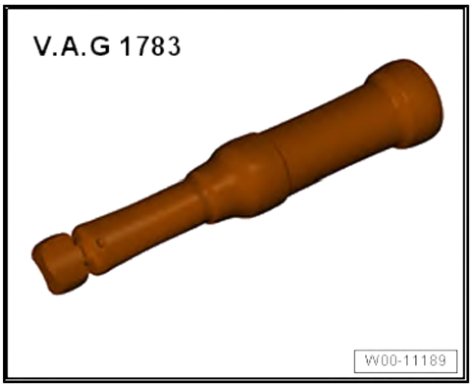
⇒ “5.14.1 Removing and installing coolant pipes on heat exchanger, LHD vehicles”, page 148

⇒ “5.14.2 Removing and installing coolant pipes on heat exchanger, RHD vehicles”, page 150

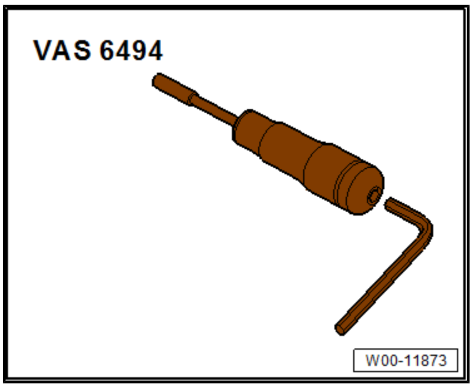
5.14.1 Removing and installing coolant pipes on heat exchanger, LHD vehicles

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1783-

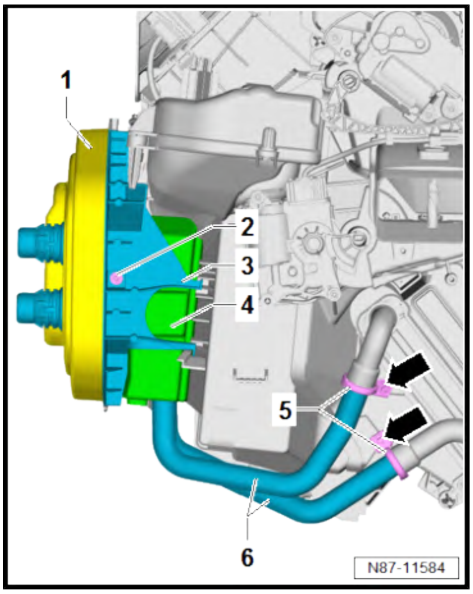


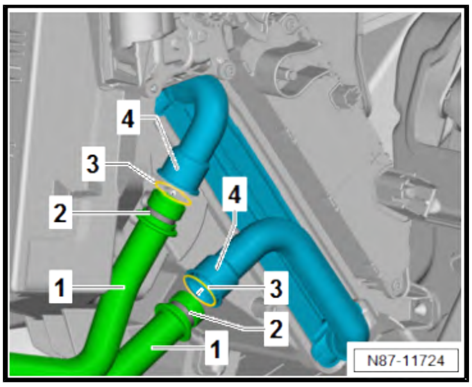
♦ Torque screwdriver - VAS 6494-



Removing

– Remove heater and air conditioning unit ⇒ page 136 .
– Pull off seal -1-.
– Unscrew bolt -2-.
– Remove retainer -3-.
– Detach the foam -4-.
– Unscrew bolts -arrows-.
– Remove clamps -5-.
– Pull off coolant pipes -6- from heat exchanger and remove.





Installing

Install in reverse order of removal, observing the following:

– Check coolant pipe connections -1- and heat exchanger connections -4- for damage and contamination. Clean or renew damaged components as necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Clean and smooth sealing surfaces -2- for seals -3-.

– Moisten new seals -3- with coolant, and slide them onto coolant pipes -1-.

NOTICE

Risk of heat exchanger malfunction due to defective seals and leaks.

– **Never pinch seals.**

– **Never cant coolant pipes.**

– **Slide on coolant pipes completely.**

– Push coolant pipe connections -1- into heat exchanger connections -4- as far as stop.

Specified torques

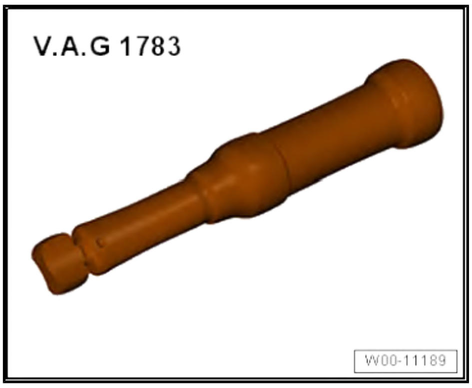
♦ ⇒ “5.1 Assembly overview - heater and air conditioning unit”, page 131

Component	Specified torque
Bolt -2- for bracket -3-	1.5 Nm

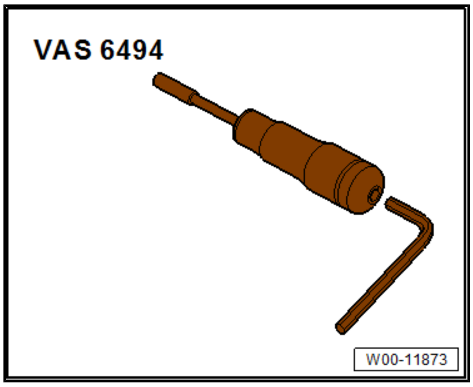
5.14.2 Removing and installing coolant pipes on heat exchanger, RHD vehicles

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1783-

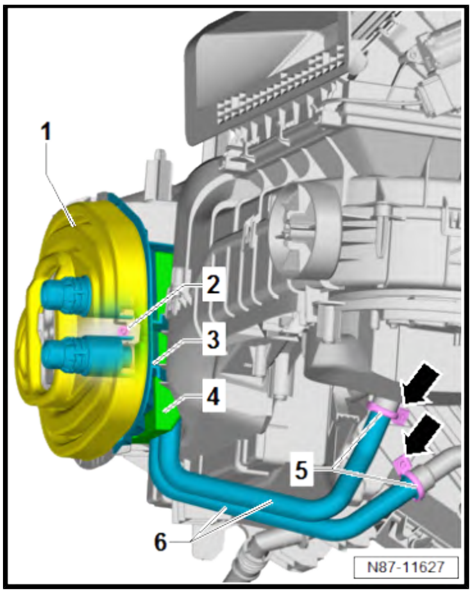


♦ Torque screwdriver - VAS 6494-



Removing

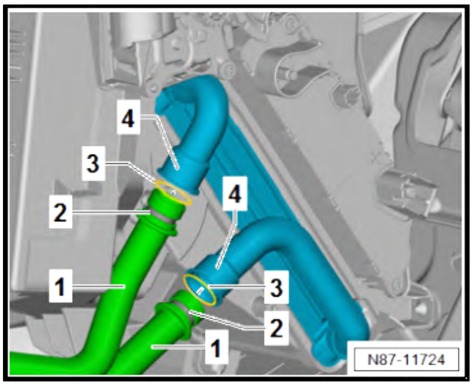
– Remove heater and air conditioning unit ⇒ page 136 .
– Pull off seal -1-.
– Unscrew bolt -2-.
– Remove retainer -3-.
– Detach the foam -4-.
– Unscrew bolts -arrows-.
– Remove clamps -5-.
– Pull off coolant pipes -6- from heat exchanger and remove.



Installing

Install in reverse order of removal, observing the following:

– Check coolant pipe connections -1- and heat exchanger connections -4- for damage and contamination. Clean or renew damaged components as necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Clean and smooth sealing surfaces -2- for seals -3-.
– Moisten new seals -3- with coolant, and slide them onto coolant pipes -1-.



NOTICE

Risk of heat exchanger malfunction due to defective seals and leaks.

– **Never pinch seals.**
– **Never cant coolant pipes.**
– **Slide on coolant pipes completely.**

– Push coolant pipe connections -1- into heat exchanger connections -4- as far as stop.

Specified torques

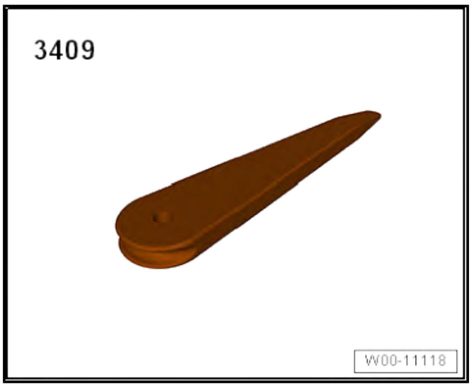
♦ ⇒ “5.1 Assembly overview - heater and air conditioning unit”, page 131

Component	Specified torque
Bolt -2- for bracket -3-	⇒ page 150

5.15 Removing and installing evaporator temperature sensor - G308-

Special tools and workshop equipment required

♦ Removal wedge - 3409-



Removing

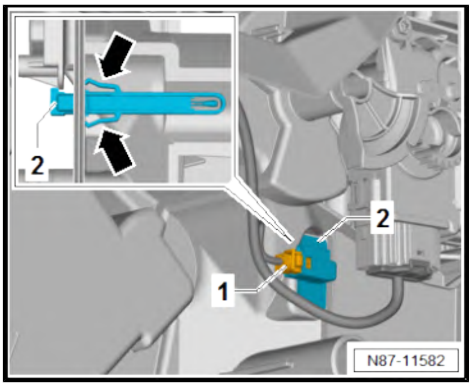
– Remove footwell vent on front passenger side ⇒ page 177 .

Right-hand drive vehicles

– Remove footwell vent on driver side ⇒ page 176 .

Continued for all vehicles

– Disconnect connector -1-.



Note

Levering with removal wedge - 3409- causes both retaining tabs -arrows- to be pushed together.

– Lever evaporator temperature sensor - G308- -2- out of evaporator housing using removal wedge - 3409- .

Installing

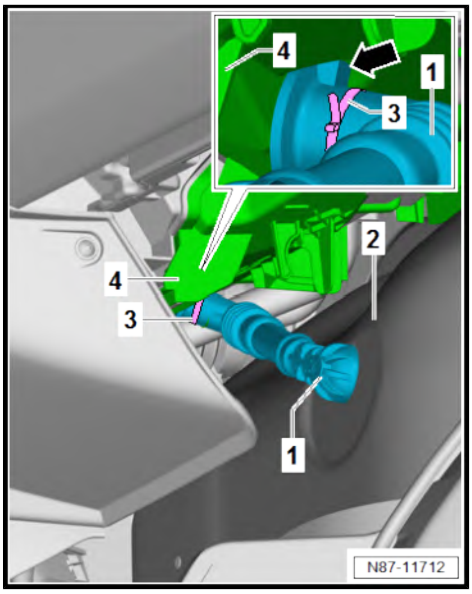
Install in reverse order of removal, observing the following:

– Check retaining tabs -arrows- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ web-MANTIS for MAN TGE.

– Check operation of heater and air conditioning system.

5.16 Removing and installing condensation drain

Removing



– Pull condensation drain hose -1- out of bulkhead -2-.

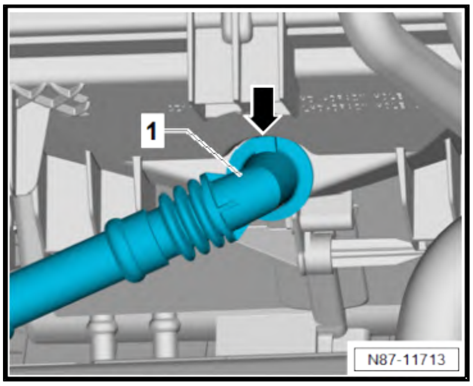
– Separate cable ties -3- if fitted.

– Unclip condensation drain hose -1- from heater and air conditioning unit -4- in area marked by -arrow-, and pull it off.

Installing

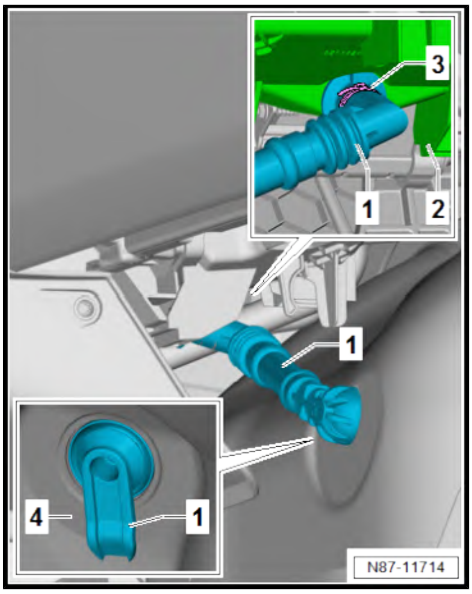
Install in reverse order of removal, observing the following:

– Check condensation drain hose ⇒ page 154 .



Note

Condensation drain hose -1- can be pushed onto heater and air conditioning unit only in one certain position -arrow-.



– Push condensation drain hose -1- onto heater and air conditioning unit -2- as far as stop.

– Fit new cable ties -3-.

– Insert condensation drain hose -1- in bulkhead opening -4- until it is heard to engage.

• Condensation drain hose -1- must be installed taut and above bulkhead insulation -4-.

– Clip condensation drain hose -1- into heater and air conditioning unit -2-.

5.17 Checking condensation drain

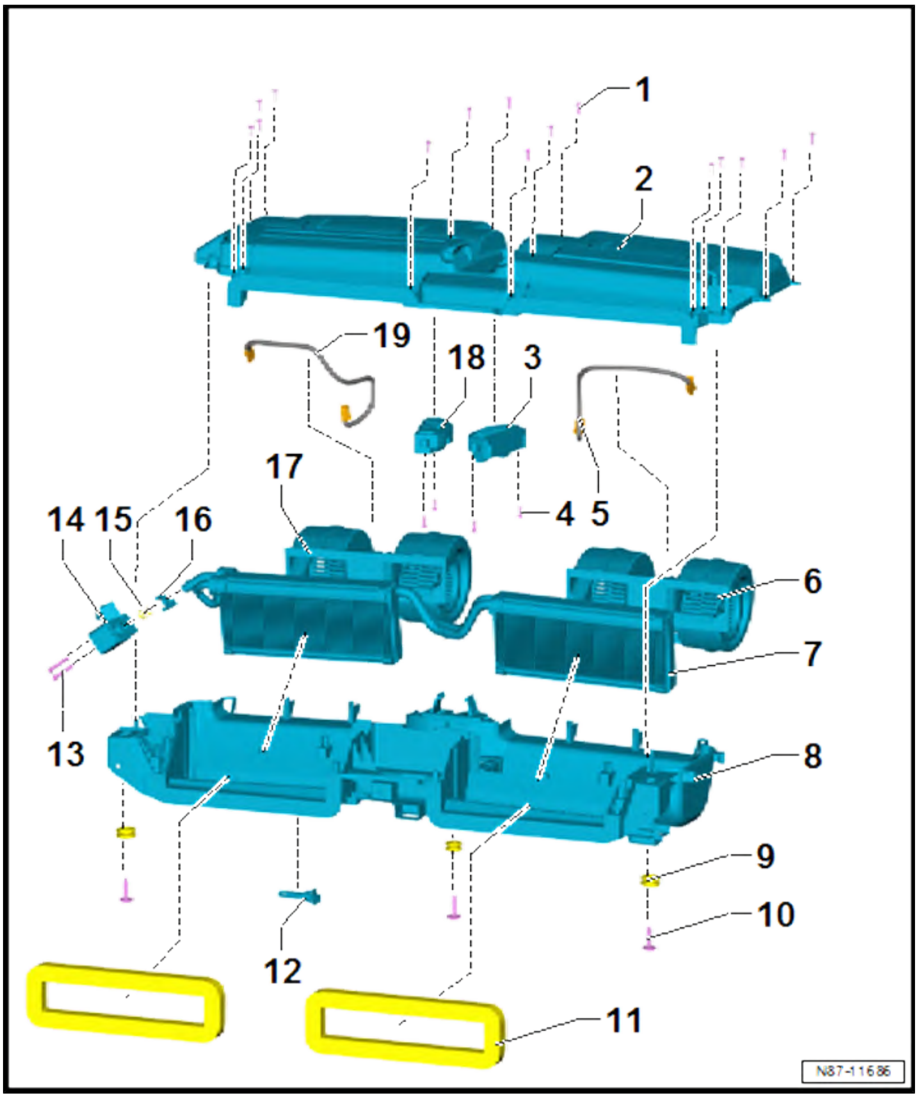
Test sequence

– Remove condensate drain hose ⇒ page 153 .

– Check condensation drain hose for narrowing, damage and blockage, clean or renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

6 Rear heater and air conditioning unit

6.1 Assembly overview - heater and air conditioning unit



1 - Bolts
- ❑ Qty. 15
- ❑ 1.5 Nm

2 - Evaporator housing
- ❑ Top
- ❑ Dismantling and assembling ⇒ page 158

3 - Control unit for supplementary blower for air conditioning - J1144-
- ❑ Removing and installing ⇒ page 167

4 - Bolts
- ❑ Qty. 4
- ❑ 1.5 Nm

5 - Adapter line

6 - Blower housing
- ❑ Right
- ❑ With supplementary blower 2 for air conditioning - V626-
- ❑ Removing and installing ⇒ page 164

7 - Evaporator
- ❑ Removing and installing ⇒ page 157

8 - Evaporator housing
- ❑ Bottom
- ❑ Dismantling and assembling ⇒ page 158

9 - Flexible joints
- ❑ Renew after removal
- ❑ Qty. 3
- ❑ Together with sleeves

10 - Bolts
- ❑ Qty. 3
- ❑ 8 Nm

11 - Seals
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ For air outlet

12 - Rear evaporator temperature sender - G464-
- ❑ Removing and installing ⇒ page 168

13 - Bolts
- ❑ Qty. 2
- ❑ For expansion valve for air conditioner refrigerant - N697-
- ❑ 4.5 Nm

14 - Expansion valve for air conditioner refrigerant - N697-
- ❑ Removing and installing ⇒ page 59

15 - Oil seals

- ❑ Renew after removal
- ❑ Qty. 2

16 - Clip

- ❑ For bolts of expansion valve for air conditioner refrigerant - N697-

17 - Blower housing

- ❑ Left
- ❑ With supplementary blower for air conditioning - V625-
- ❑ Removing and installing ⇒ page 164

18 - Control unit for supplementary blower 2 for air conditioning - J1145-

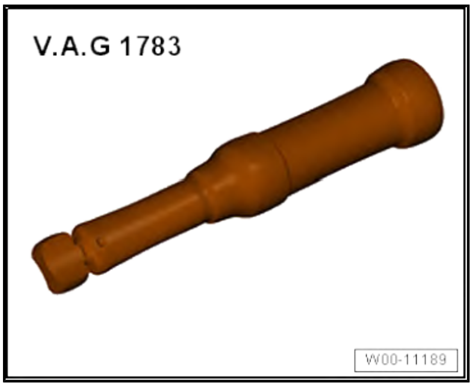
- ❑ Removing and installing ⇒ page 167

19 - Adapter line

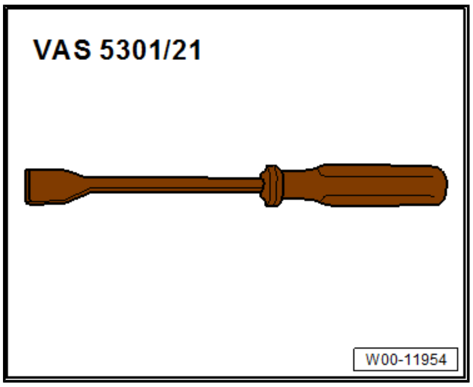
6.2 Removing and installing evaporator

Special tools and workshop equipment required

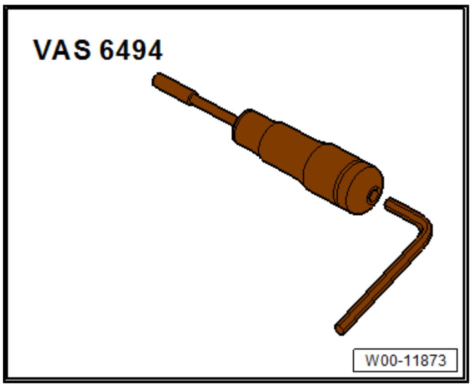
- ♦ Torque wrench - V.A.G 1783-



- ♦ Flat scraper - VAS 5301/21-

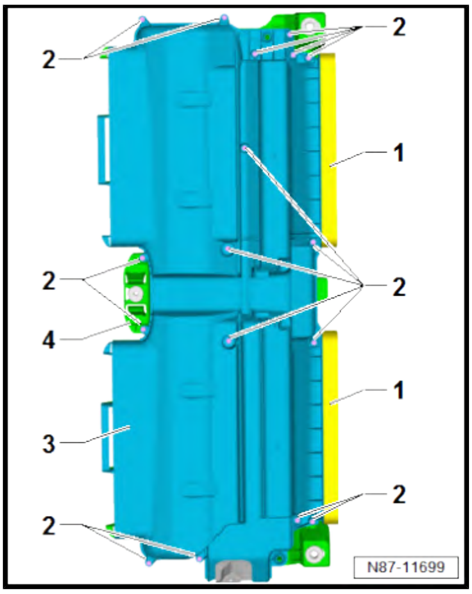


- ♦ Torque screwdriver - VAS 6494-



Removing

– Remove rear heater and air conditioning unit ⇒ page 161 .



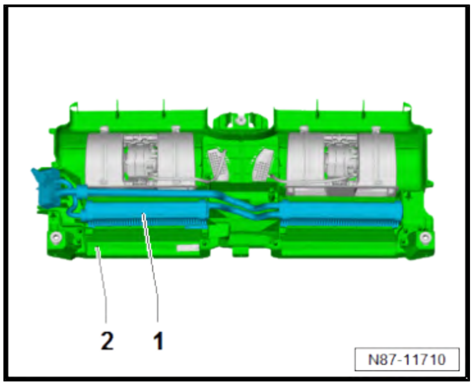
Note

The air outlet seals -1- cannot be removed intact.

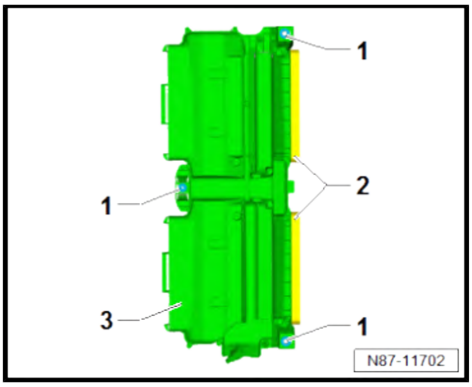
– Remove air outlet seals -1- from evaporator housing using flat scraper - VAS 5301/21- .

– Unscrew bolts -2-.

– Separate top evaporator housing -3- from bottom evaporator housing -4-.



– Remove evaporator together with expansion valve for air conditioner refrigerant - N697- -1- upwards from lower evaporator housing -2-.



Installing

Install in reverse order of removal, observing the following:

– Renew flexible joints -1- together with sleeves ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Clean bonding surface of air outlet seals -2- on evaporator housing -3-.

– Renew air outlet seals -2- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

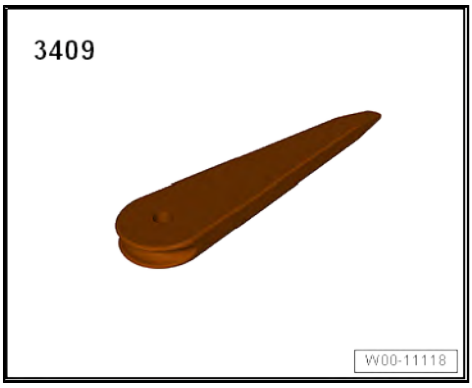
Specified torques

♦ ⇒ “6.1 Assembly overview - heater and air conditioning unit”, page 155

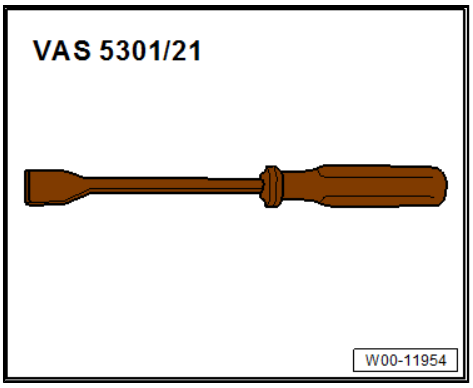
6.3 Dismantling and assembling evaporator housing

Special tools and workshop equipment required

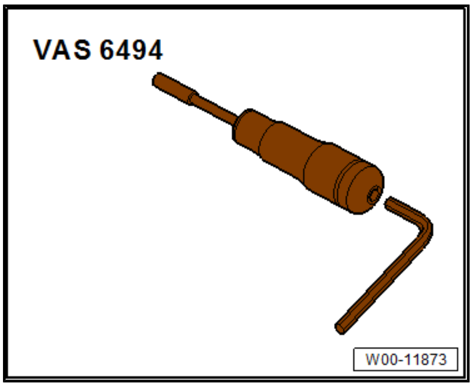
- Removal wedge - 3409-



- Scraper - VAS 5301/21-



- Torque screwdriver - VAS 6494-



Dismantling

– Remove rear heater and air conditioning unit ⇒ page 161 .

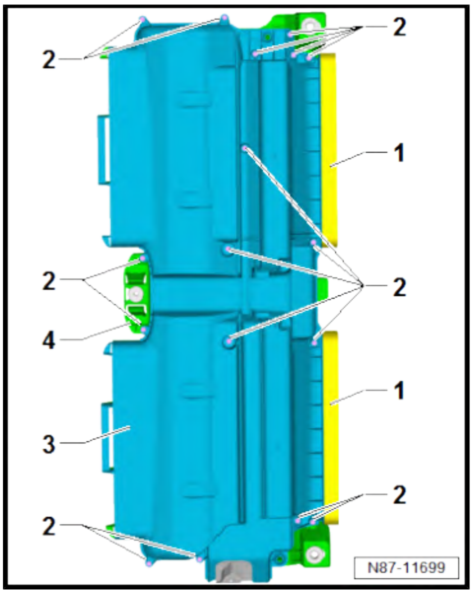
Note

The air outlet seals -1- cannot be detached intact.

– Remove air outlet seals -1- from evaporator housing using flat scraper - VAS 5301/21- .

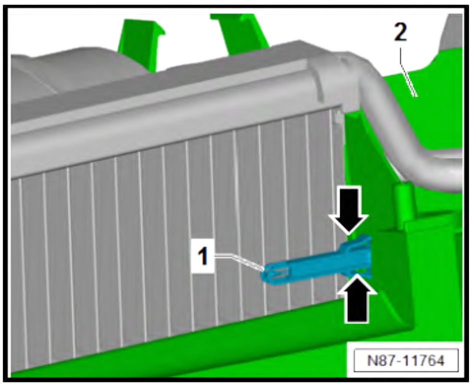
– Unscrew bolts -2-.

– Separate top evaporator housing -3- from bottom evaporator housing -4-.



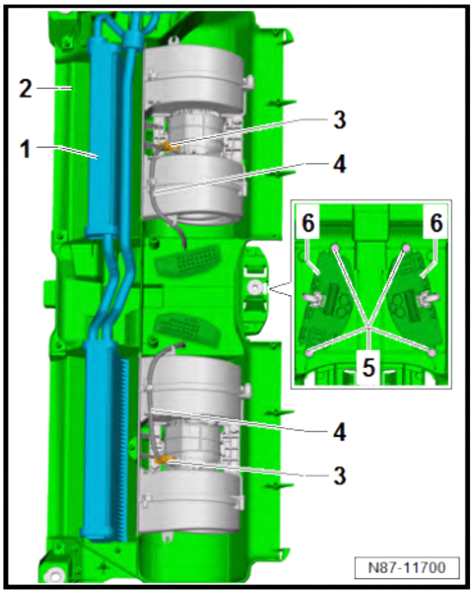
Note

Levering with removal wedge - 3409- causes both retaining tabs -arrows- to be pushed together.



– Lever rear evaporator temperature sensor - G464- -1- out of evaporator housing -2- using removal wedge - 3409- .

– Pull rear evaporator temperature sensor - G464- -1- out of evaporator housing -2-.



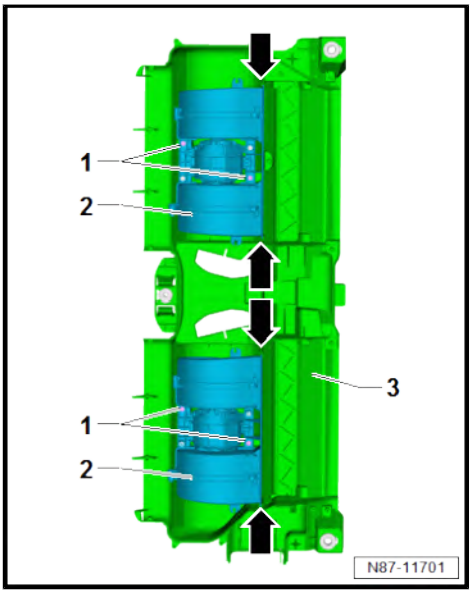
– Remove evaporator -1- together with expansion valve for air conditioner refrigerant - N697- upwards from bottom evaporator housing -2-.

– Separate electrical connectors -3-.

– Move clear adapter wire -4-.

– Unscrew bolts -5-.

– Remove control unit for supplementary blower for air conditioning - J1144- and control unit for supplementary blower 2 for air conditioning - J1145- -6- together with adapter cable -4-.

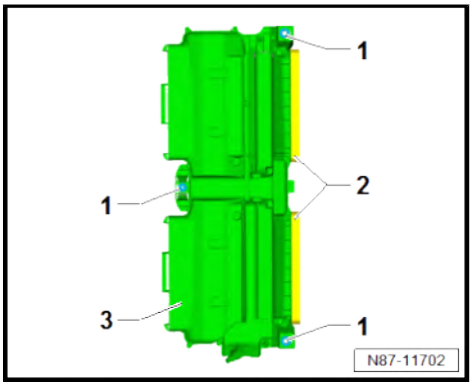


– Unscrew bolts -1-.

– Pull supplementary blower for air conditioning - V625- and supplementary blower 2 for air conditioning - V626- -2- out of bottom evaporator housing -3- upwards off guides -arrows-.

Installing

Assembly is carried out in reverse sequence; note the following:



– Renew flexible joints -1- together with sleeves ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Clean bonding surface of air outlet seals -2- on evaporator housing -3-.

– Renew air outlet seals -2- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

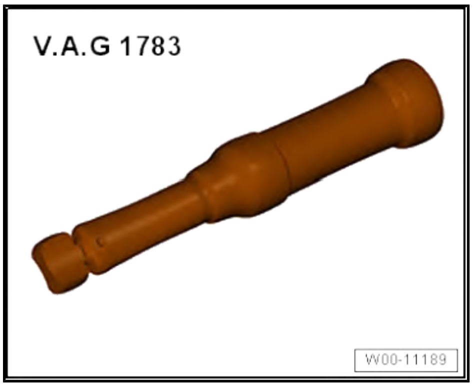
Specified torques

♦ ⇒ “6.1 Assembly overview - heater and air conditioning unit”, page 155

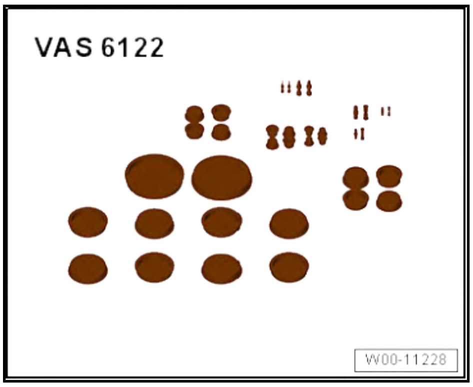
6.4 Removing and installing heater and air conditioning unit

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1783-



♦ Set of plugs for engine - VAS 6122-



Removing

– Remove moulded headliner in area of driver seat ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .

– Remove front air duct on second evaporator housing ⇒ page 180 .

Vehicles with roof end strip

– Remove roof end strip ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing roof end strip .

Vehicles with rear air duct on second evaporator housing

– Remove rear air duct on second evaporator housing ⇒ page 181 .

Vehicles with R134a refrigerant

– Drain refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

Vehicles with R1234yf refrigerant

– Drain refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Draining refrigerant circuit .

Continued for all vehicles

CAUTION

Risk of freezing injury caused by escaping pressurised refrigerant.

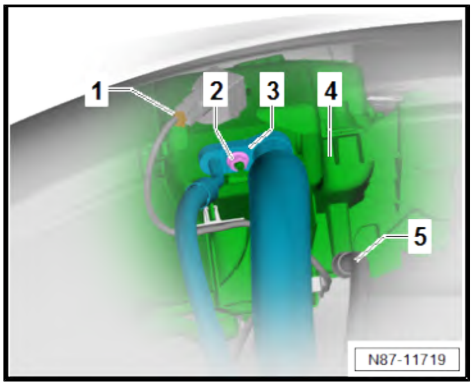
There is a risk of injury to the skin and parts of the body due to freezing.

- **Wear protective gloves.**
- **Wear protective goggles.**
- **Extract refrigerant and open the refrigerant circuit immediately afterwards.**
- **If more than 10 minutes have passed since the refrigerant was extracted, repeat the extraction process before opening the refrigerant circuit. Pressure could build up in the refrigerant circuit from continued evaporation.**

– For any further work, immediately seal open lines and connections with clean plugs from engine bung set - VAS 6122- .

– Disconnect connector -1-.

– Unscrew nut -2-.

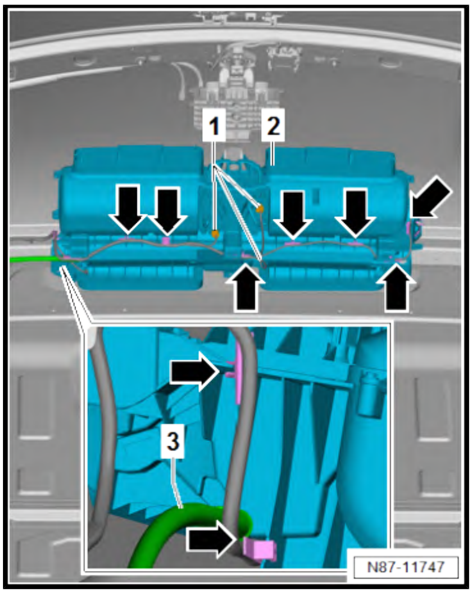


NOTICE

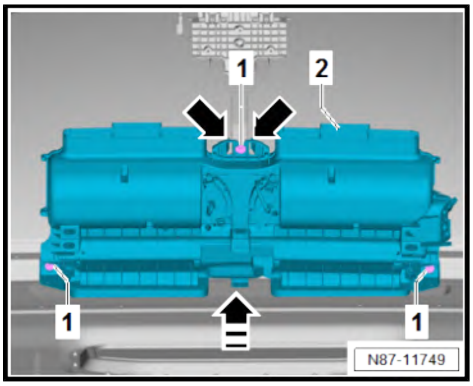
Risk of damage to refrigerant lines from rupture of inner foil.

- Never bend refrigerant lines to a radius less than 100 mm.

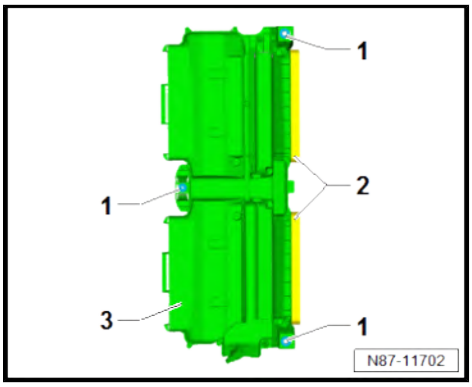
– Detach refrigerant lines -3- together with condensation drain hose -5- from evaporator housing -4-, and lay them aside.



– Separate electrical connectors -1-.
– Unclip wiring harness from evaporator housing -2- in area indicated by -arrows-.
– Pull off condensation drain hose -3- on evaporator housing -2-.



– Unscrew bolts -1-.
– Remove evaporator housing -2- in direction of -arrow-. Press retaining tabs apart in area marked with -arrows- when doing this.

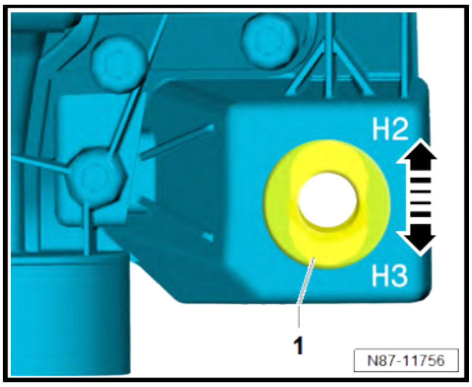


Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Renew flexible joints -1- together with sleeves ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Check air outlet seals -2- for damage, renew seals if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Check evaporator housing -3- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

Vehicles with PR numbers 3R0 or 3R2



Note

Only one side is described as follows. The description for the remaining flexible joints is basically the same.

– Ensure that flexible joint -1- is seated correctly by checking markings -H2- and -H3-. If necessary, move flexible joint -1- to correct position by pushing in direction of -arrow-.

Continued for all vehicles

NOTICE

Risk of damage to air conditioner compressor if refrigerant circuit is empty.

– **Never start the engine if the refrigerant circuit is empty.**

Vehicles with R134a refrigerant

– Charge refrigerant circuit ⇒ Air conditioning system with R134a refrigerant; Rep. gr. 00 ; Working with air conditioner service station .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning system with refrigerant R134a; Rep. gr. 00 ; Detecting leaks in refrigerant circuit .

Vehicles with R1234yf refrigerant

– Charge refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Working with air conditioner service station; Charging refrigerant circuit .

– Perform leakage test on re-established line connections of refrigerant circuit ⇒ Air conditioning systems with refrigerant R1234yf - general information; Rep. gr. 87 ; Refrigerant circuit; Detecting leaks .

Continued for all vehicles

– Commissioning air conditioning system after charging refrigerant circuit ⇒ page 75 .

– Check operation of heater and air conditioning system.

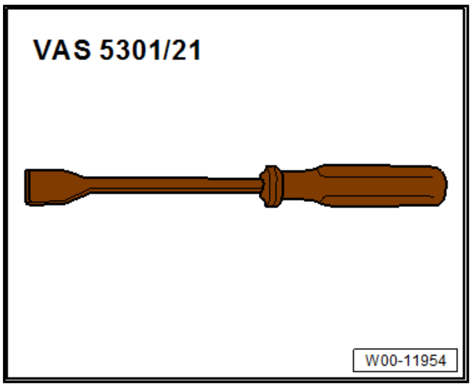
Specified torques

♦ ⇒ “6.1 Assembly overview - heater and air conditioning unit”, page 155

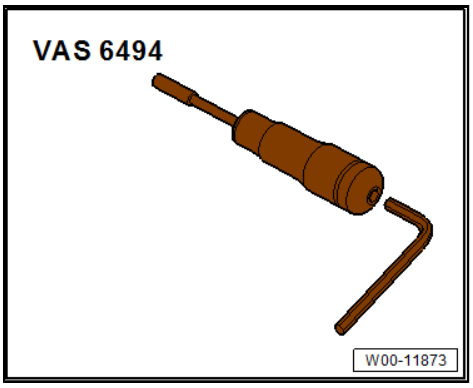
6.5 Removing and installing supplementary blower for air conditioning - V625- / supplementary blower 2 for air conditioning - V626-

Special tools and workshop equipment required

- Flat scraper - VAS 5301/21-



- Torque screwdriver - VAS 6494-



Note

- *Removal and installation are described for the supplementary blower for air conditioning - V625- .*
- *The removal and installation procedure for the supplementary blower 2 for air conditioning - V626- is the same.*

Removing

– Remove rear heater and air conditioning unit ⇒ page 161 .

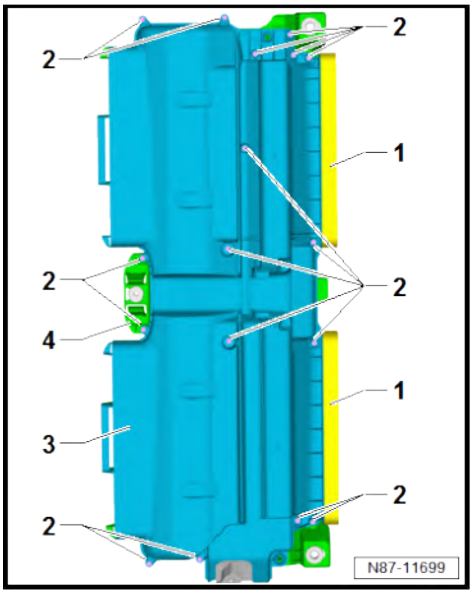
Note

The air outlet seals -1- cannot be removed intact.

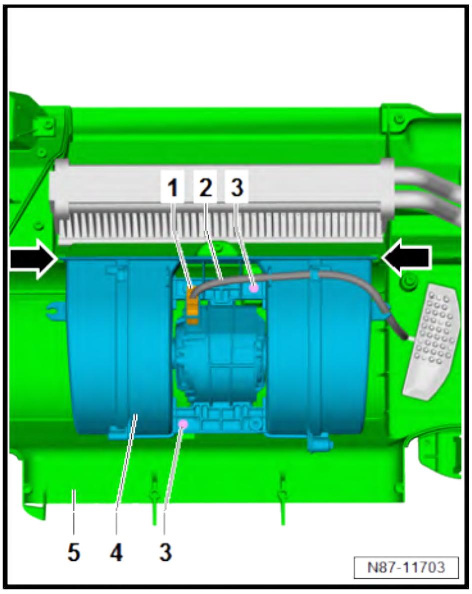
– Remove air outlet seals -1- from evaporator housing using flat scraper - VAS 5301/21- .

– Unscrew bolts -2-.

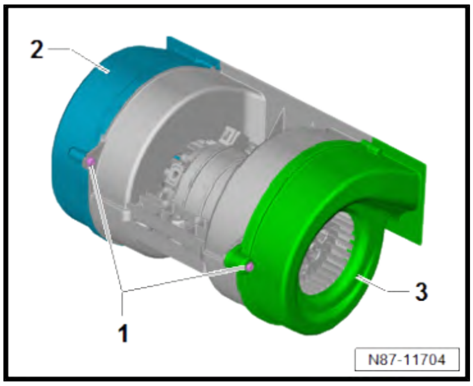
– Separate top evaporator housing -3- from bottom evaporator housing -4-.



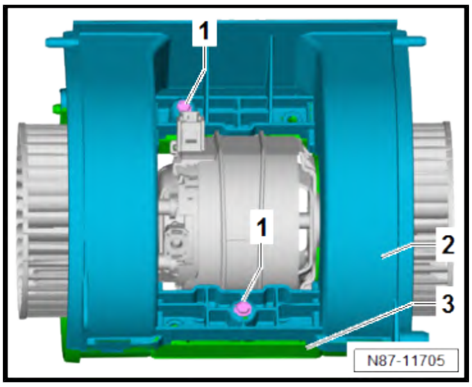
– Disconnect connector -1-.
– Move clear adapter wire -2-.
– Unscrew bolts -3-.
– Pull supplementary blower for air conditioning - V625- -4- out of bottom evaporator housing -5- upwards off guides -arrows-.



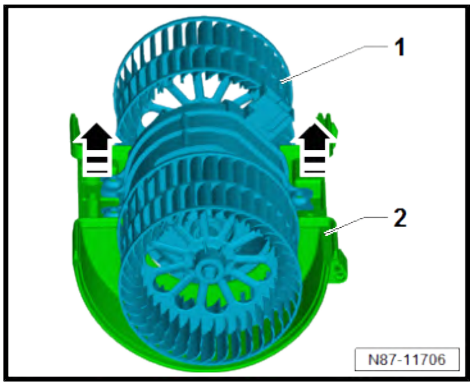
– Unscrew bolts -1-.
– Remove left blower housing -2- and right blower housing -3- from middle section.



– Unscrew bolts -1-.
– Remove top blower housing -2- from bottom blower housing -3-.



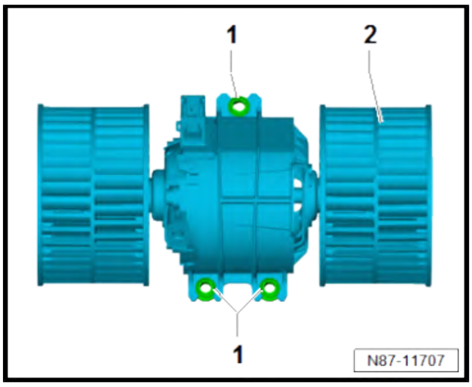
– Remove supplementary blower for air conditioning - V625- -1- in direction of -arrow- from bottom blower housing -2-.



Installing

Install in reverse order of removal, observing the following:

– Renew flexible joints -1- on supplementary blower for air conditioning - V625- -2- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



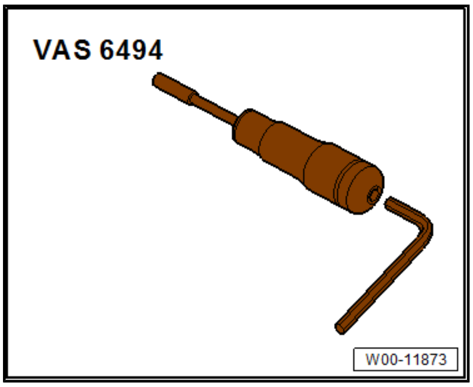
Specified torques

♦ ⇒ “6.1 Assembly overview - heater and air conditioning unit”, page 155

6.6 Removing and installing control unit for supplementary blower for air conditioning - J1144- / control unit for supplementary blower 2 for air conditioning - J1145-

Special tools and workshop equipment required

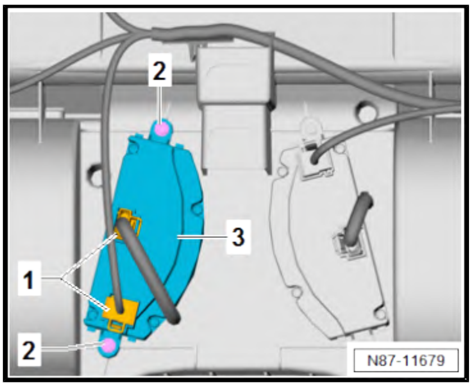
♦ Torque screwdriver - VAS 6494-



Note

♦ *Removal and installation are described for the control unit for supplementary blower for air conditioning - J1144- .*

♦ *The removal and installation procedure for the control unit for supplementary blower 2 for air conditioning - J1145- is the same.*



Removing

– Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .

CAUTION

Risk of burns when touching hot cooling surface of control unit.

Risk of burns to the hands.

– Wear protective gloves.

– Separate electrical connectors -1-.

– Unscrew bolts -2-. While doing this, remove control unit for supplementary blower for air conditioning - J1144- -3-.

Installing

Install in reverse order of removal, observing the following:

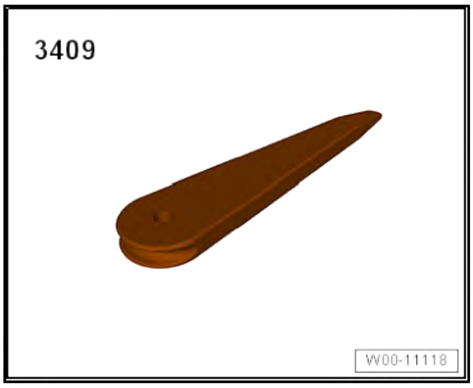
Specified torques

♦ ⇒ “6.1 Assembly overview - heater and air conditioning unit”, page 155

6.7 Removing and installing rear evaporator temperature sensor - G464-

Special tools and workshop equipment required

♦ Removal wedge - 3409-



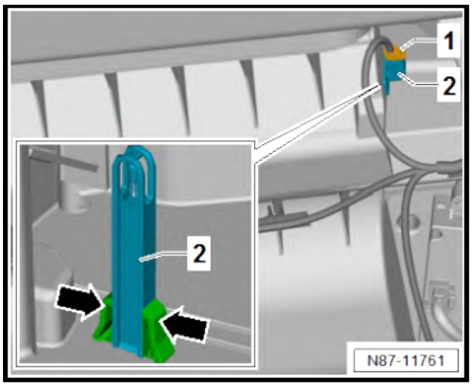
Removing

Vehicles with roof end strip

– Remove roof end strip ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing roof end strip .

Vehicles with rear air duct on second evaporator housing

– Remove rear air duct on second evaporator housing ⇒ page 181 .



Continued for all vehicles

– Disconnect connector -1-.

Note

Levering with removal wedge - 3409- causes both retaining tabs -arrows- to be pushed together.

– Lever rear evaporator temperature sensor - G464- -2- out of evaporator housing using removal wedge - 3409- .

Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs -arrows- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check operation of heater and air conditioning system.

6.8 Removing and installing condensation drain hose

⇒ “6.8.1 Removing and installing condensation drain hose, left side of vehicle”, page 169

⇒ “6.8.2 Removing and installing condensation drain hose, right side of vehicle”, page 170

6.8.1 Removing and installing condensation drain hose, left side of vehicle

Special tools and workshop equipment required

♦ Velcro strap according to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

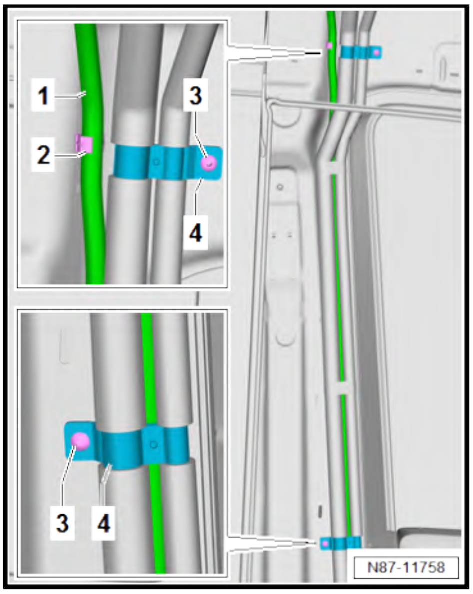
Removing

– Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .

– Remove left step ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing step; Removing and installing front step .

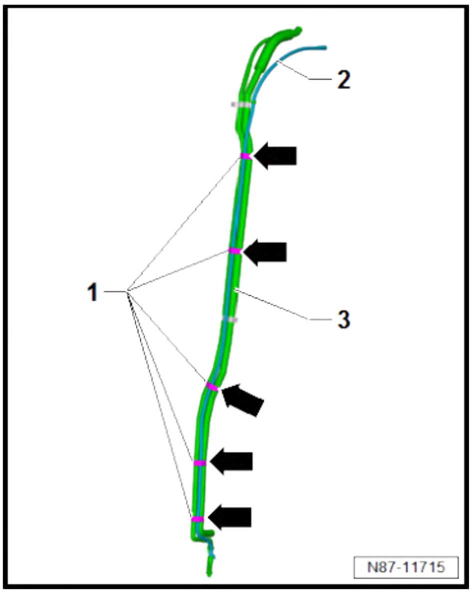
– Unclip condensation drain hose -1- from hose retainer -2-.

– Remove spreader rivets -3- for refrigerant line retainer -4-.



Note

For reasons of clarity, the refrigerant lines -3- are shown removed in the illustration.



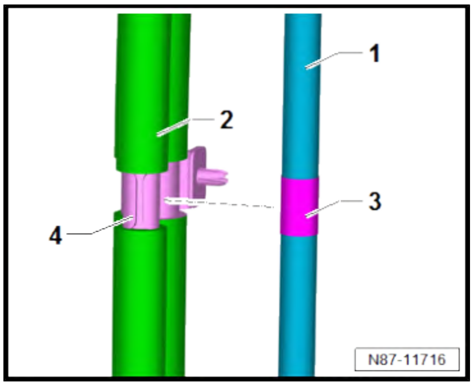
– If present, remove fleece tape -1- in area marked by -arrows-.

– Remove condensation drain hose -2- from refrigerant lines -3-.

Installing

Install in reverse order of removal, observing the following:

– Check condensation drain hose ⇒ page 172 .

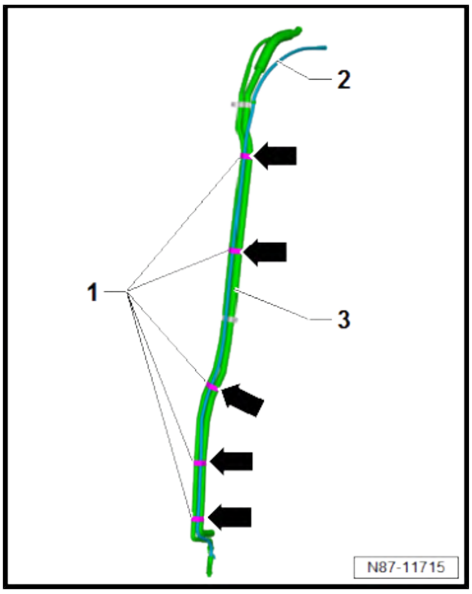


– Position condensation drain hose -1- centrally on refrigerant lines -2-.

– Align adhesive tape mark -3- on condensation drain hose -1- to height of refrigerant line retainer -4-.

Note

For reasons of clarity, the refrigerant lines -3- are shown removed in the illustration.



– Secure condensation drain hose -2- to refrigerant lines -3- using fleece tape -1- in area marked by -arrows-.

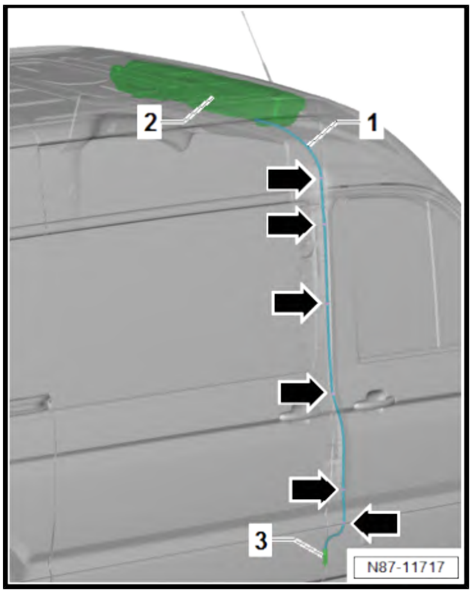
6.8.2 Removing and installing condensation drain hose, right side of vehicle

Removing

– Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .

– Remove right step ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing step; Removing and installing front step .

– Unclip condensation drain hose from hose retainer.



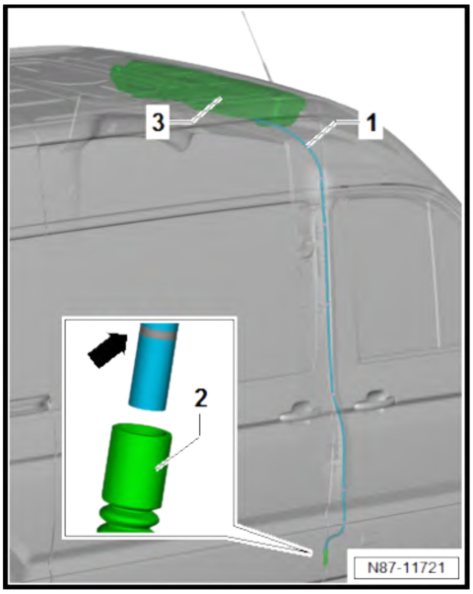
– Pull off condensation drain hose -1- on evaporator housing -2-.

– Pull out condensation drain hose -1- from water drain valve -3-.

– Unclip condensation drain hose -1- in area marked by -arrows- and remove.

Installing

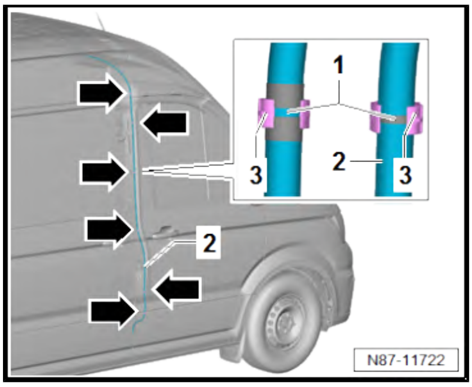
Install in reverse order of removal, observing the following:



– Check condensation drain hose ⇒ page 172 .

– Push condensation drain hose -1- into water drain valve -2- as far as marking -arrow-.

– Push condensation drain hose -3- on evaporator housing -1- as far as stop.



Note

♦ *There may be discrepancies in the location of the markings -1- on the condensation drain hose -2- depending on where it is attached.*

♦ *The condensation drain hose -2- must be secured with clips in area of narrow markings -1- on hose retainer -3-.*

– Secure condensation drain hose -2- with clips in area marked by -arrows-.

6.9 Checking condensation drain hose

Test sequence

– Remove condensate drain hose ⇒ page 169 .

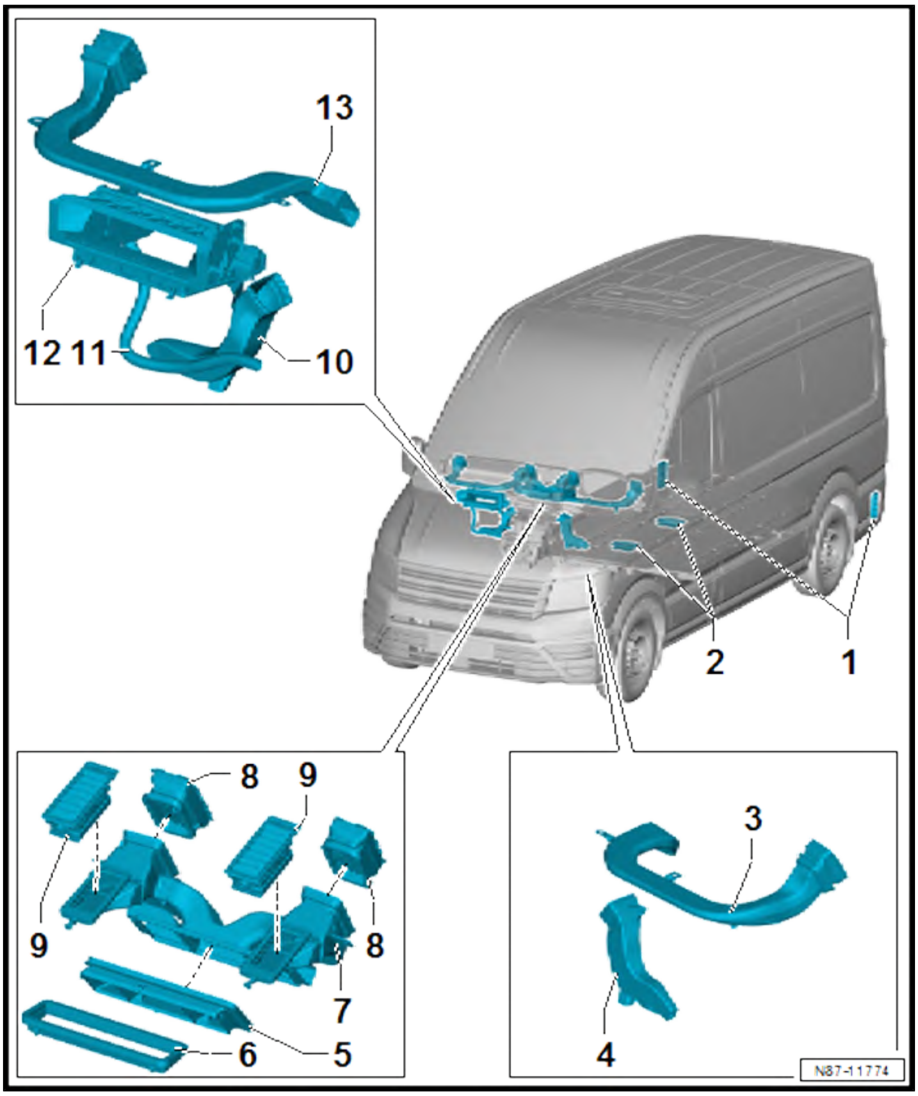
– Check condensation drain hose for damage and blockage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check water drain valve for damage and blockage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7 Air duct

7.1 Overview of fitting locations - air ducts and air distribution in passenger compartment

7.1.1 Overview of fitting locations - air ducts and air distribution in passenger compartment



1 - Forced ventilation
- ❑ Passenger compartment
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 182
- ❑ Checking ⇒ page 184

2 - Floor vents
- ❑ In passenger compartment
- ❑ Removing and installing ⇒ page 33
- ❑ Cleaning ⇒ page 34

3 - Air duct
- ❑ For left dash panel vent
- ❑ Removing and installing ⇒ page 179

4 - Footwell vent
- ❑ Driver side
- ❑ Removing and installing ⇒ page 176

5 - Intermediate piece
- ❑ Removing and installing ⇒ page 190

6 - Air duct
- ❑ For vent defrost function
- ❑ Removing and installing ⇒ page 189

7 - Air distribution
- ❑ Centre vent
- ❑ Removing and installing ⇒ page 178

8 - Centre vent
- ❑ Qty. 2
- ❑ Removing and installing ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing centre vents .

9 - Centre vent
- ❑ Top
- ❑ Qty. 2
- ❑ Depending on equipment
- ❑ Removing and installing ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing centre vents .

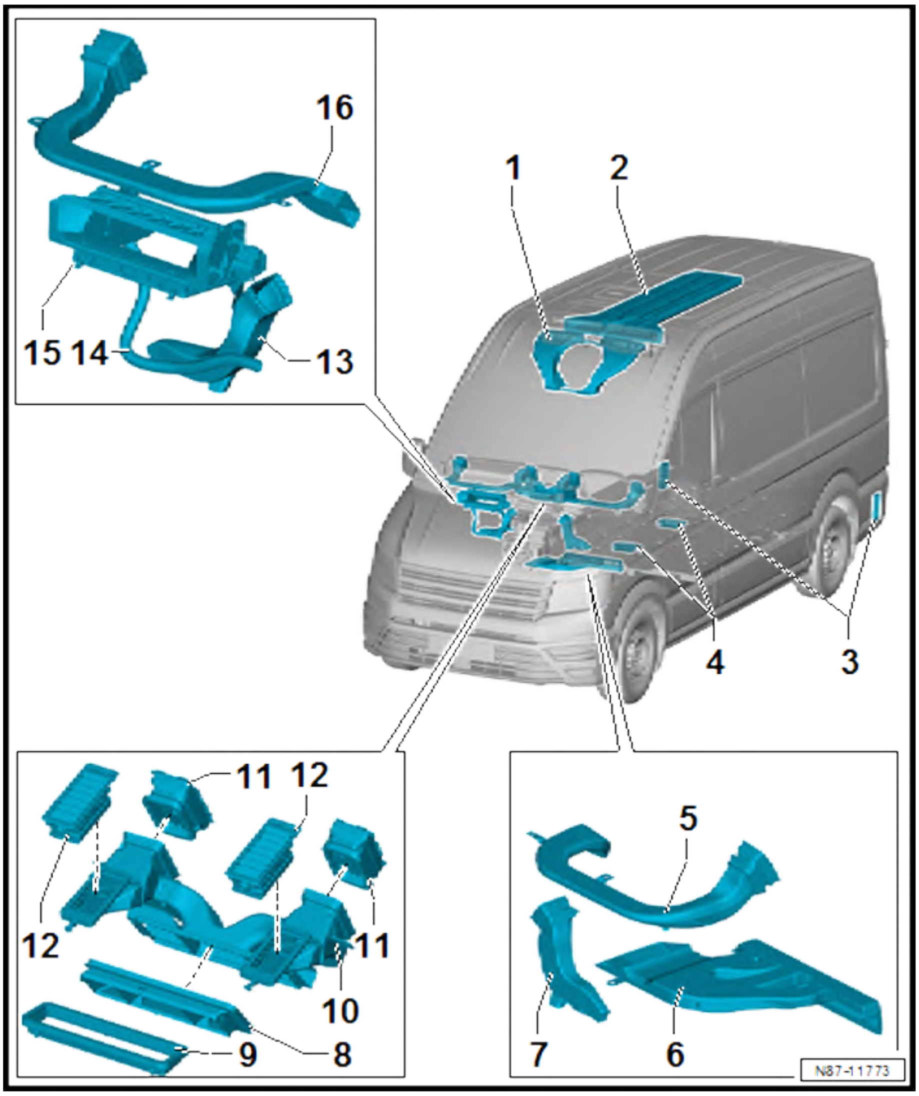
10 - Footwell vent
- ❑ Front passenger side
- ❑ Removing and installing ⇒ page 177

11 - Cold-air hose, glove compartment
- ❑ Removing and installing ⇒ page 190

12 - Air intake duct
- ❑ Removing and installing ⇒ page 189

13 - Air duct
- ❑ For right dash panel vent
- ❑ Removing and installing ⇒ page 179

7.1.2 Overview of fitting locations - air duct and air distribution in passenger compartment, vehicles with second heat exchanger and/or second evaporator



1 - Air duct
- ❑ At front of second evaporator housing
- ❑ Removing and installing ⇒ page 180

2 - Air duct
- ❑ At rear of second evaporator housing
- ❑ Removing and installing ⇒ page 181

3 - Forced ventilation
- ❑ Passenger compartment
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 182
- ❑ Checking ⇒ page 184

4 - Floor vents
- ❑ In passenger compartment
- ❑ Removing and installing ⇒ page 33
- ❑ Cleaning ⇒ page 34

5 - Air duct
- ❑ For left dash panel vent
- ❑ Removing and installing ⇒ page 179

6 - Air duct
- ❑ On second heater unit
- ❑ Removing and installing ⇒ page 33

7 - Footwell vent
- ❑ Driver side
- ❑ Removing and installing ⇒ page 176

8 - Intermediate piece
- ❑ Removing and installing ⇒ page 190

9 - Air duct
- ❑ For vent defrost function
- ❑ Removing and installing ⇒ page 189

10 - Air distribution
- ❑ Centre vent
- ❑ Removing and installing ⇒ page 178

11 - Centre vent

- ❑ Qty. 2
- ❑ Removing and installing ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing centre vents .

12 - Centre vent

- ❑ Top
- ❑ Qty. 2
- ❑ Depending on equipment
- ❑ Removing and installing ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing centre vents .

13 - Footwell vent

- ❑ Front passenger side
- ❑ Removing and installing ⇒ page 177

14 - Cold-air hose, glove compartment

- ❑ Removing and installing ⇒ page 190

15 - Air intake duct

- ❑ Removing and installing ⇒ page 189

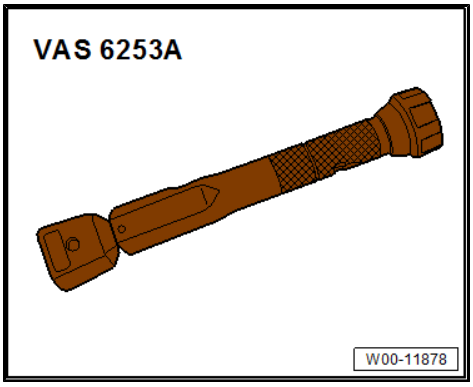
16 - Air duct

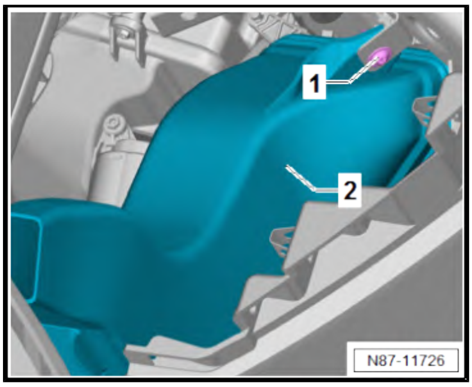
- ❑ For right dash panel vent
- ❑ Removing and installing ⇒ page 179

7.2 Removing and installing driver side foot-well vents

Special tools and workshop equipment required

- ♦ Torque wrench - VAS 6253A-





Removing

– Remove lower steering column trim ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments and covers; Removing and installing lower steering column trim .

– Remove dash panel trim on driver side ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing dash panel trim on driver side .

– Unscrew bolt -1-.

– Swivel down driver-side footwell vent -2- off air distribution housing.

Installing

Install in reverse order of removal, observing the following:

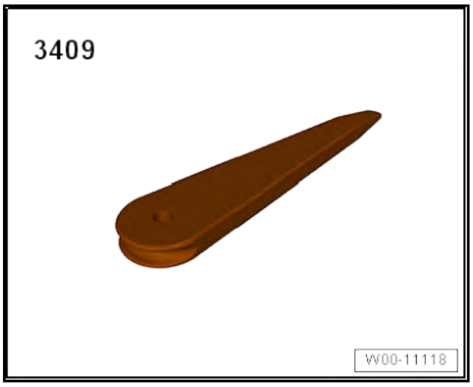
Specified torques

Component	Specified torque
Bolt -1-	1.5 Nm

7.3 Removing and installing front passenger side footwell vents

Special tools and workshop equipment required

◆ Removal wedge - 3409-



Removing

– Move glove compartment lid to service position ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments and covers; Glove compartment lid, service position .

Vehicles with control unit 1 for information electronics - J794-

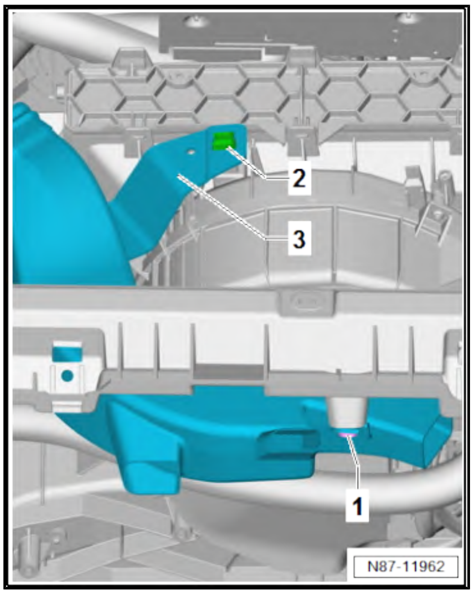
– Remove installation frame for control unit 1 for information electronics - J794- ⇒ General body repairs, interior; Rep. gr. 68 ; Storage compartments/covers; Removing and installing installation frame for information electronics .

Continued for all vehicles

– If present, remove spreader rivet -1- using removal wedge - 3409- .

– Press together locking lug -2-. While doing so, swivel front passenger-side footwell vent -3- in direction of seat.

– Pull off front passenger side footwell vent -3- from air distribution housing.



Installing

Install in reverse order of removal, observing the following:

– If previously not present, fit spreader rivet -1- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.4 Removing and installing air ducts

⇒ "7.4.1 Removing and installing dash panel vent air distribution air duct", page 178

⇒ "7.4.2 Removing and installing left and right air ducts of dash panel vent", page 179

⇒ "7.4.3 Removing and installing air duct of top centre vent", page 180

⇒ "7.4.4 Removing and installing front air duct on second evaporator housing", page 180

⇒ "7.4.5 Removing and installing rear air duct on second evaporator housing", page 181

⇒ "7.4.6 Removing and installing rear air duct vent on second evaporator housing", page 181

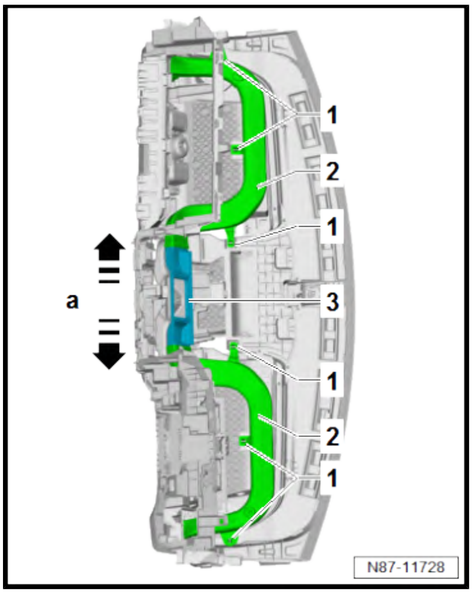
7.4.1 Removing and installing dash panel vent air distribution air duct

Removing

– Remove dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .

– Compress locking lugs -1-. While doing so, detach dash panel vent air ducts -2- along with air distribution -3- from dash panel.

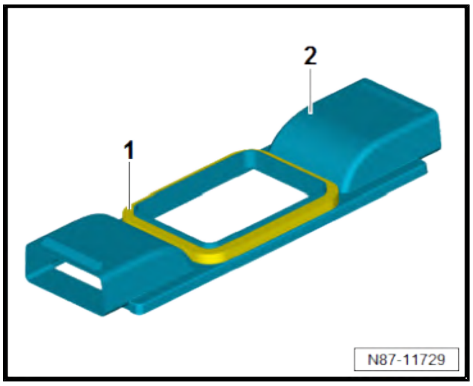
– Pull dash panel vent air ducts -2- in direction of -arrow a- off air distribution -3-.



Installing

Install in reverse order of removal, observing the following:

– Ensure seal -1- is seated correctly on air distribution for air duct of dash panel vent -2- and check for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



7.4.2 Removing and installing left and right air ducts of dash panel vent

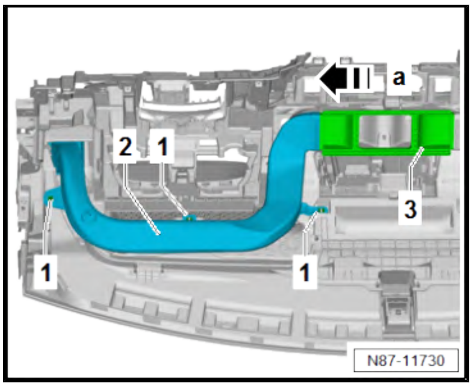
Removing

– Remove dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .

Dash panel vent air duct, driver side

– Compress locking lugs -1-. While doing so, detach dash panel vent air duct -2- along with air distribution -3- from dash panel.

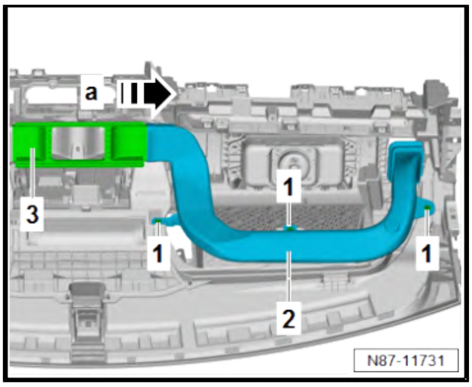
– Pull dash panel vent air duct -2- in direction of -arrow a- off air distribution -3-.



Dash panel vent air duct, front passenger side

– Compress locking lugs -1-. While doing so, detach dash panel vent air duct -2- along with air distribution -3- from dash panel.

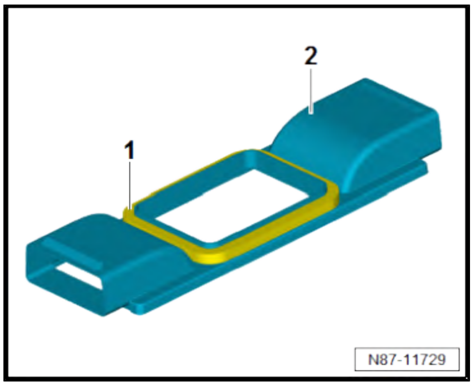
– Pull dash panel vent air duct -2- in direction of -arrow a- off air distribution -3-.



Installing

Install in reverse order of removal, observing the following:

– Ensure seal -1- is seated correctly on air distribution for air duct of dash panel vent -2- and check for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



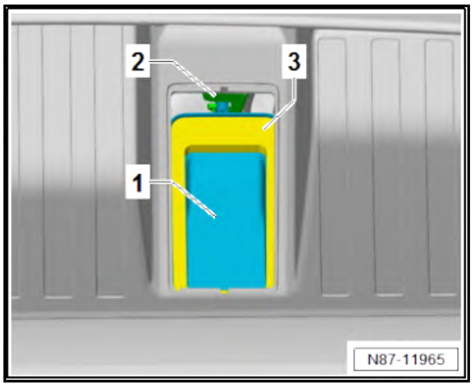
7.4.3 Removing and installing air duct of top centre vent

Removing

– Remove top centre vent on affected side ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing centre vent .

– Detach air duct -1- from bracket -2-.

– Pull down air duct -1- from air distribution. When doing this, remove through opening in dash panel.



Installing

Install in reverse order of removal, observing the following:

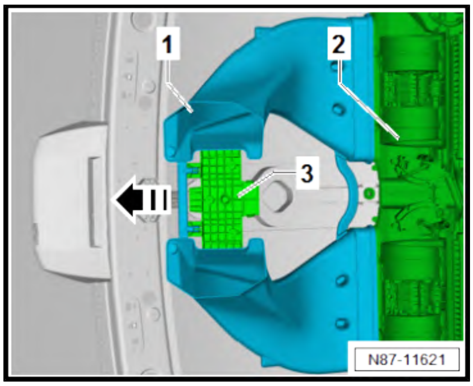
– Ensure seal -3- on air duct is seated correctly and check for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.4.4 Removing and installing front air duct on second evaporator housing

Removing

– Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .

– Pull off air duct -1- in direction of -arrow- from evaporator housing -2- and air duct carrier -3-.



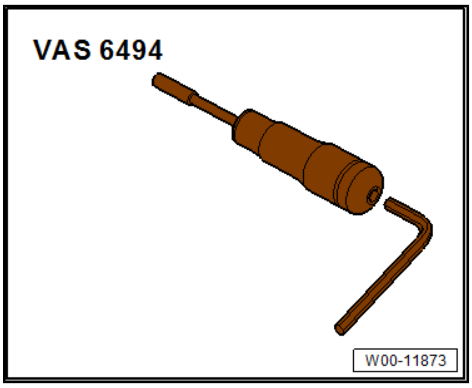
Installing

Install in reverse order of removal.

7.4.5 Removing and installing rear air duct on second evaporator housing

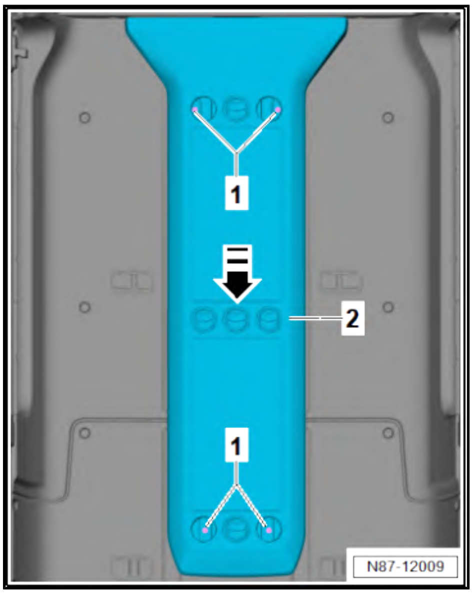
Special tools and workshop equipment required

- ♦ Torque screwdriver - VAS 6494-



Removing

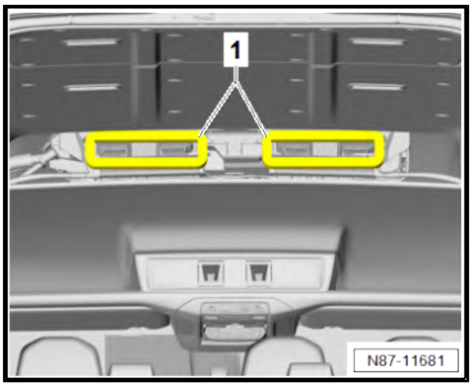
- – Remove front and rear vents on left and right for air duct ⇒ page 181 .
- – Unscrew bolts -1-.
- – Pull off air duct -2- off in direction of -arrow-.



Installing

Install in reverse order of removal, observing the following:

- – Ensure seals -1- are seated correctly and check for damage; renew if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



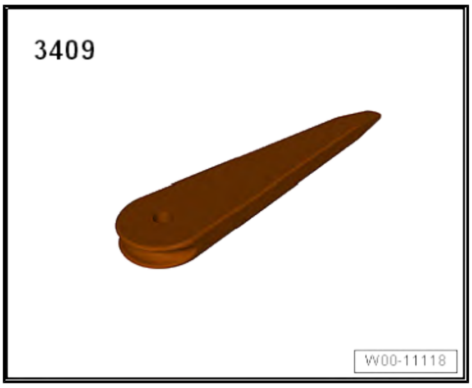
Specified torques

Component	Specified torque
Bolts -1-	1.5 Nm

7.4.6 Removing and installing rear air duct vent on second evaporator housing

Special tools and workshop equipment required

♦ Removal wedge - 3409-

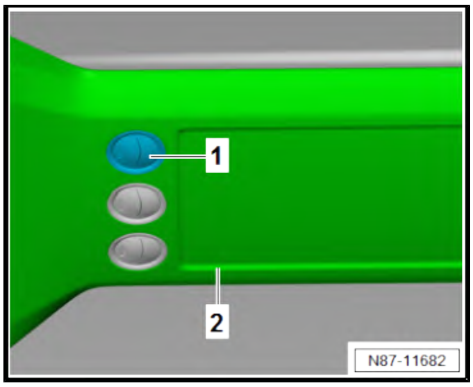


Removing

– Lever vent -1- out of air duct -2- using removal wedge - 3409- .

Installing

Install in reverse order of removal.



7.5 Removing and installing forced ventilation for passenger compartment

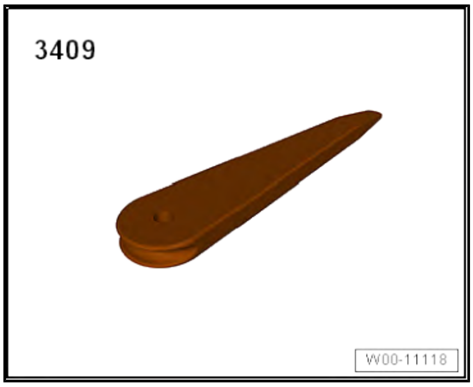
⇒ “7.5.1 Removing and installing forced ventilation for passenger compartment, van body”, page 182

⇒ “7.5.2 Removing and installing forced ventilation for passenger compartment, drop-side”, page 183

7.5.1 Removing and installing forced ventilation for passenger compartment, van body

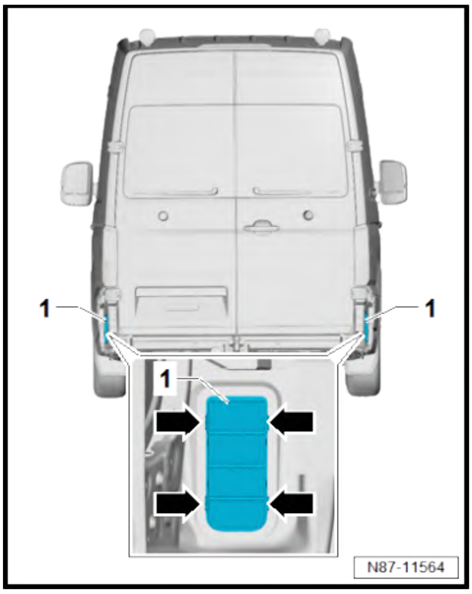
Special tools and workshop equipment required

♦ Removal wedge - 3409-



Removing

- Remove rear bumper cover ⇒ General body repairs, exterior; Rep. gr. 63 ; Rear bumper; Removing and installing bumper cover .
- Use removal wedge - 3409- to release locking lugs -arrows-.
- Remove vent frames -1- from side panel.



Installing

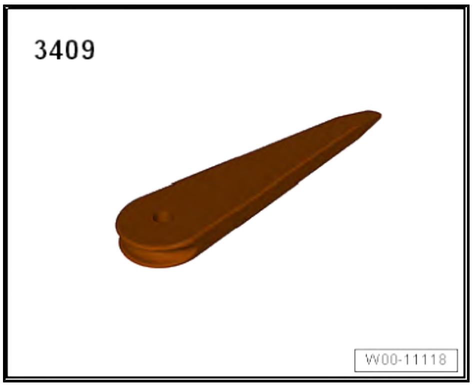
Install in reverse order of removal, observing the following:

- Check forced ventilation for passenger compartment ⇒ page 184 .

7.5.2 Removing and installing forced ventilation for passenger compartment, dropside

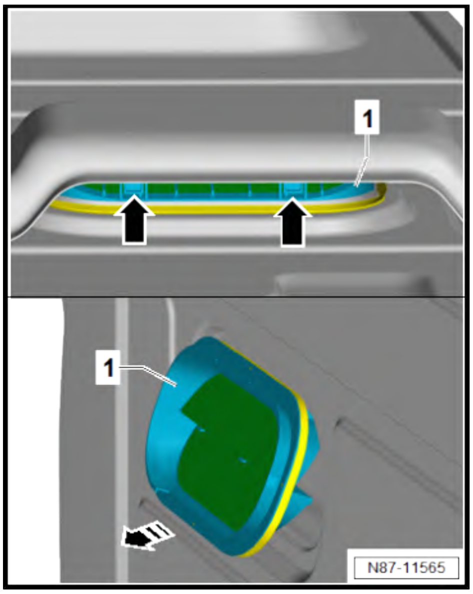
Special tools and workshop equipment required

- ♦ Removal wedge - 3409-



Removing

- Remove rear bench seat ⇒ General body repairs, interior; Rep. gr. 72 ; Rear seats; Removing and installing bench seat / individual seats .
- Use removal wedge - 3409- to release locking lugs -arrows-.
- Tilt ventilation frame -1- in direction of -arrow-. Remove ventilation frame -1- between dropside body and back panel.



Installing

Install in reverse order of removal, observing the following:

- Check forced ventilation for passenger compartment ⇒ page 184 .

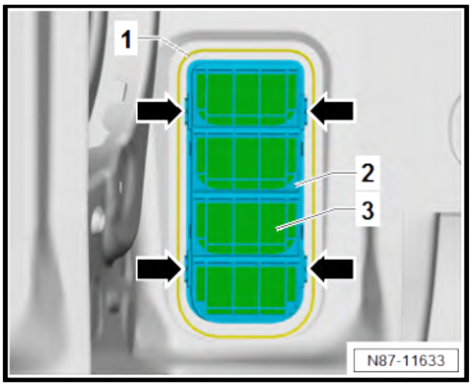
7.6 Checking forced ventilation for passenger compartment

Test sequence

Note

- *The used air escapes through vent openings.*
- *If the ventilation is to work properly, the vent openings must not be covered.*

Vehicles with enclosed body



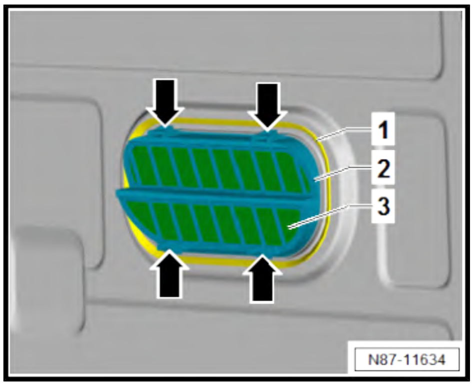
– Remove forced ventilation for passenger compartment ⇒ page 182 .

– Check seal -1- on ventilation frame -2- for damage.

– Check sealing lips -3- in ventilation frame -2- to ensure they have freedom of movement and close automatically.

– Check retaining tabs -arrows- for damage.

Vehicles with dropside body



– Remove forced ventilation for passenger compartment ⇒ page 183 .

– Check seal -1- on ventilation frame -2- for damage.

– Check sealing lips -3- in ventilation frame -2- to ensure they have freedom of movement and close automatically.

– Check retaining tabs -arrows- for damage.

– Check vent trim for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.7 Removing and installing fresh air intake

⇒ “7.7.1 Removing and installing fresh air intake, bonnet”, page 184

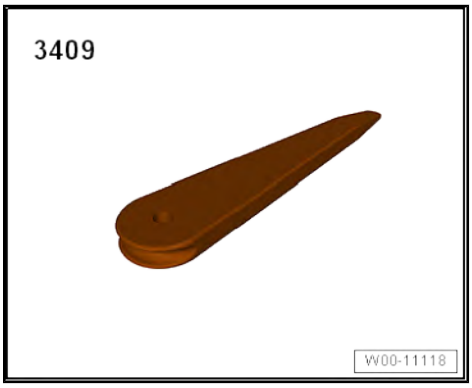
⇒ “7.7.2 Removing and installing fresh air intake, bulkhead, left-hand drive vehicles”, page 186

⇒ “7.7.3 Removing and installing fresh air intake, bulkhead, right-hand drive vehicles”, page 187

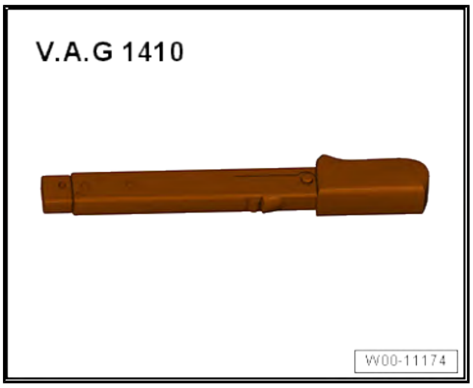
7.7.1 Removing and installing fresh air intake, bonnet

Special tools and workshop equipment required

- Removal wedge - 3409-



- Torque wrench - V.A.G 1410-



Removing

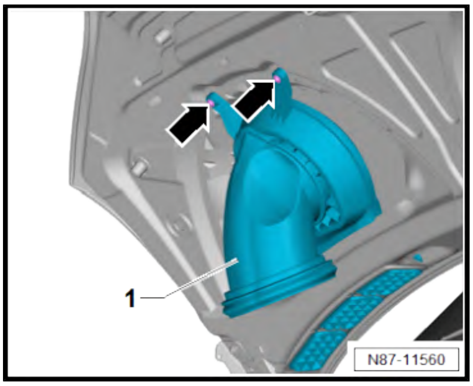
Vehicles with bonnet insulation

– Remove bonnet insulation ⇒ General body repairs, exterior; Rep. gr. 55 ; Bonnet; Removing and installing insulation; Removing and installing insulation .

Continued for all vehicles

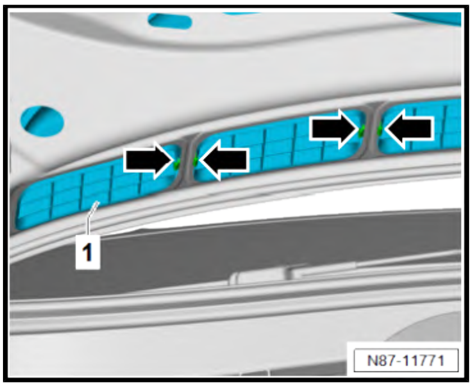
– Unscrew bolts -arrows-.

– Detach fresh air intake duct for bonnet -1- from bonnet.



– Use removal wedge - 3409- to release locking lugs -arrows-.

– Guide fresh air intake duct for bonnet -1- out of bonnet.



Installing

Install in reverse order of removal, observing the following:

– Ensure seal of fresh air intake duct on bonnet is seated correctly and check for damage; renew if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check fresh air intake duct for bonnet for contamination (e.g. foliage), and clean it if necessary.

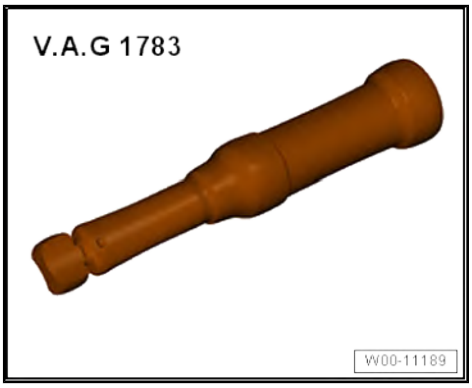
Specified torques

Component	Specified torque
Bolts -arrows-	8 Nm

7.7.2 Removing and installing fresh air intake, bulkhead, left-hand drive vehicles

Special tools and workshop equipment required

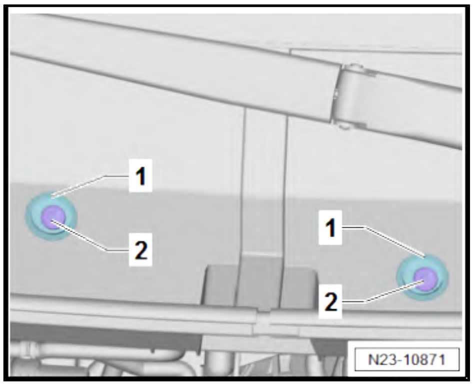
- ♦ Torque wrench - V.A.G 1783-



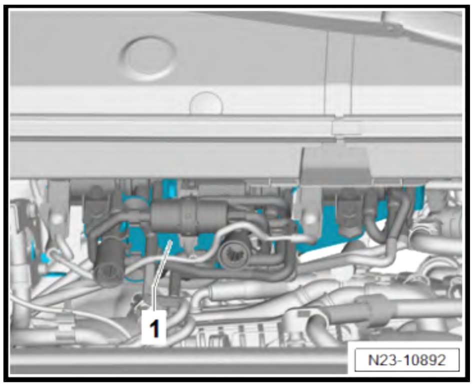
Removing

– Remove plug -1-.

– Unscrew bolts -2-.

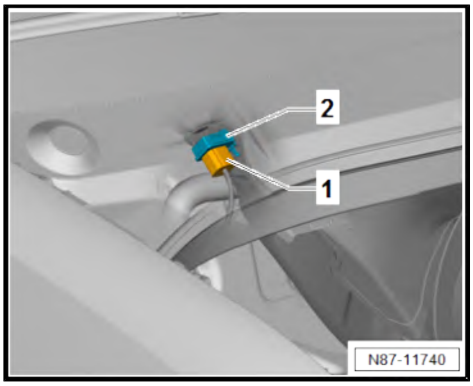


– Detach and lay aside holder for solenoid valves -1-.



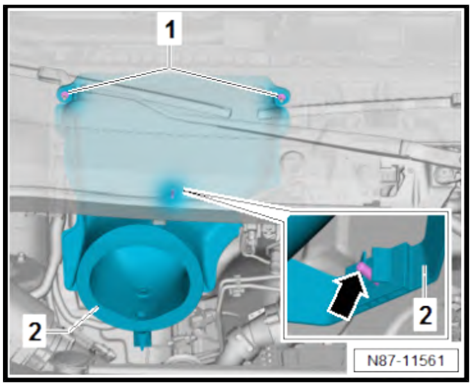
Vehicles with Climatronic, to calendar week 45/2017

– Disconnect electrical connector -1- on air quality sensor - G238- -2-.



Continued for all vehicles

– Unscrew nuts -1-.
– Pull off fresh air intake duct of bulkhead -2- from stud in area marked with -arrow-.



Installing

Install in reverse order of removal, observing the following:

– Ensure seal of fresh air intake duct on bulkhead is seated correctly and check for damage; renew if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Check water drain valve in water drain hose for contamination (e.g. foliage), and clean it if necessary.

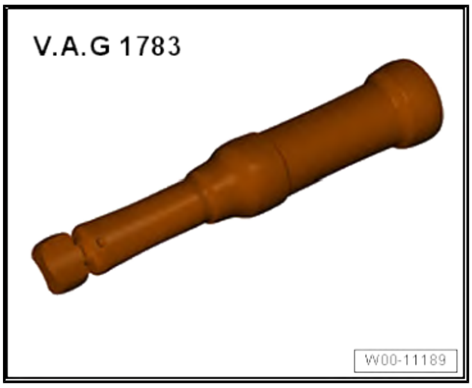
Specified torques

Component	Specified torque
Nuts -1-	8 Nm

7.7.3 Removing and installing fresh air intake, bulkhead, right-hand drive vehicles

Special tools and workshop equipment required

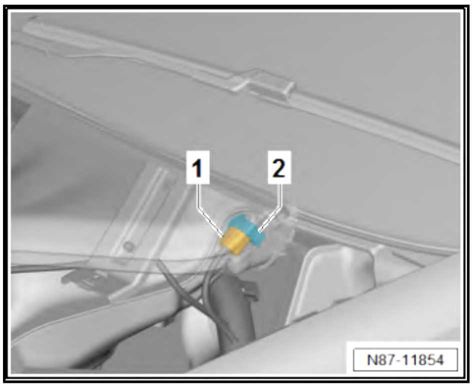
♦ Torque wrench - V.A.G 1783-



Removing

Vehicles with Climatronic, to calendar week 45/2017

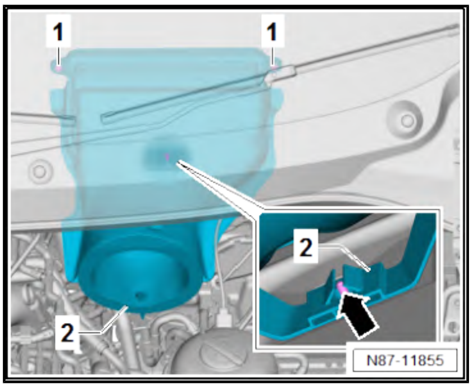
– Disconnect electrical connector -1- on air quality sensor -G238- -2-.



Continuation all vehicles

– Unscrew nuts -1-.

– Pull off fresh air intake duct of bulkhead -2- from stud in area marked with -arrow-.



Installing

Install in reverse order of removal, observing the following:

– Ensure seal of fresh air intake duct on bulkhead is seated correctly and check for damage; renew if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check water drain valve in water drain hose for contamination (e.g. foliage), and clean it if necessary.

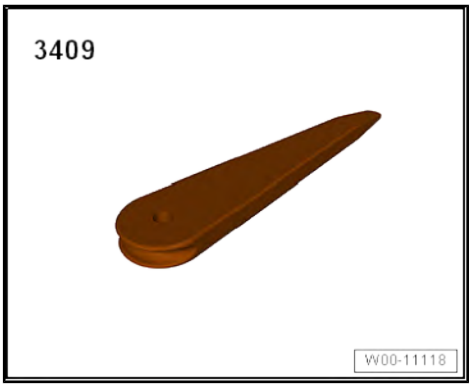
Specified torques

Component	Specified torque
Nuts -1-	⇒ page 187

7.8 Removing and installing cover for fresh air intake

Special tools and workshop equipment required

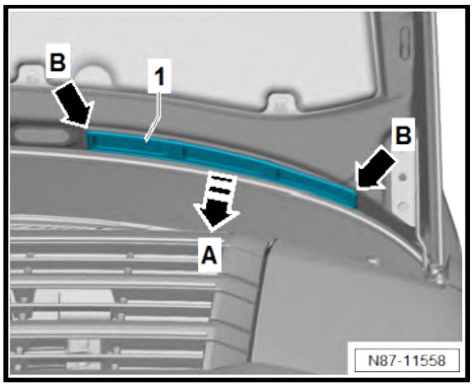
♦ Removal wedge - 3409-



Removing

– Release cover for fresh air intake duct -1- with removal wedge - 3409- in direction of -arrow A-.

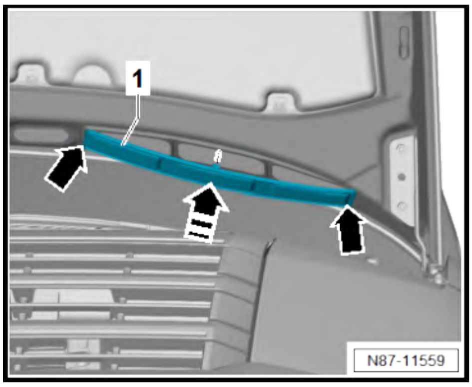
– Release cover for fresh air intake duct -1- in area of -arrows B- from bonnet and remove.



Installing

Install in reverse order of removal, observing the following:

– Check fresh air intake duct cover -1- for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



Note

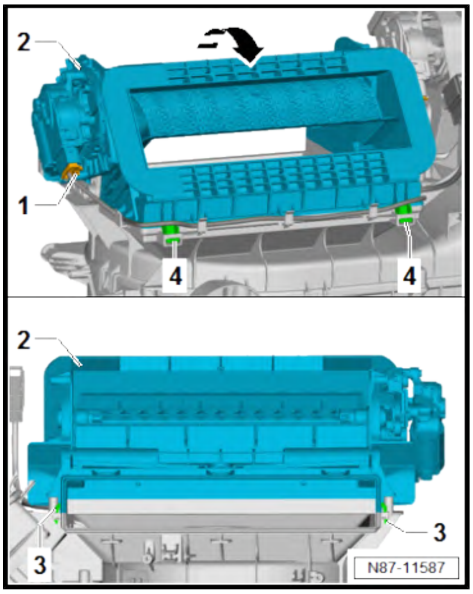
Install cover for fresh air intake duct -1- so that lettering points downwards.

– Insert cover for fresh air intake duct -1- in bonnet in area of -arrows-.

– Push cover for fresh air intake duct -1- in direction of -arrow-.

7.9 Removing and installing air intake duct

Removing



– Remove dust and pollen filter ⇒ page 144 .

– Remove air intake duct ⇒ page 192 .

– Separate electrical connector -1- and lay wiring harness on intake duct -2- to one side.

– Release locking lugs -3-.

– Swivel intake duct -2- in direction of -arrow- out of mountings -4-.

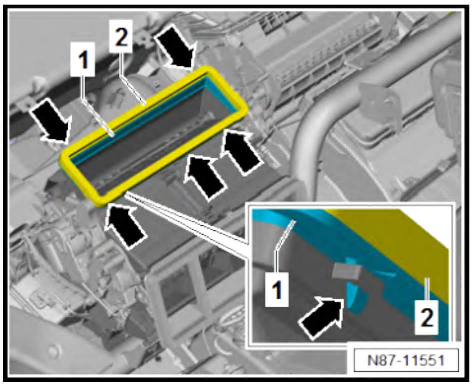
Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs -3- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.10 Removing and installing air duct for defroster vent

Removing



– Remove dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .

– Release locking lugs -arrows-. While doing this, detach air duct for defroster vent -1-.

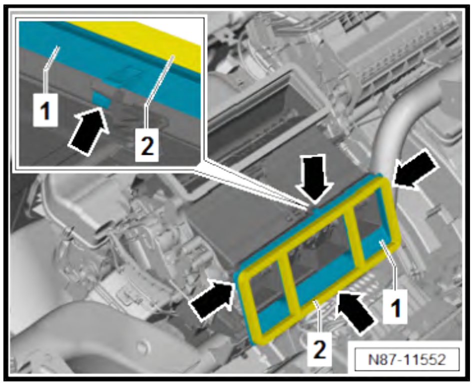
Installing

Install in reverse order of removal, observing the following:

– Ensure seal -2- is seated correctly and check for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check retaining tabs -arrows- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.11 Removing and installing intermediate piece



Removing

– Remove dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .

– Release locking lugs -arrows-. While doing this, remove intermediate piece -1-.

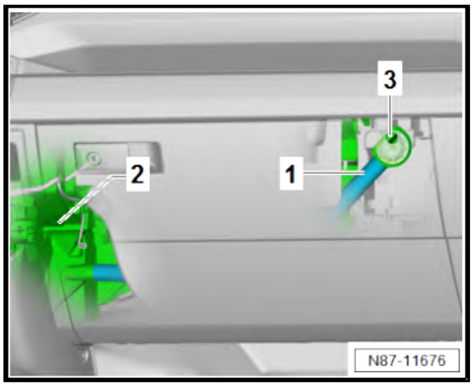
Installing

Install in reverse order of removal, observing the following:

– Ensure seal -2- is seated correctly and check for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Check retaining tabs -arrows- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.12 Removing and installing cold air hose of glove compartment



Removing

– Pull off cold air hose of glove compartment -1- from heater and air conditioning unit -2-.

– Pull off cold air hose of glove compartment -1- from connection -3- and remove.

Installing

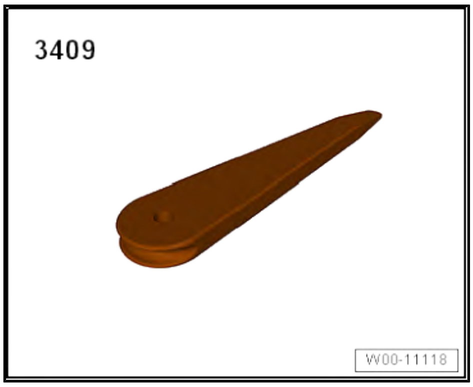
Install in reverse order of removal, observing the following:

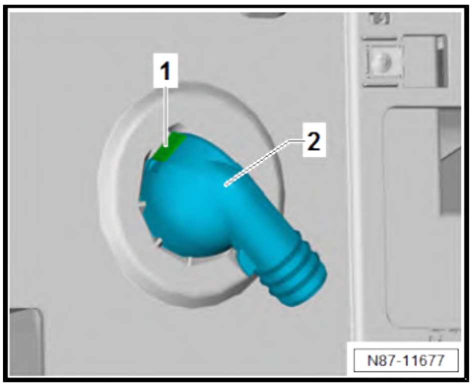
– Check cold air hose in glove box -1- for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.13 Removing and installing connection for cold air hose of glove compartment

Special tools and workshop equipment required

♦ Removal wedge - 3409-





Removing

– Remove glove compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing glove compartment .

– Release locking lug -1- using removal wedge - 3409- . When doing this, swing cold air hose of glove compartment -2- inwards and remove.

Installing

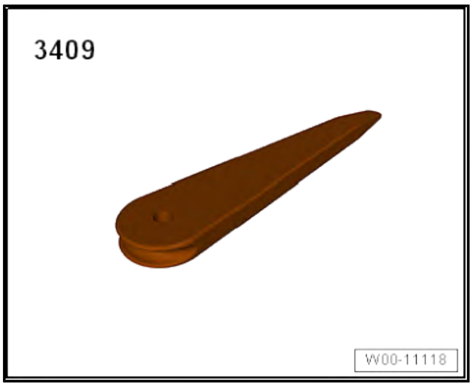
Install in reverse order of removal, observing the following:

– Check connection for cold air hose in glove box -2- for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.14 Removing and installing valve for cold air hose of glove compartment

Special tools and workshop equipment required

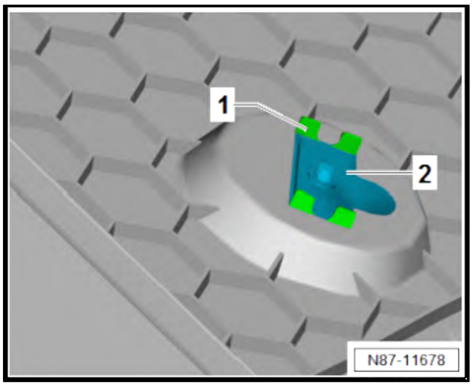
♦ Removal wedge - 3409-



Removing

– Move glove compartment lid to service position ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments and covers; Glove compartment lid, service position .

– Release locking lugs -1- using removal wedge - 3409- . When doing this, remove valve for cold air hose of glove compartment -2-.



Installing

Install in reverse order of removal, observing the following:

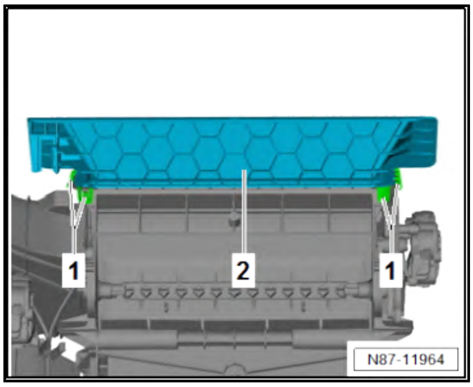
– Check retaining tabs -1- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7.15 Removing and installing air intake duct

Removing

– Remove heater and air conditioning unit ⇒ page 136 .

– Release locking lugs -1-. Remove air intake duct -2- when doing so.



Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs -1- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

8 Operating and display unit

⇒ “8.1 Overview of operating and display unit”, page 193

⇒ “8.2 Removing and installing operating and display unit”, page 196

8.1 Overview of operating and display unit

⇒ “8.1.1 Overview of operating and display unit, vehicles with Climatic”, page 193

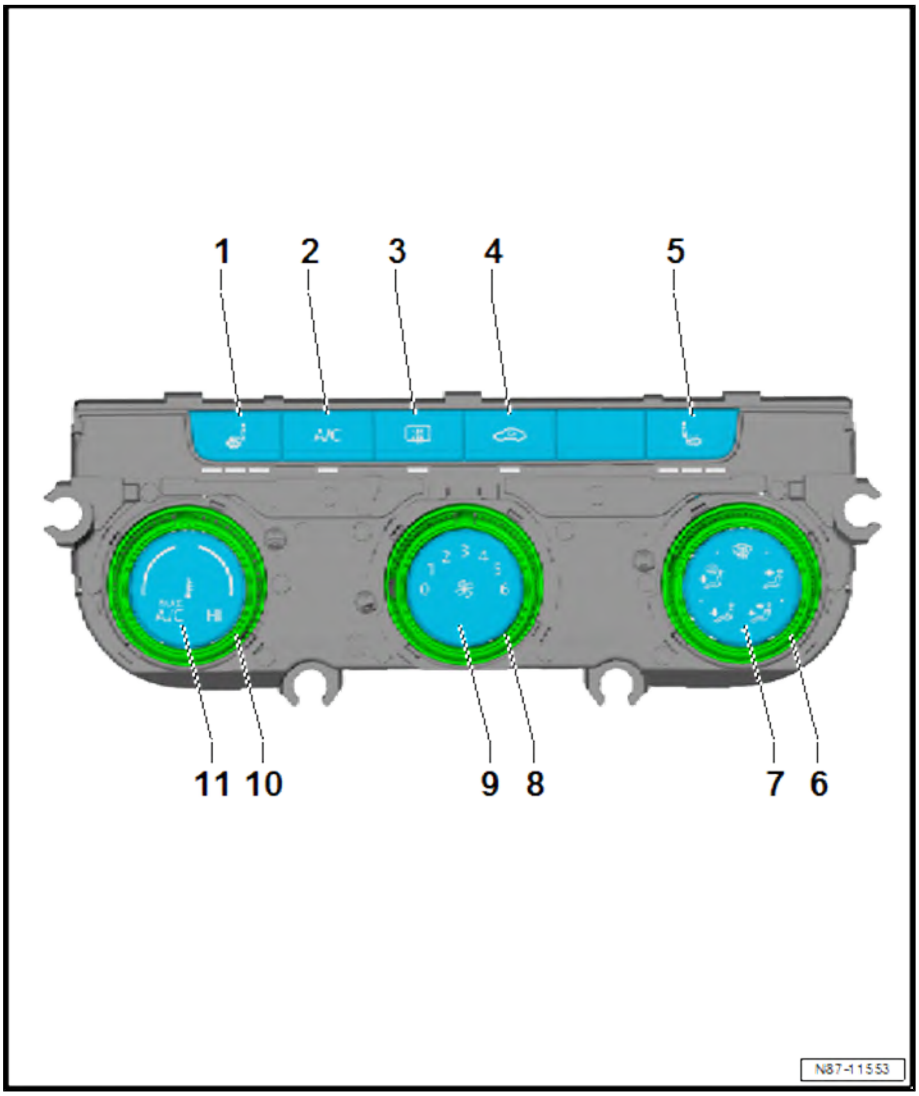
⇒ “8.1.2 Overview of operating and display unit, vehicles with Climatronic”, page 194

⇒ “8.1.3 Overview of operating and display unit in roof console”, page 196

8.1.1 Overview of operating and display unit, vehicles with Climatic

Note

- *Additional display and operating unit 1 - E857- with air conditioning system control unit - J301-*
- *A warning lamp in the instrument panel controls will indicate that the selected function is active.*



1 - Button for left seat heating - E653-
- ❑ Depending on equipment
- ❑ Seat heating can be set in 3 stages; current setting is displayed via LED

2 - A/C button
- ❑ If the A/C button is deactivated, the air conditioner compressor will be set to almost zero delivery.

3 - Heated rear window button
- ❑ Works only while engine is running and switches off automatically after 10 minutes at the latest

4 - Air recirculation button
- ❑ Pressing the air recirculation button will prevent polluted air from entering the interior.

5 - Button for right seat heating - E654-
- ❑ Depending on equipment
- ❑ Seat heating can be set in 3 stages; current setting is displayed via LED

6 - Rotary knob for setting air distribution

7 - Display for air distribution

8 - Rotary knob for setting blower speed

- ❑ Anticlockwise: reduces blower speed
- ❑ Clockwise: increases speed

9 - Display for blower speed

10 - Rotary knob for setting temperature

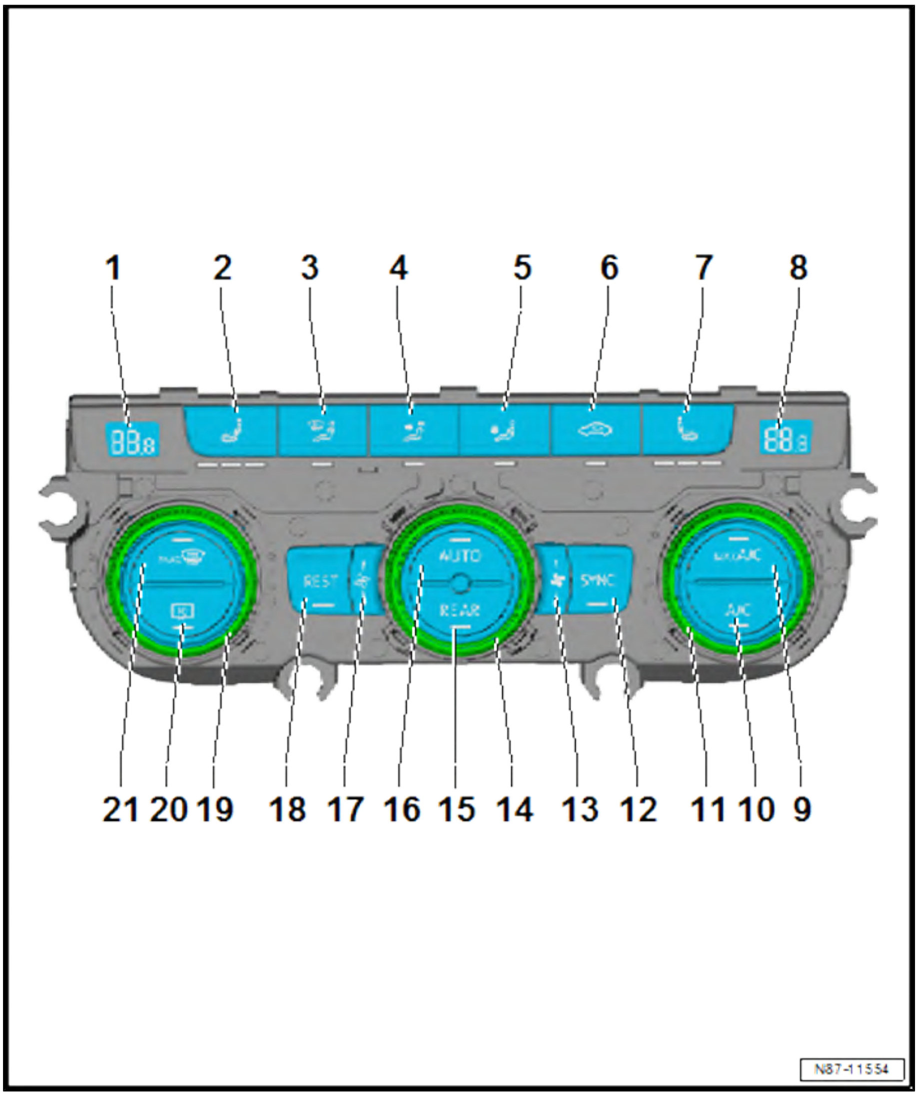
- ❑ Anticlockwise: lowers temperature
- ❑ Clockwise: increases temperature

11 - Temperature setting display

8.1.2 Overview of operating and display unit, vehicles with Climatronic

Note

- ♦ *Additional display and operating unit 1 - E857- with Climatronic control unit - J255-*
- ♦ *A warning lamp in the instrument panel controls will indicate that the selected function is active.*



1 - Left temperature indicator

- ❑ Indicates temperature setting on driver side.

2 - Button for left seat heating - E653-

- ❑ Depending on equipment
- ❑ Seat heating can be set in 3 stages; current setting is displayed via LED

3 - Button to regulate windscreen air distribution

4 - Button for centre air distribution

- ❑ Air distributed to upper body via middle vents

5 - Button for bottom air distribution

- ❑ Air distribution to footwell

6 - Air recirculation button

- ❑ Pressing the air recirculation button will prevent polluted air from entering the interior.

7 - Button for right seat heating - E654-

- ❑ Depending on equipment
- ❑ Seat heating can be set in 3 stages; current setting is displayed via LED

8 - Right temperature indicator
- ❑ Indicates temperature setting on front passenger side.

9 - A/C button
- ❑ Air conditioning system is operated with maximum cooling output

10 - A/C button
- ❑ If the A/C button is deactivated, the air conditioner compressor will be set to almost zero delivery.

11 - Rotary knob for setting right temperature
- ❑ Set temperature is indicated in display.

12 - SYNC button
- ❑ Synchronisation of climate zones to driver value

13 - Reduce blower speed
- ❑ Display only, no function

14 - Rotary knob for fan speed adjustment
- ❑ Turning the rotary knob enables individual adjustment of the fan speed
- ❑ If the fan speed is changed individually, the AUTO function is deactivated automatically.

15 - REAR button
- ❑ Pressing the button enables regulation of the air conditioning for the rear seats from this operating unit

16 - AUTO function button
- ❑ In automatic mode, the Climatronic maintains the selected interior temperature automatically

17 - Increase blower speed
- ❑ Display only, no function

18 - REST button
- ❑ Pressing the button causes the circulation pump to pump the residual heat of the engine to the heat exchanger

19 - Rotary knob for setting left temperature
- ❑ Set temperature is indicated in display.

20 - Heated rear window button
- ❑ Works only while engine is running and switches off automatically after 10 minutes at the latest

21 - Max. defrost function button
- ❑ Air drawn in from outside is channelled to windscreen, and, if activated, air recirculation mode is automatically switched off
- ❑ To demist windscreen as quickly as possible, air is dried at temperatures above +1.5°C.
- ❑ Air conditioner compressor is switched on and blower is set to a high speed

8.1.3 Overview of operating and display unit in roof console

Note

- *Additional display and operating unit 1 - E857- with Climatronic control unit - J255-*
- *A warning lamp in the instrument panel controls will indicate that the selected function is active.*

1 - Button for setting blower speed

- ❑ Increase blower speed

2 - Button for setting blower speed

- ❑ Decrease blower speed

3 - Blower speed display

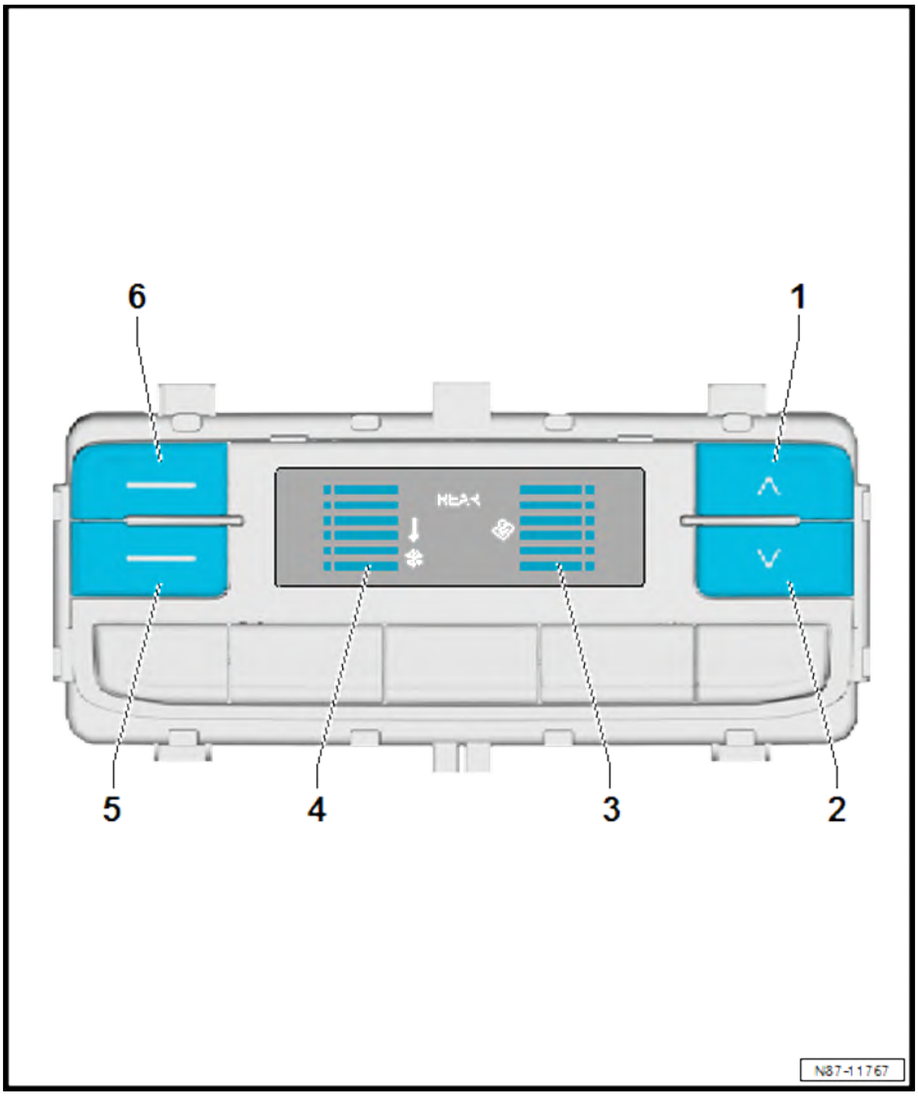
4 - Temperature range display

5 - Button for setting temperature range

- ❑ Reduce temperature range

6 - Button for setting temperature range

- ❑ Increase temperature range



8.2 Removing and installing operating and display unit

⇒ “8.2.1 Removing and installing operating and display unit for front air conditioning system E87 ”, page 196

⇒ “8.2.2 Removing and installing operating and display unit for rear air conditioning system E857 ”, page 198

8.2.1 Removing and installing operating and display unit for front air conditioning system - E87-

Special tools and workshop equipment required

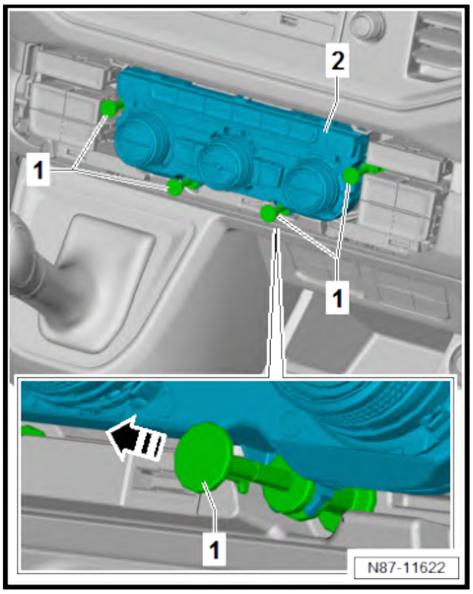
♦ Vehicle diagnostic tester

Removing

– Remove trim for operating and display unit for front air conditioning system - E87- ⇒ General body repairs, interior; Rep. gr. 68 ; Centre console; Removing and installing trim for operating and display unit for front air conditioning system .

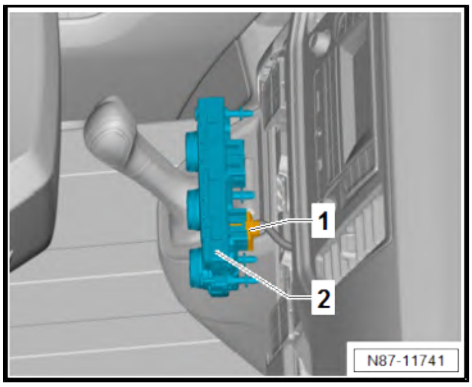
– Pull spreader rivets -1- in direction of -arrow-.

– Remove front operating and display unit -2- from dash panel as far as possible until electrical connector/s can be separated.



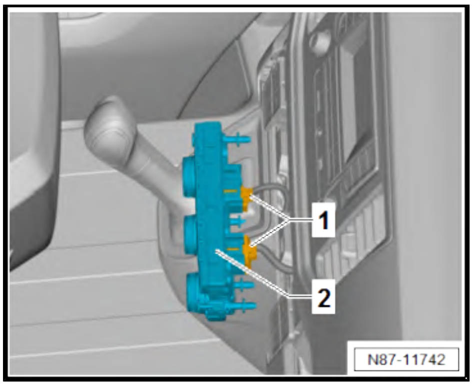
Vehicles with heater

– Disconnect electrical connector -1- of additional display and operating unit 1 - E857- with heater control unit - J65- -2-.

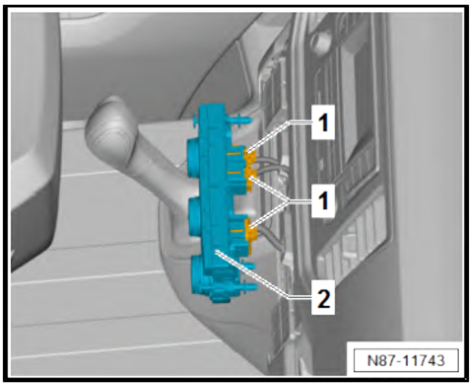


Vehicles with Climatic

– Disconnect electrical connectors -1- of additional display and operating unit 1 - E857- with air conditioning system control unit - J301- -2-.



Vehicles with Climatronic



– Disconnect electrical connectors -1- of additional display and operating unit 1 - E857- with Climatronic control unit - J255- -2-.

Continued for all vehicles

– Remove front operating and display unit -2-.

Installing

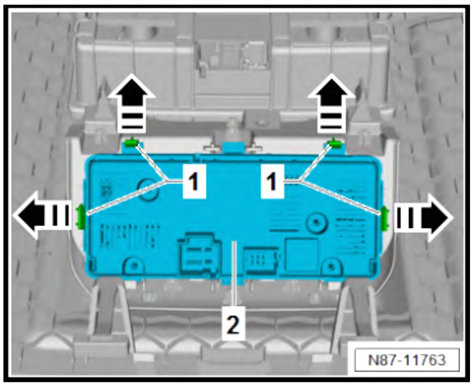
Install in reverse order of removal, observing the following:

Procedure for replacement of front operating and display unit

– Perform adaptation using ⇒ Vehicle diagnostic tester.

8.2.2 Removing and installing operating and display unit for rear air conditioning system - E857-

Removing



– Remove storage compartment trim ⇒ General body repairs, interior; Rep. gr. 68 ; Storage compartments/covers; Removing and installing headliner storage compartment; Removing and installing storage compartment trim .

– Press locking lugs -1- in direction of -arrows-. Remove operating and display unit for rear air conditioning system - E857- -2- when doing this.

Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs -1- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

9 Other controlling and regulating components

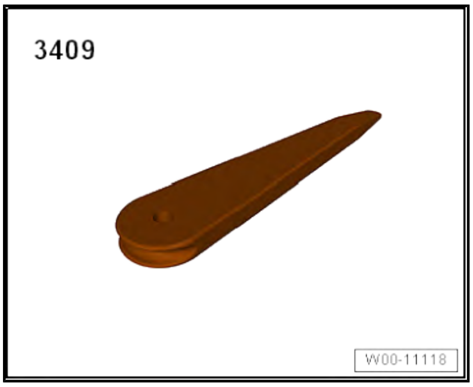
9.1 Removing and installing sunlight penetration photosensor - G107-

Note

Only on vehicles with Climatronic.

Special tools and workshop equipment required

- Removal wedge - 3409-

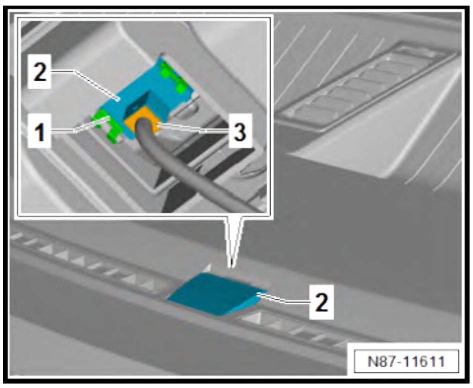


Removing

– Release locking lugs -1- by lifting sunlight penetration photosensor - G107- -2- with removal wedge - 3409- .

– Remove sunlight penetration photosensor - G107- -2- upwards out of dash panel.

– Separate electrical connector -3-.



Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs -1- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

9.2 Principle of operation of air quality sensor - G238-

- The air quality sensor detects noxious substances in the ambient air (primarily petrol and/or diesel fumes).
- The Climatronic control unit - J255- evaluates the signal from the air quality sensor - G238- . The air conditioning system is actuated depending on the degree and manner of ambient air pollution.
- At an ambient temperature higher than approx. +2°C, air recirculation is actuated even when there is a minimal increase in noxious substances in the ambient air.
- At an ambient temperature between approx. +2°C and approx. -5°C, air recirculation is not switched on until there is a strong increase in noxious substances in the ambient air. The air conditioner compressor is also switched on at the same time.
- At an ambient temperature less than approx. -5°C, air recirculation is not actuated until there is a strong increase in noxious substances in the ambient air, but only for approx. 15 seconds (the air conditioner compressor is not actuated). When the concentration decreases, the air conditioning system is switched back to fresh air mode.
- The automatic recirculation mode can be switched off at any time. If the function is active, the air conditioner compressor will be switched on even at an ambient temperature below +2° C when automatic recirculation mode is required. However, even with this function, it is not possible to operate the air conditioning compressor at temperatures below -5°C.
- On vehicles with automatic recirculation mode, the air conditioner compressor can also be switched on at temperatures down to approx. –5 °C even if the air recirculation mode has been activated manually.
- So that the air conditioning system does not operate continually in air recirculation mode in areas with consistently high levels of pollutants, the "intelligent" sensor adapts its sensitivity to the prevailing environmental pollution.
- If the level of noxious substances in the ambient air remains relatively high over a long period of time, the intelligent sensor starts to adapt to the change in environmental conditions so that, generally, the demand for recirculated air lasts less than 12 minutes in areas where the ambient air exhibits a constant level of pollution. If a series of pollution peaks occur, the air conditioning system may operate over a longer period of time in air recirculation mode.
- A certain amount of time is required for repositioning of the air conditioning system flaps. To prevent noxious substances from entering the passenger compartment while the flaps are closing (e.g. when driving through a cloud of diesel smoke), a dust and pollen filter with an activated charcoal layer is installed. A saturated filter cannot perform this task and should be renewed.
- To prevent frequent operation of the recirculation/fresh air flap, the flap is not actuated immediately if there is a nominal increase in harmful substances in the outside air. The sensor does not send a request to the Climatronic control unit - J255- . The effect of the activated charcoal filter in the dust and pollen filter is adequate for this.
- To prevent the air recirculation/fresh air flap from switching too frequently, the sensor's request for automatic recirculation mode continues for at least 25 seconds (minimum waiting pe-

riod) even if the air pollutants decrease to a level that no longer requires air recirculation.

- If the air conditioner compressor is switched off (A/C button is deactivated), the maximum duration for automatic recirculation is limited to approx. 15 seconds by the Climatronic control unit - J255- so that condensation does not develop on the windows.
- To clear condensation from windows as quickly as possible, the Climatronic control unit - J255- does not permit air recirculation when the defrost function is activated.
- The air quality sensor - G238- requires approx. 30 seconds to become operational once the ignition has been switched on (warm-up period). During this time, the sensor can send no request for automatic recirculation mode to the Climatronic control unit - J255- .
- The air quality sensor - G238- is a highly sensitive electronic component which direct contact with solvents, fuels and certain chemical compositions could damage beyond repair. For this reason, do not install sensors that may have come into contact with these substances.

9.3 Removing and installing air quality sensor - G238-

Note

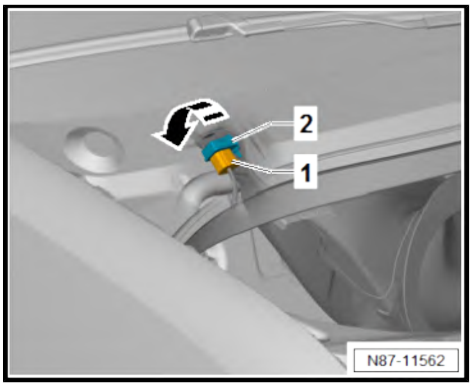
- *The air quality sensor - G238- is a highly sensitive electronic component which could be destroyed by direct contact with solvents, fuels and certain chemical compounds.*
- *Only installed up to week 45/2017.*

Removing

– Separate connector -1-.

– Turn air quality sensor - G238- -2- 90° in direction of -arrow-.

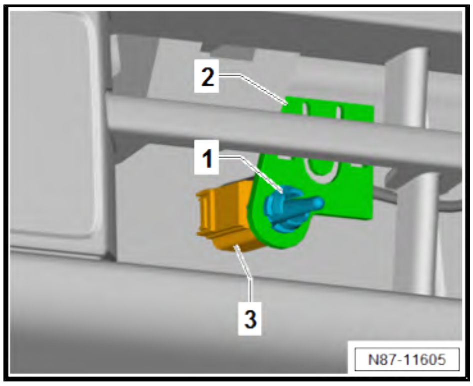
– Remove air quality sensor - G238- -2- from fresh air intake duct of bulkhead.



Installing

Install in reverse order of removal.

9.4 Removing and installing ambient temperature sensor - G17-



Removing

– Unclip ambient temperature sensor - G17- -1- from bracket -2-.

– Separate electrical connector -3-.

Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs on ambient temperature sensor - G17- -1- for damage; renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

9.5 Removing and installing humidity sender for air conditioning system - G260-

The humidity sender for air conditioning system - G260- and the light and rain sensor - G397- form one component and are installed depending on vehicle equipment.

Removal and installation procedure is described in ⇒ Electrical system; Rep. gr. 92 ; Windscreen wiper system; Removing and installing rain and light sensor .

9.6 Removing and installing footwell vent temperature sender - G192-

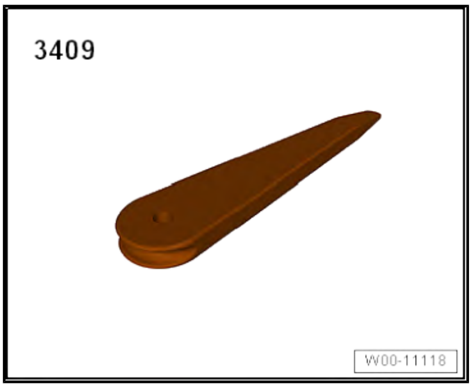
Note

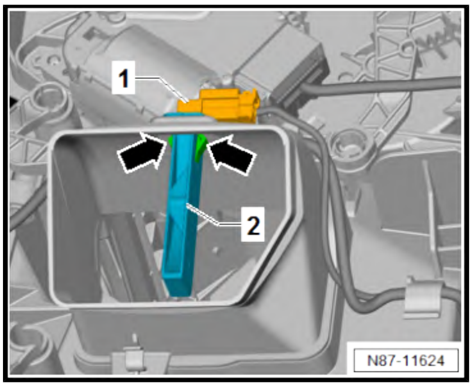
♦ *Only on vehicles with Climatronic.*

♦ *On vehicles with Climatic, apertures in air duct system are sealed with plugs.*

Special tools and workshop equipment required

♦ Removal wedge - 3409-





Removing

– Remove footwell vent on driver side ⇒ page 176 .

– Disconnect connector -1-.

Note

Levering with removal wedge - 3409- causes both retaining tabs -arrows- to be pushed together.

– Lever evaporator temperature sensor - G308- -2- out of evaporator housing using removal wedge - 3409- .

Installing

Install in reverse order of removal, observing the following:

– Check retaining tabs -arrows- for damage, renew component if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

9.7 Removing and installing chest vent temperature sensors at front - G385- / - G386-

⇒ “9.7.1 Removing and installing front left chest vent temperature sensor G385 ”, page 203

⇒ “9.7.2 Removing and installing front right chest vent temperature sensor G386 ”, page 204

9.7.1 Removing and installing front left chest vent temperature sensor - G385-

Note

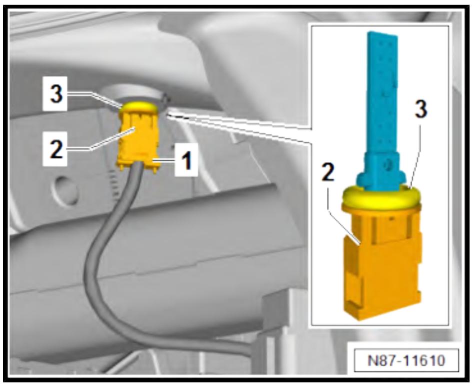
- *Only on vehicles with Climatronic.*
- *On vehicles with Climatic, apertures in air duct system are sealed with plugs.*

Removing

– Remove cover for central electrics ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing cover for central electrics .

– Disconnect connector -1-.

– Turn front left chest vent temperature sensor - G385- -2- by 90°. While doing this, remove it from air duct.



Installing

Install in reverse order of removal, observing the following:

– Check seal -3- for damage; renew seal if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

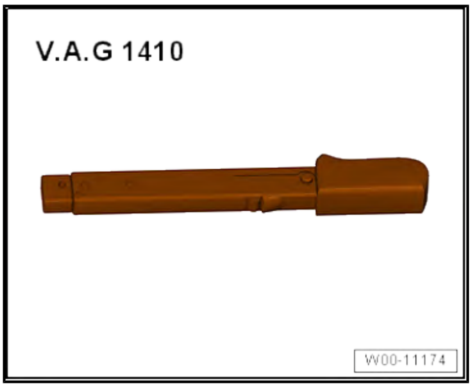
9.7.2 Removing and installing front right chest vent temperature sensor - G386-

Note

- *Only on vehicles with Climatronic.*
- *On vehicles with Climatic, apertures in air duct system are sealed with plugs.*

Special tools and workshop equipment required

- Torque wrench - V.A.G 1410-



Removing

– Remove glove compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments/covers; Removing and installing glove compartment .

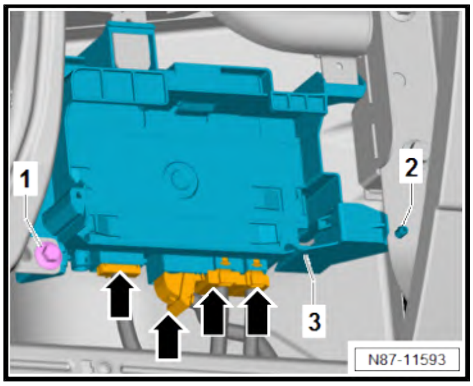
Vehicles with special vehicle control unit - J608-

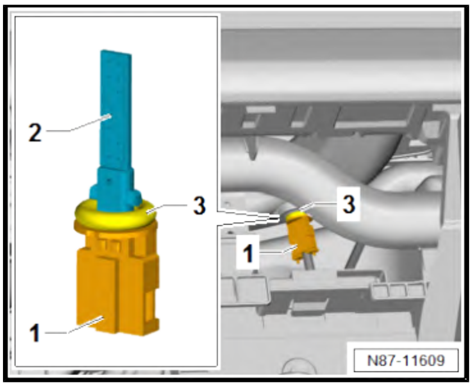
– Disconnect electrical connectors -arrows-.

– Unscrew bolt -1-.

– Press together spreader clip -2-. While doing so, push special vehicle control unit - J608- -3- in direction of engine.

– Remove bracket for special vehicle control unit - J608- -3- in direction of footwell.





Continued for all vehicles

– Disconnect connector -1-.

– Turn front right chest vent temperature sensor - G386- -2- by 90°. While doing this, remove it from air duct.

Installing

Install in reverse order of removal, observing the following:

– Check seal -3- for damage; renew seal if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

Specified torques

♦ Control units; Overview of fitting locations - control units ⇒ Electrical system; Rep. gr. 97 ; Control units; Overview of fitting locations - control units